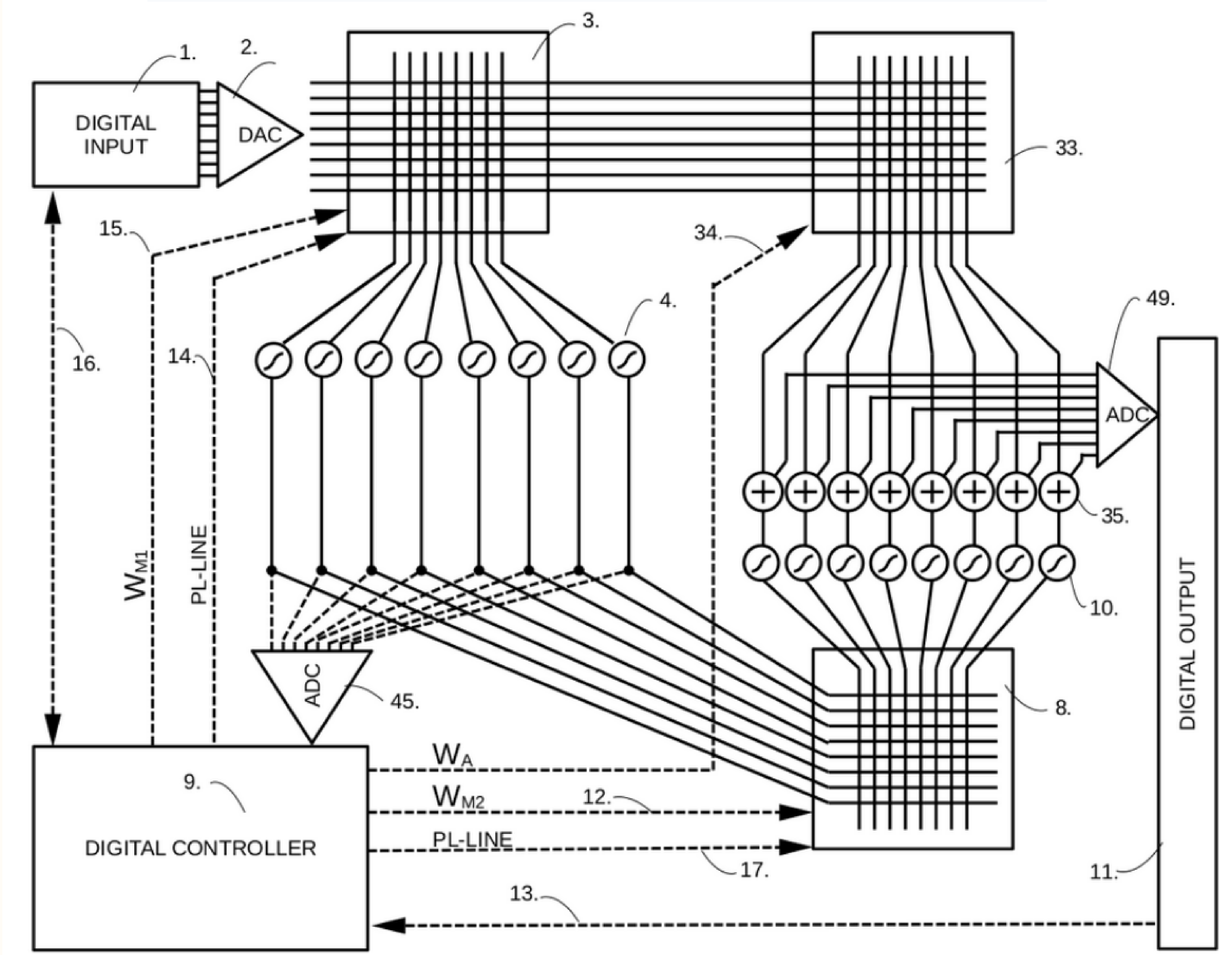
TITLE: INVERTED MAML FOR ANALOG IMC MESH

PATENT APP NO: SE 2630397-4

STATUS: PATENT PENDING

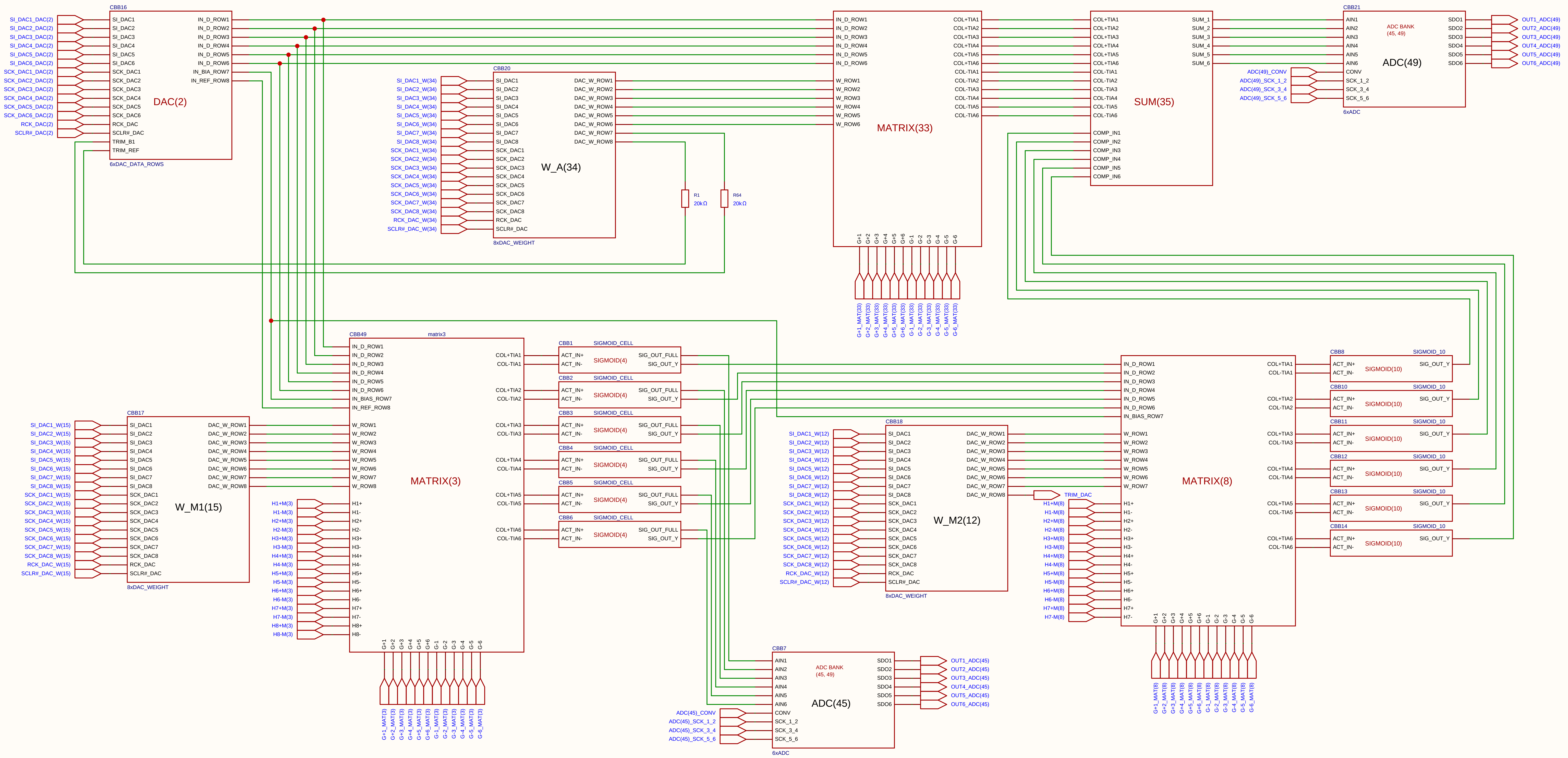
INVENTOR: OLLE WELIN

PRIORITY DATE: 2026-06-03



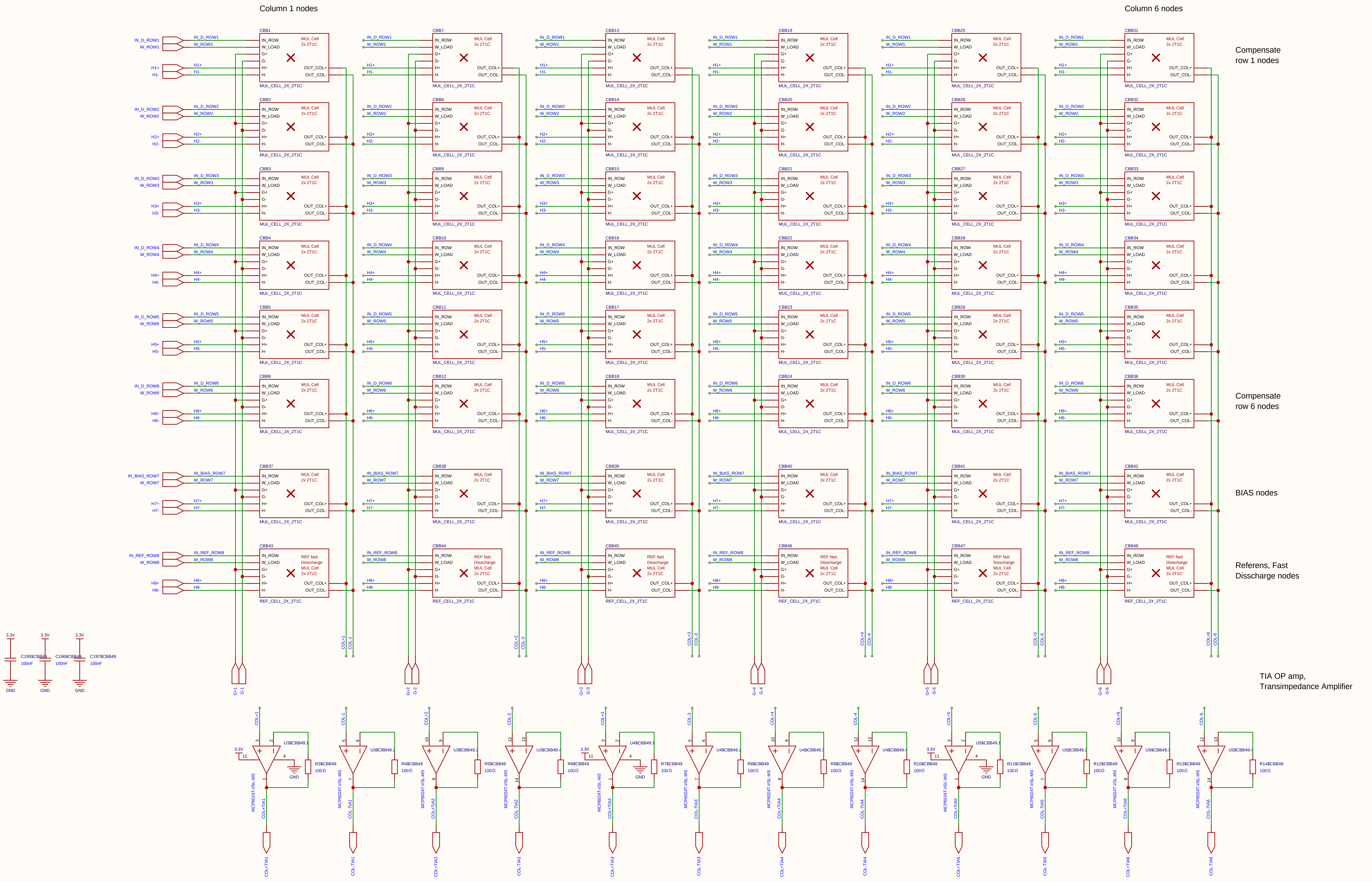
Schematic	Inverted_MAML	Create at	2026-06-19
		Update at	2026-06-19
Board	Analog_Matrix	Page	cover
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
Version	Size	Page 1 Total 2	
EasyEDA	V1.0	A4	EasyEDA.com

The "ATOMIC TRIAD"



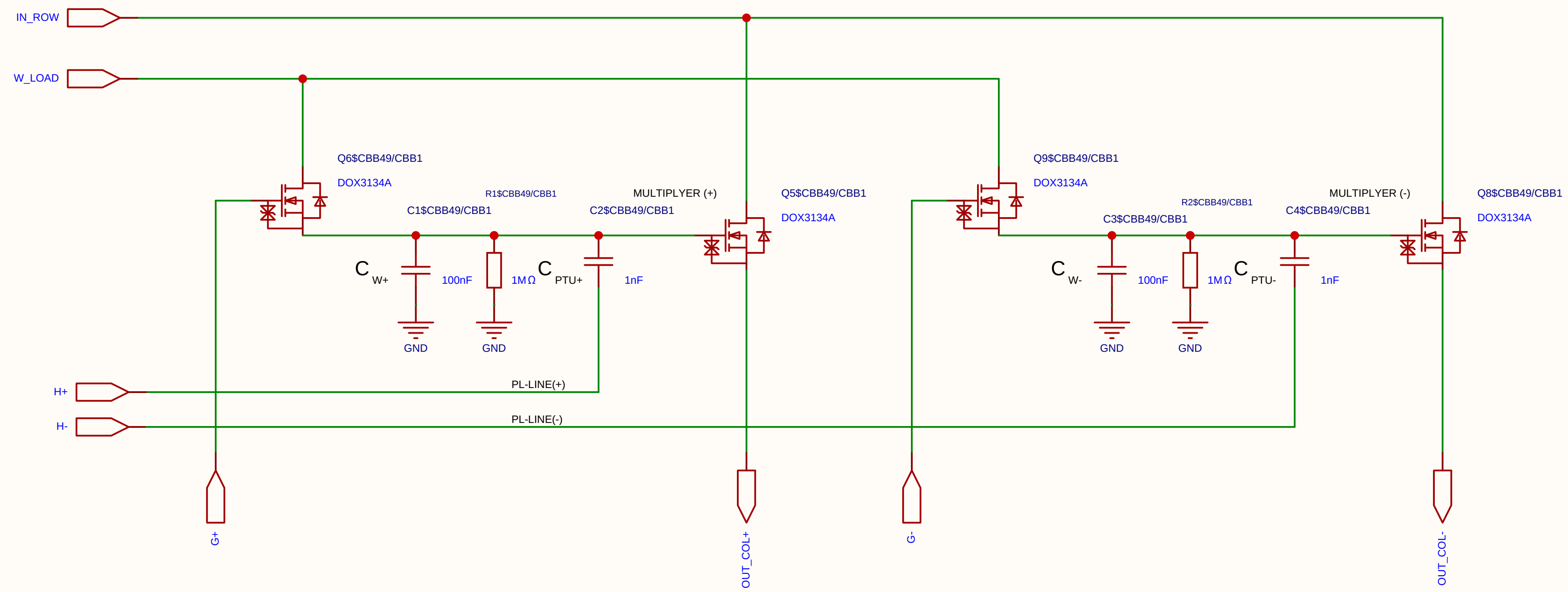
Schematic	Inverted_MAML	Create at	2026-06-19
		Update at	2026-06-20
Board	Analog_Matrix	Page	Atomic_Triad
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 2 Total 2
	V1.0	A4	EasyEDA.com

$$V_{ut\_TIA} = 1.65\ \mathrm{V} - (5\ \mathrm{mA} \times 100\ \Omega) = 1.65\ \mathrm{V} - 0.5\ \mathrm{V} = \mathbf{1.15}\ \mathrm{V}$$



Schematic	matrix3		Create at	2026-06-19
			Update at	2026-06-19
Board			Page	matx3
Drawn		Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

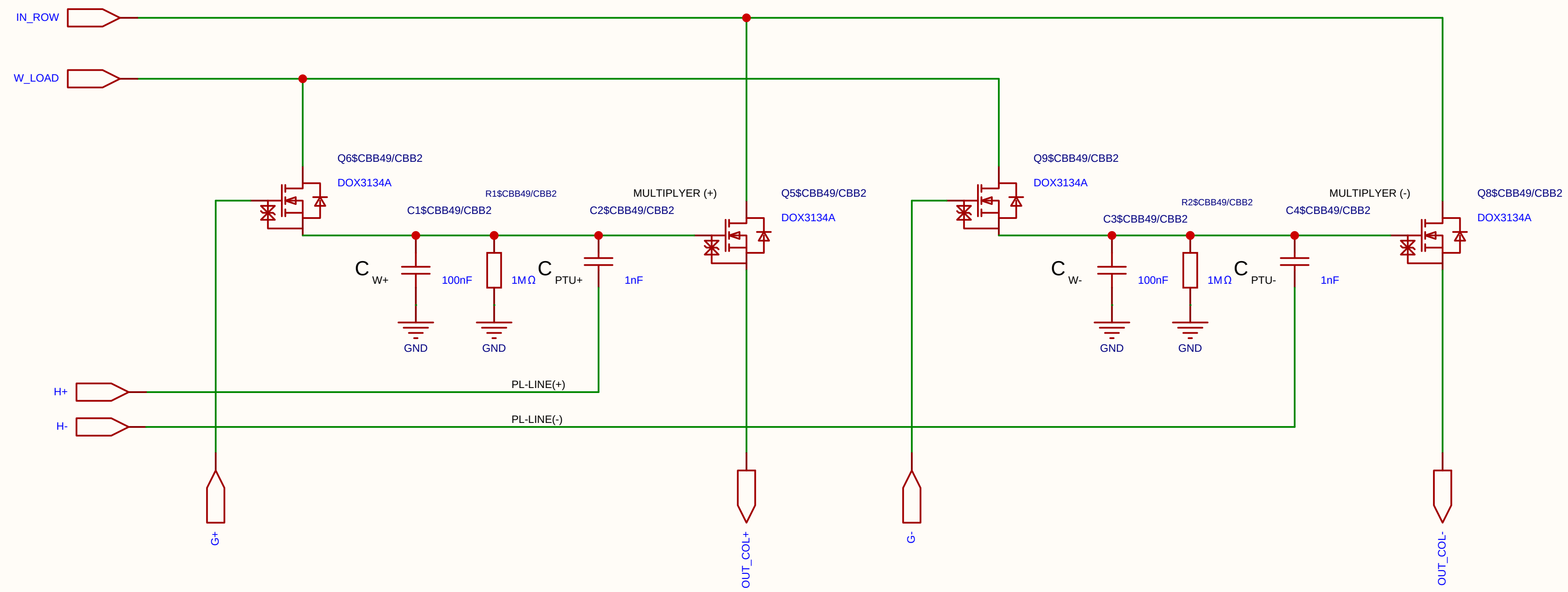
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
Version	Size	Page 1 Total 1	
V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
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Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

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Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4

Page 1 Total 1

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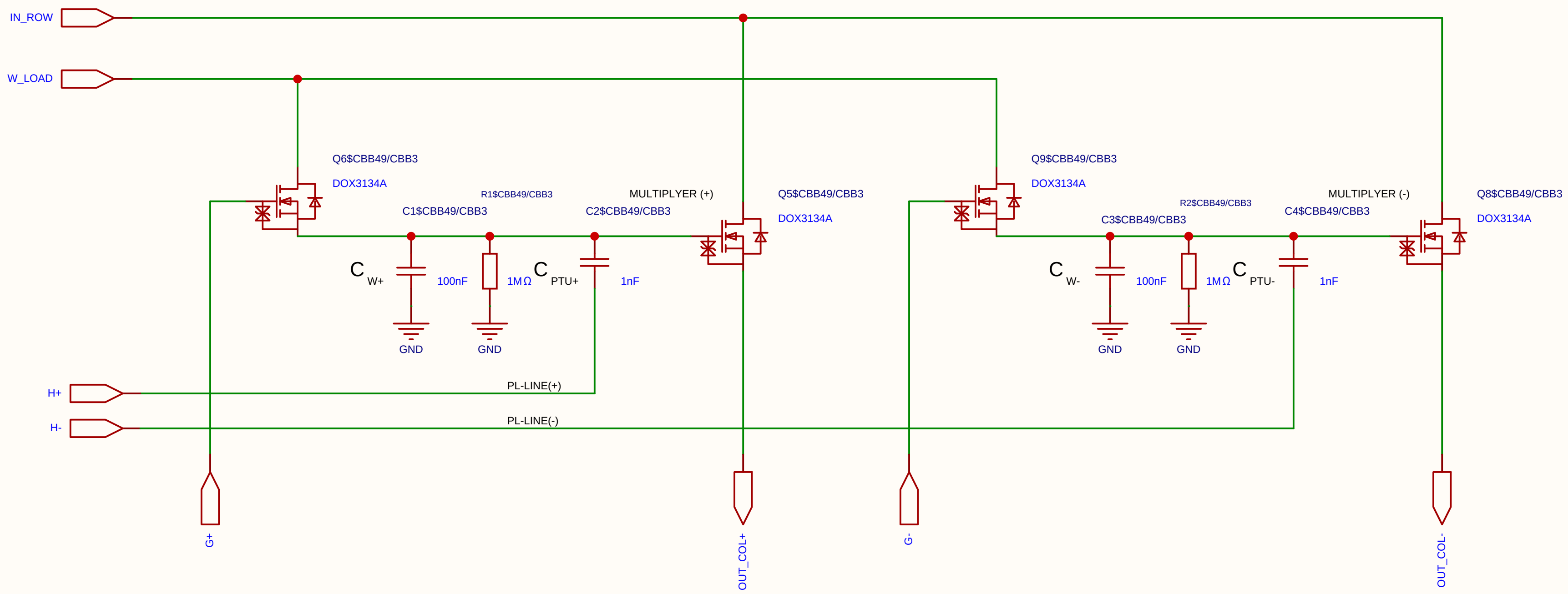
Analog Multiplication CELL Kernel
Differential 2x 2T1C



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Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
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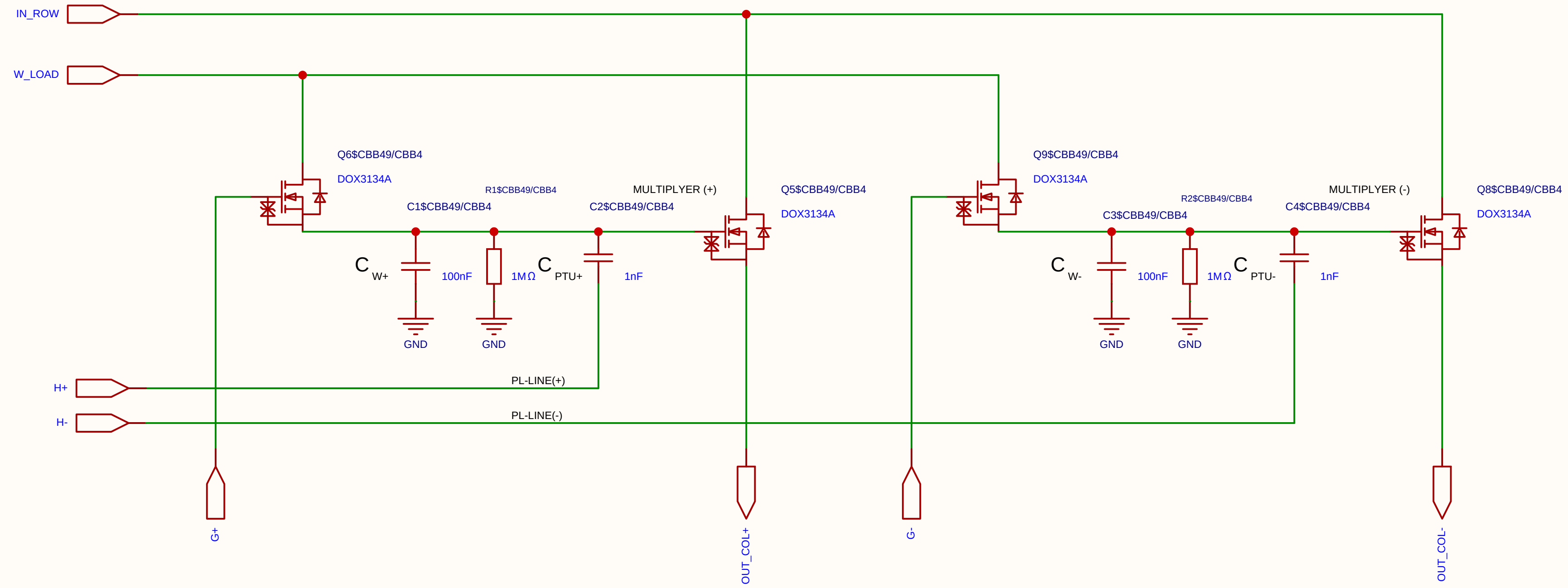
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
Version	Size	Page 1 Total 1	
V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

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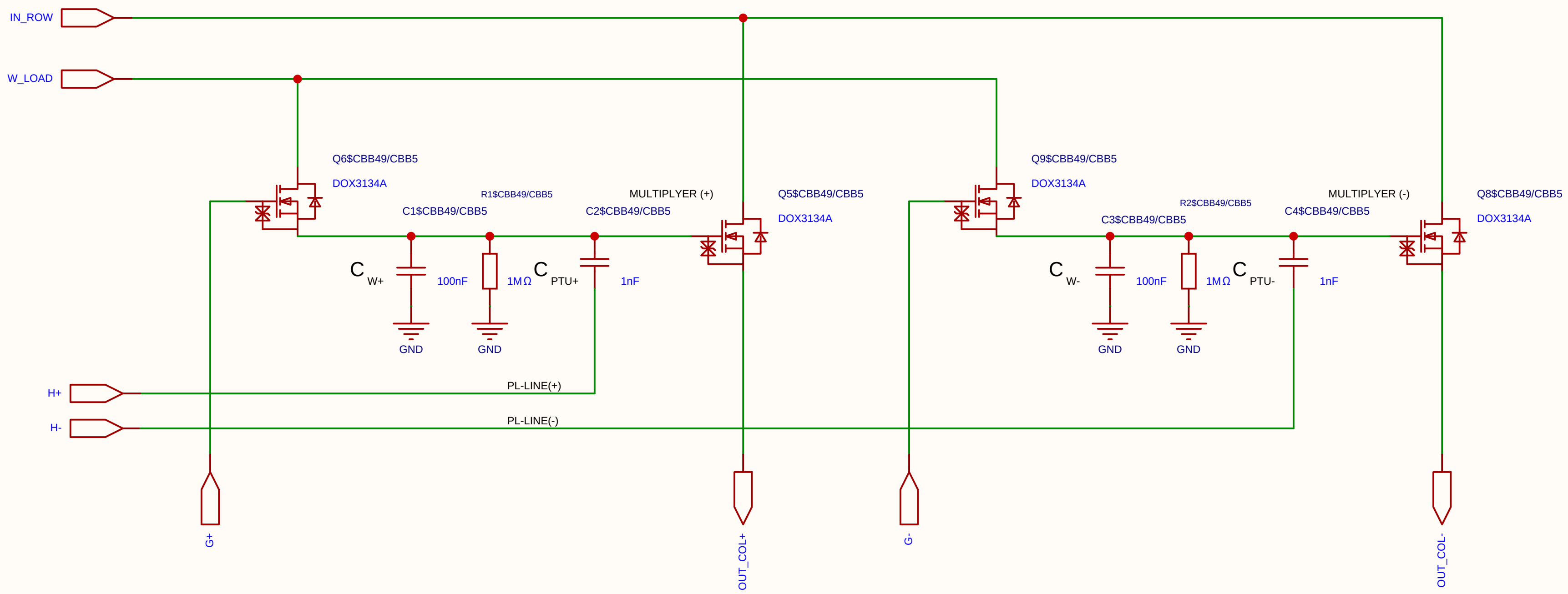
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
Version	Size	Page 1 Total 1	
V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

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4. Voltage Levels on the PL Row

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		Version: V1.0	Size: A4
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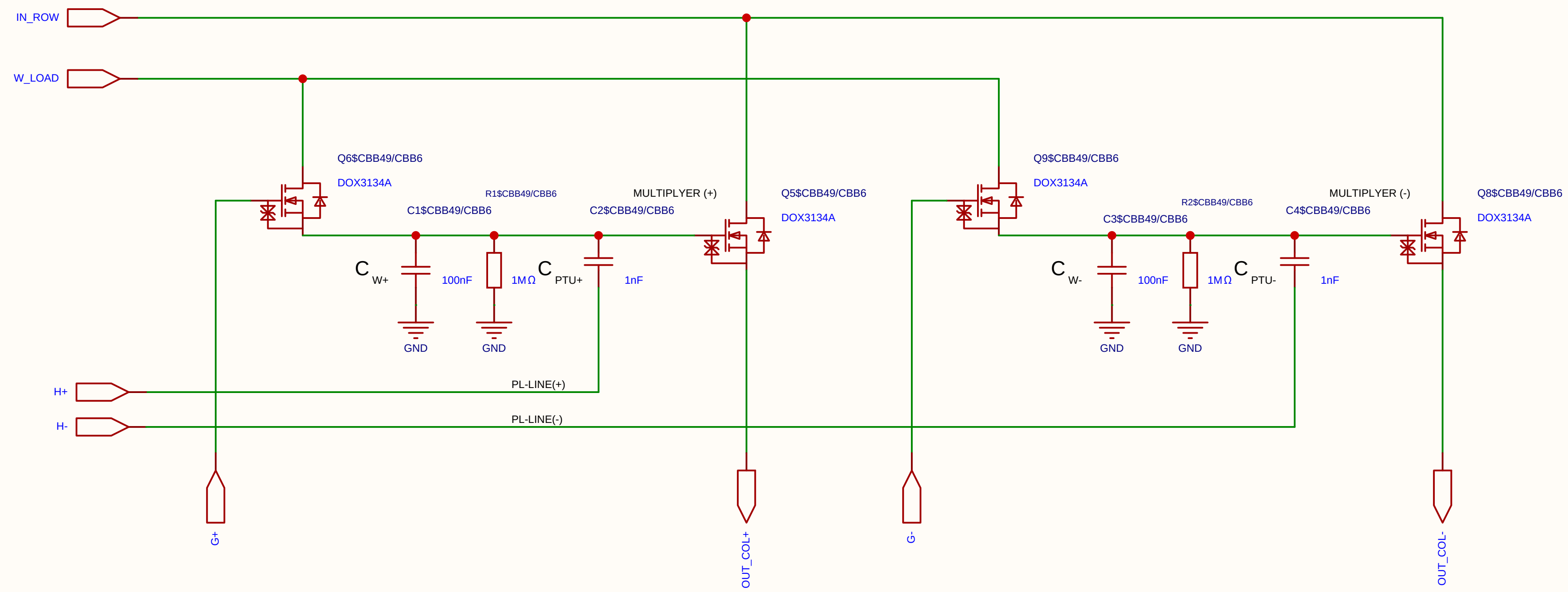
Analog Multiplication CELL Kernel
Differential 2x 2T1C



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	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
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Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

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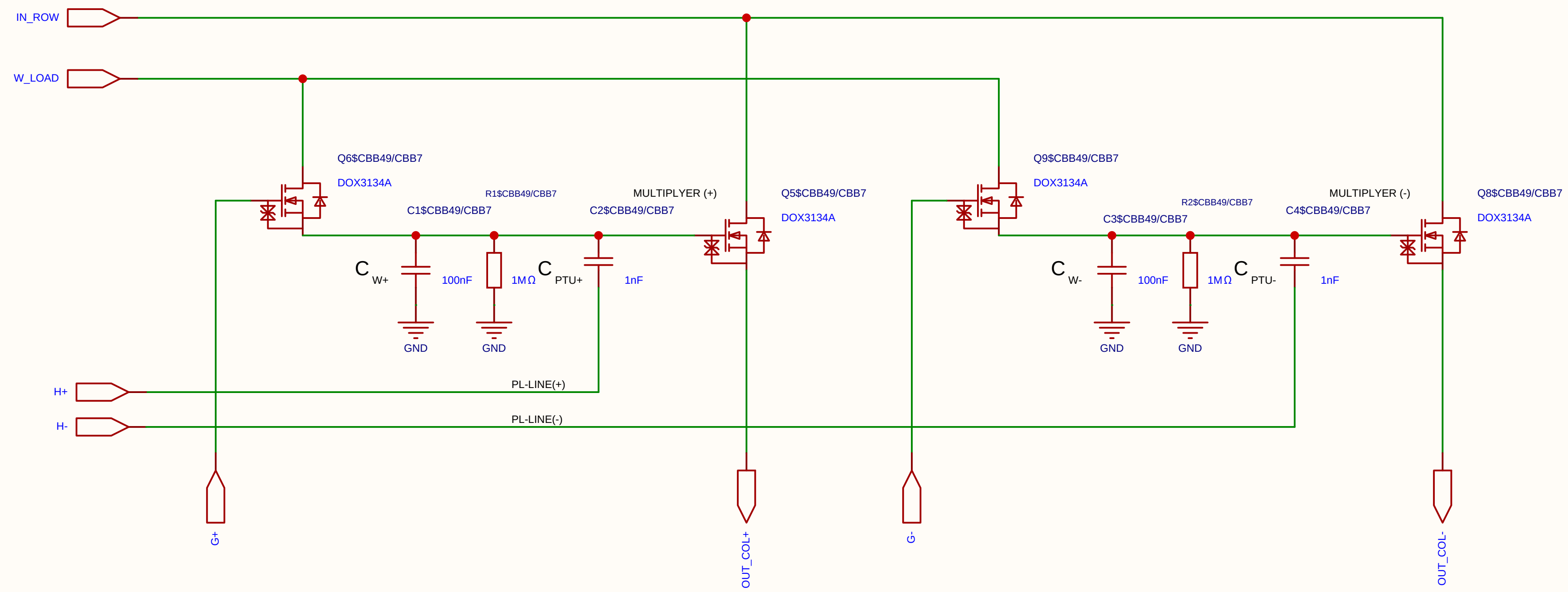

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Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
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This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
		Version: V1.0	Size: A4
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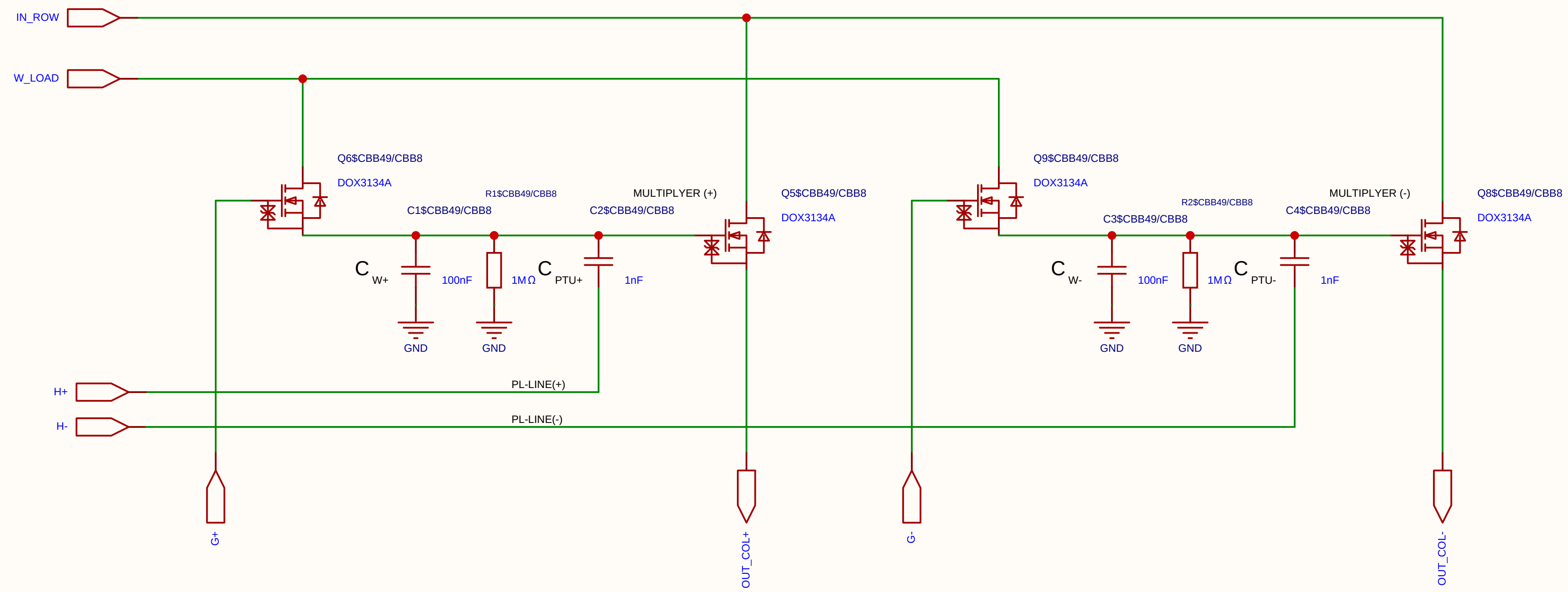
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

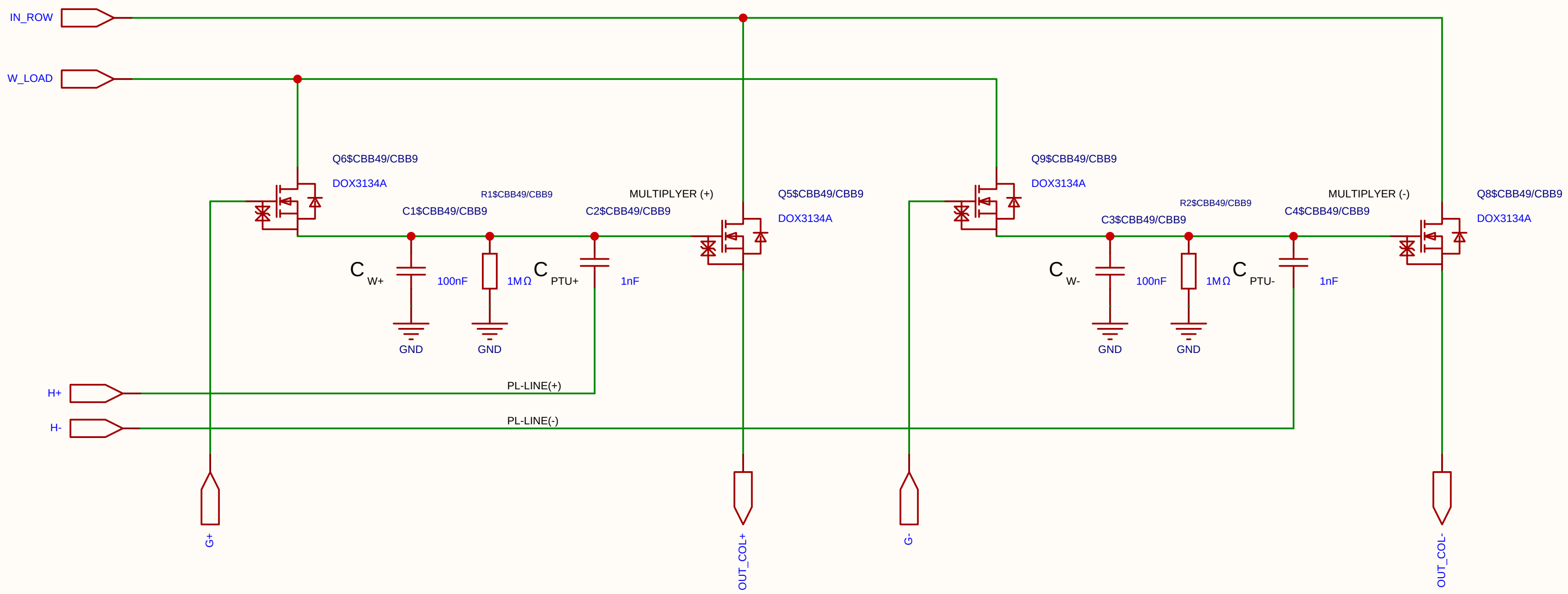
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
EasyEDA	Version: V1.0	Size: A4	Page 1 Total 1 — EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
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	Version	Size	Page 1 Total 1
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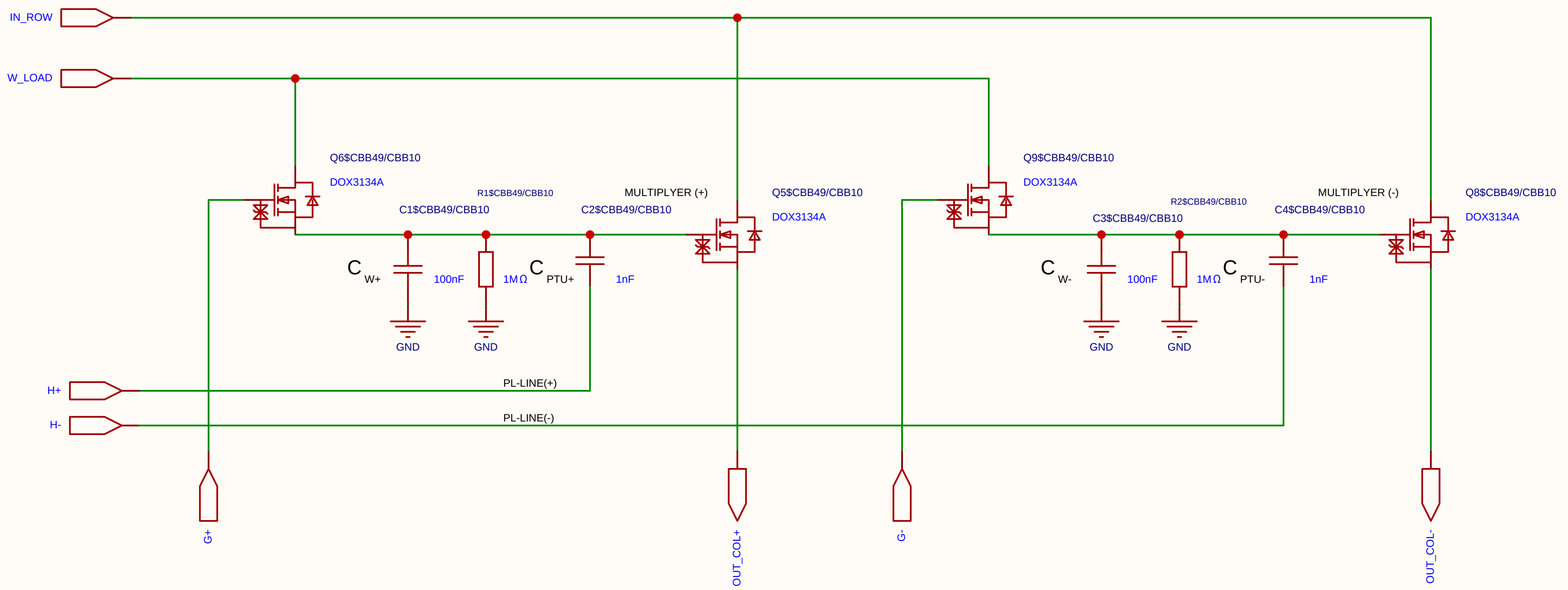
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
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		Version	Size
EasyEDA		V1.0	A4

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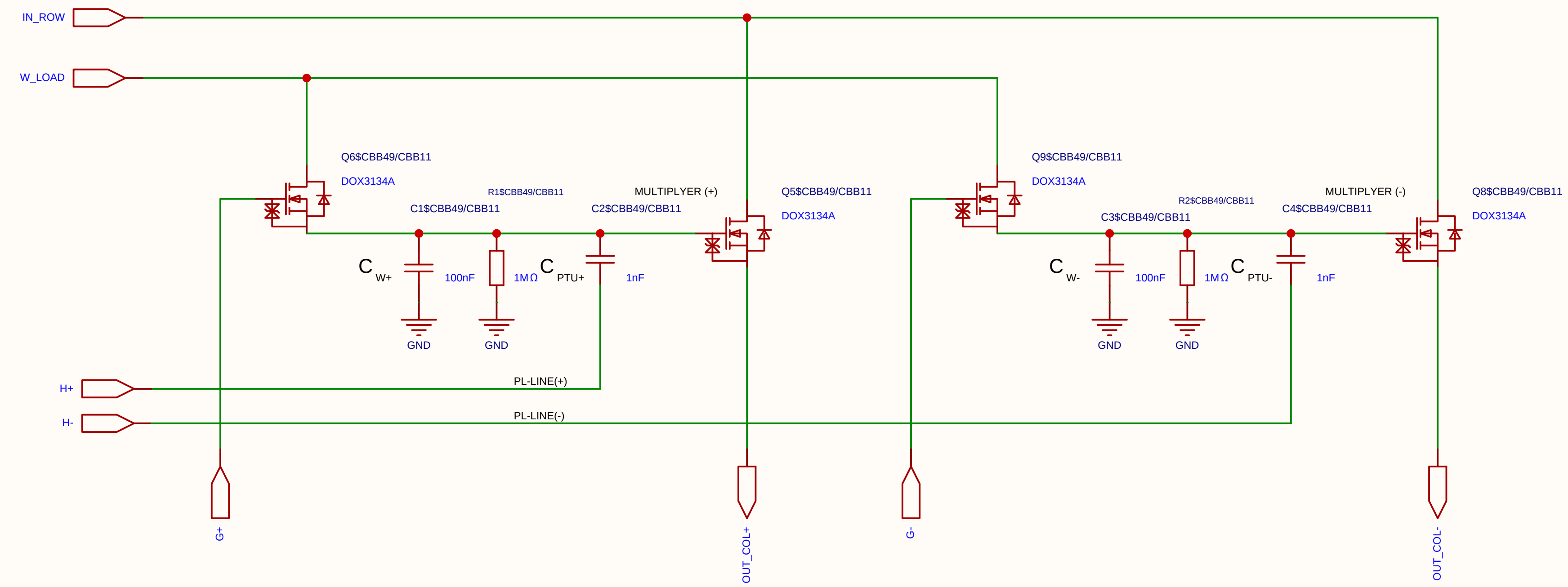
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

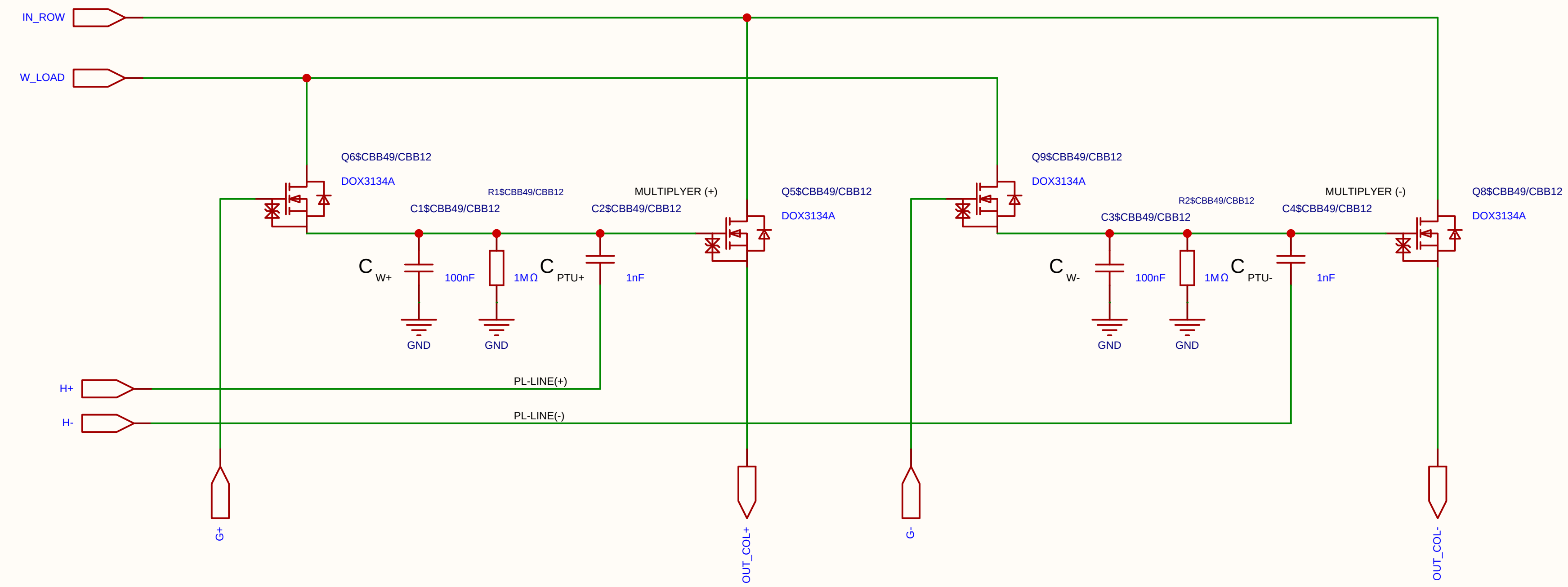
This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4

Version	Size	
V1.0	A4	Page 1 Total 1
		EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
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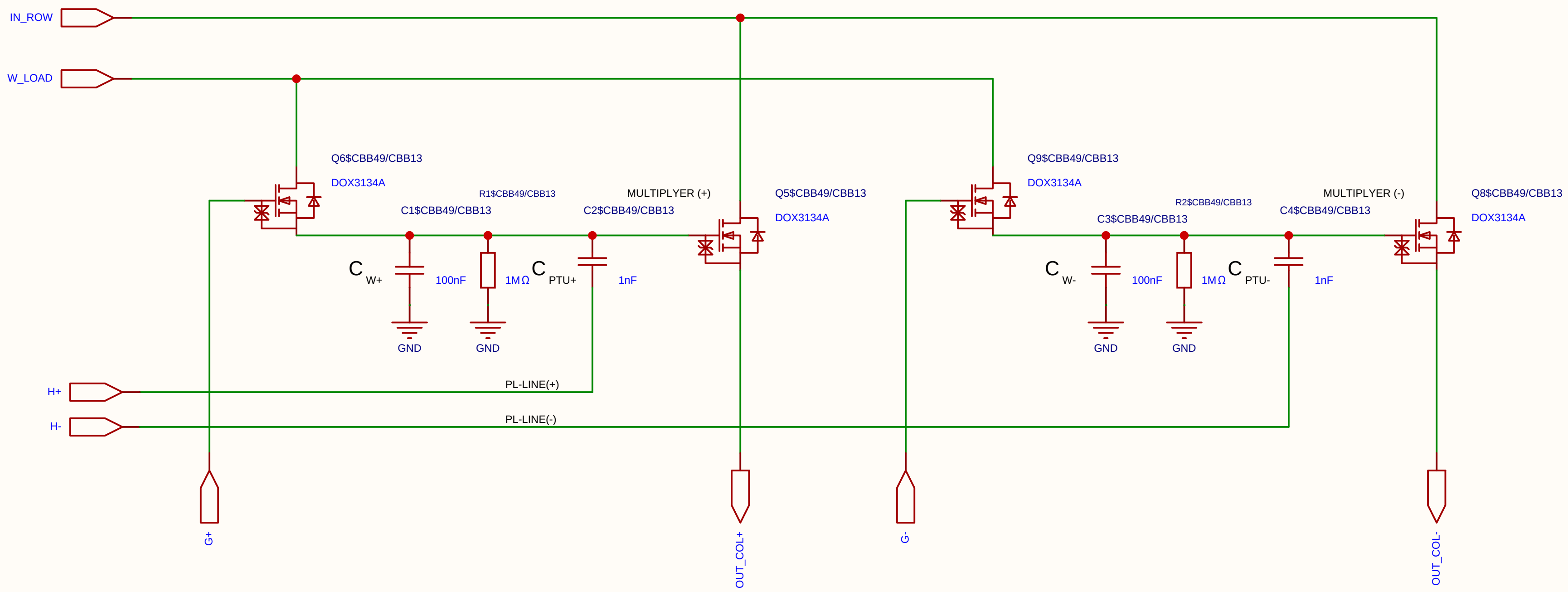
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
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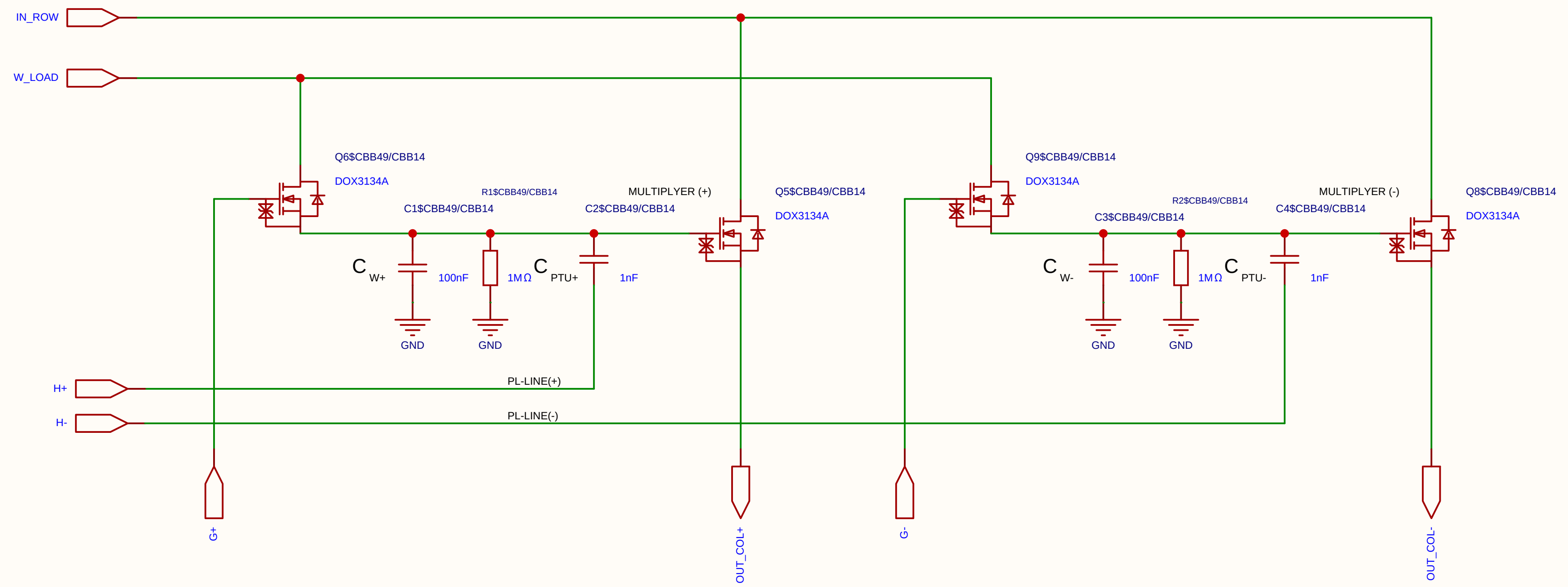
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
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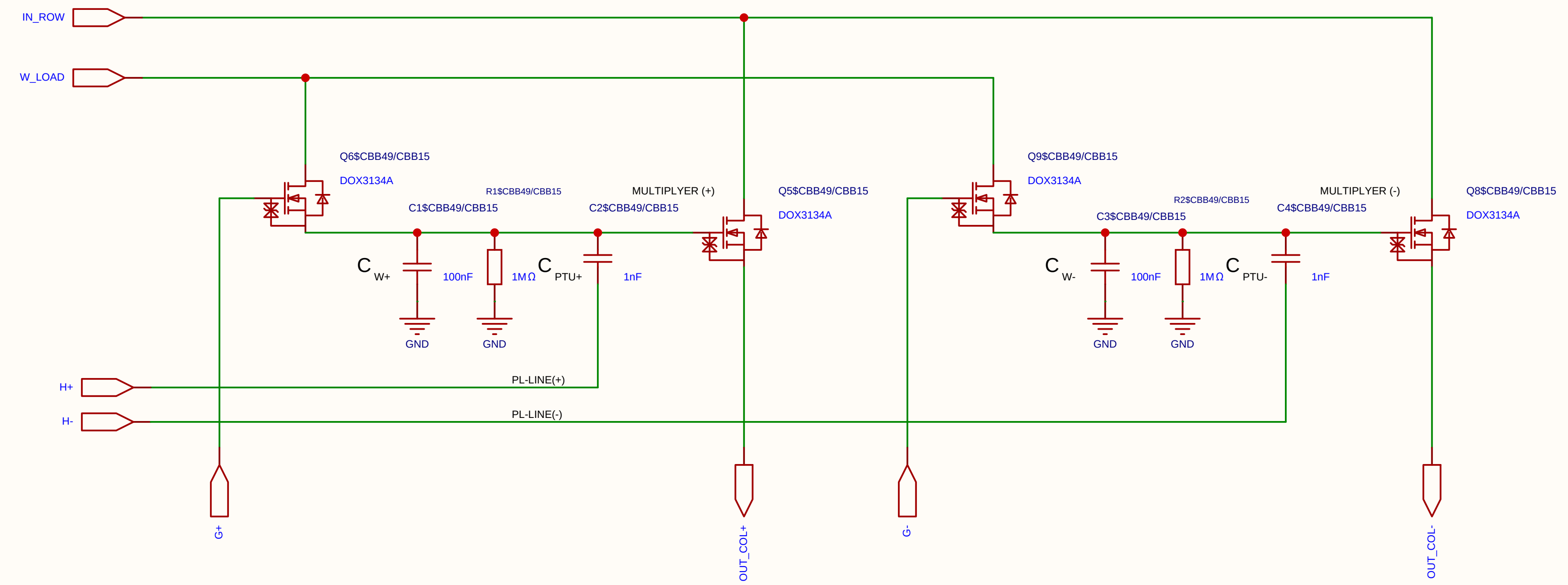
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

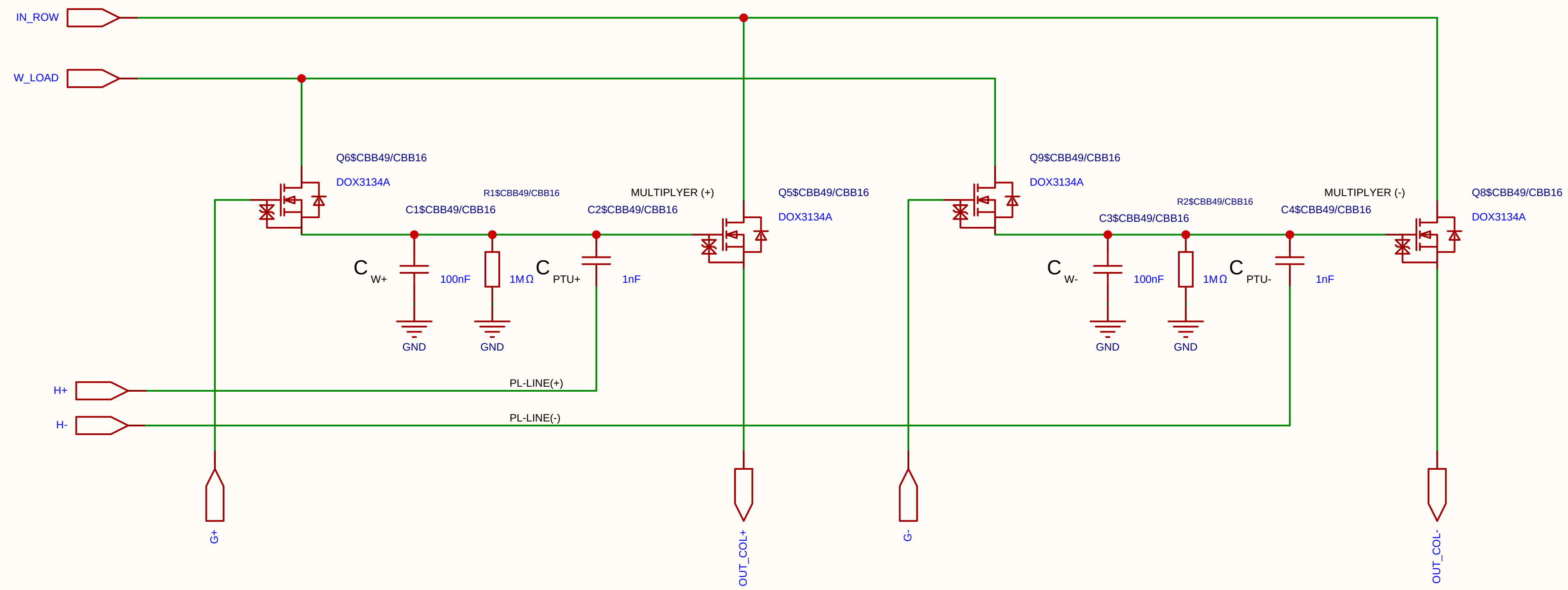
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

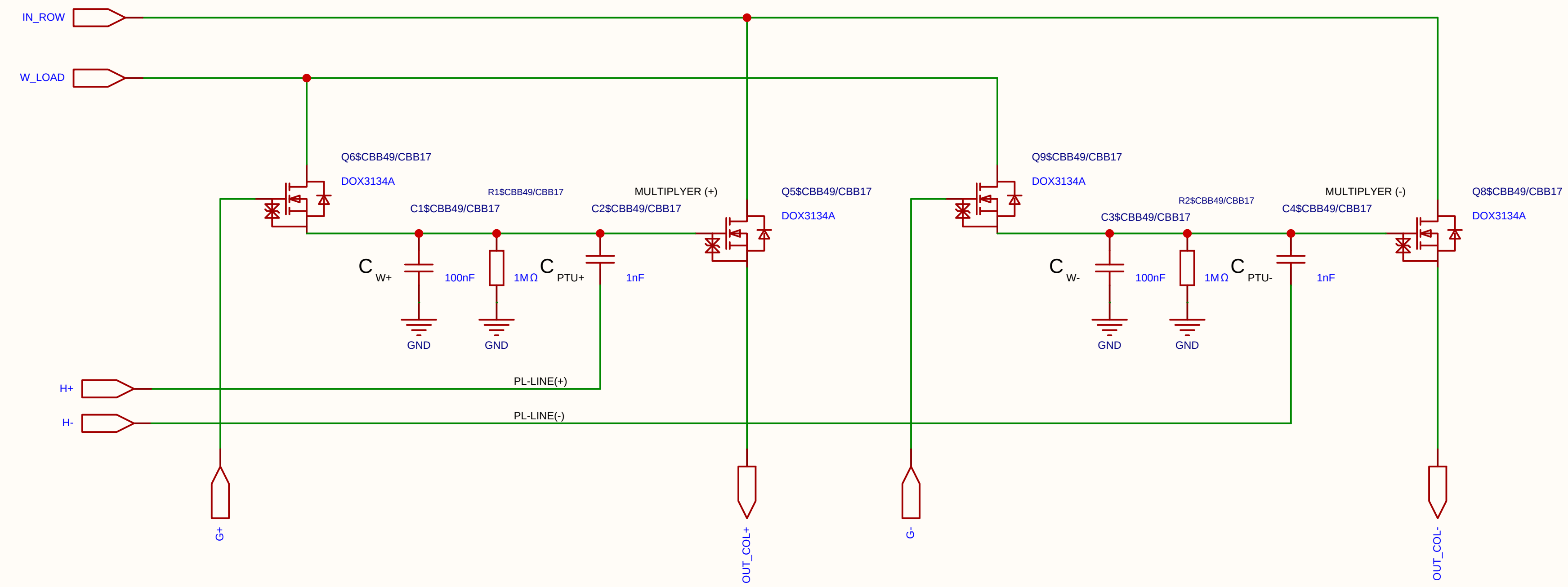
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

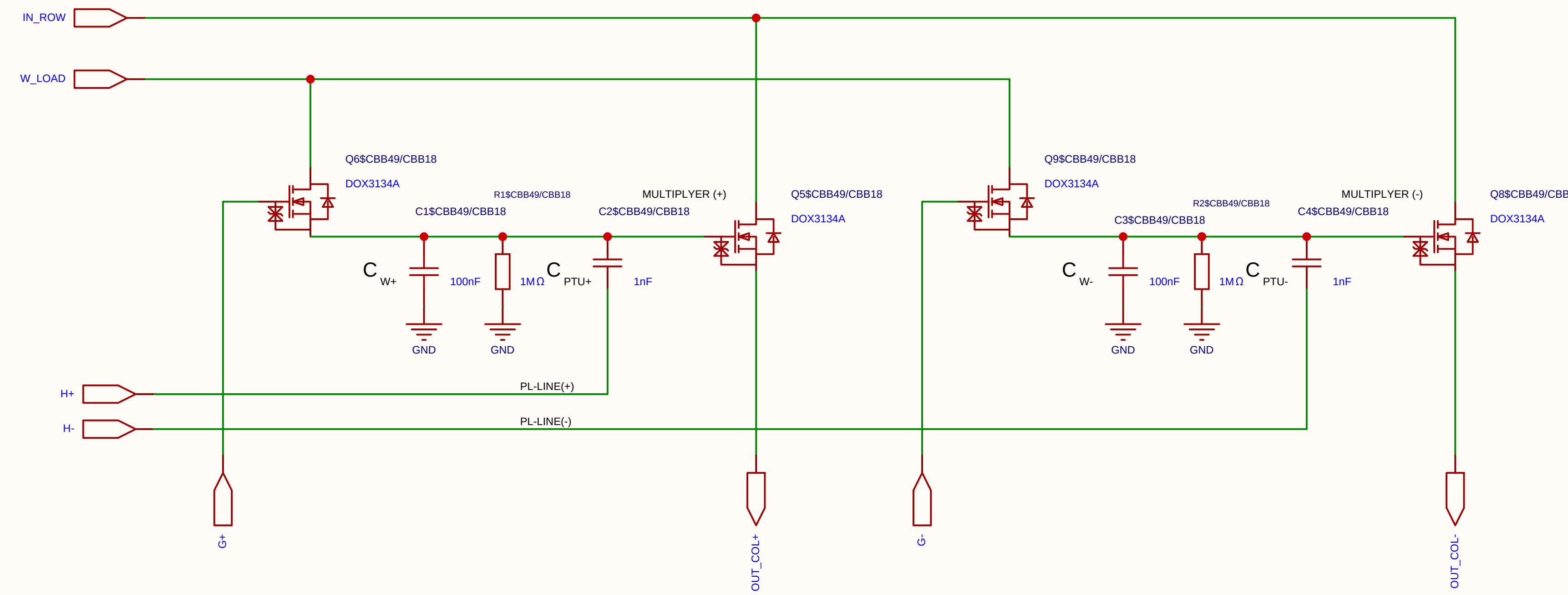
This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4

Version	Size	Page 1 Total 1
V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

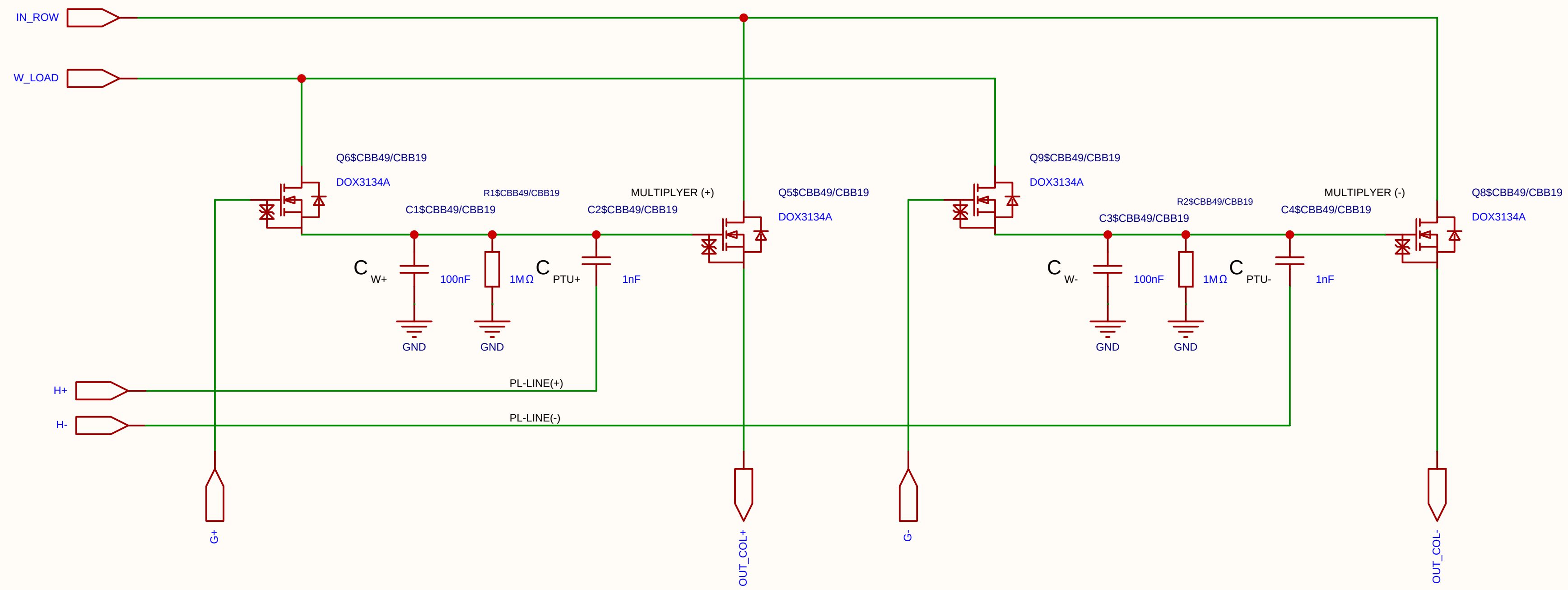
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

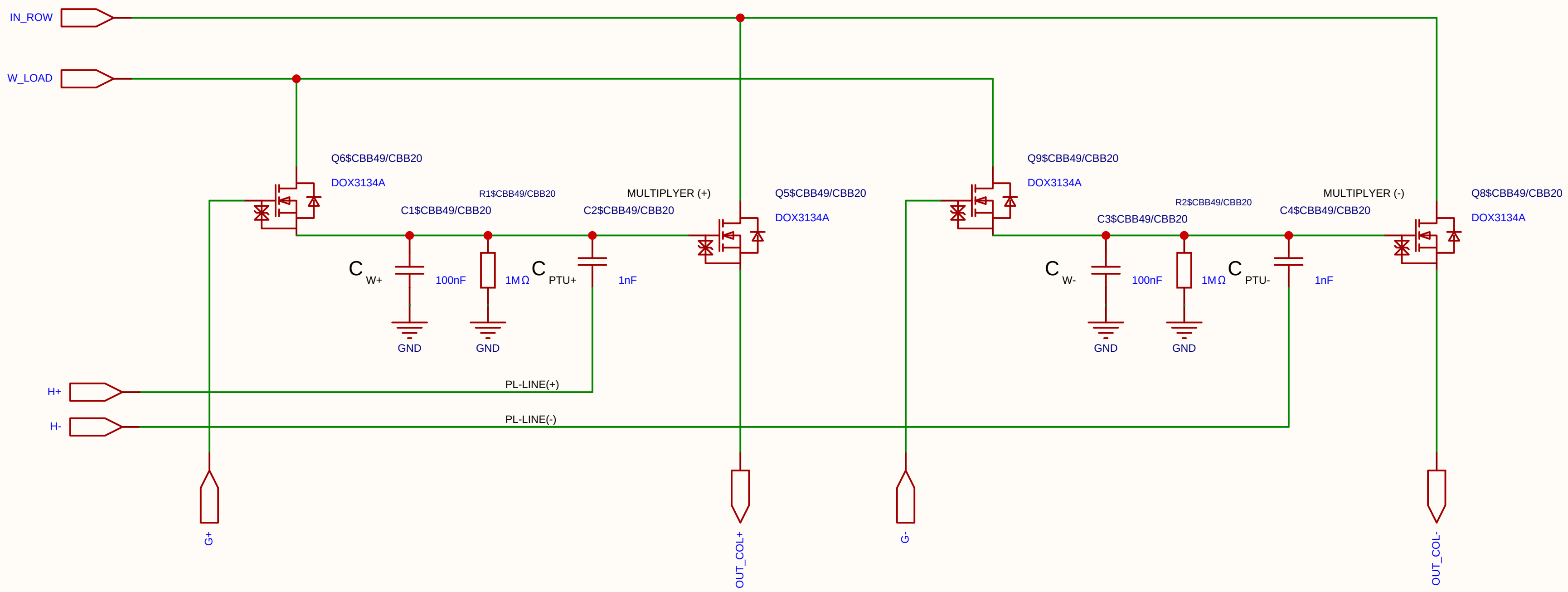
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
EasyEDA	Version: V1.0	Size: A4	Page 1 Total 1
			EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

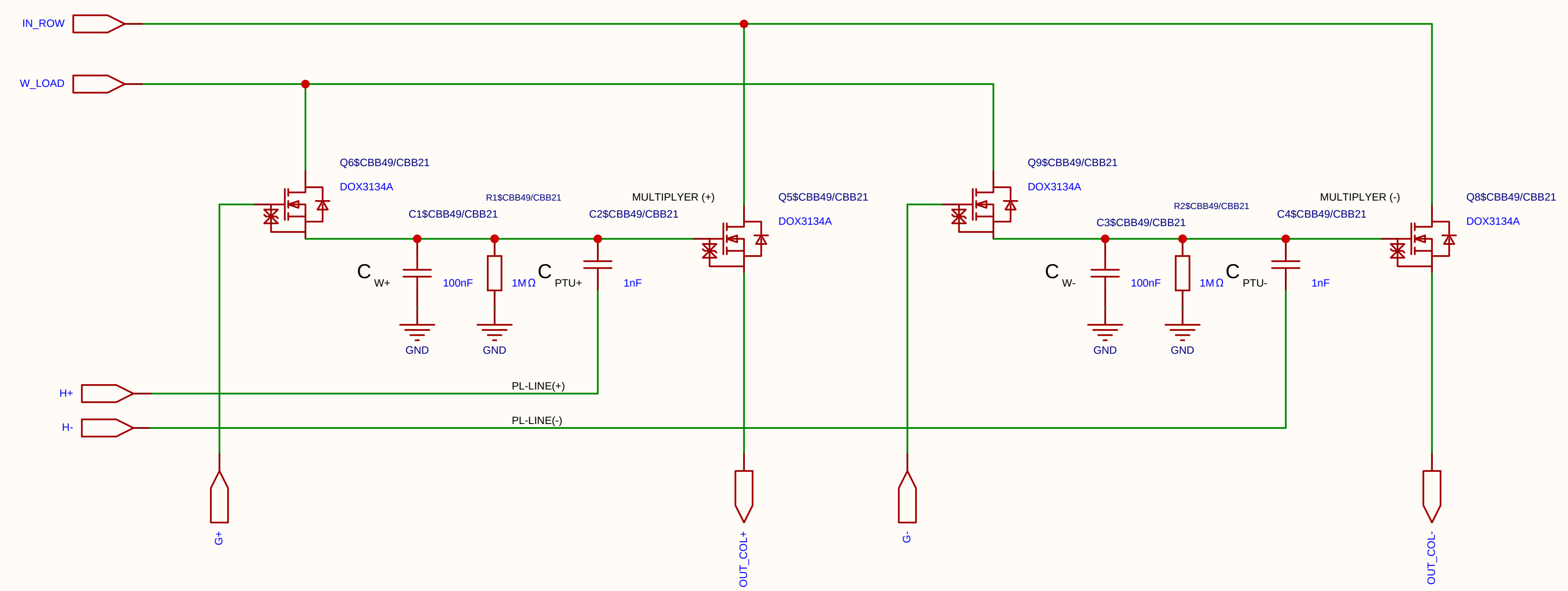
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

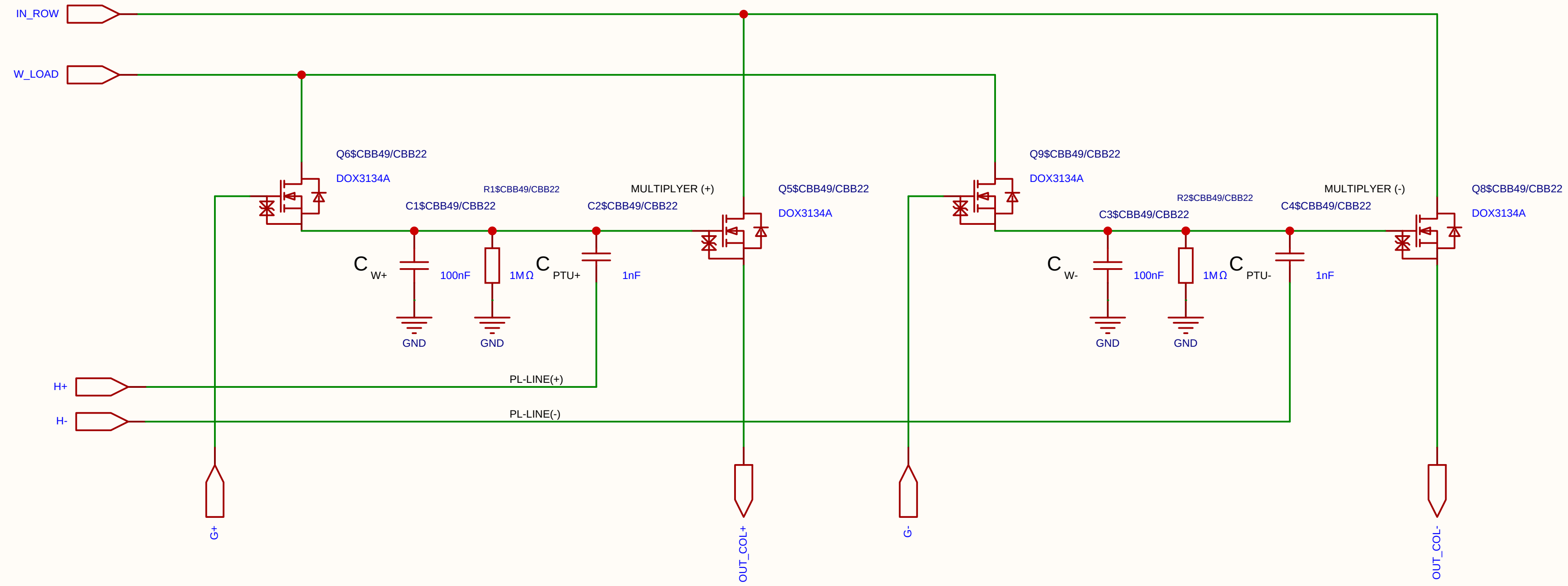
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
		Version	Size
EasyEDA		V1.0	A4
		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

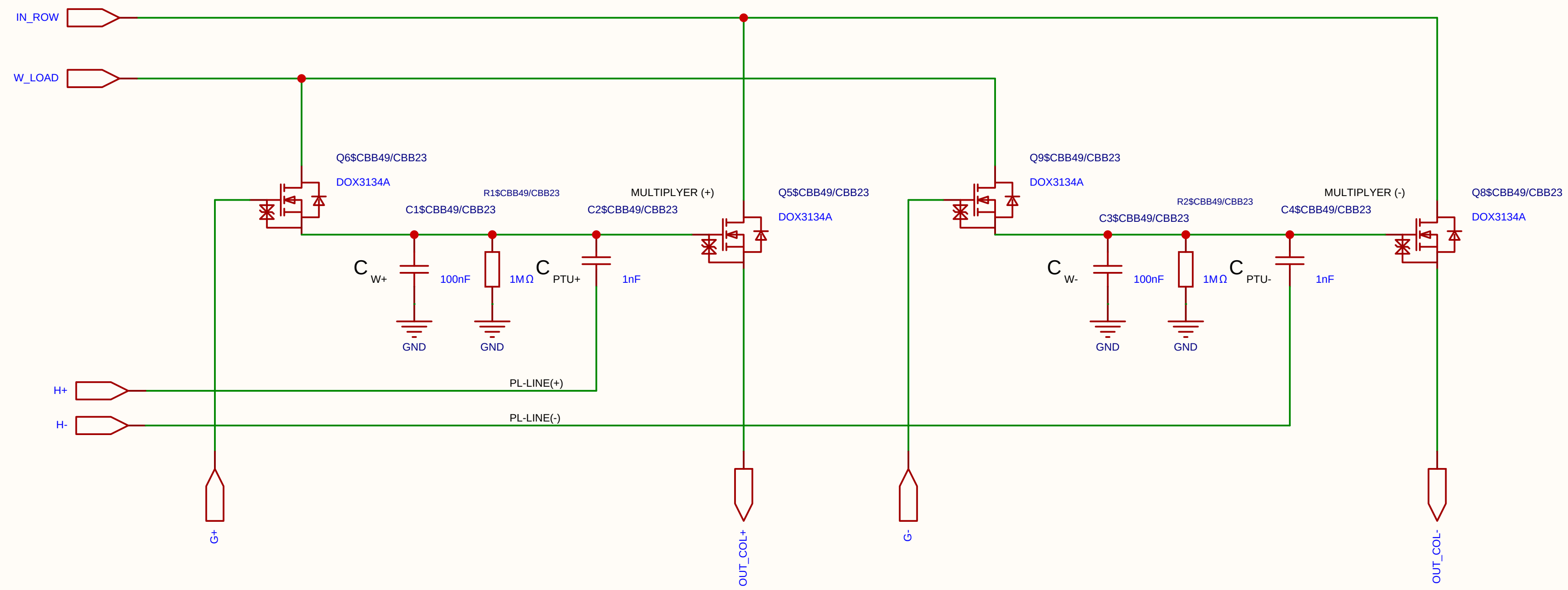
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4
		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

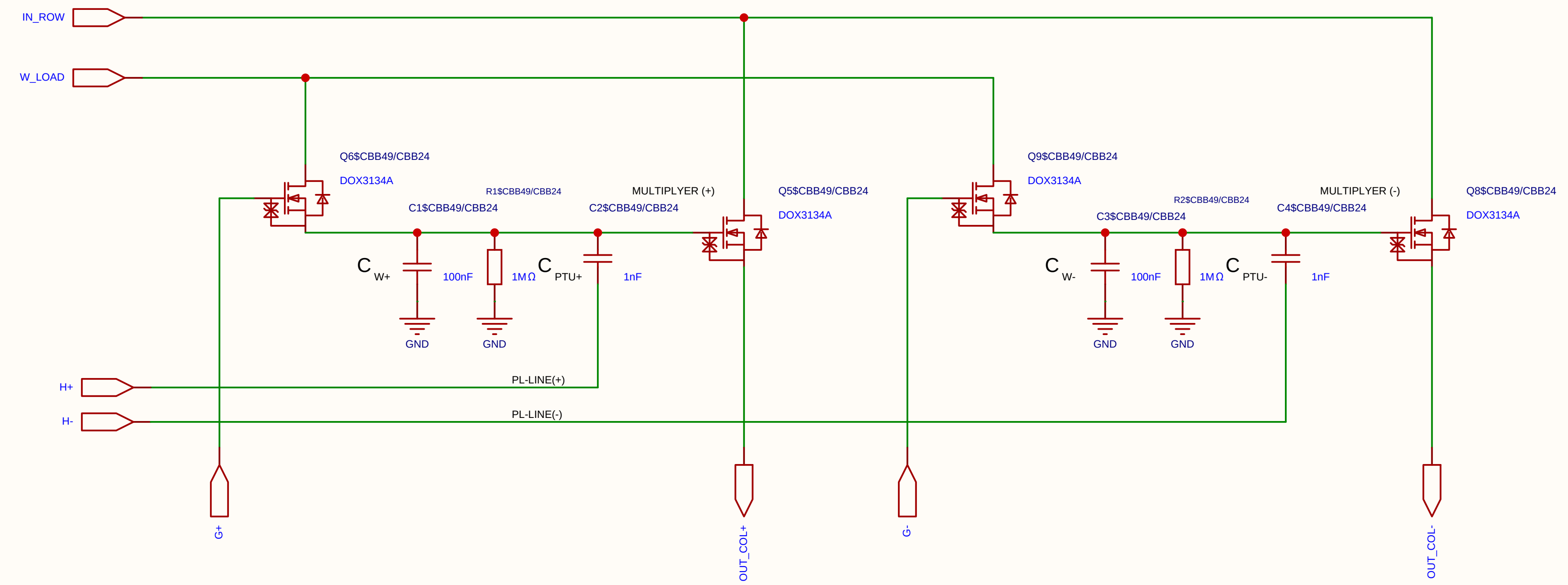
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

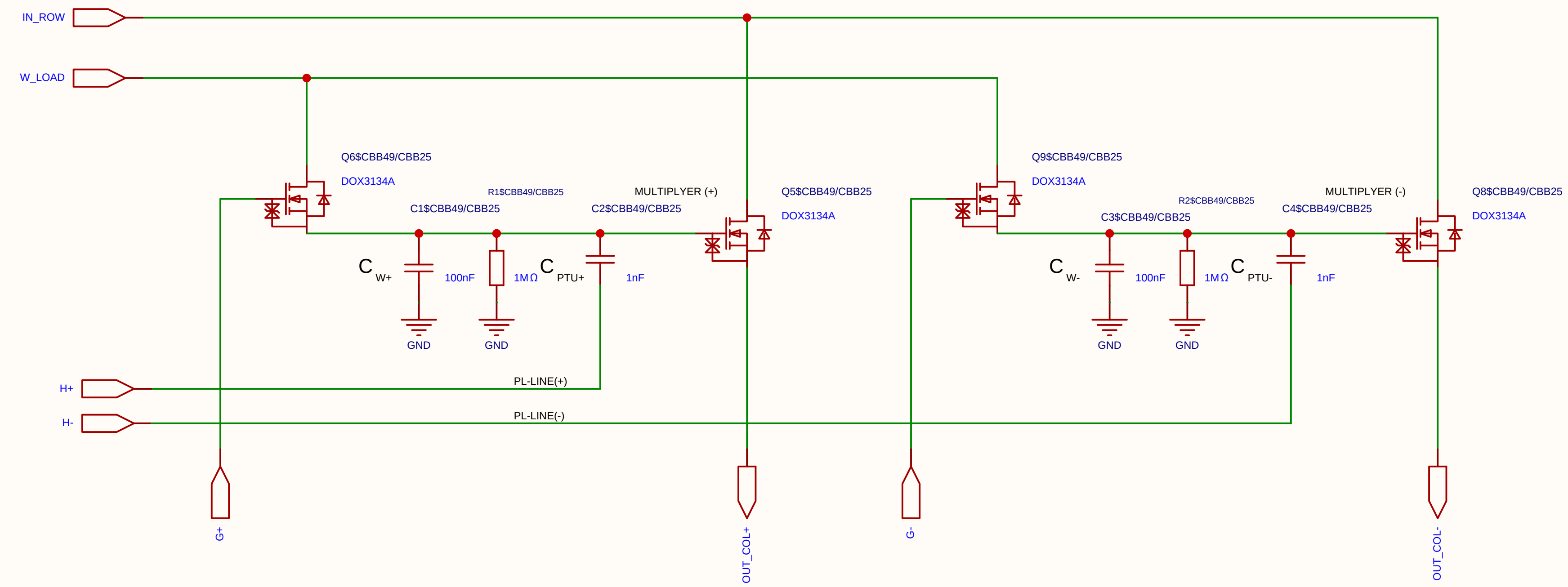
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

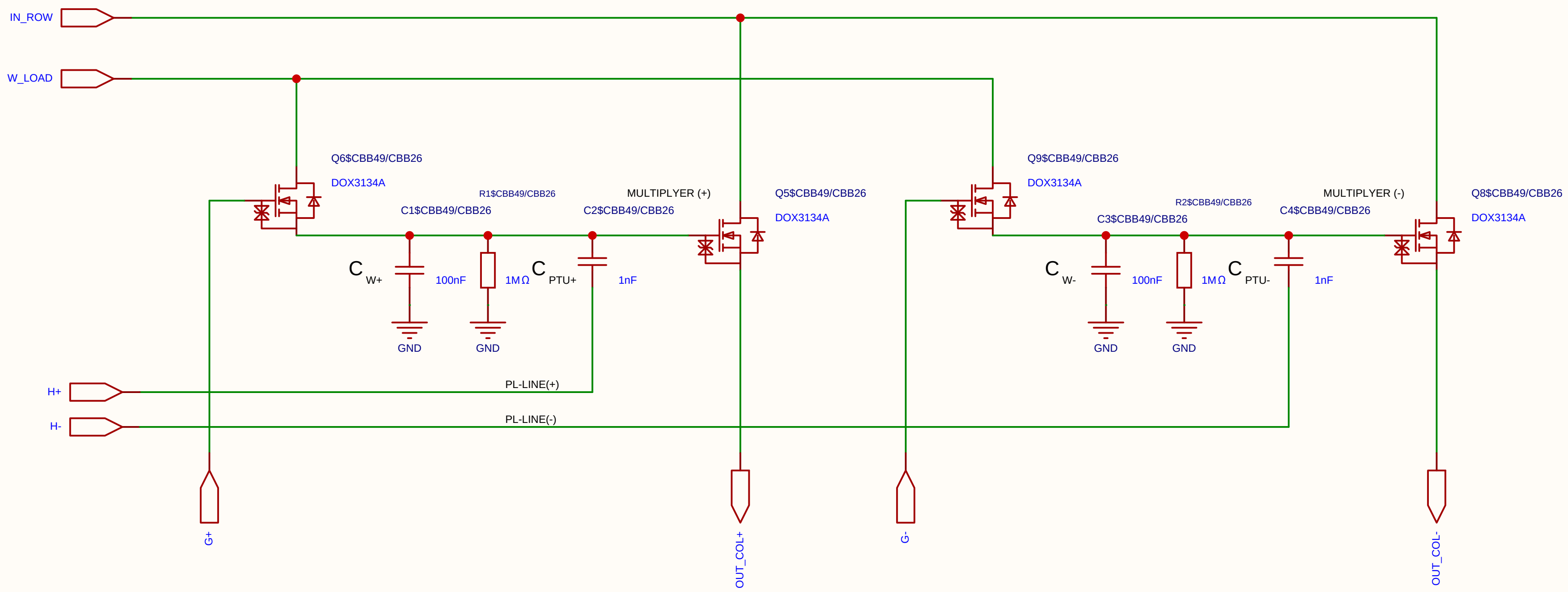
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
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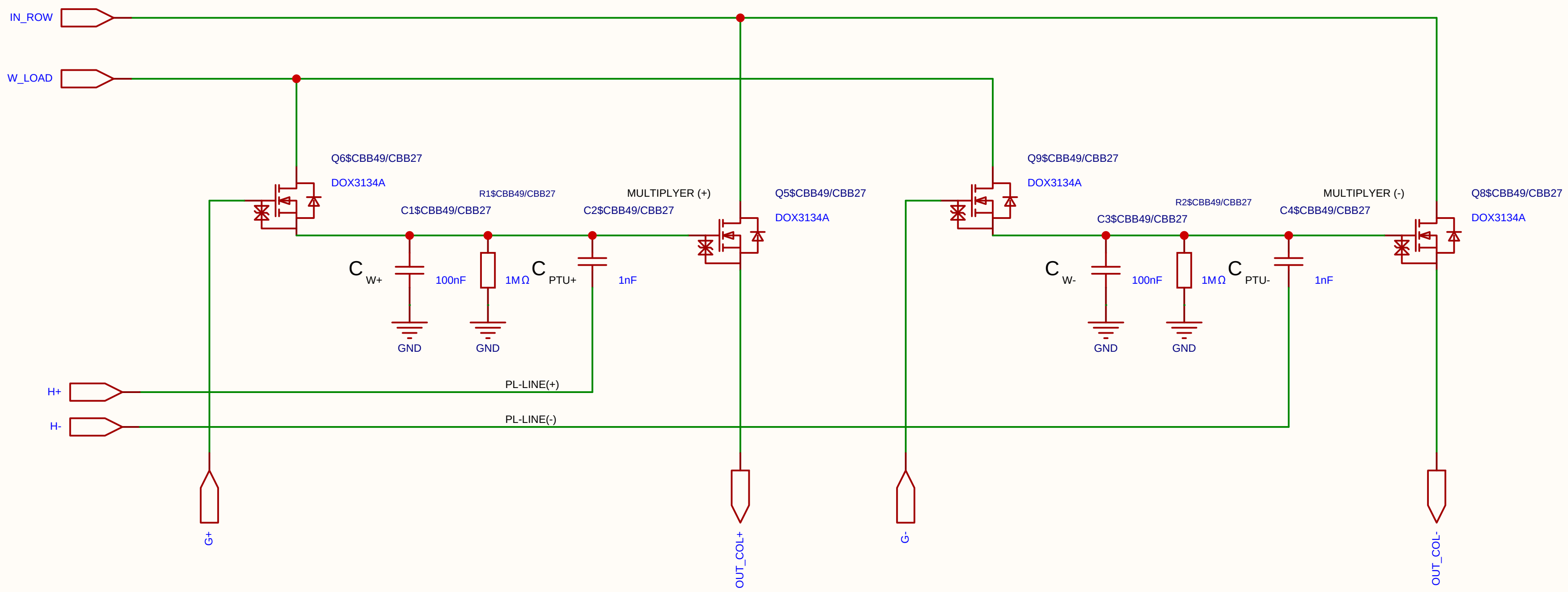
Version	Size	Page 1 Total 1	
V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

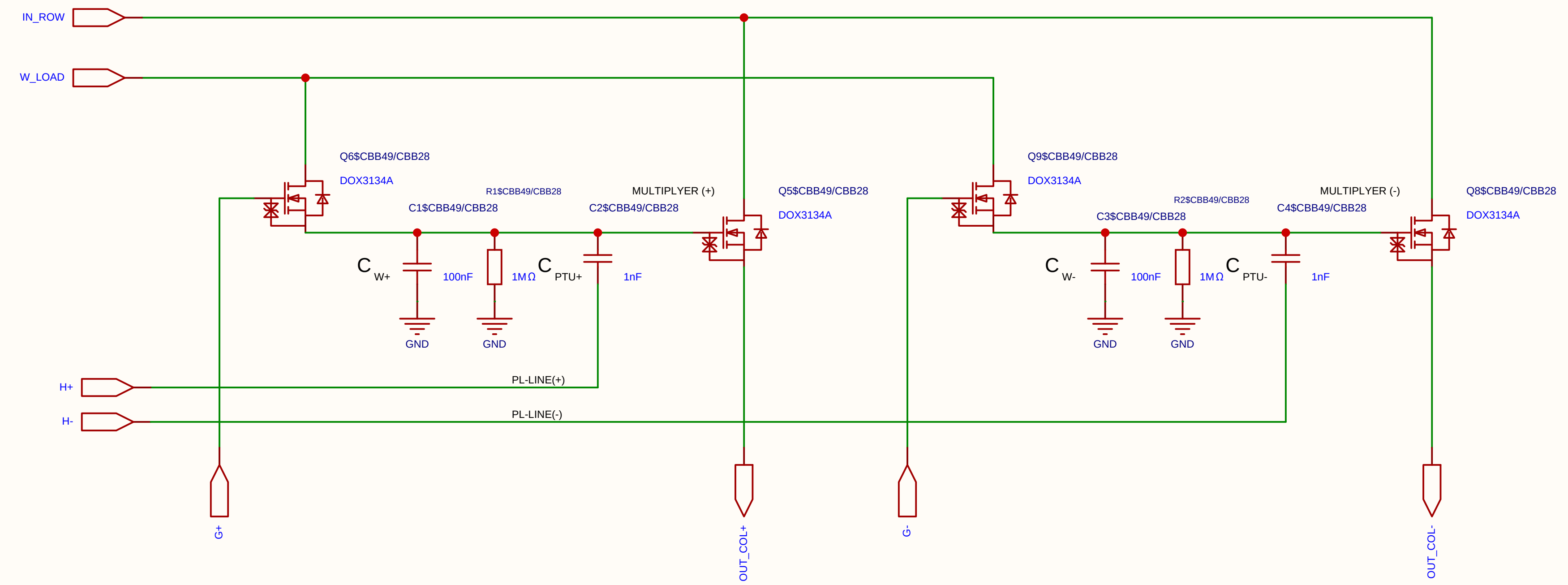
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	V1.0
		Size	A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

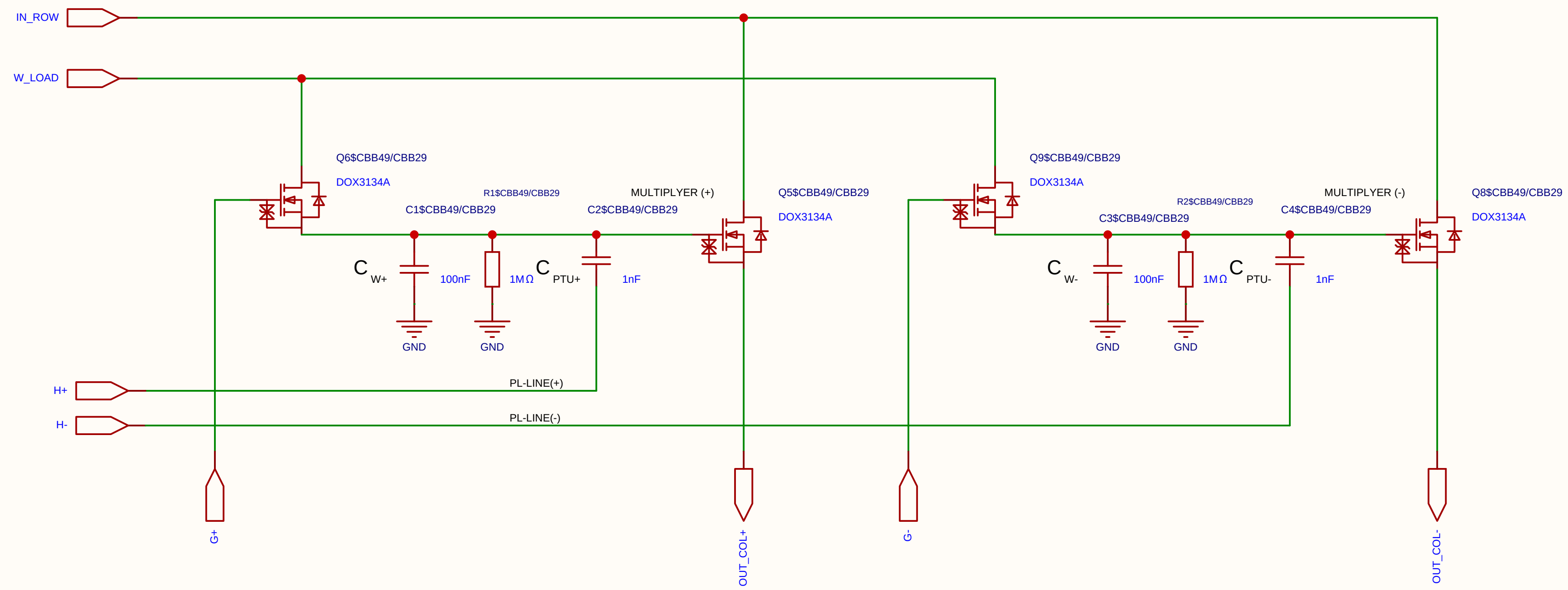
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

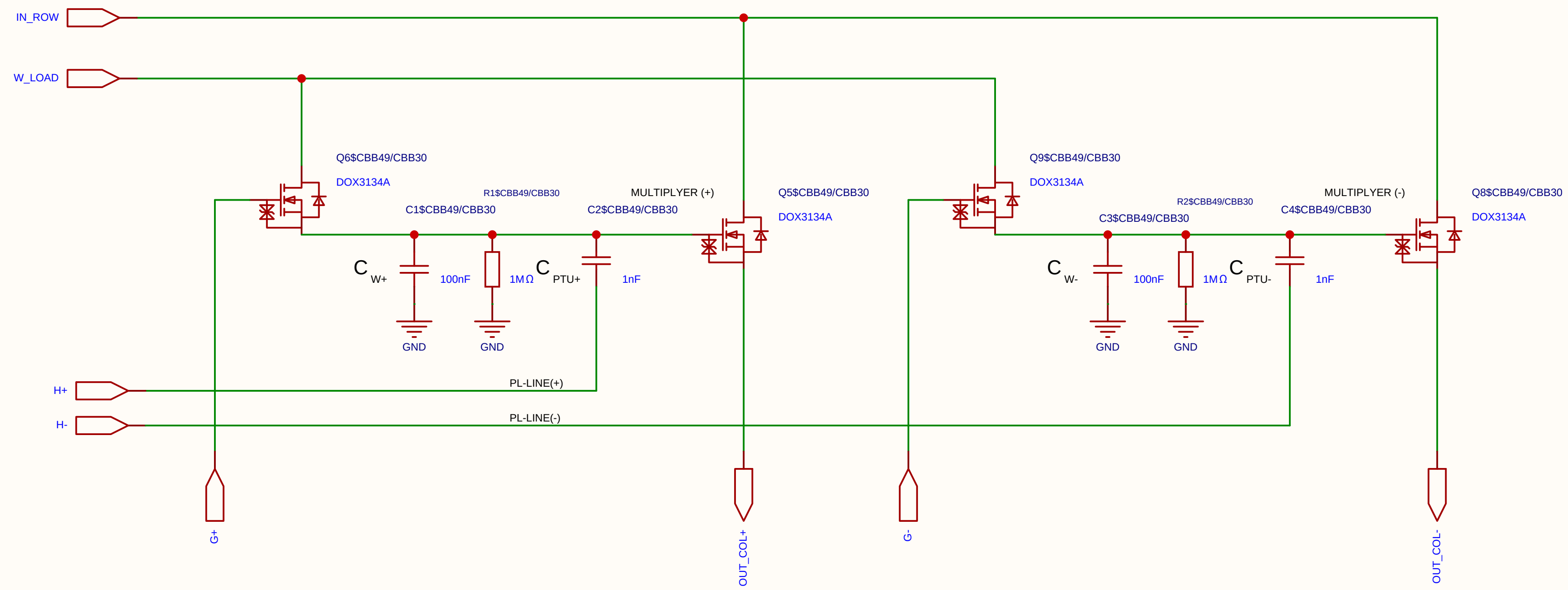
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version / V1.0	Size / A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version / Size	Page 1 Total 1
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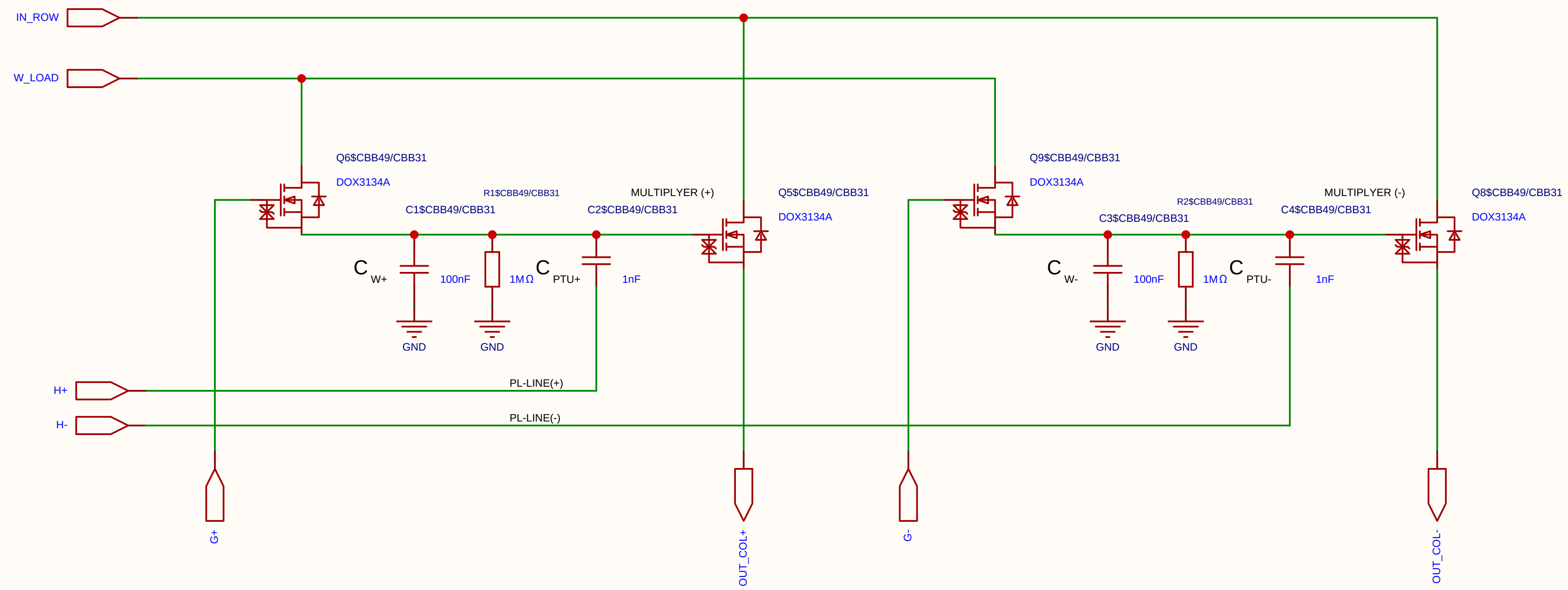
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
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		Version	Size
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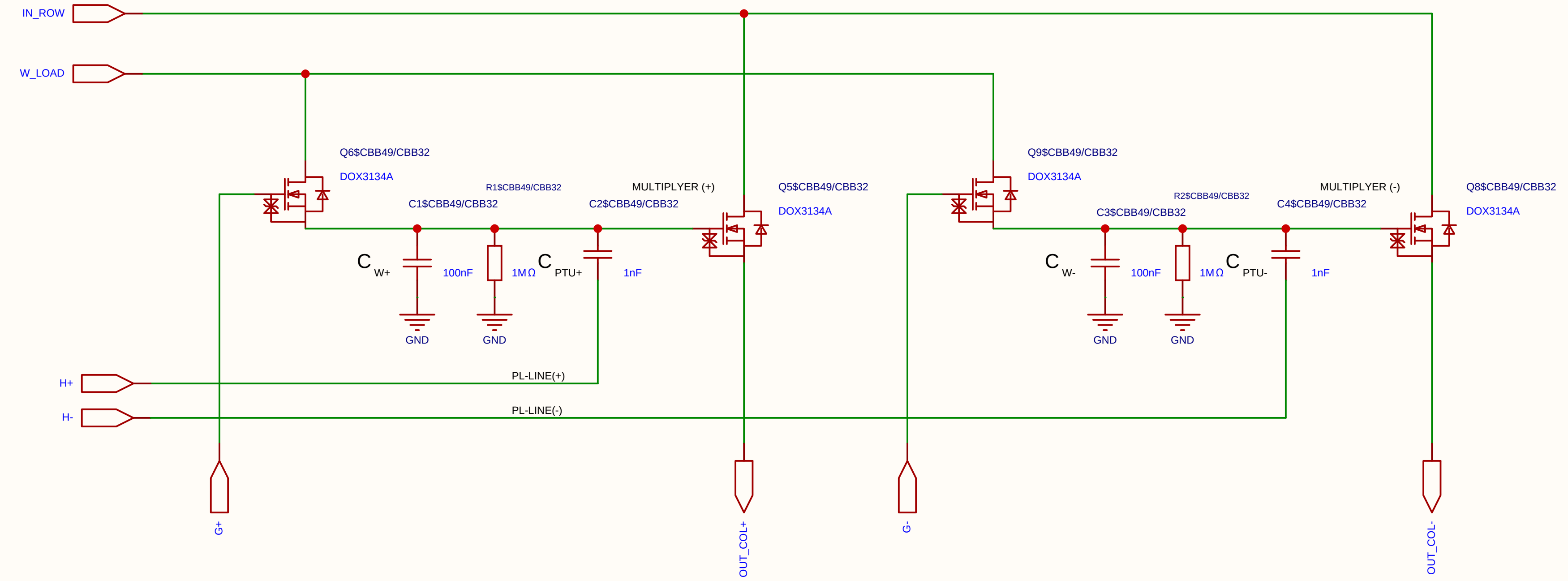
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	

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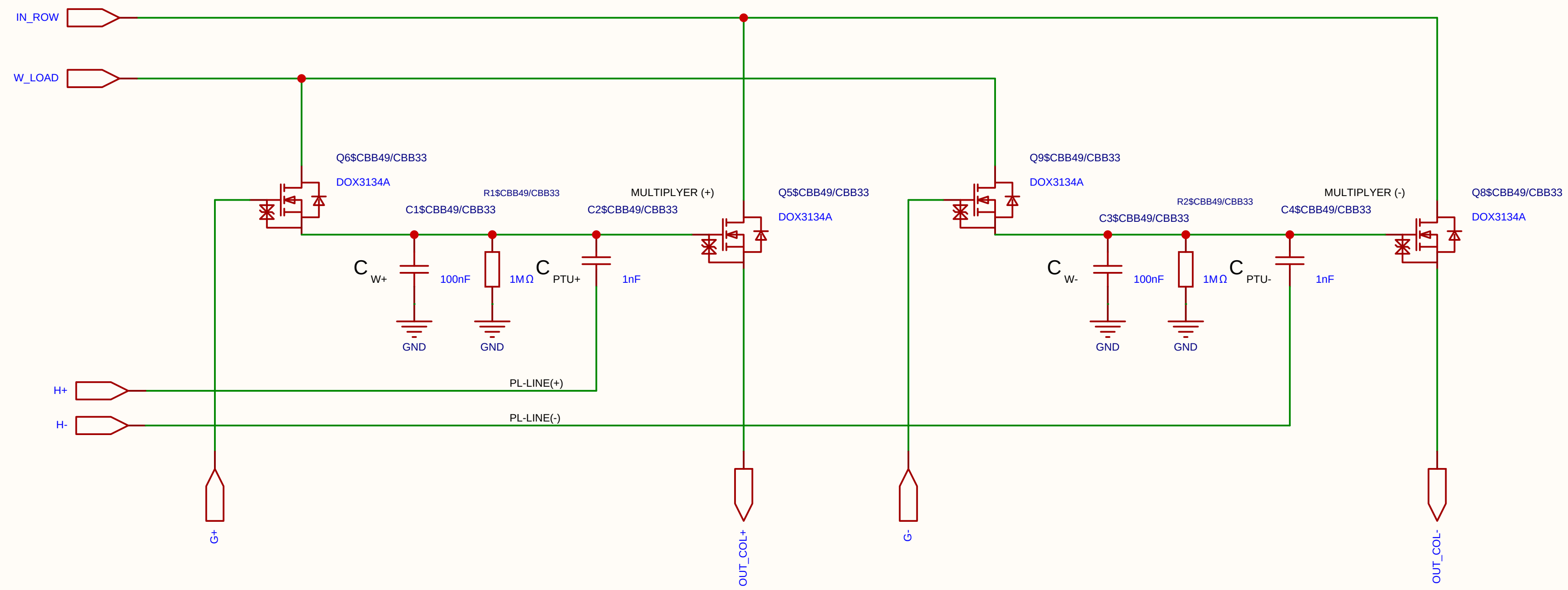
V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

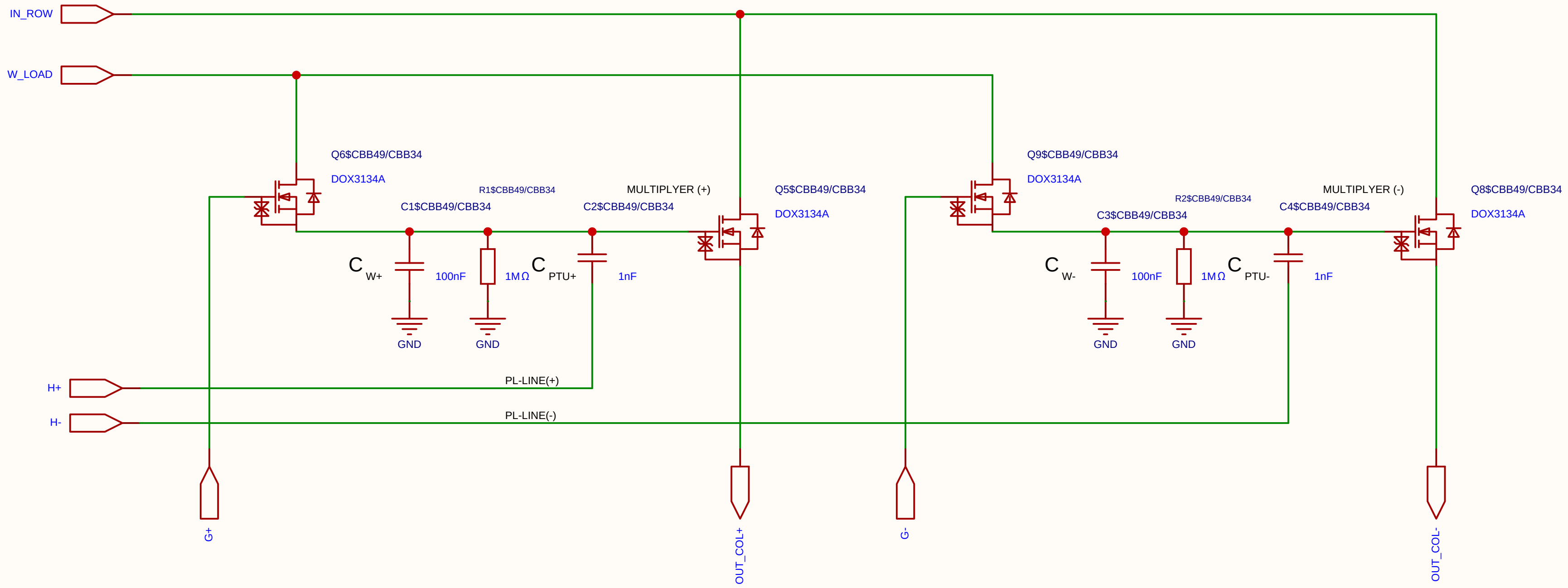
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
		Version / Size / Page 1 Total 1	
EasyEDA		V1.0 / A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
		Version	Size
EasyEDA		V1.0	A4

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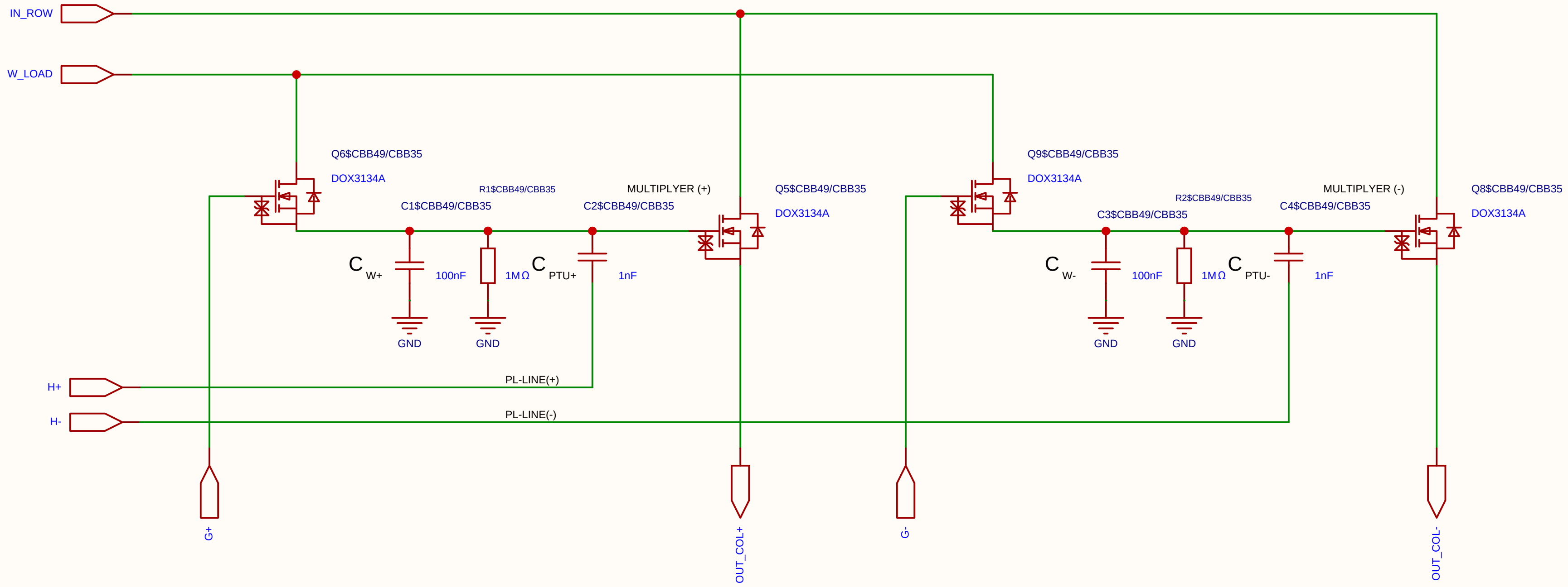
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4

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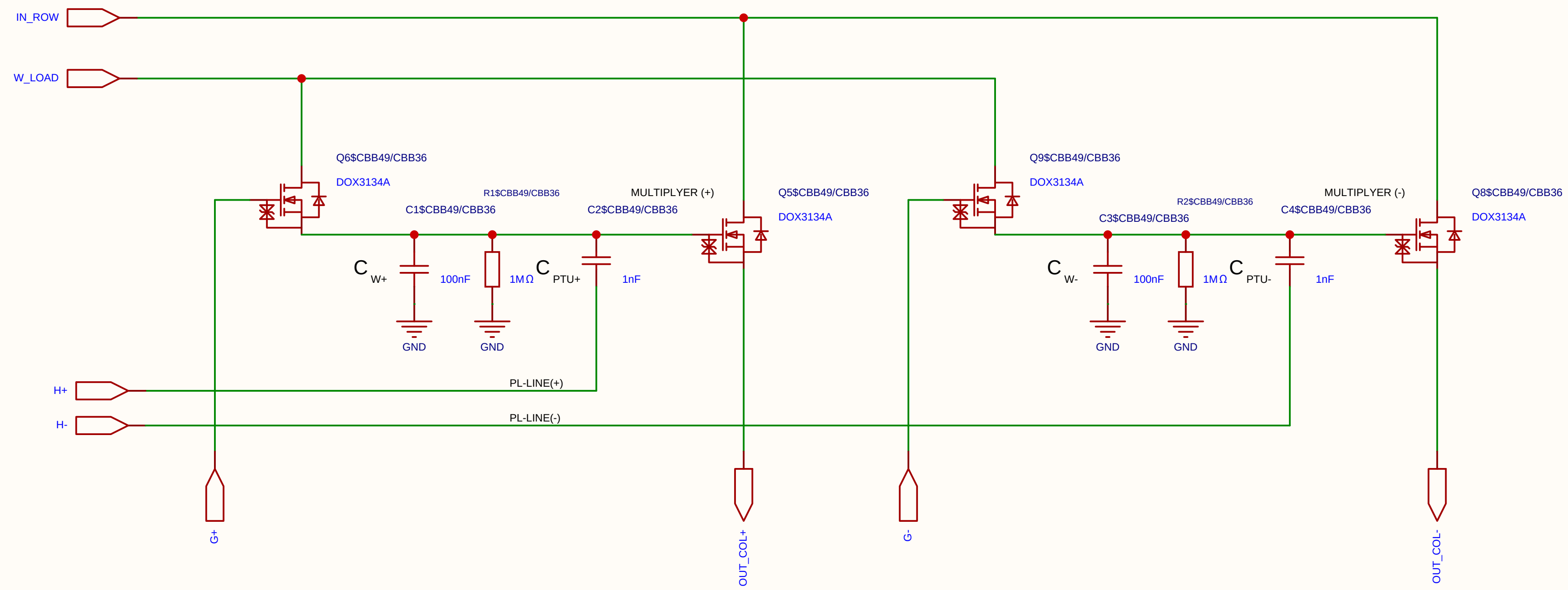
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
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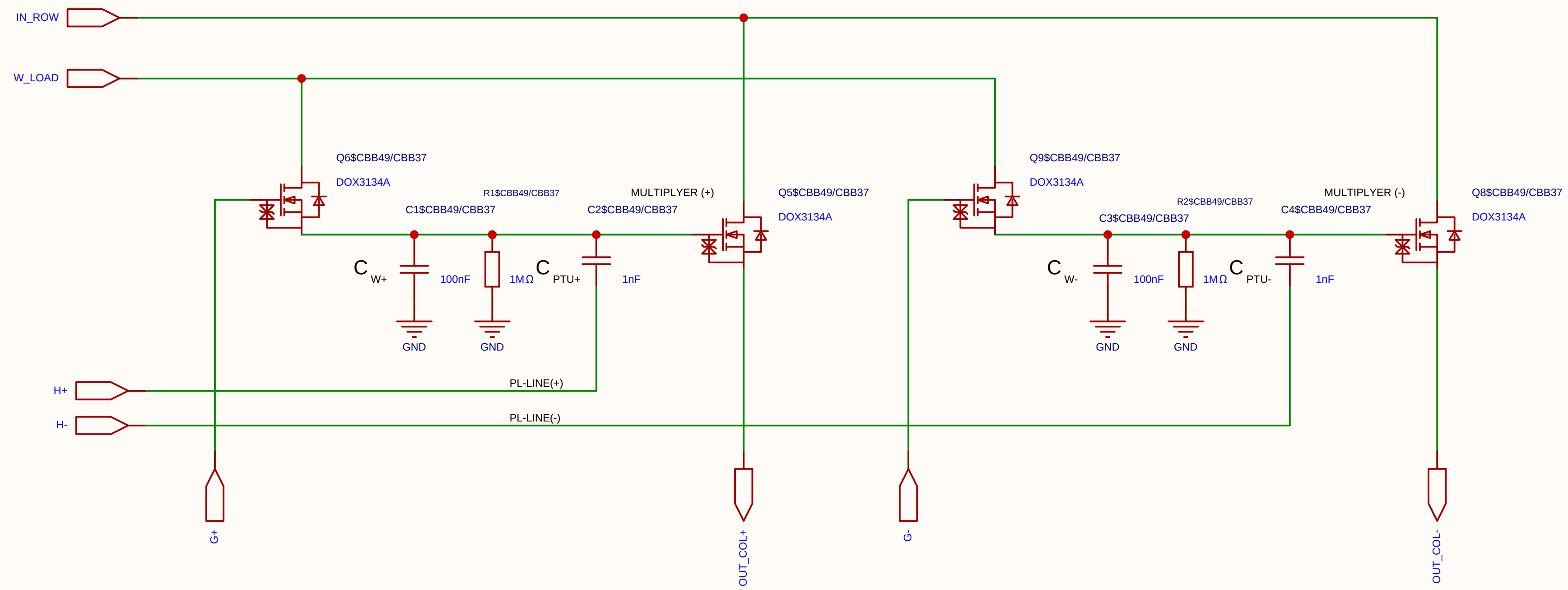
		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

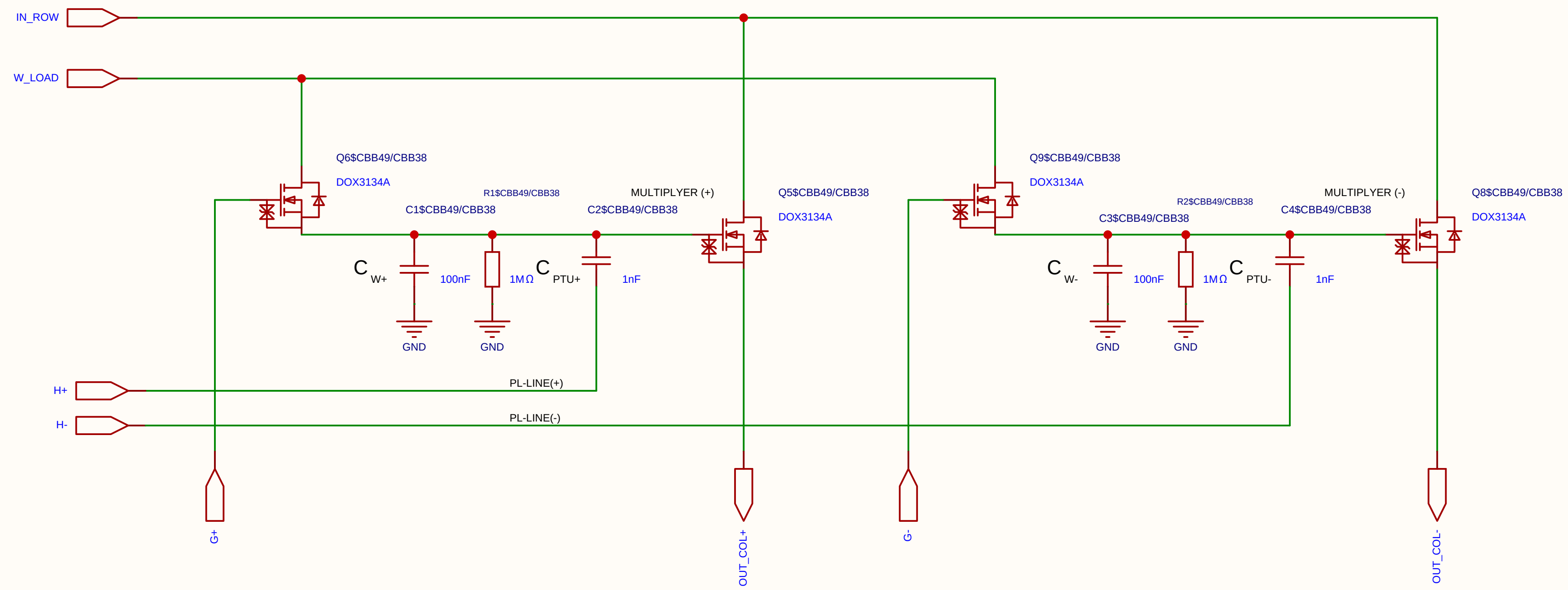
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

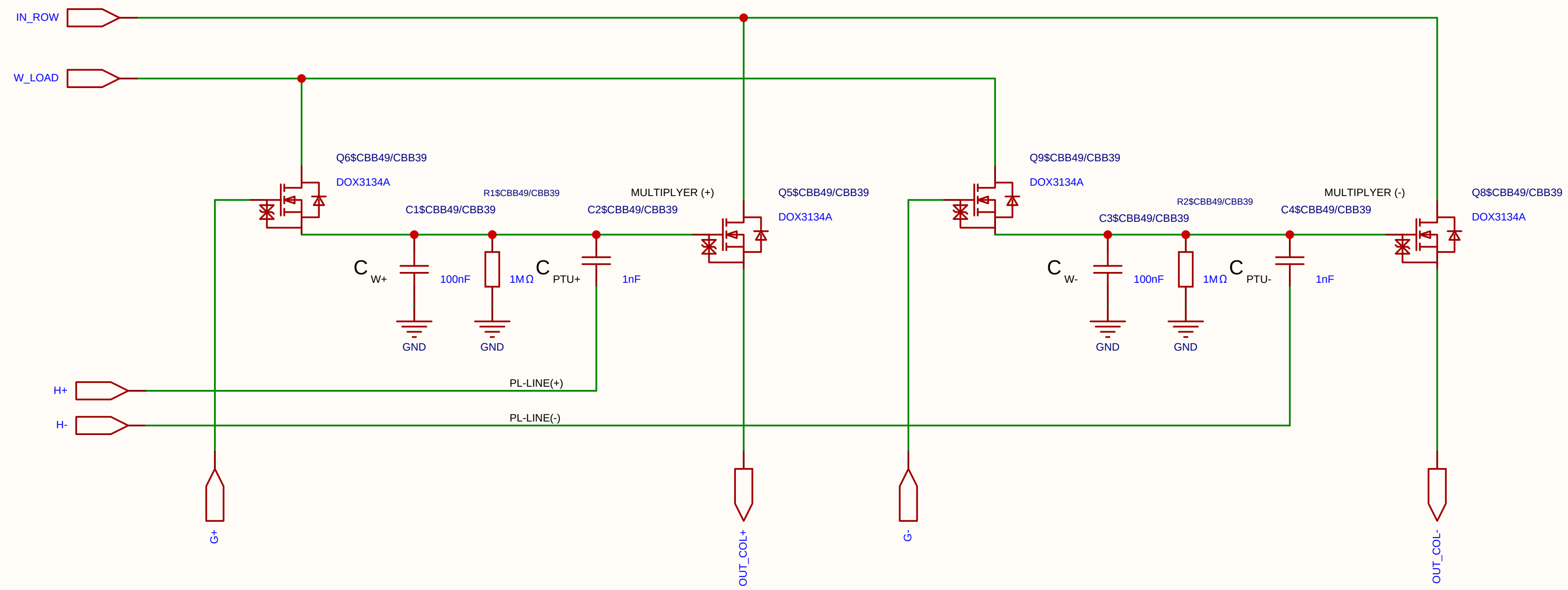
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
EasyEDA	Version: V1.0	Size: A4	Page 1 Total 1
			EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

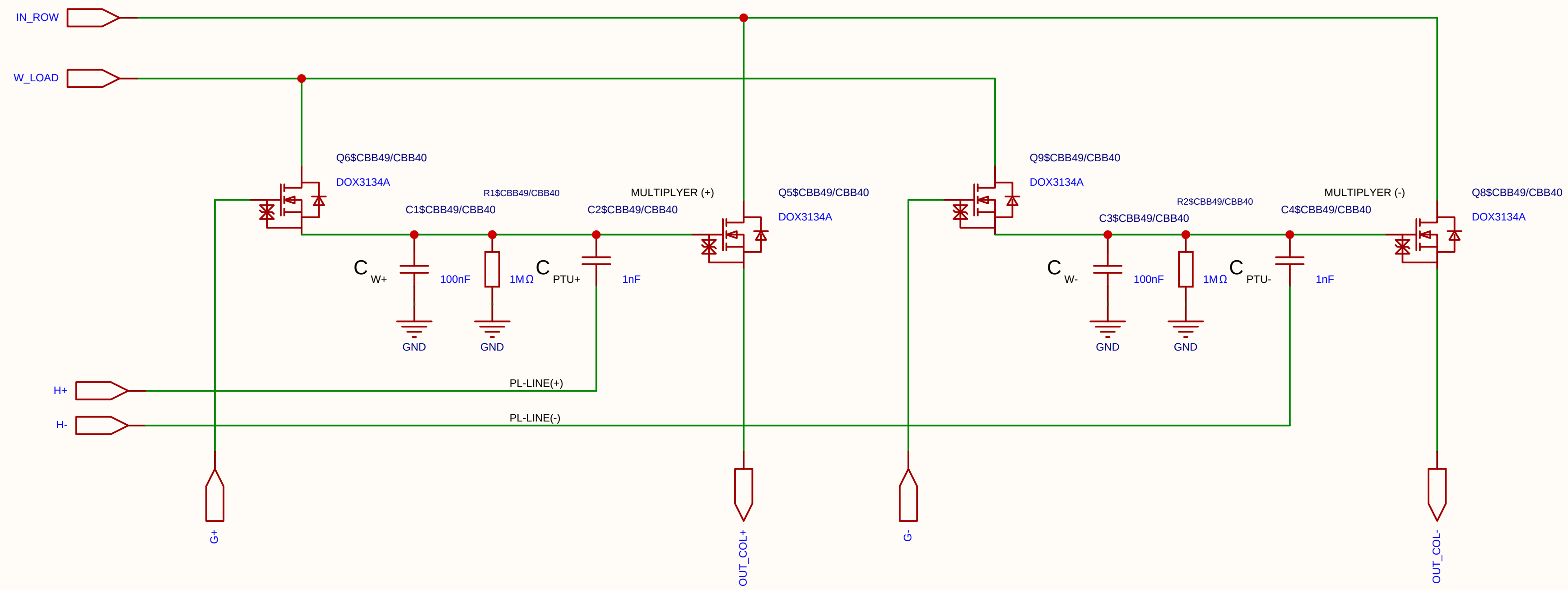
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C		Create at	2026-06-14
			Update at	2026-06-19
Board			Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

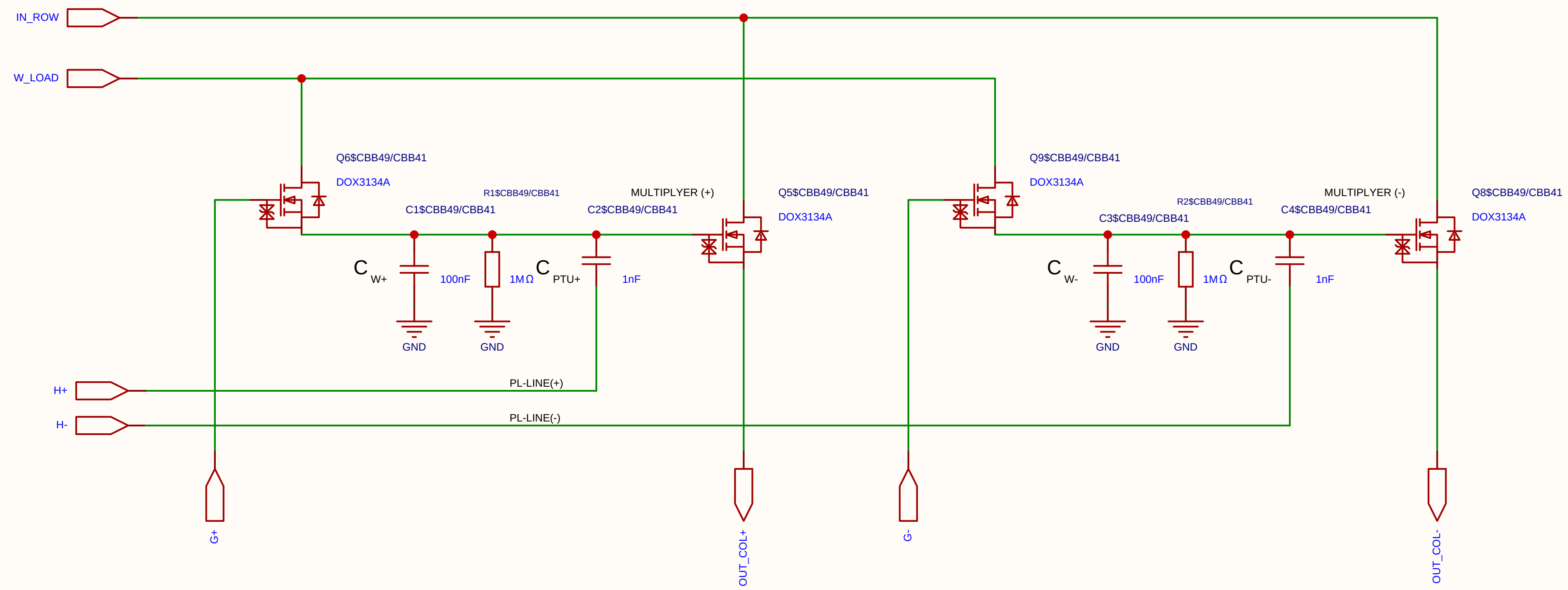
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version / V1.0	Size / A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

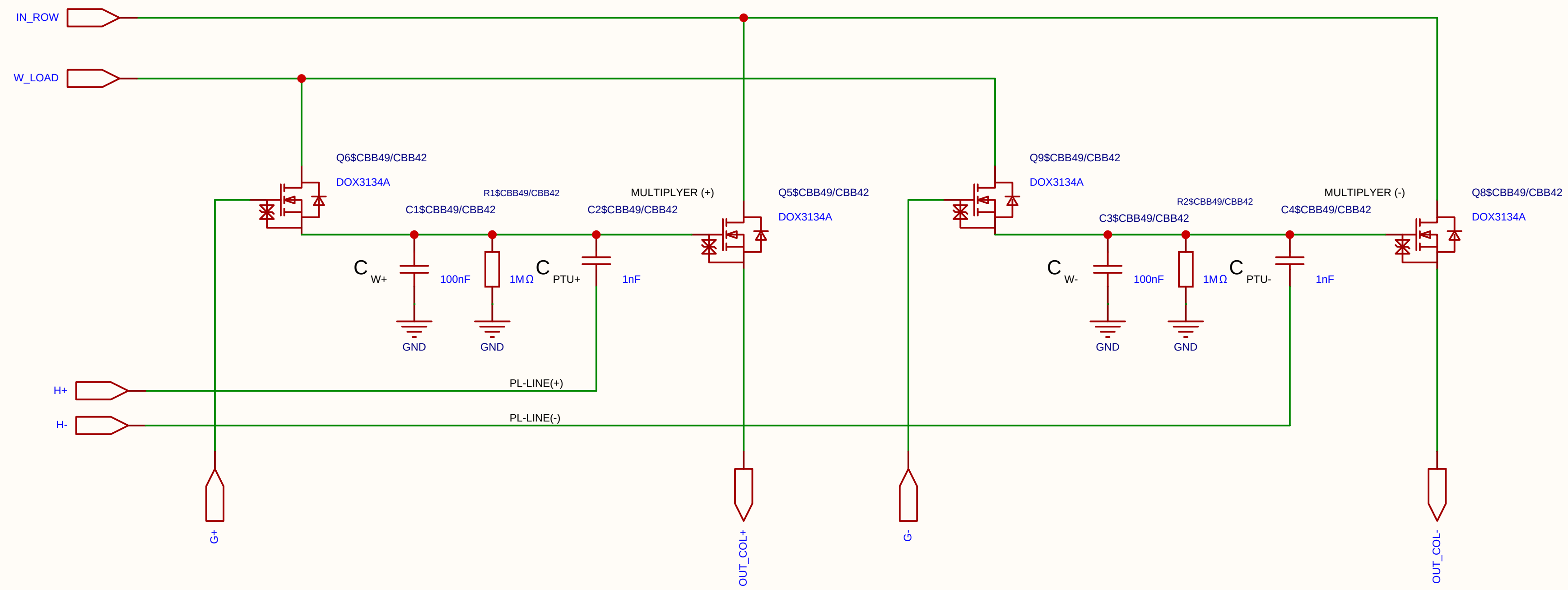
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

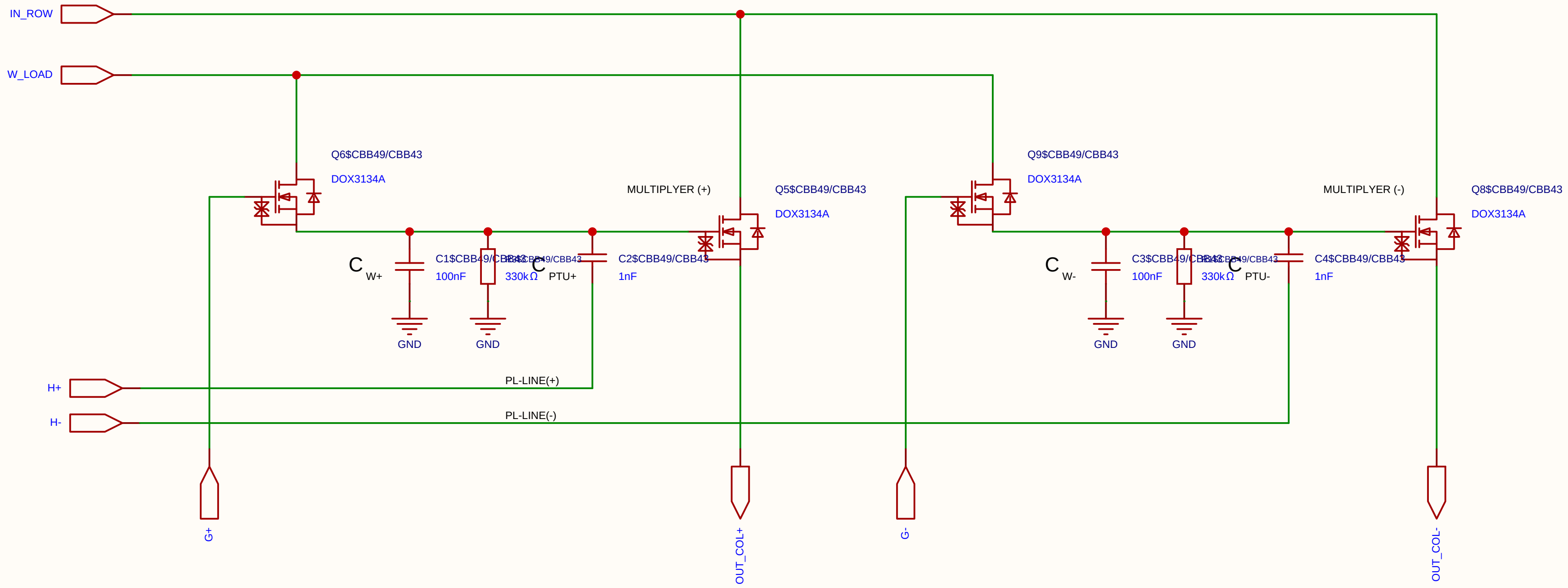
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
		V1.0	A4
		Page 1 Total 1	EasyEDA.com

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

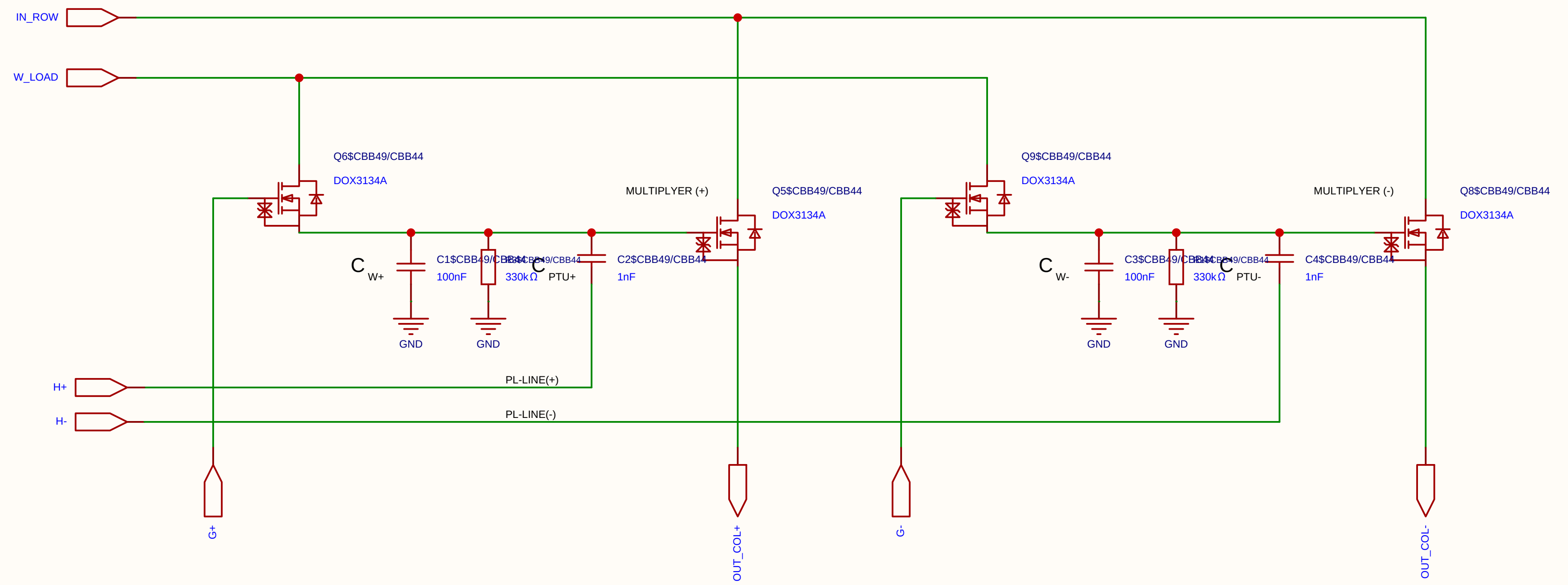
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C	Create at	2026-06-11
		Update at	2026-06-19
Board		Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

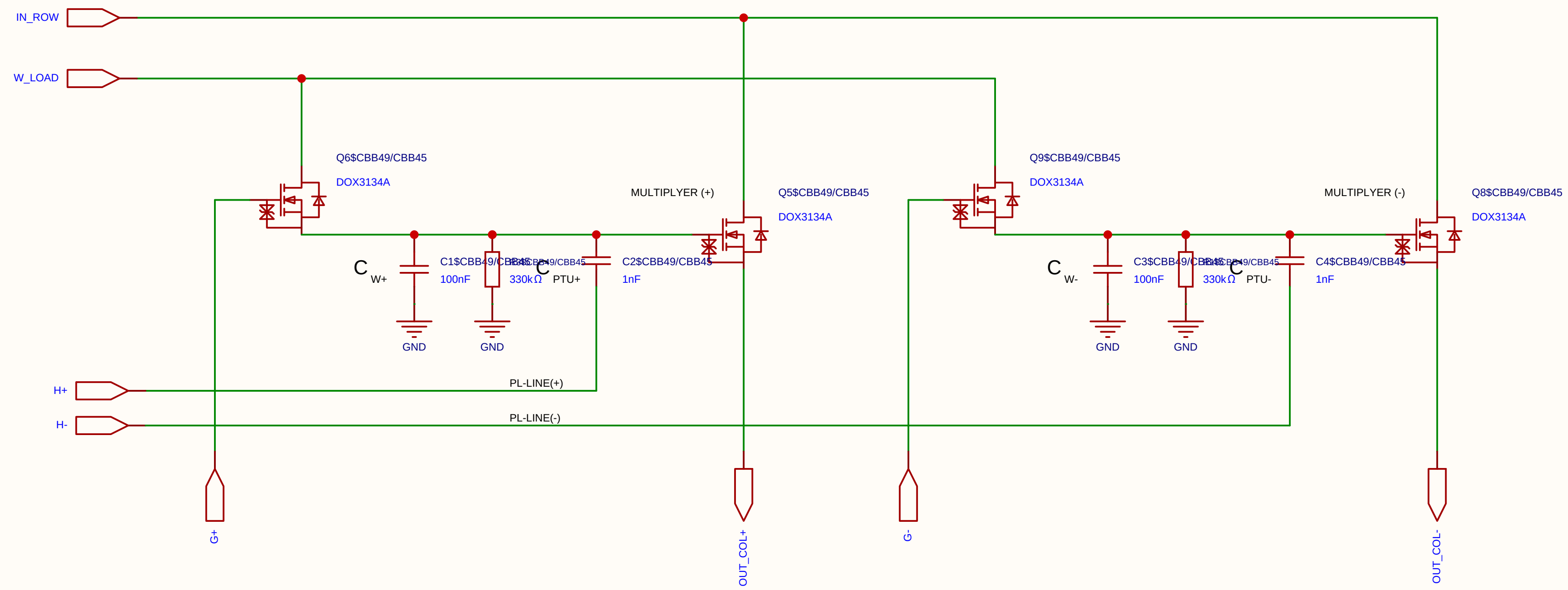
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C	Create at	2026-06-11
		Update at	2026-06-19
Board		Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	V1.0
		Size	A4
		Page 1 Total 1	EasyEDA.com

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

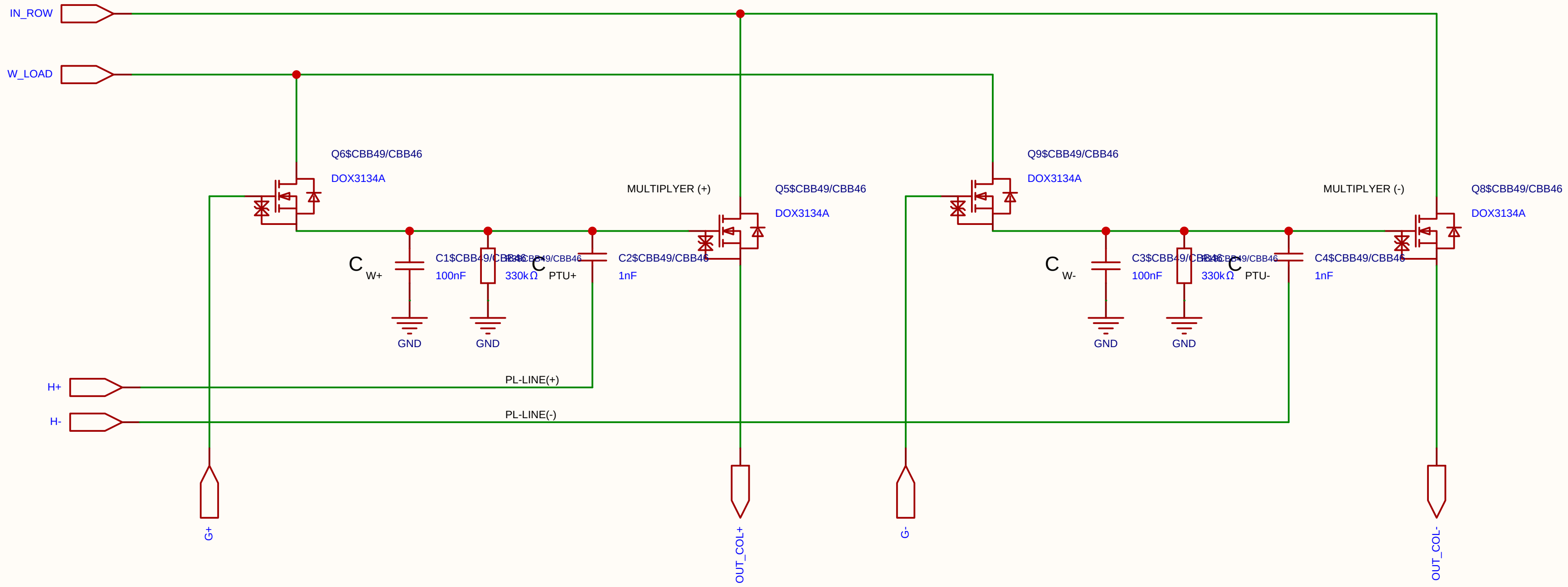
COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C	Create at	2026-06-11
		Update at	2026-06-19
Board		Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4

Page 1 Total 1

EasyEDA.com

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

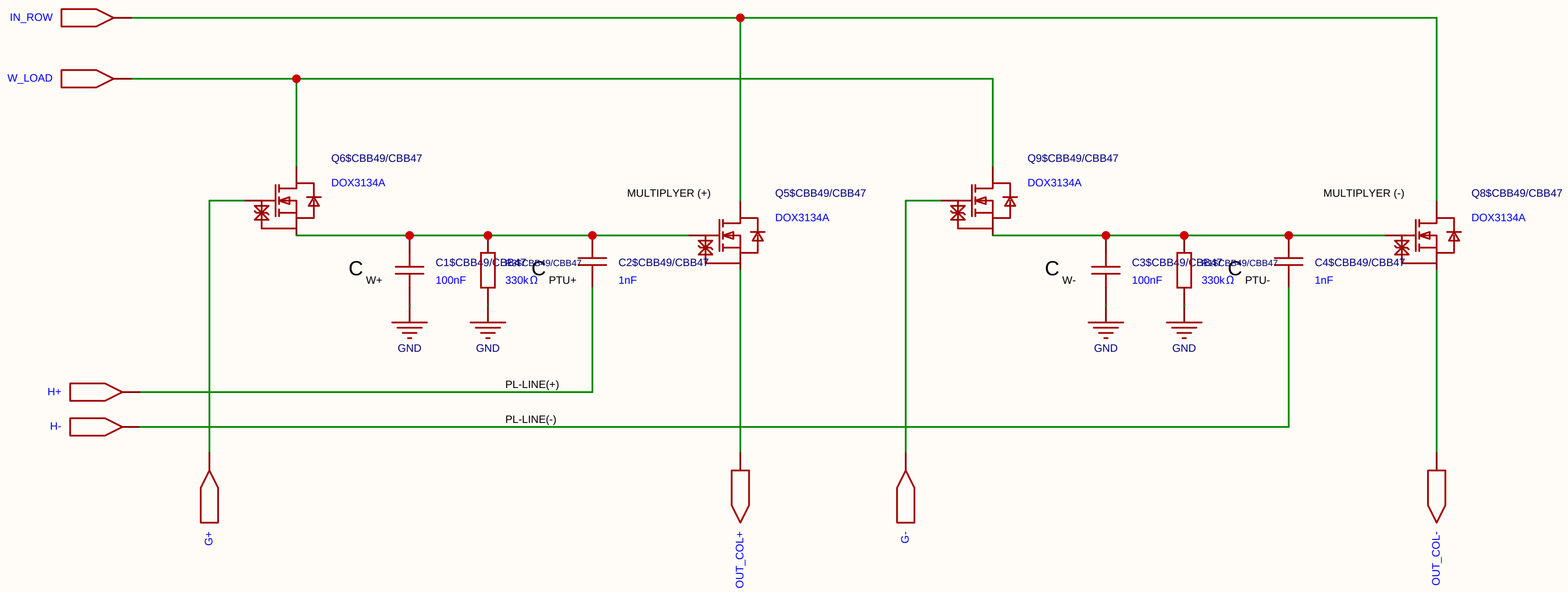
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C	Create at	2026-06-11
		Update at	2026-06-19
Board		Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size / Page 1 Total 1
EasyEDA		V1.0	A4 / EasyEDA.com

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

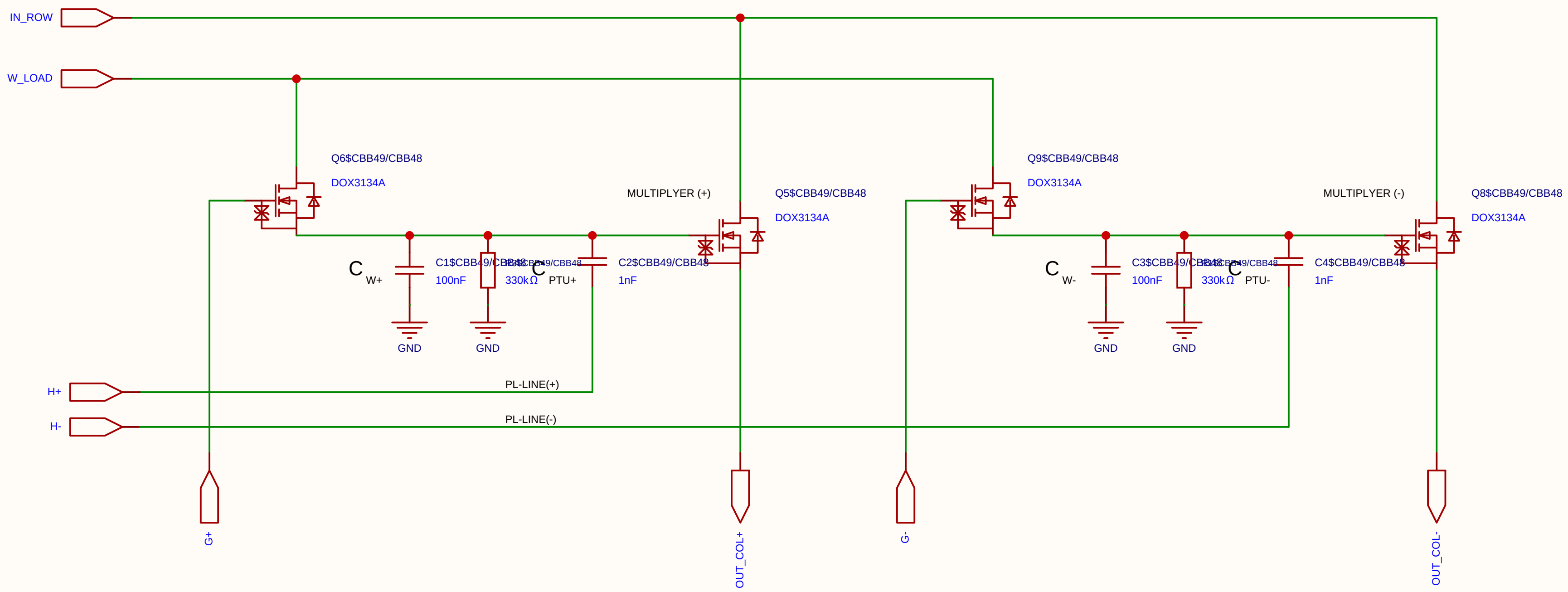
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C		Create at	2026-06-11
			Update at	2026-06-19
Board			Page	matrix_3_REF_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog REF DISSCHARGE
Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
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Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	REF_CELL_2X_2T1C	Create at	2026-06-11
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Reviewed			
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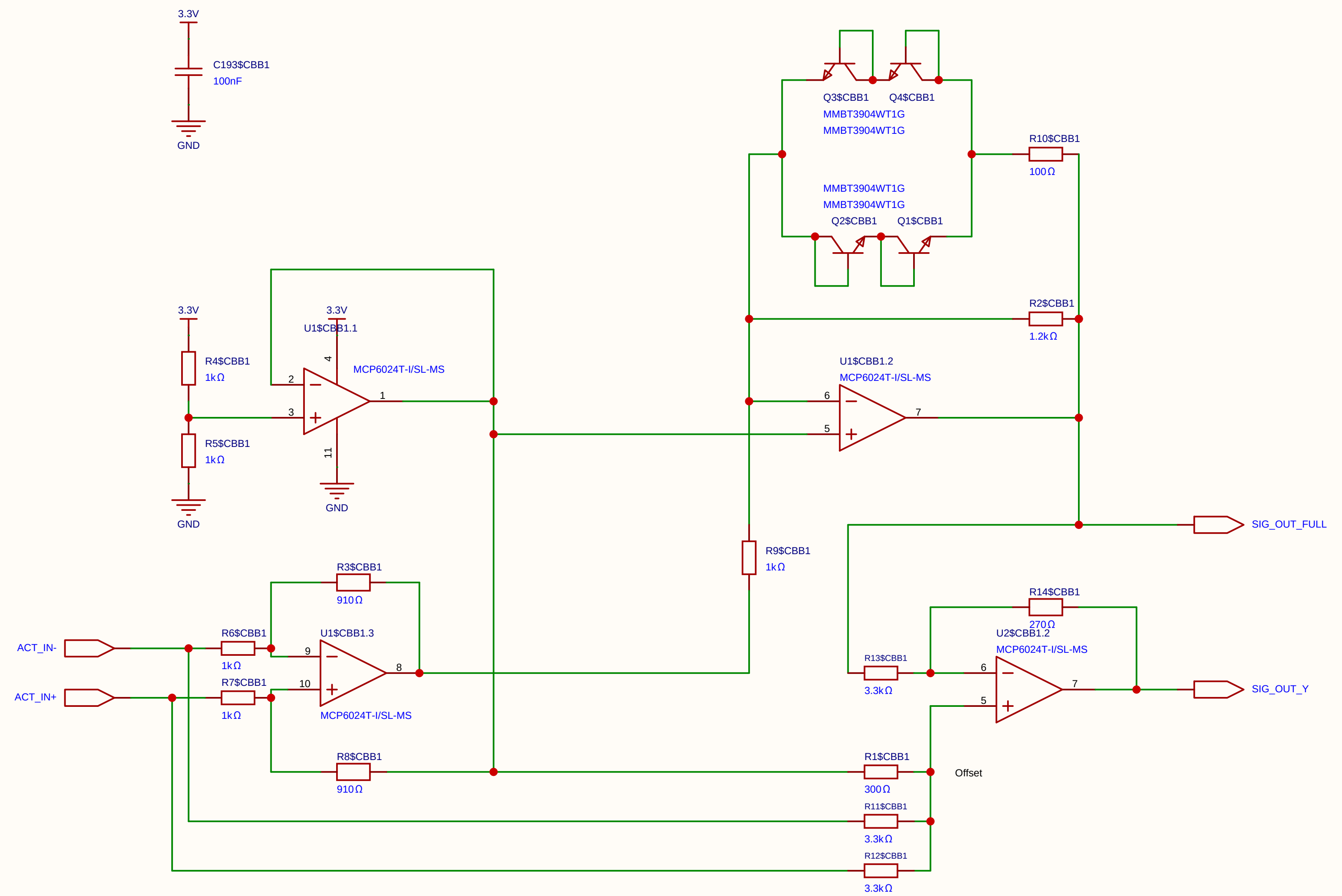
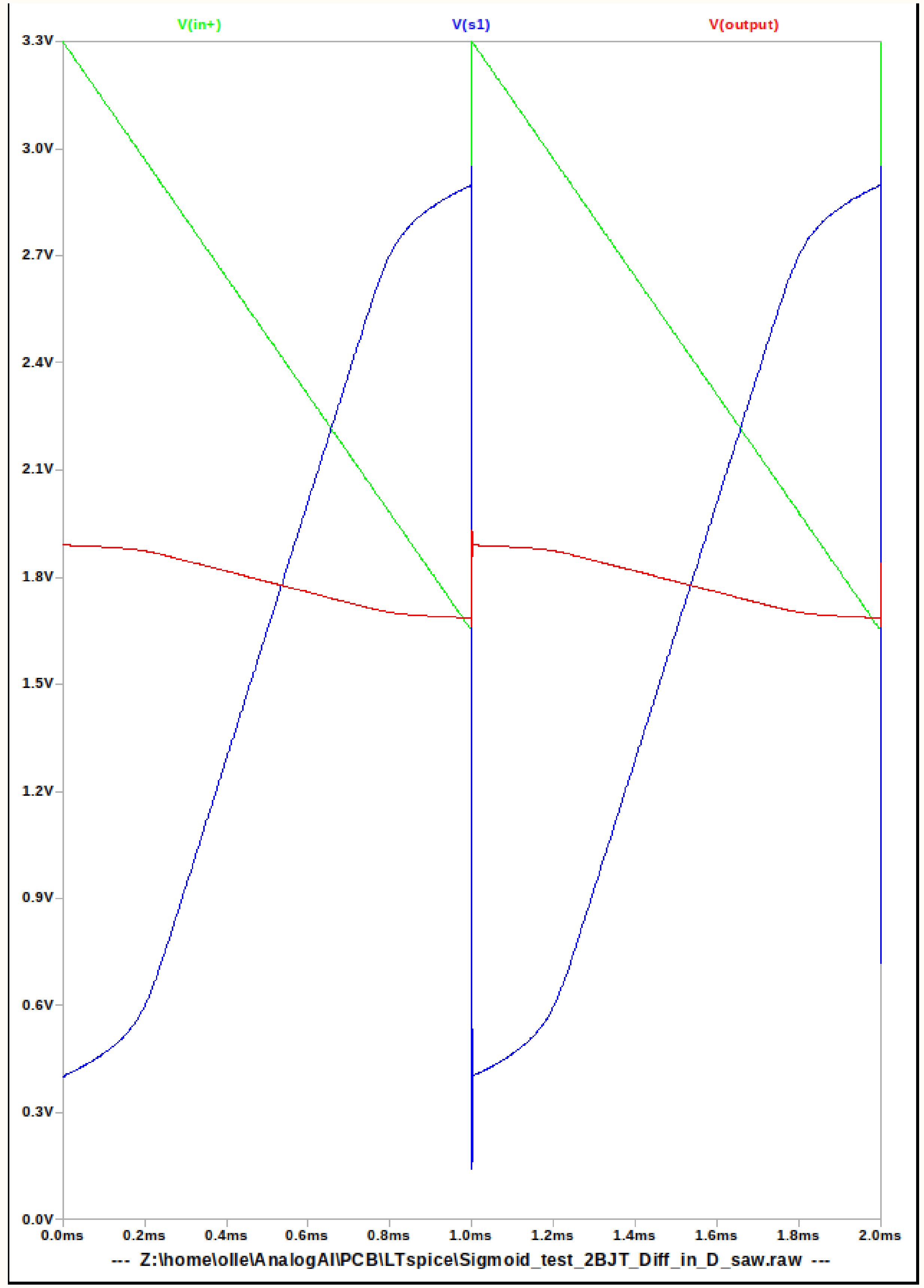
EasyEDA		V1.0	A4
		Page 1 Total 1	EasyEDA.com





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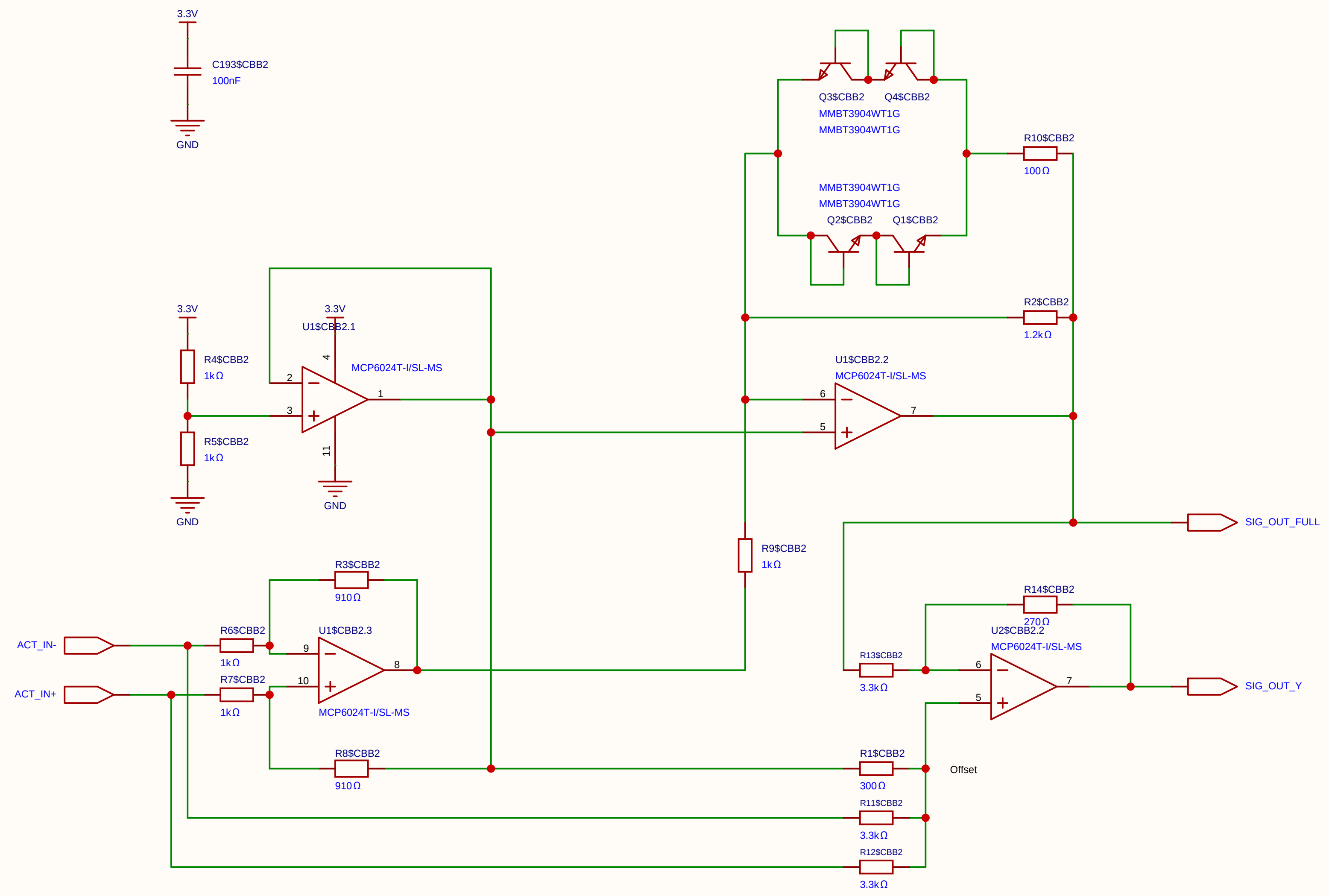

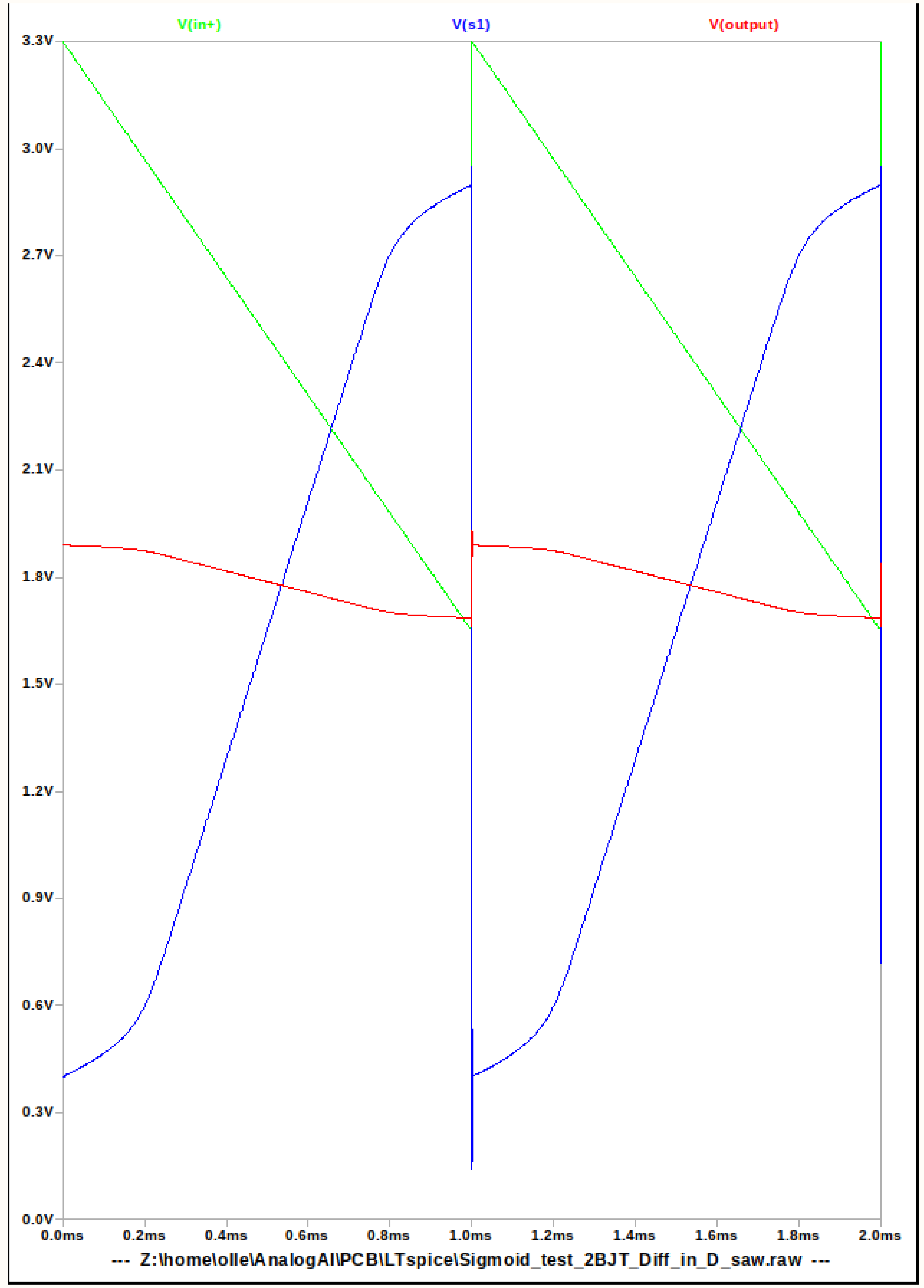

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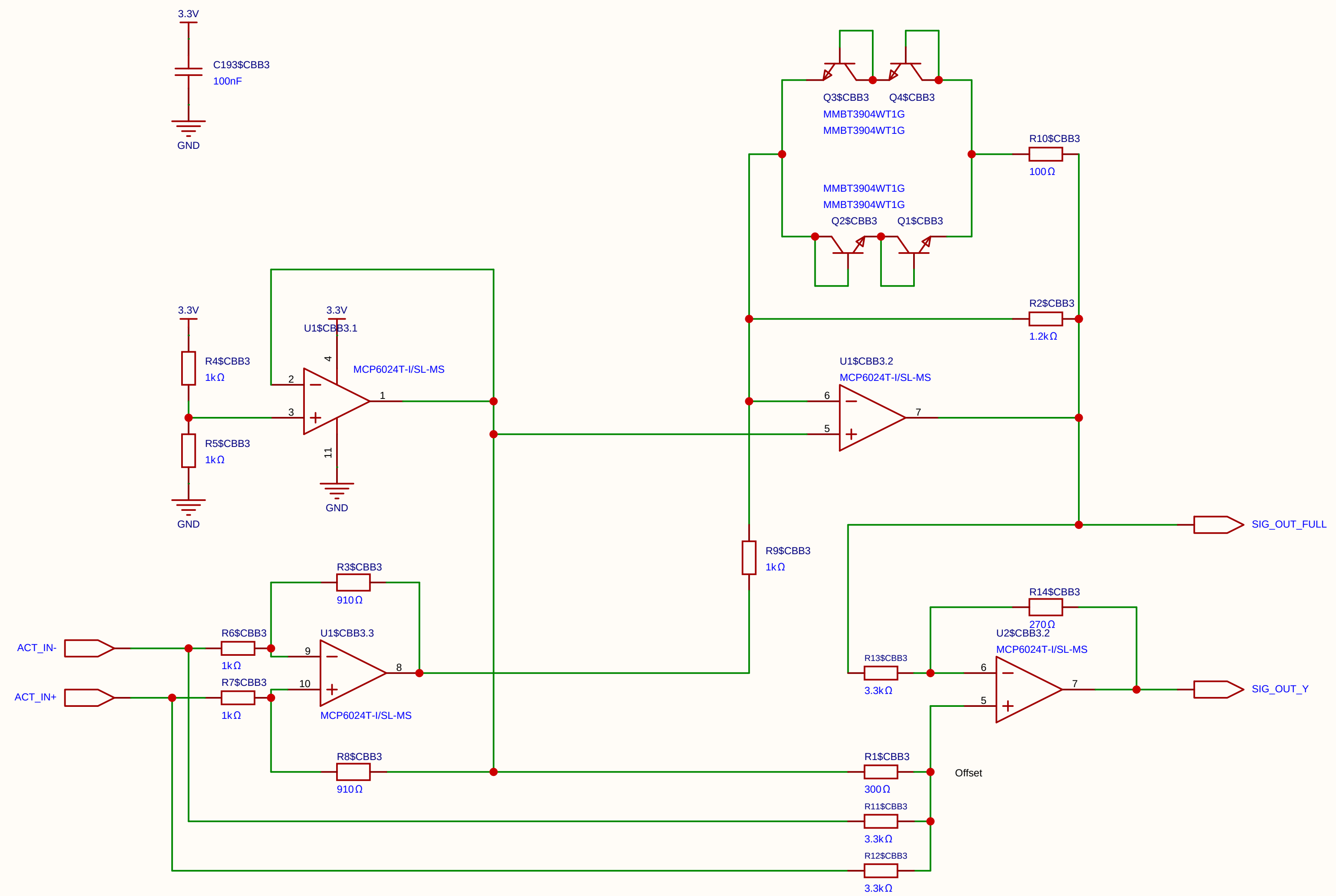

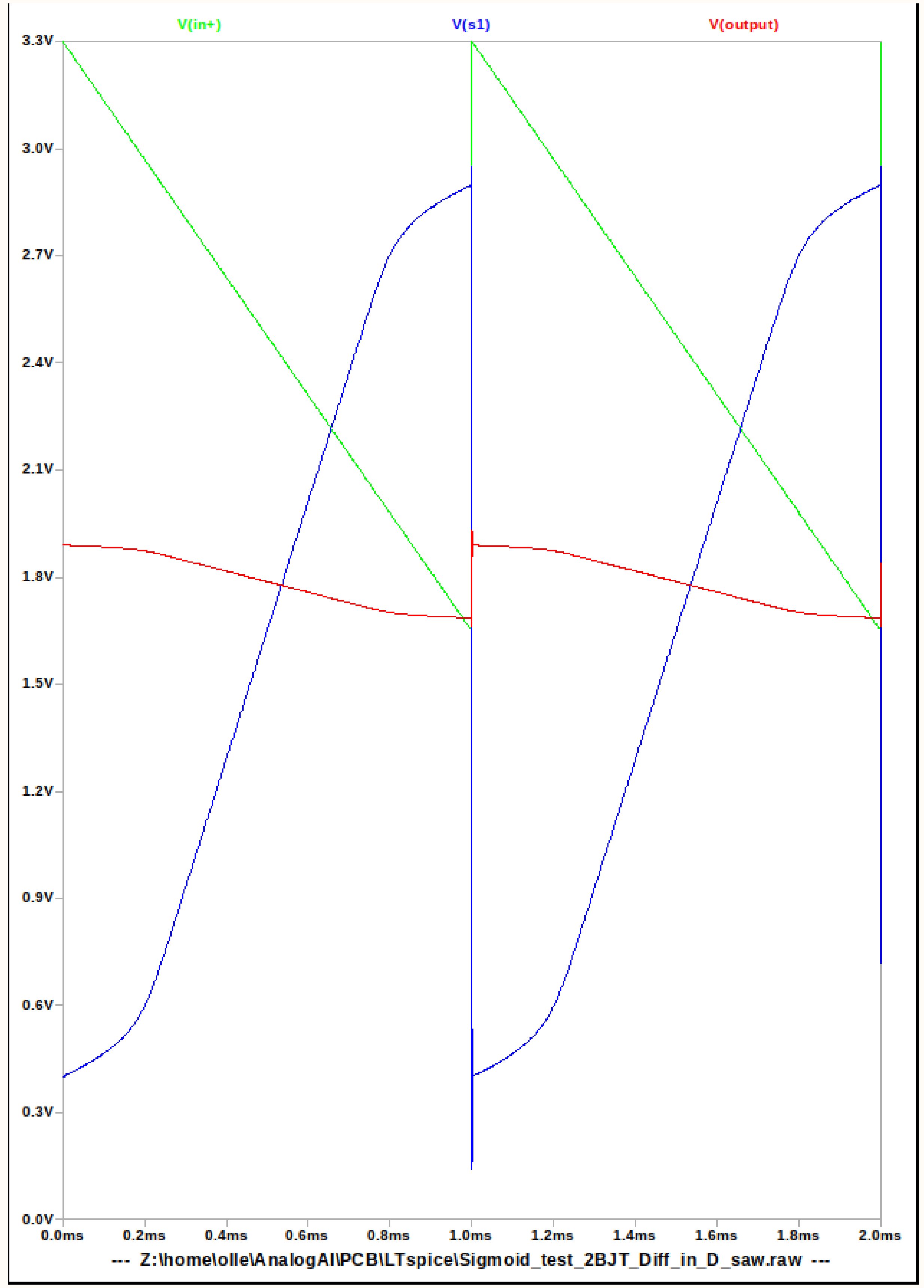

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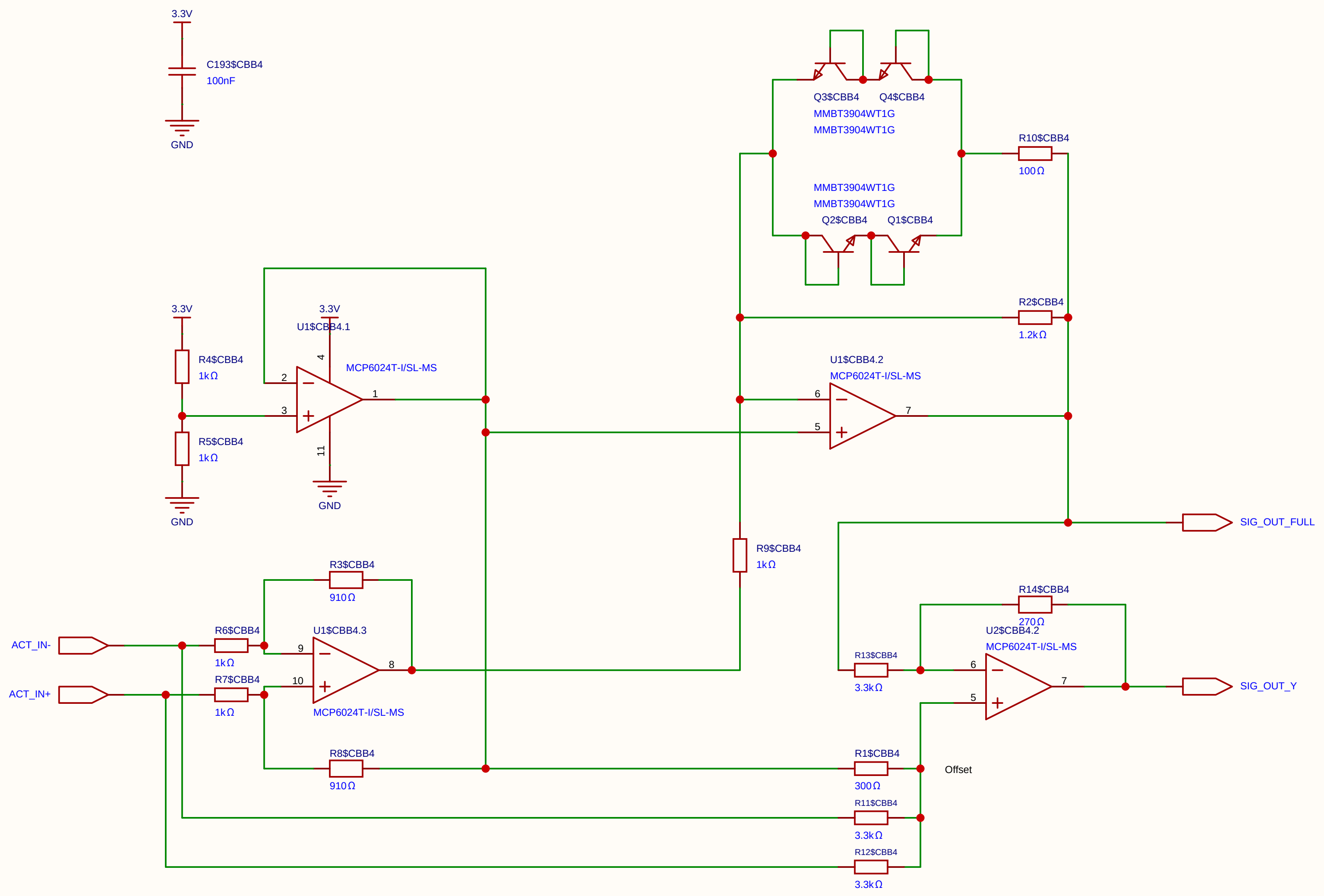

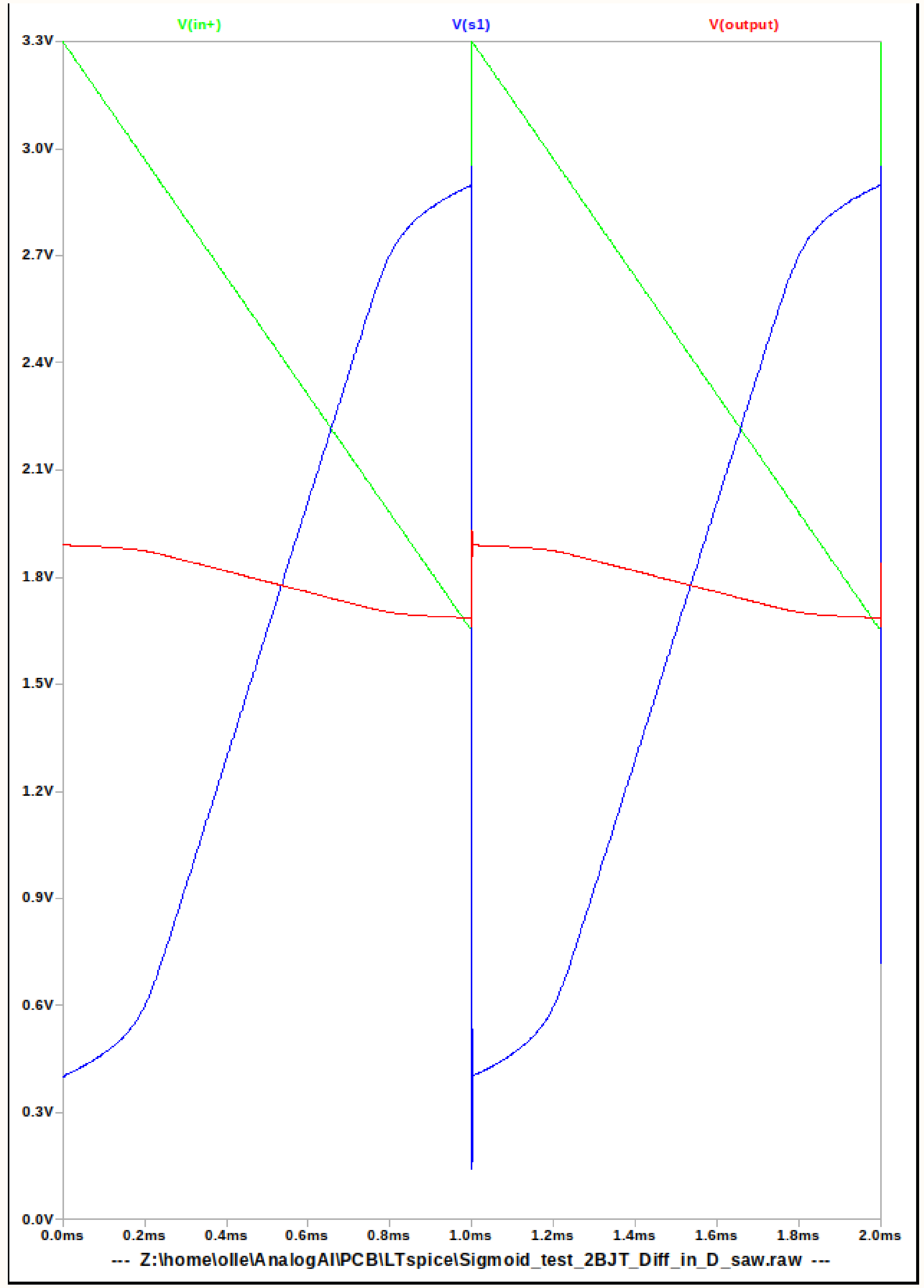

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Board				Page	SIMOID(4)
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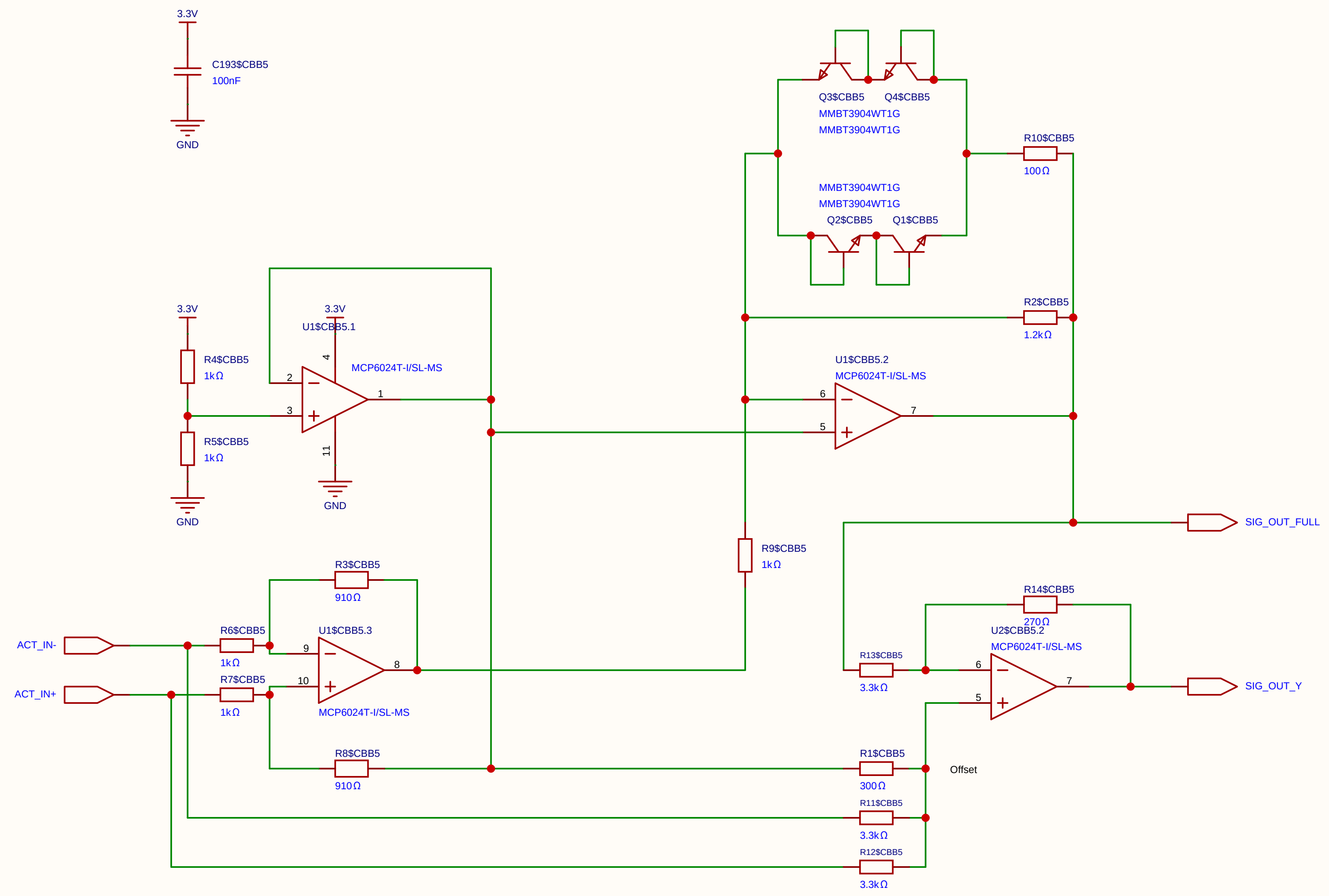
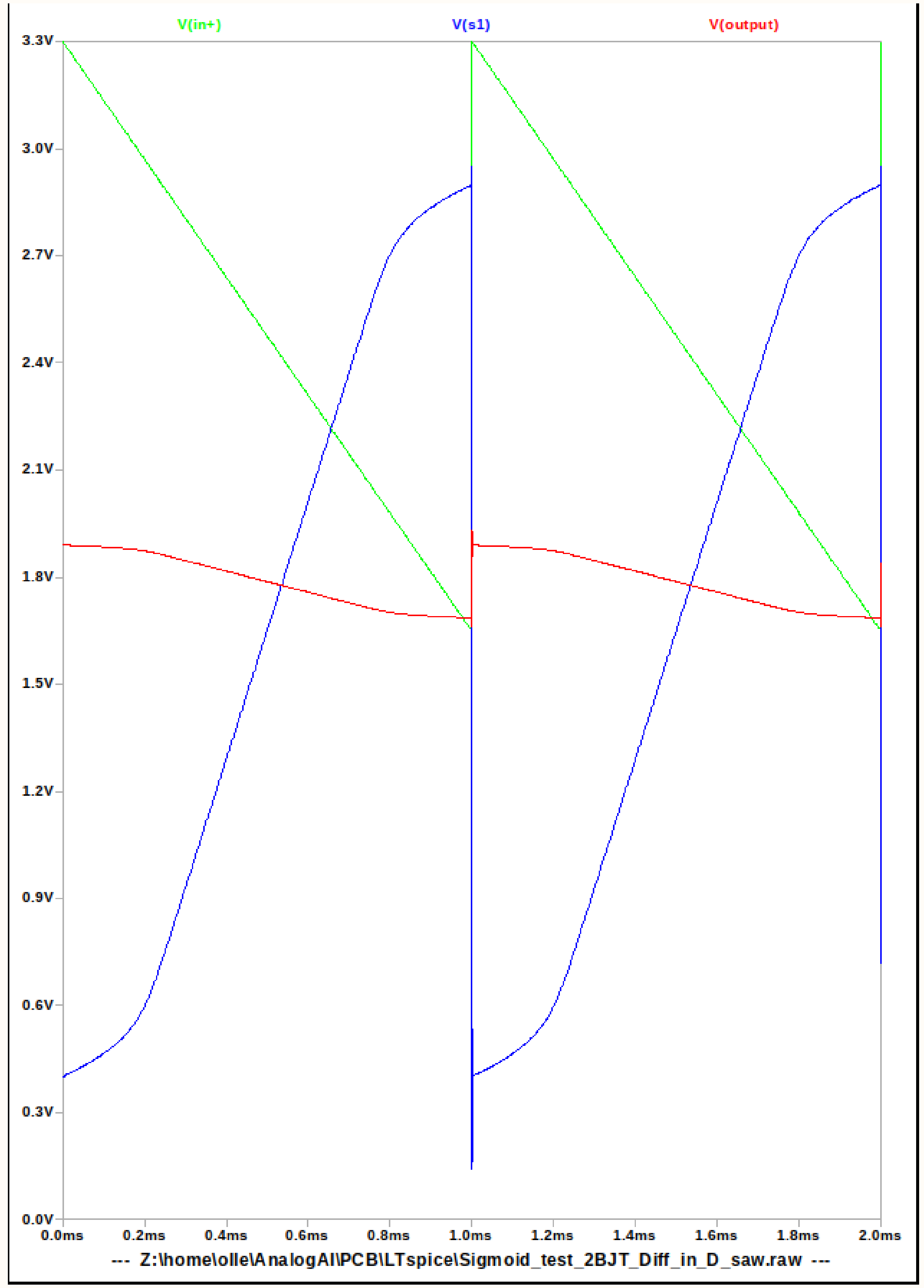
EasyEDA		V1.0	A4	EasyEDA.com	





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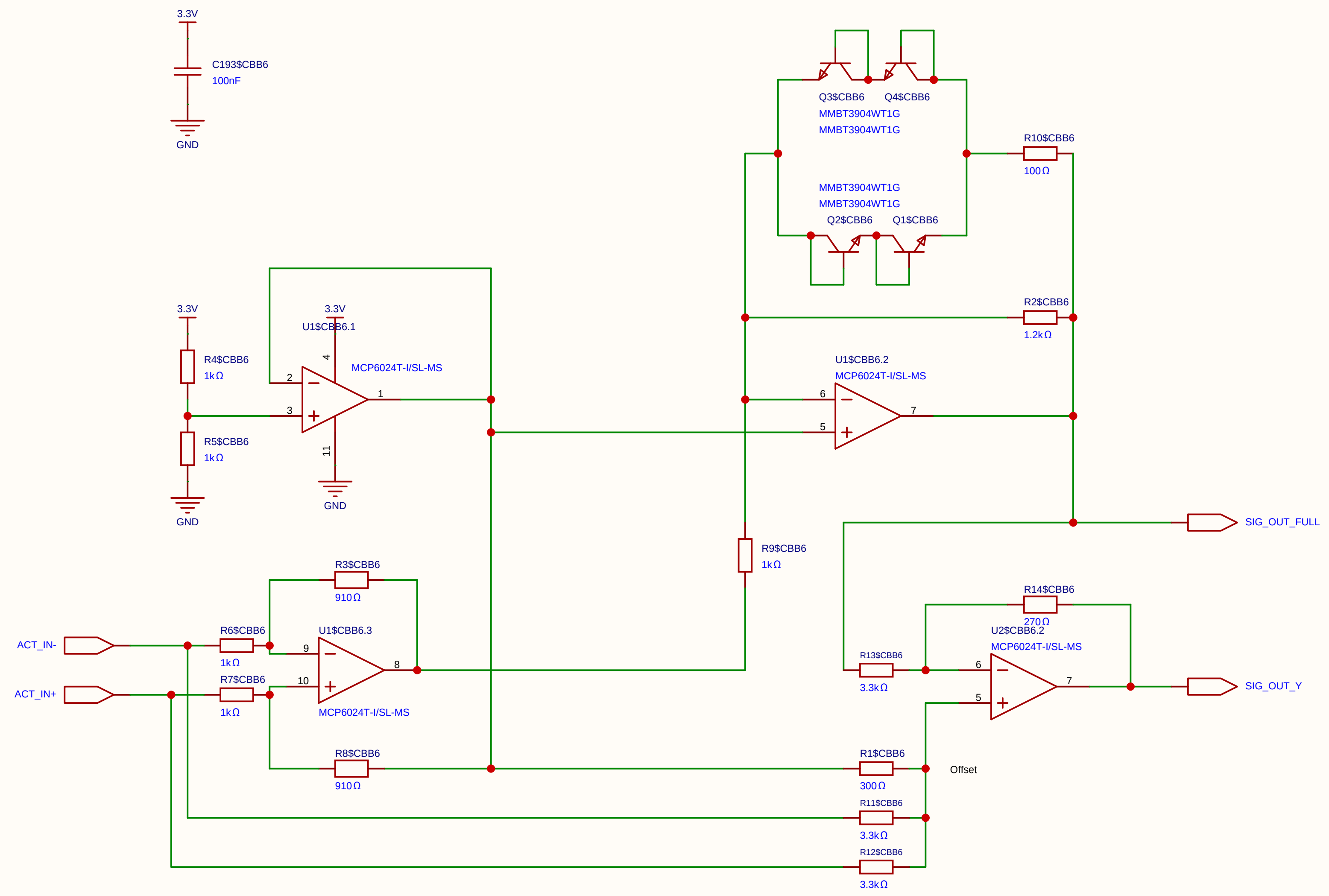

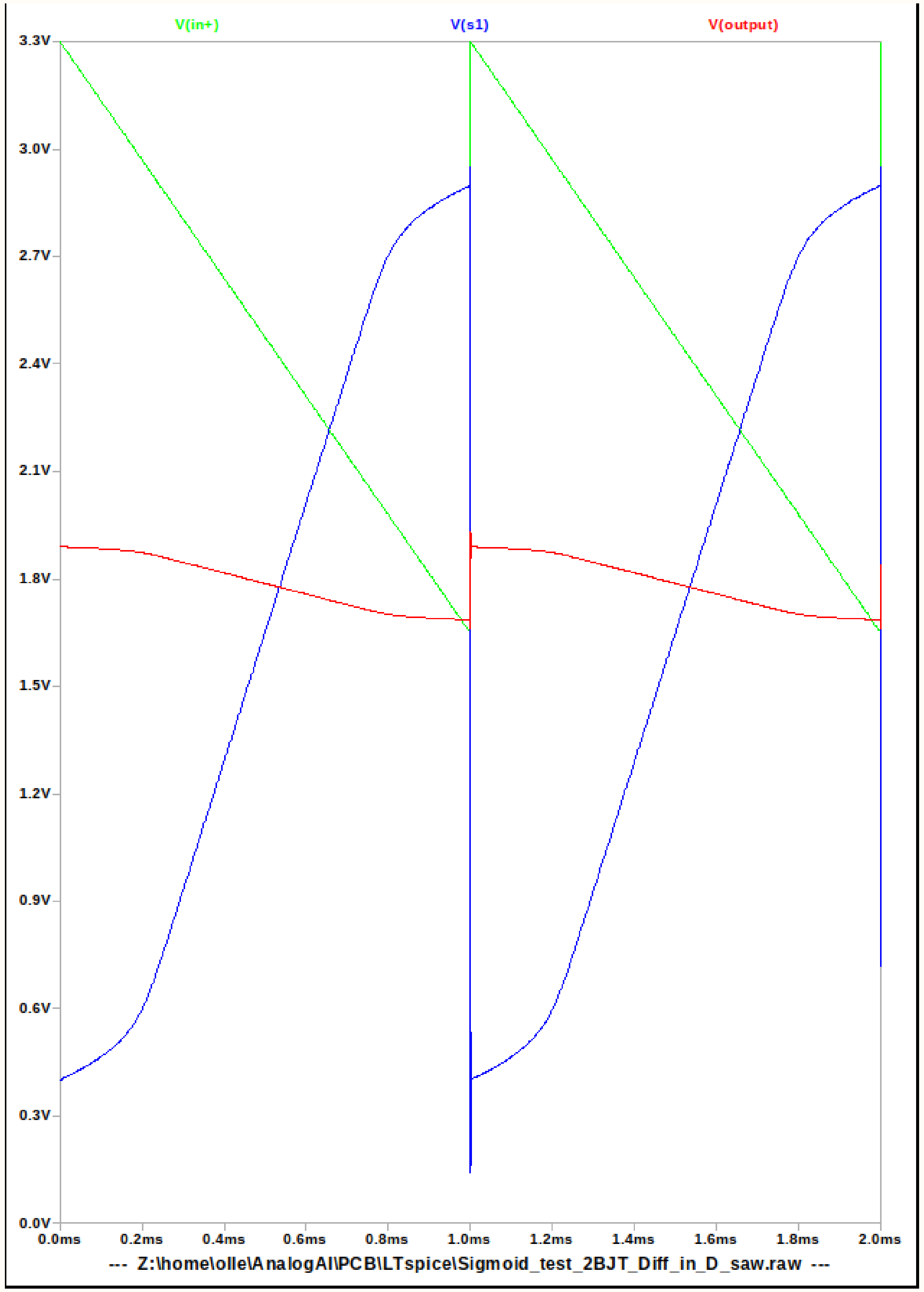

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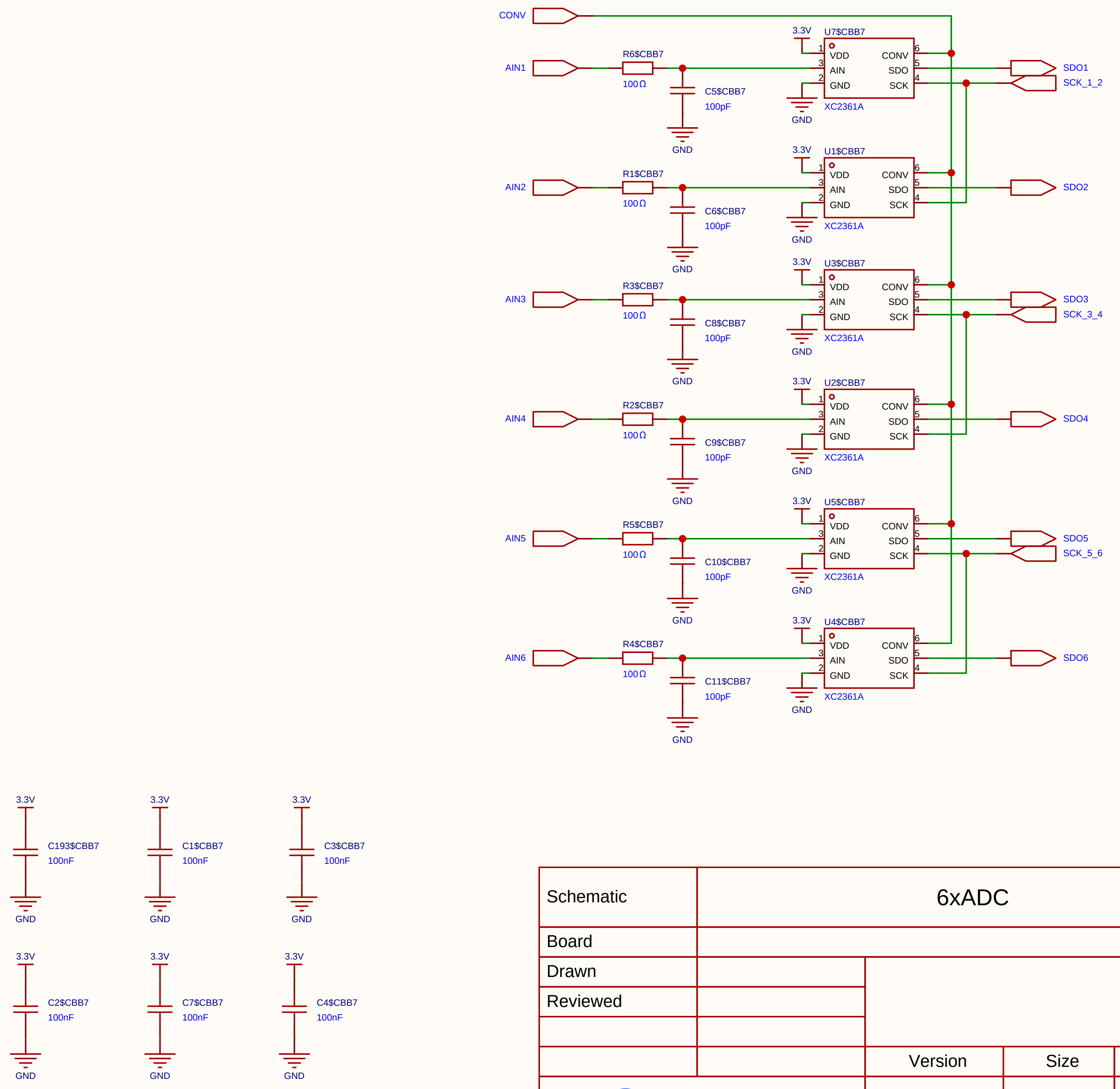

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Reviewed			
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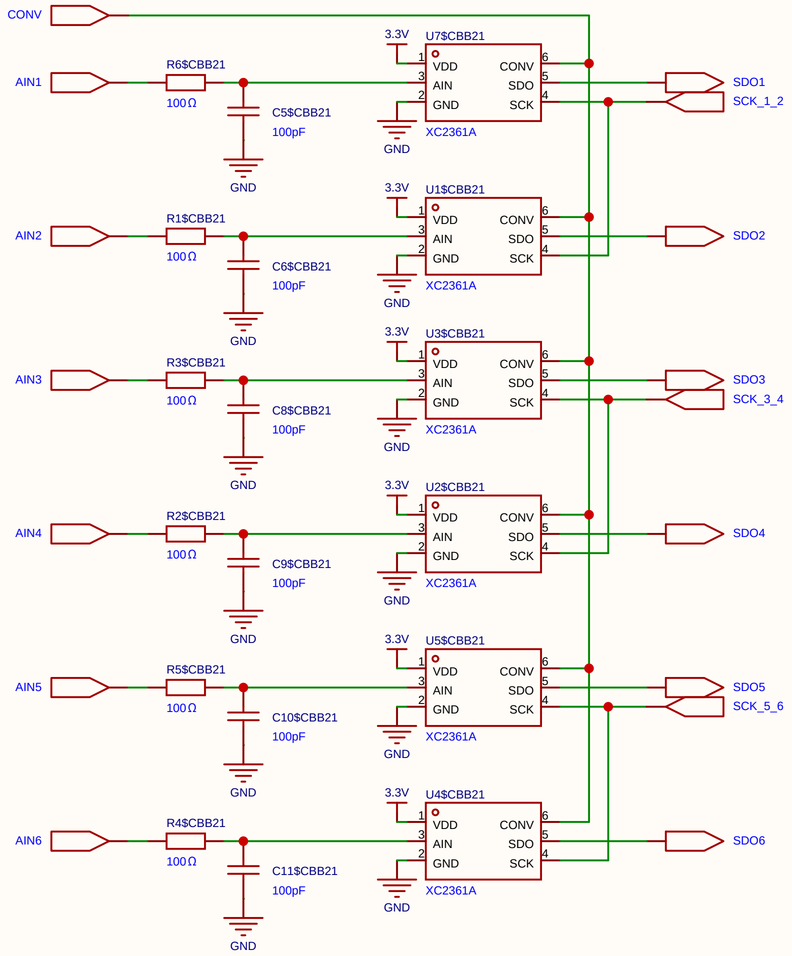
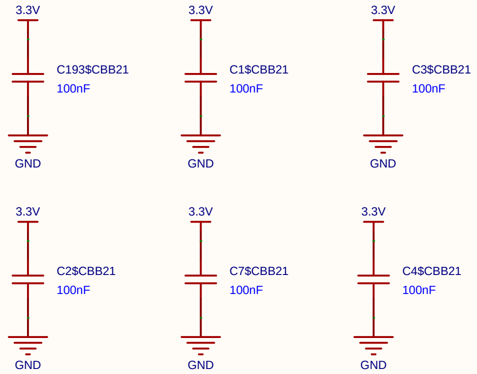
EasyEDA		V1.0	A4	EasyEDA.com





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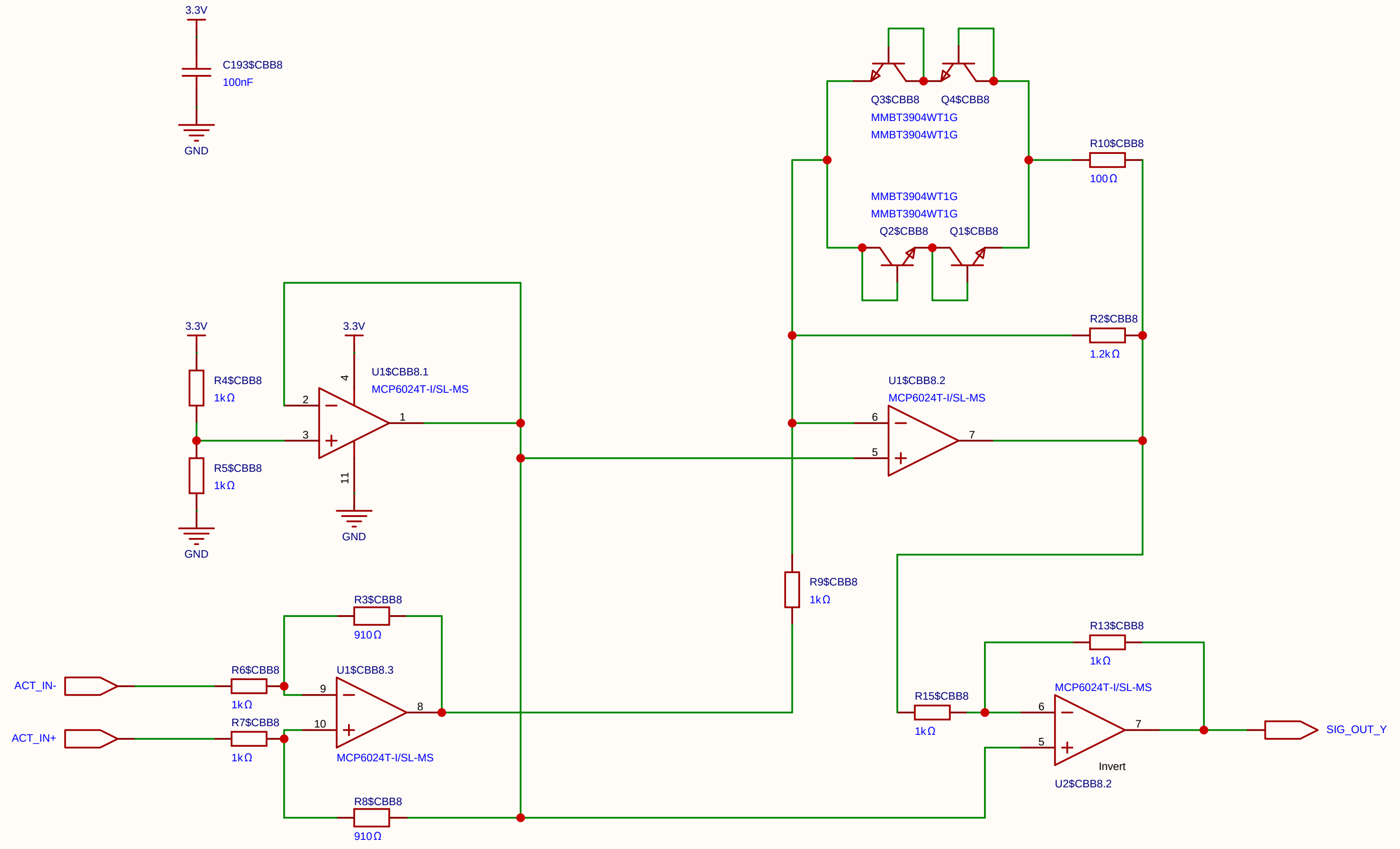

Schematic	SIGMOID_10	Create at	2026-06-19
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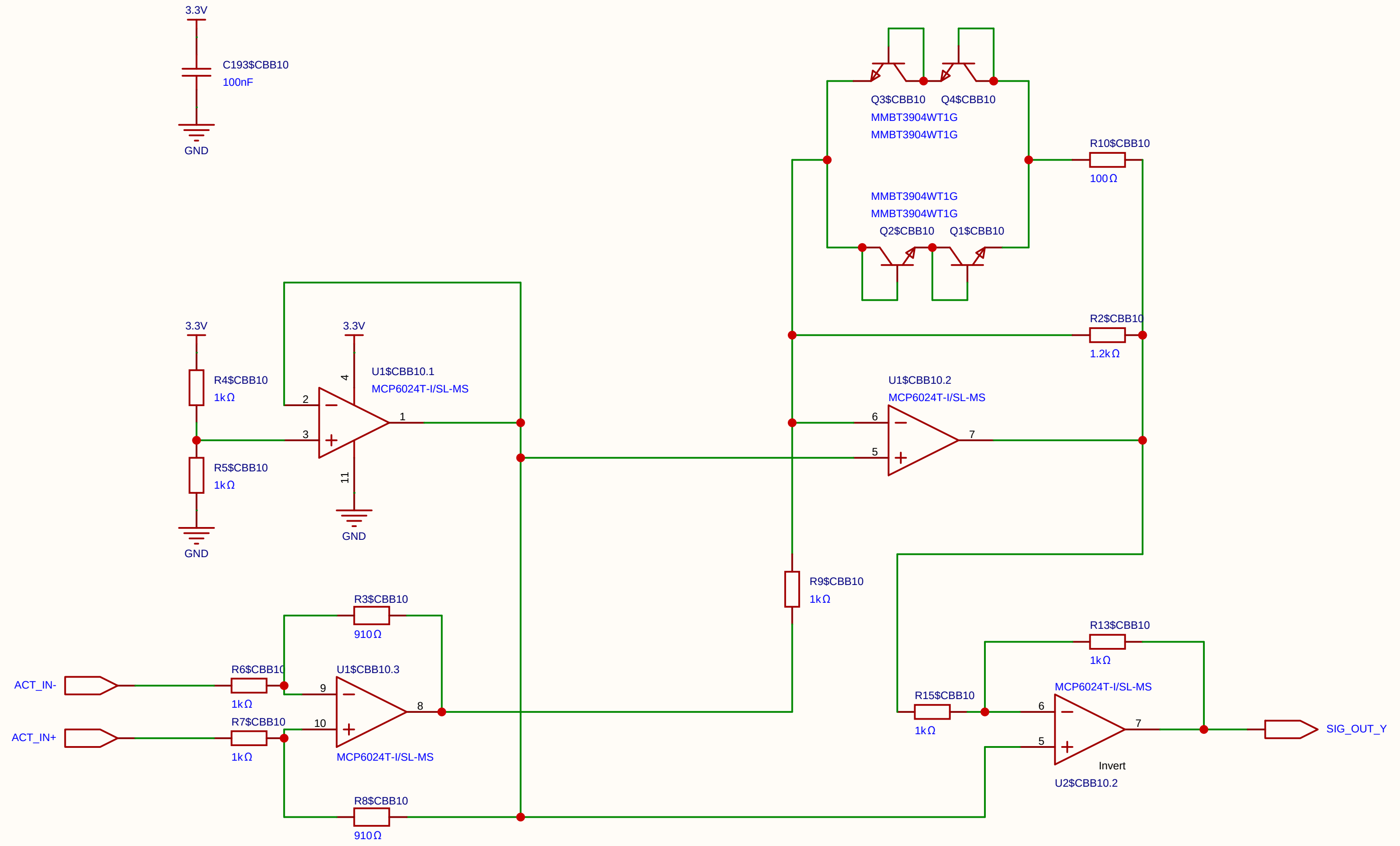
EasyEDA		V1.0 / A4 / EasyEDA.com	



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Reviewed				
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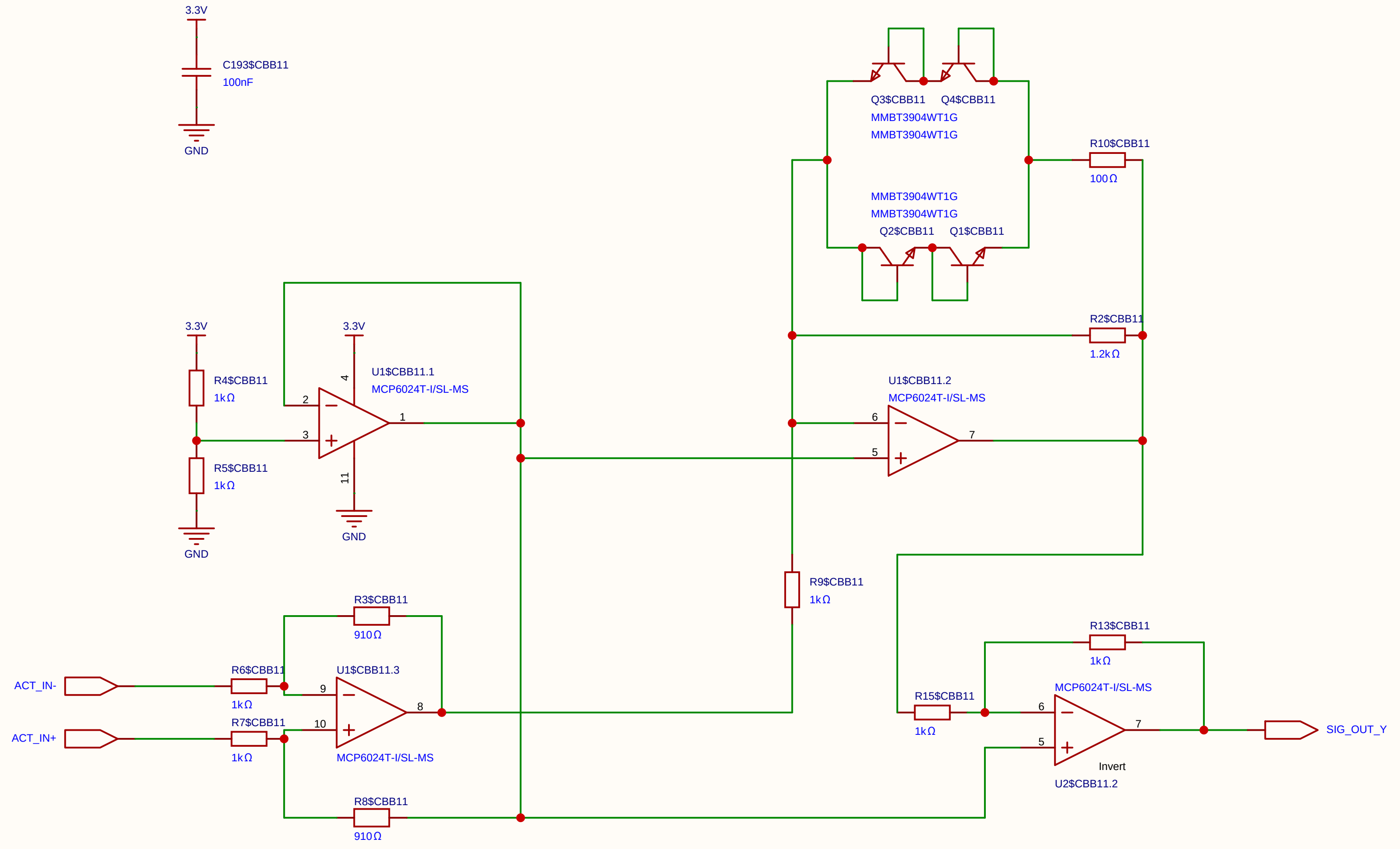

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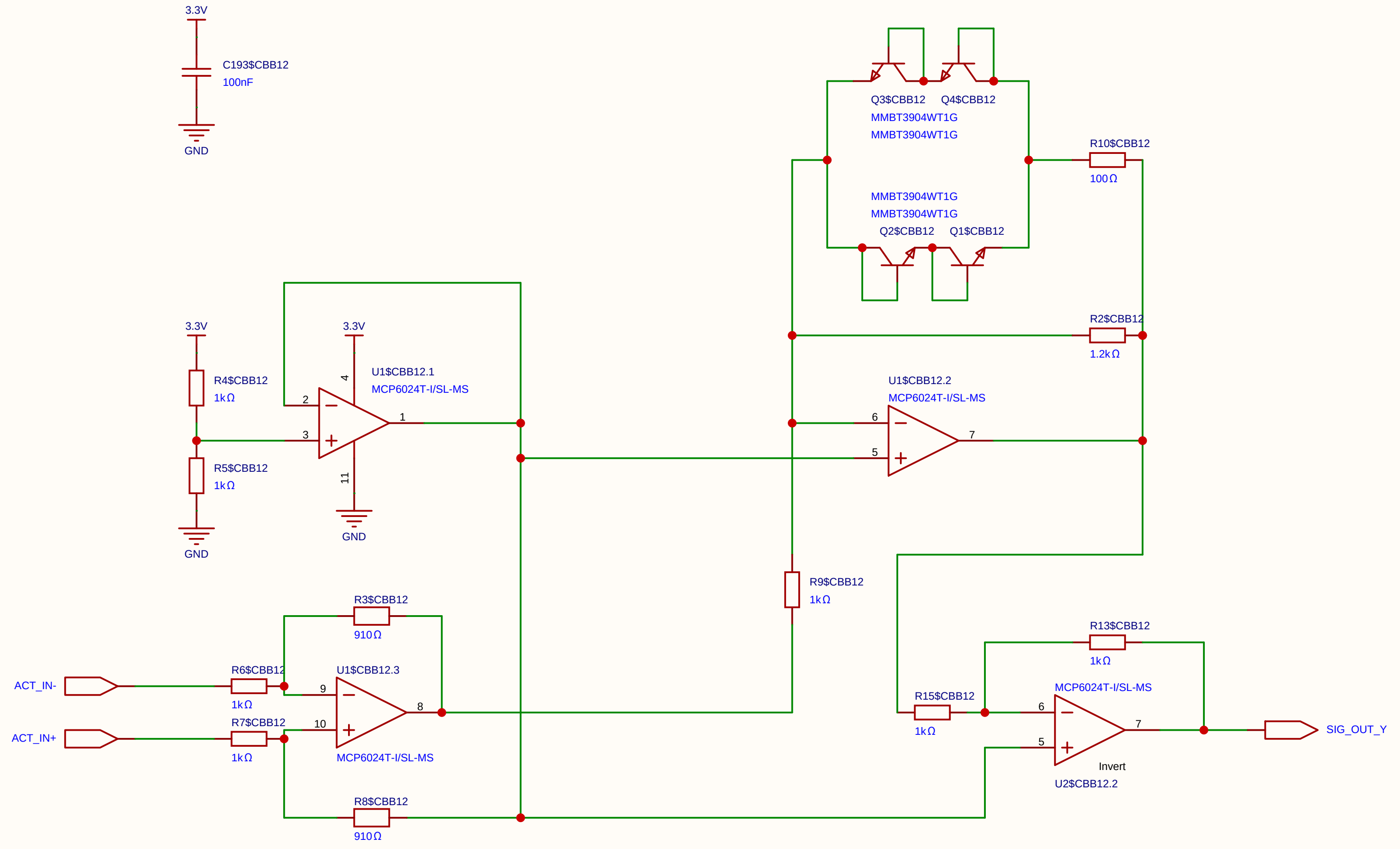

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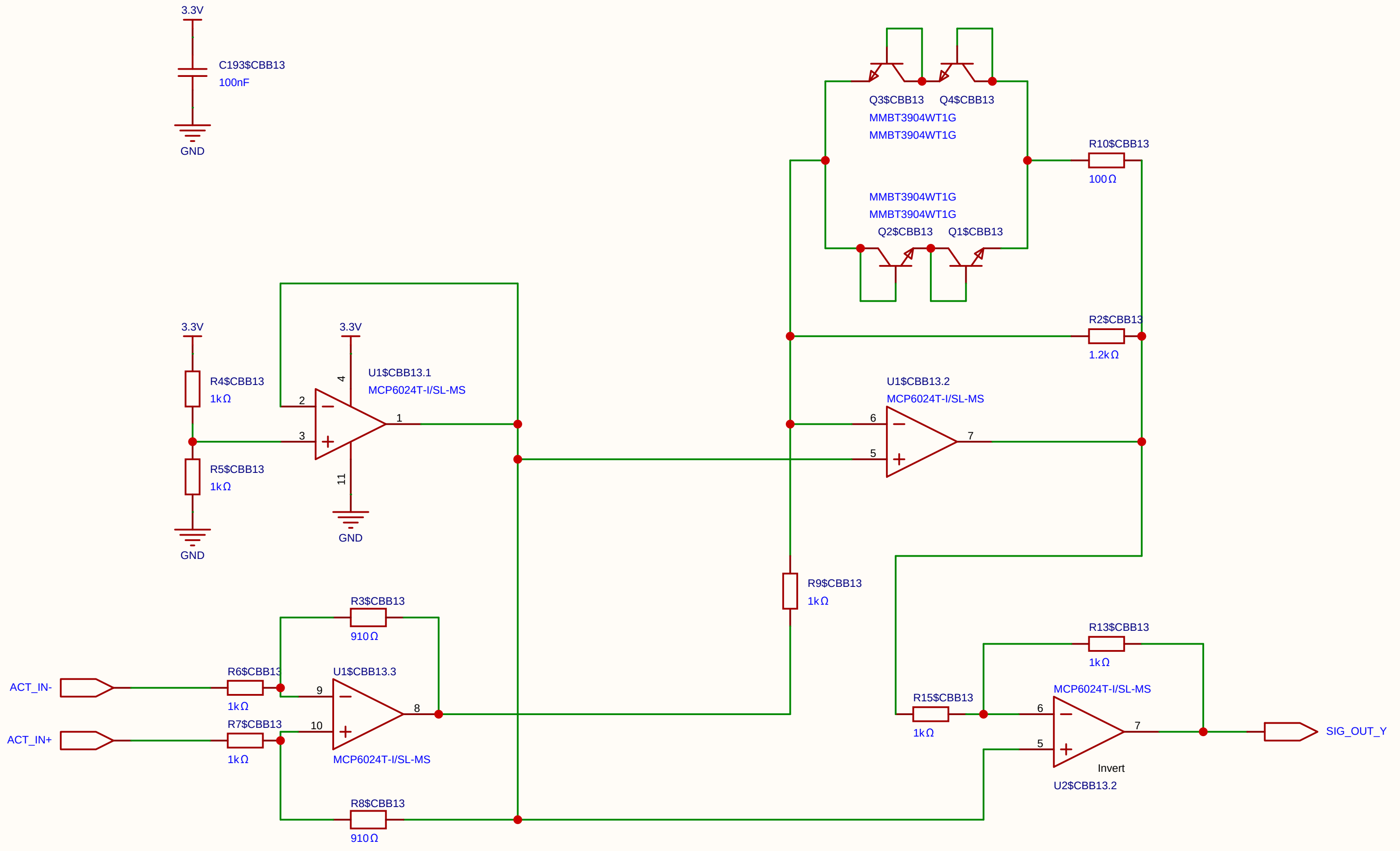
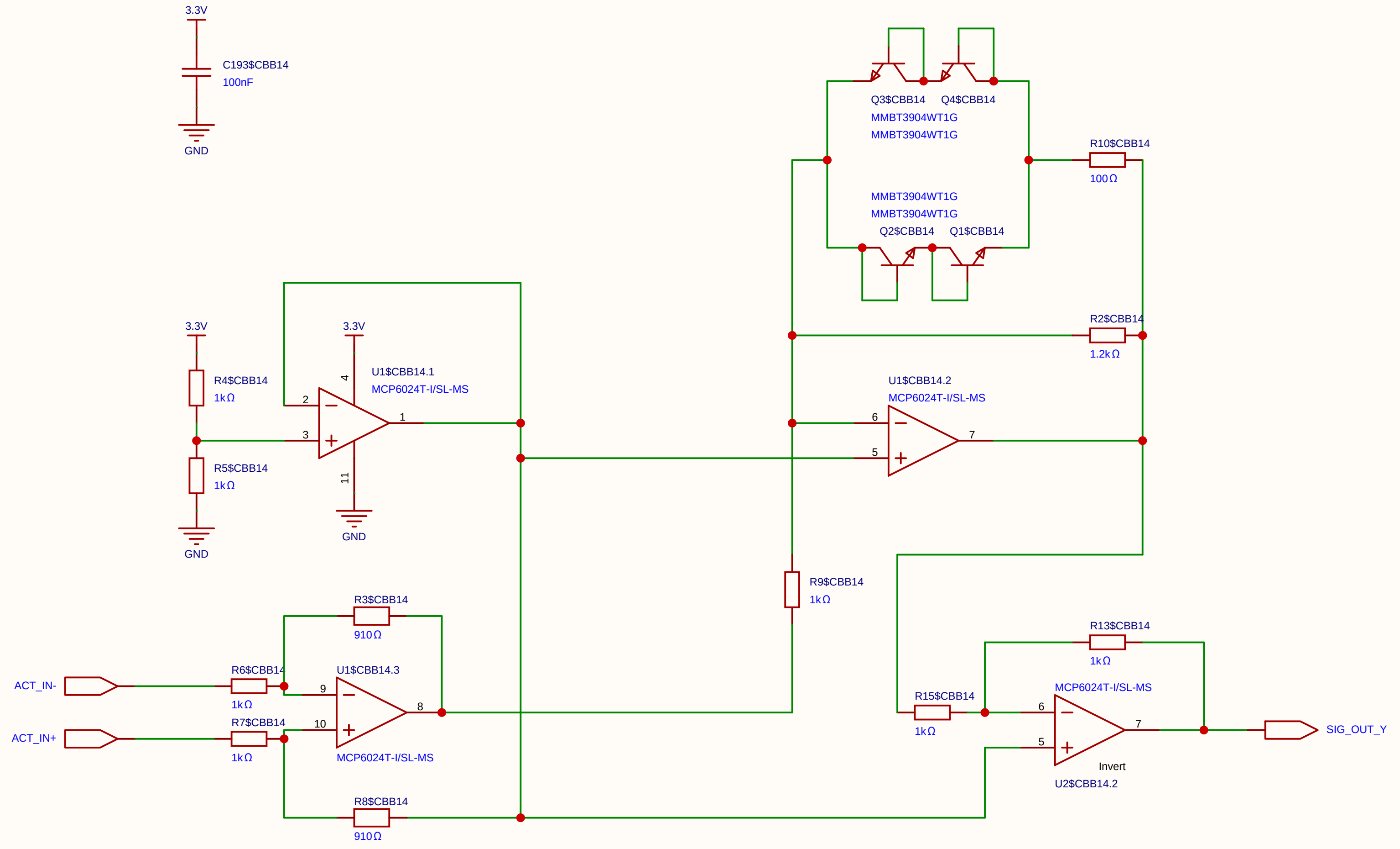
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com



Schematic	SIGMOID_10	Create at	2026-06-19
		Update at	2026-06-19
Board		Page	SIGMOID(10)
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com



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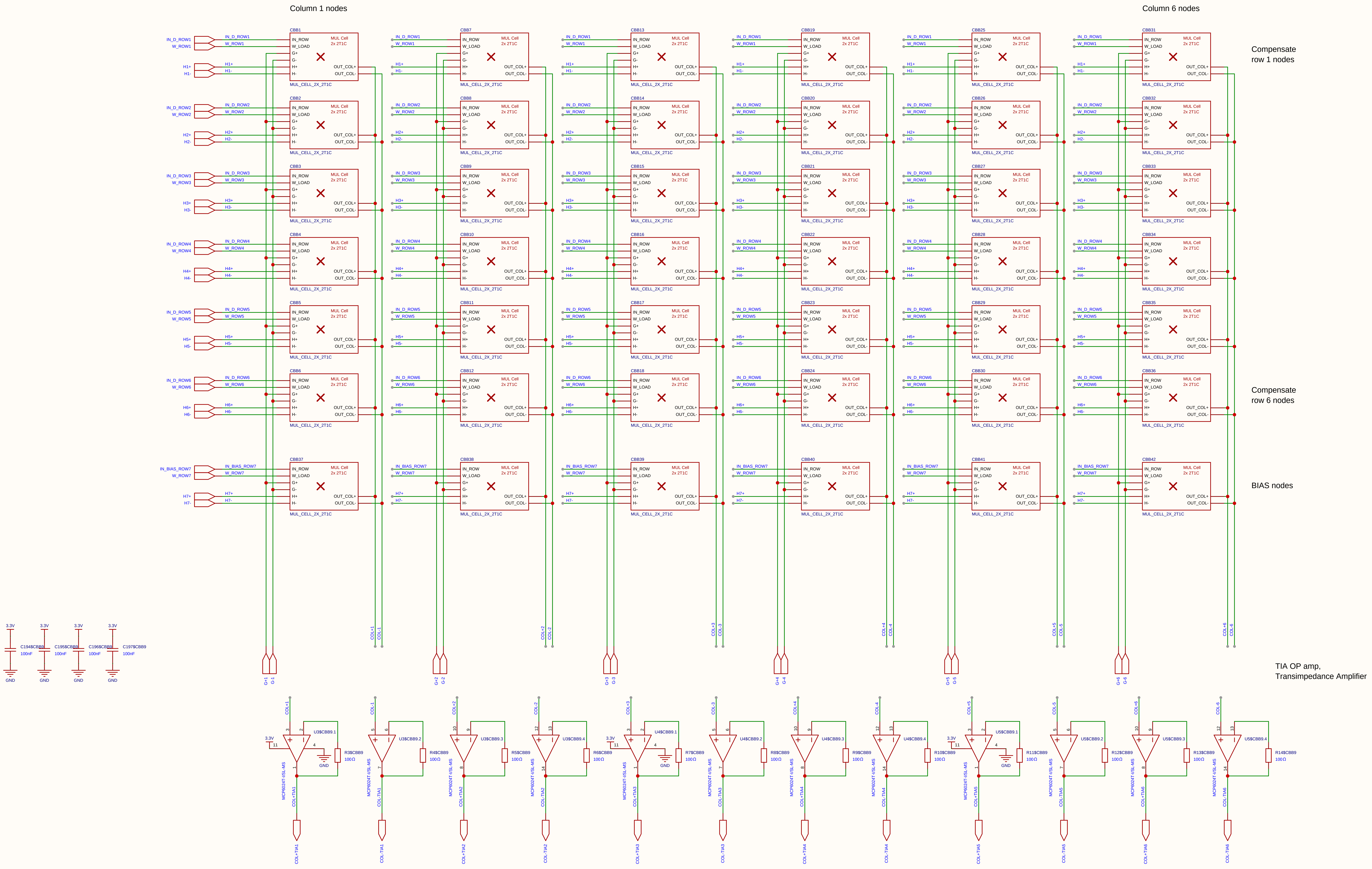
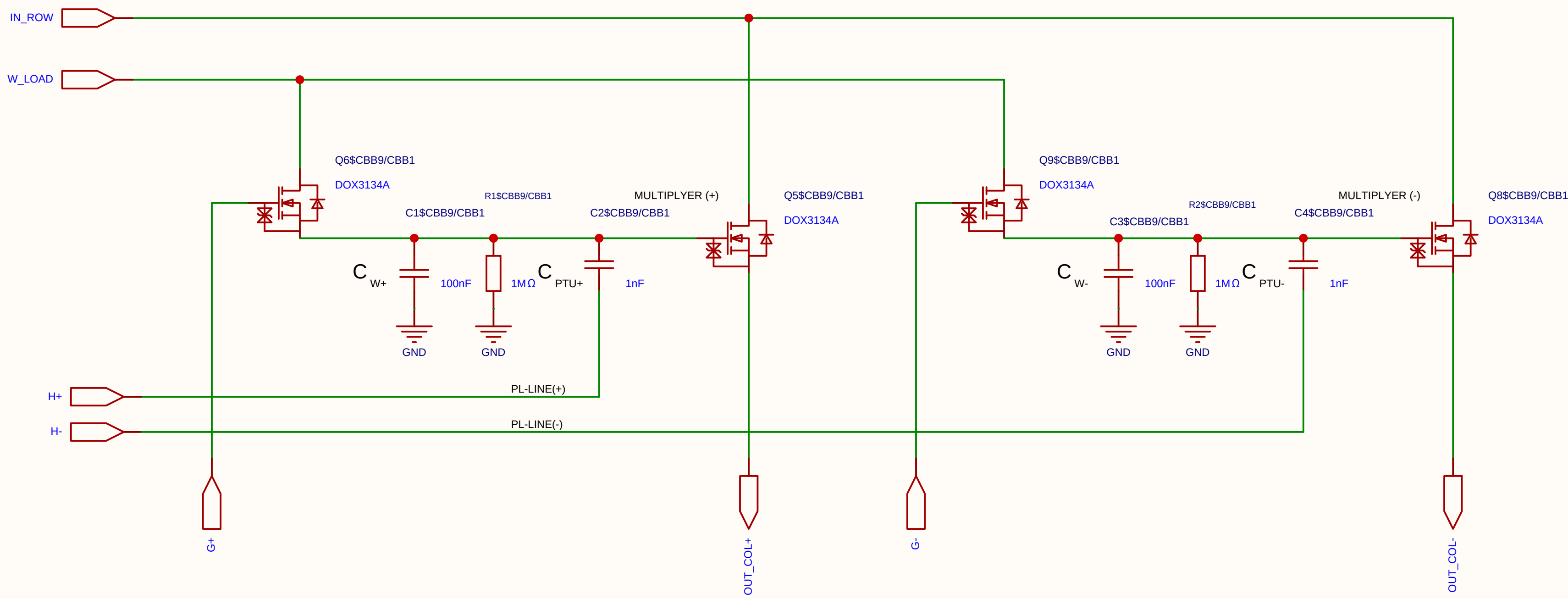
Board		Page	SIGMOID(10)
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
		V1.0	A4
EasyEDA		Page 1 Total 1	EasyEDA.com

$$V_{ut\_TIA} = 1.65\ \mathrm{V} - (5\ \mathrm{mA} \times 100\ \Omega) = 1.65\ \mathrm{V} - 0.5\ \mathrm{V} = \mathbf{1.15}\ \mathrm{V}$$



Schematic	matrix8	Create at	2026-06-19
		Update at	2026-06-19
Board		Page	matx8
Drawn		Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 μs**	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	**2 μs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

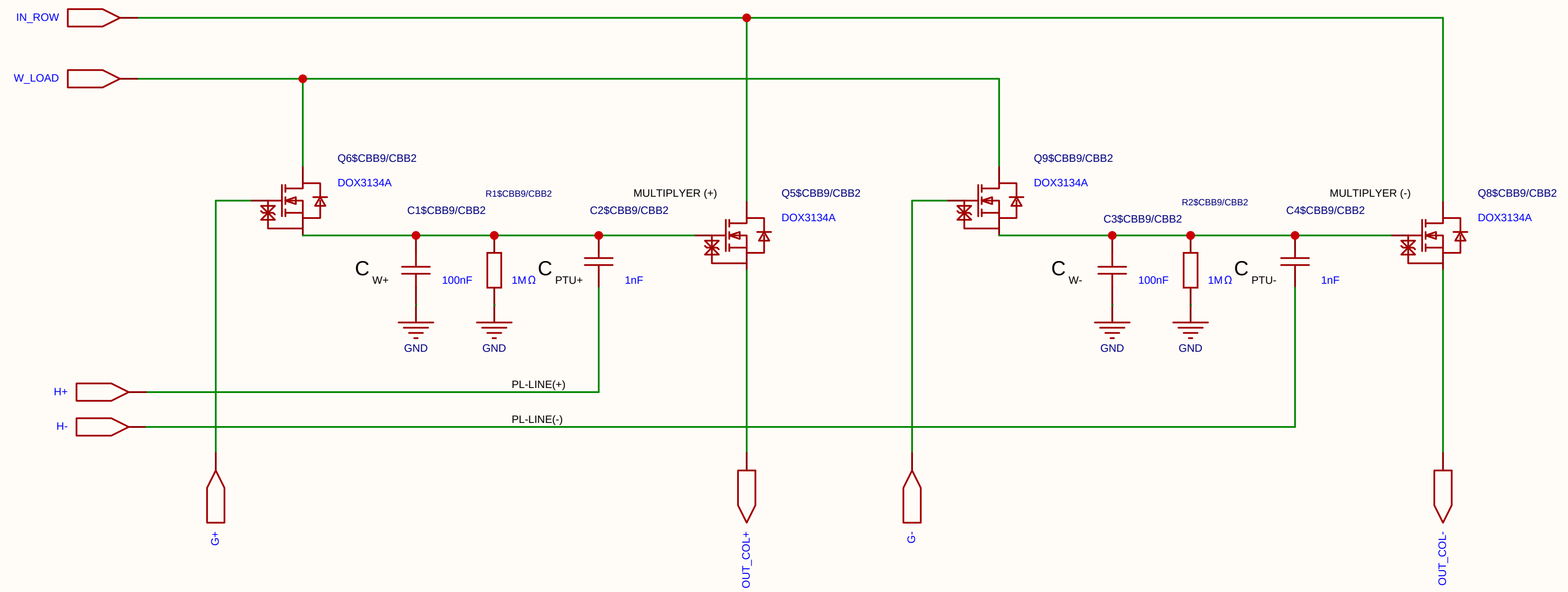
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
Version	Size	Page 1 Total 1	
V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

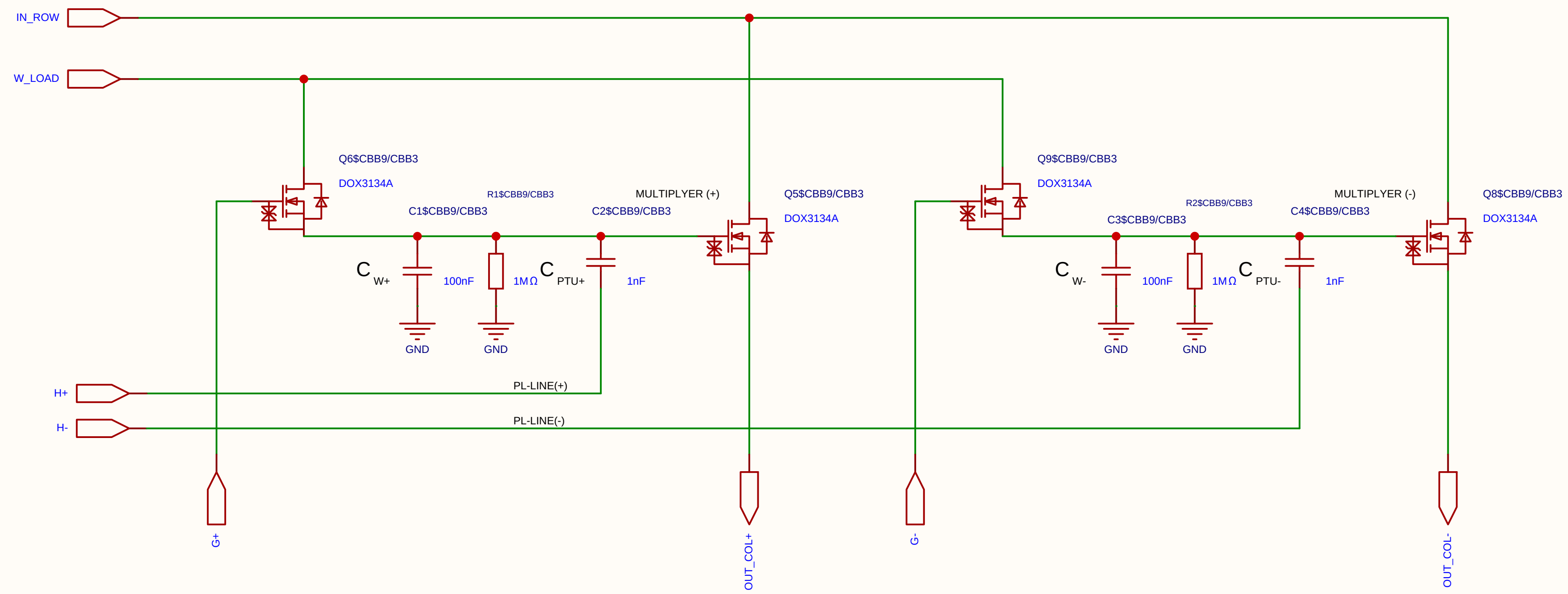
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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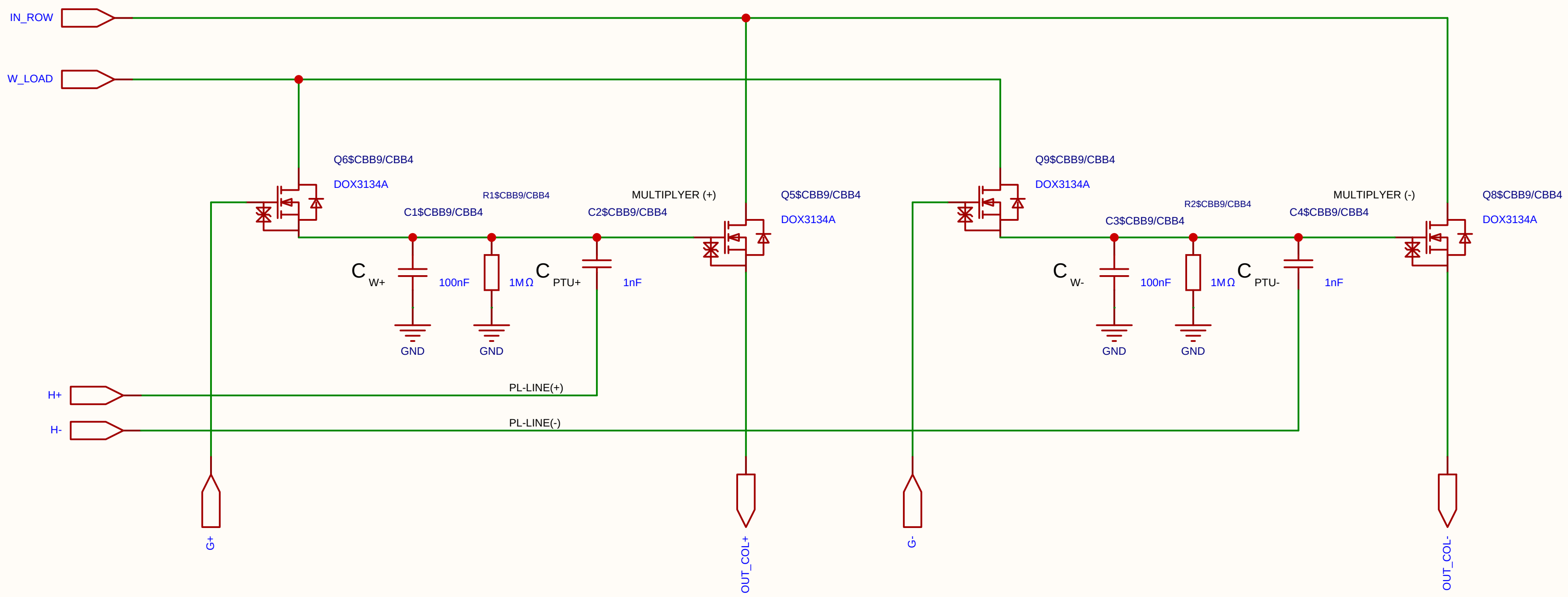
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4

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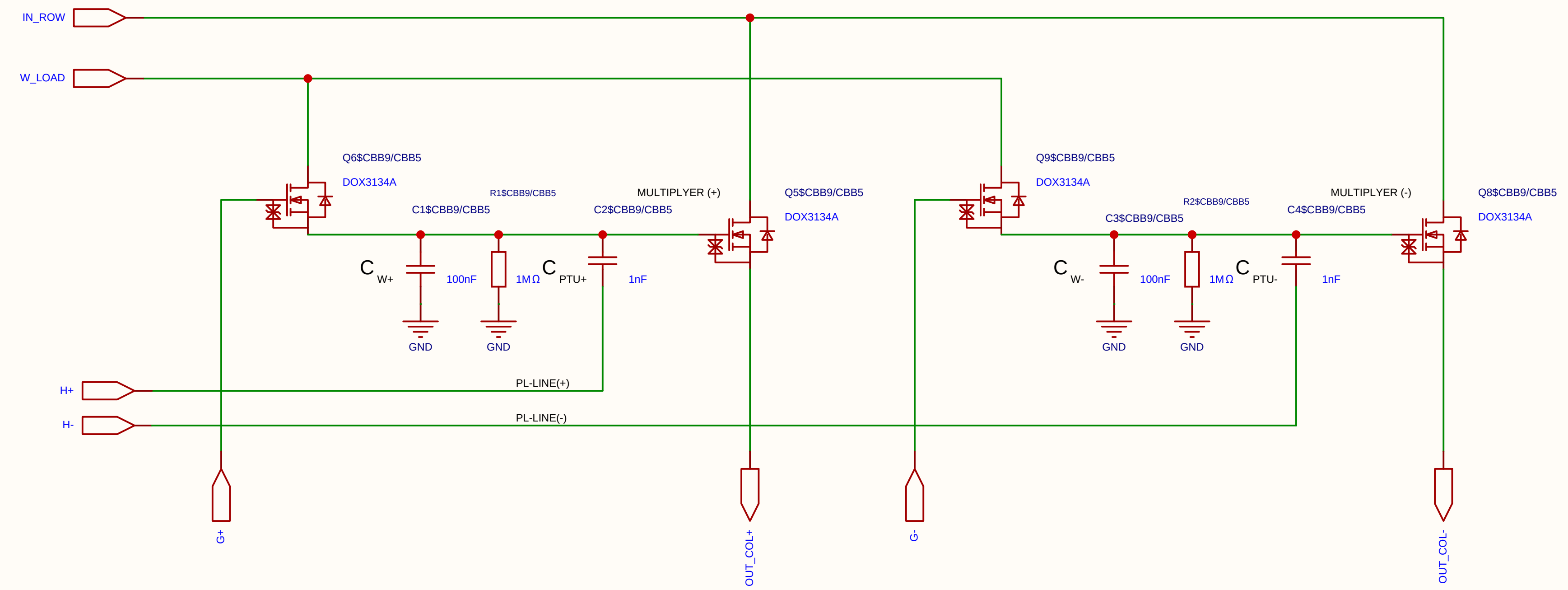
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

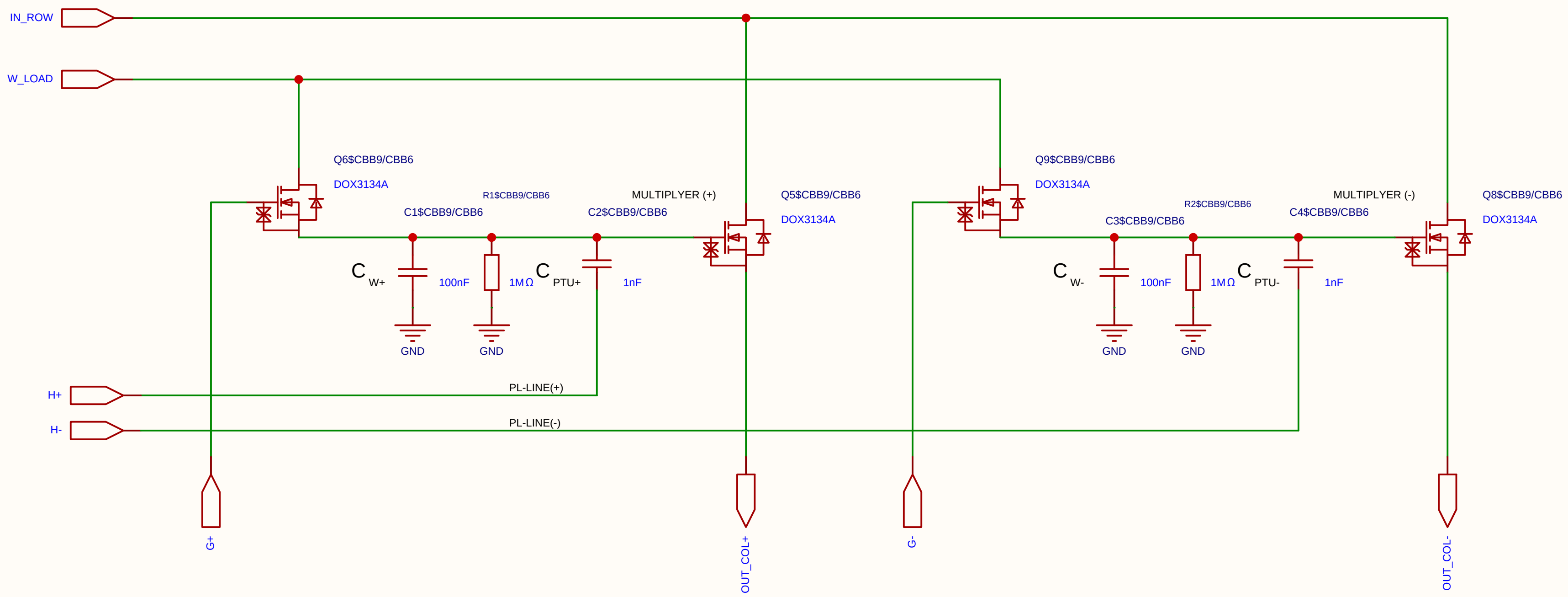
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

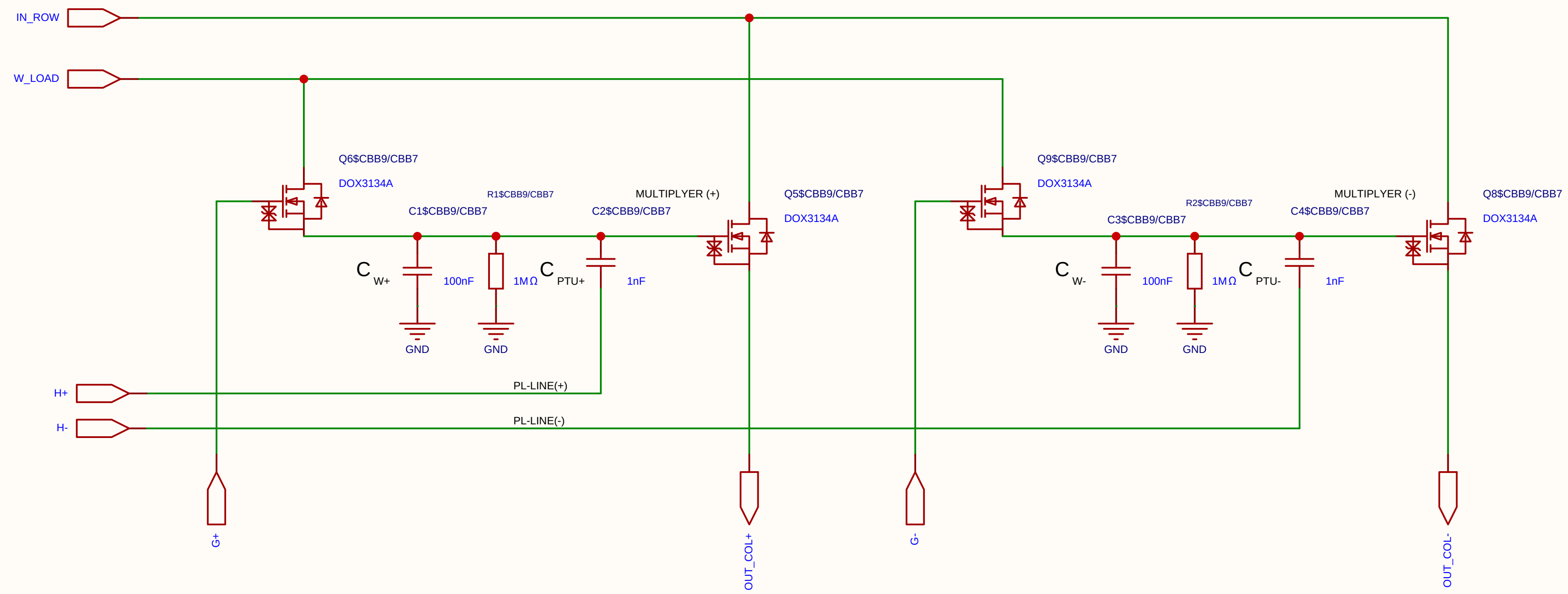
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4

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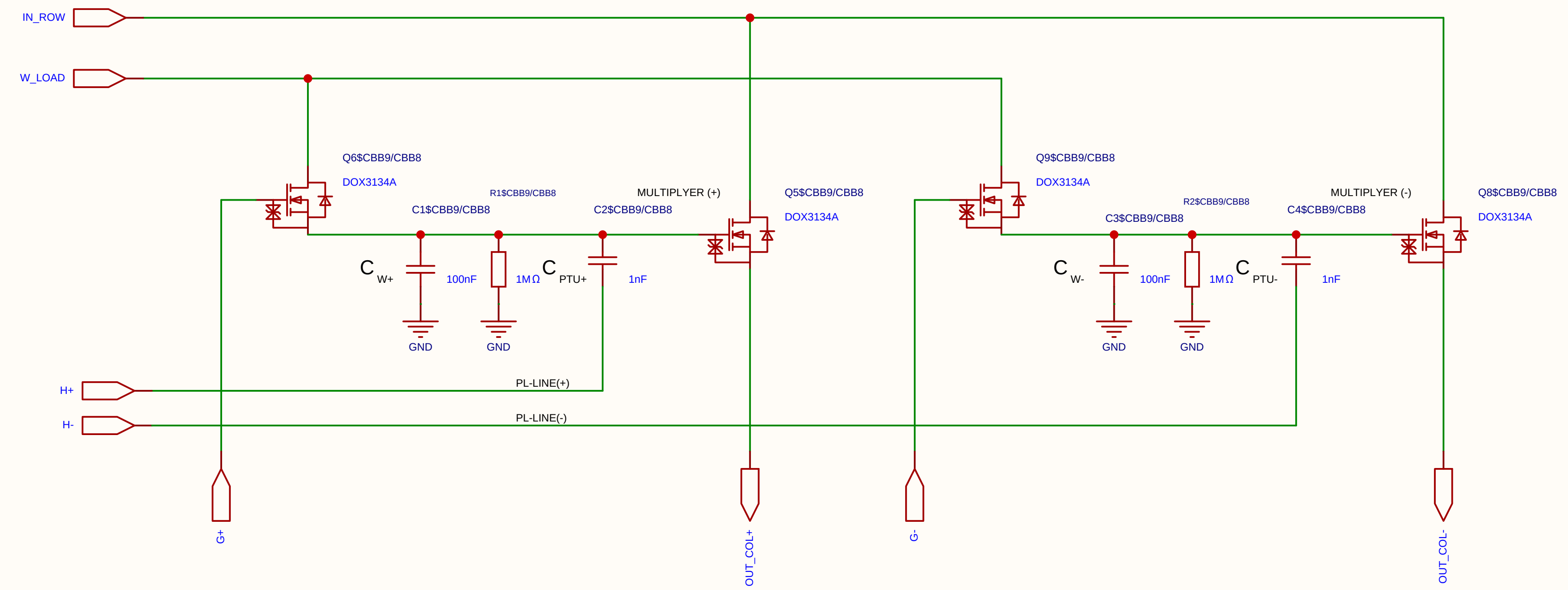
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

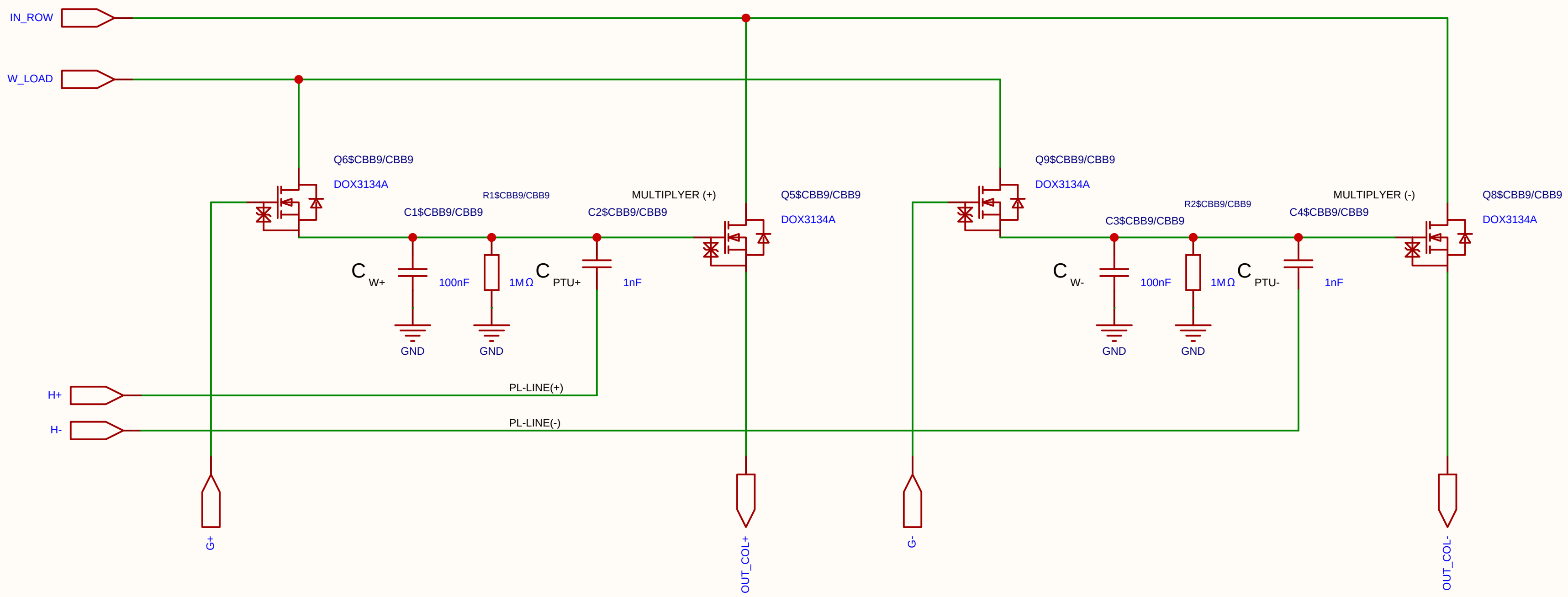
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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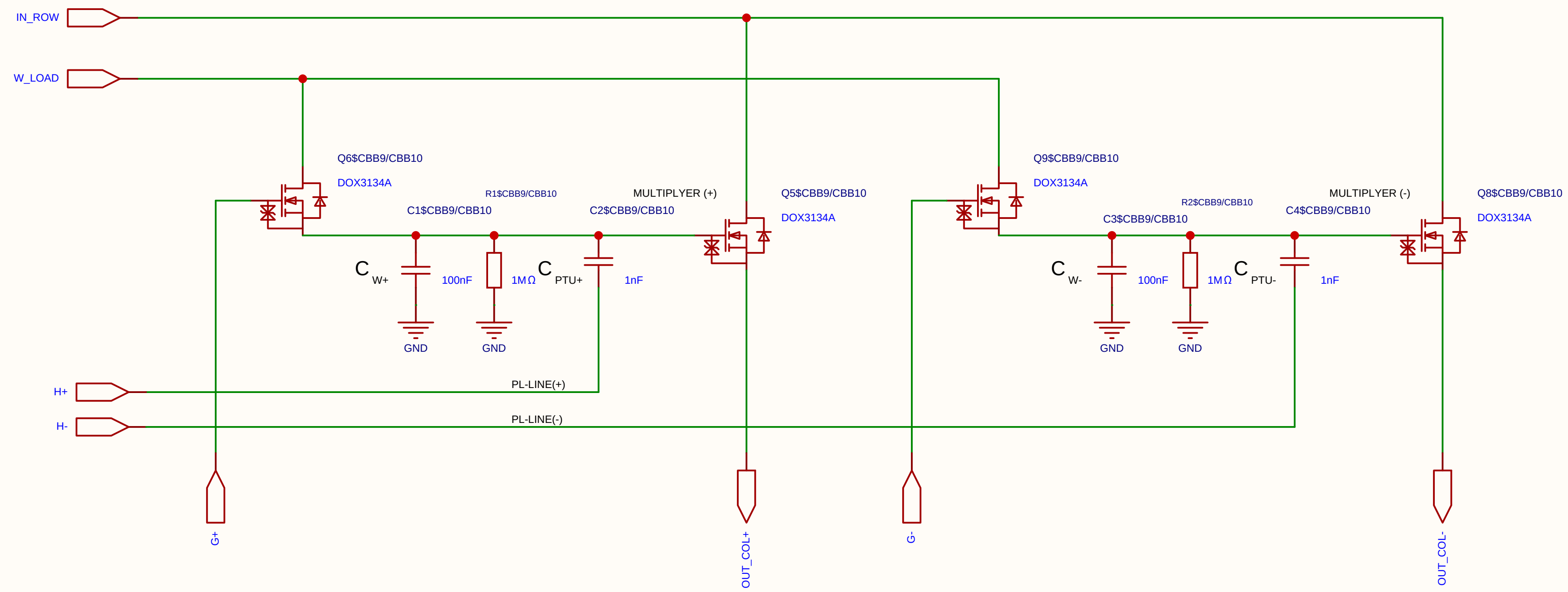
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4

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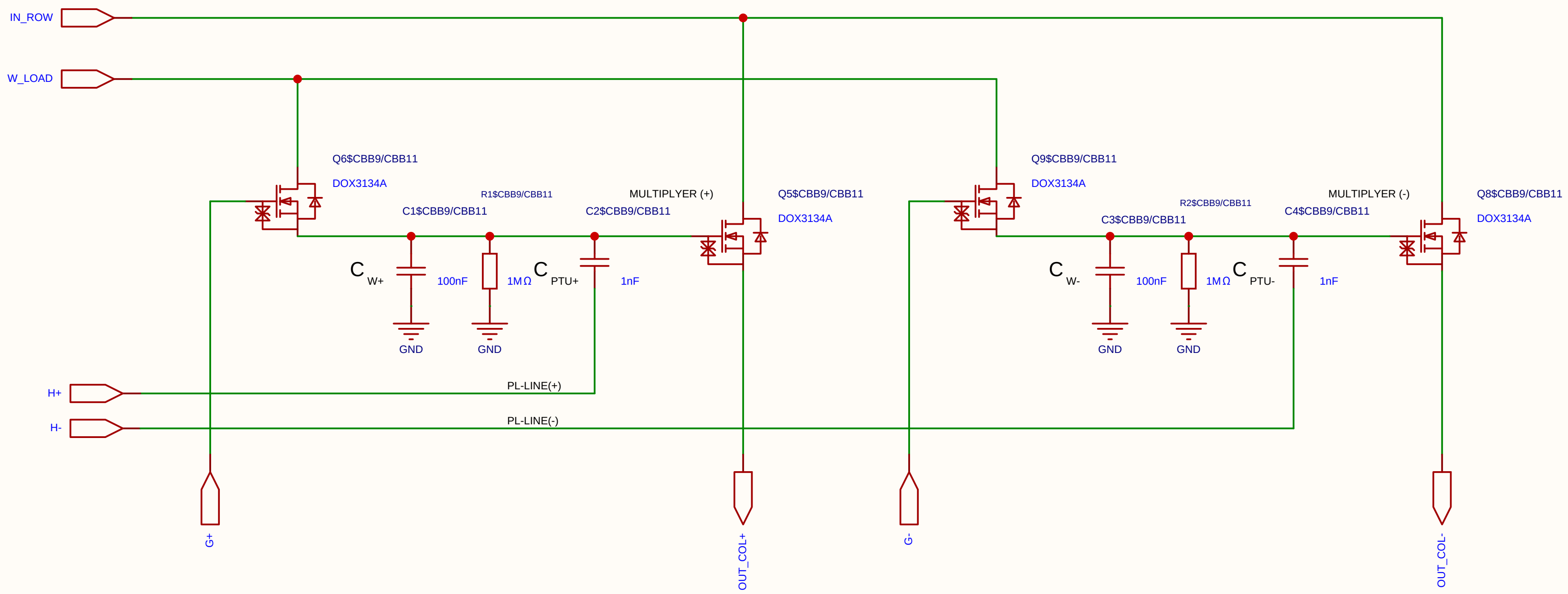
Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

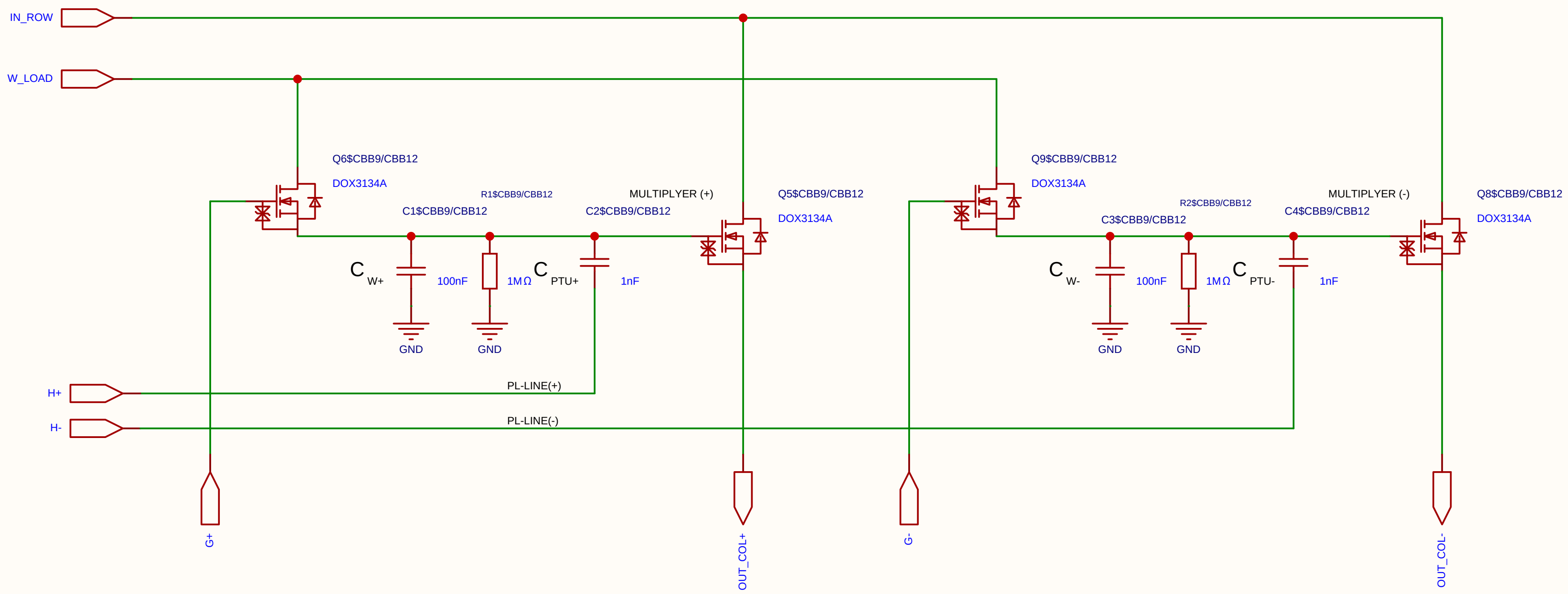
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

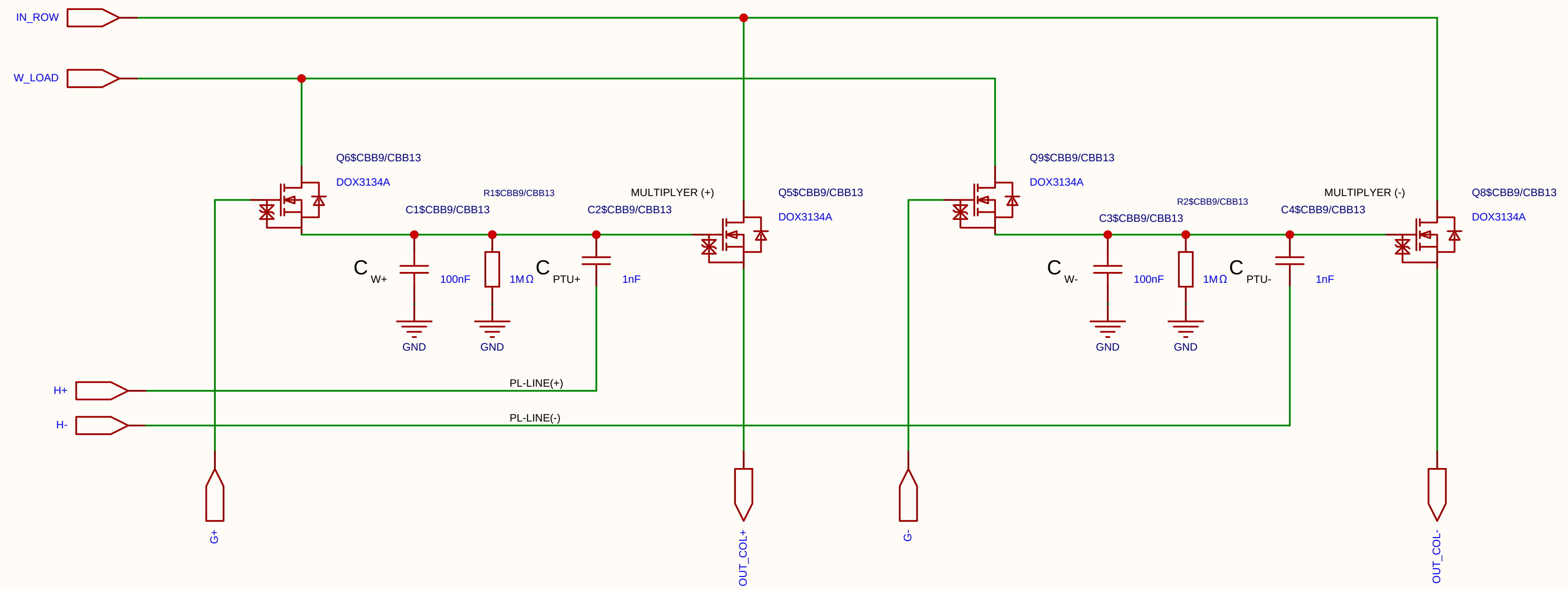
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
		Version / Size	Page 1 Total 1
EasyEDA		V1.0 / A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
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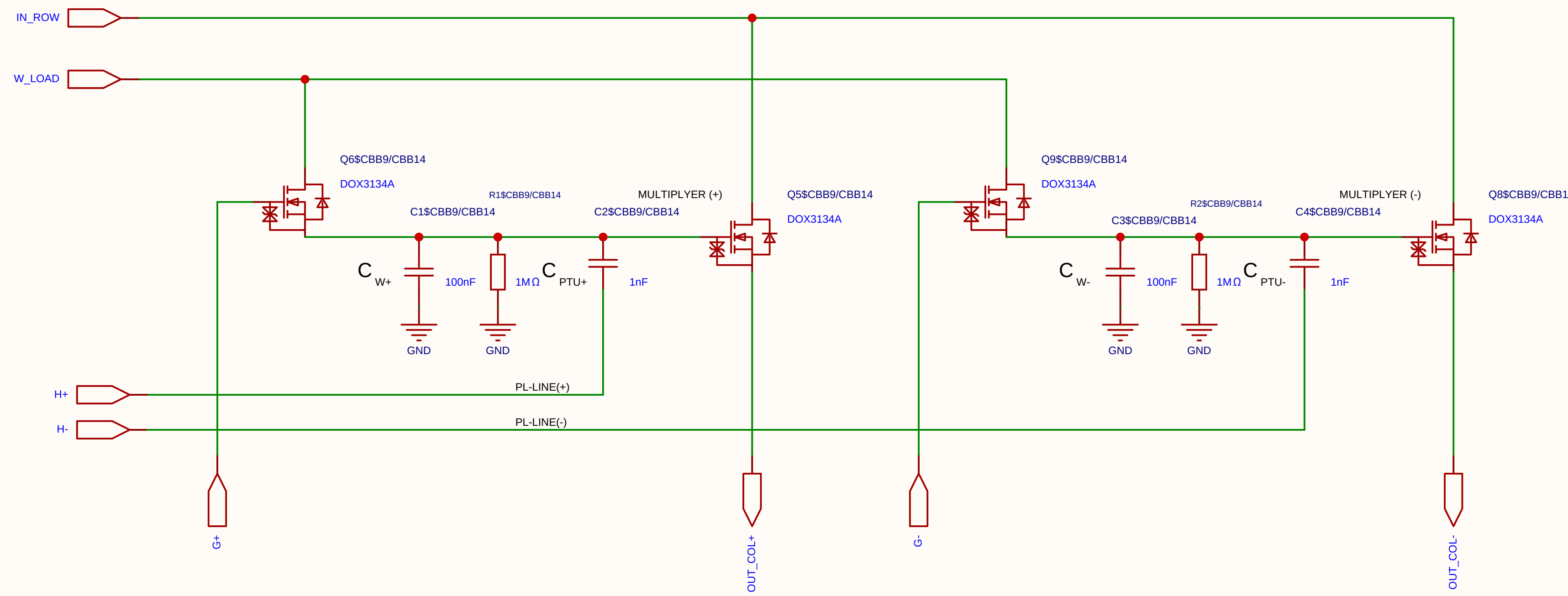
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
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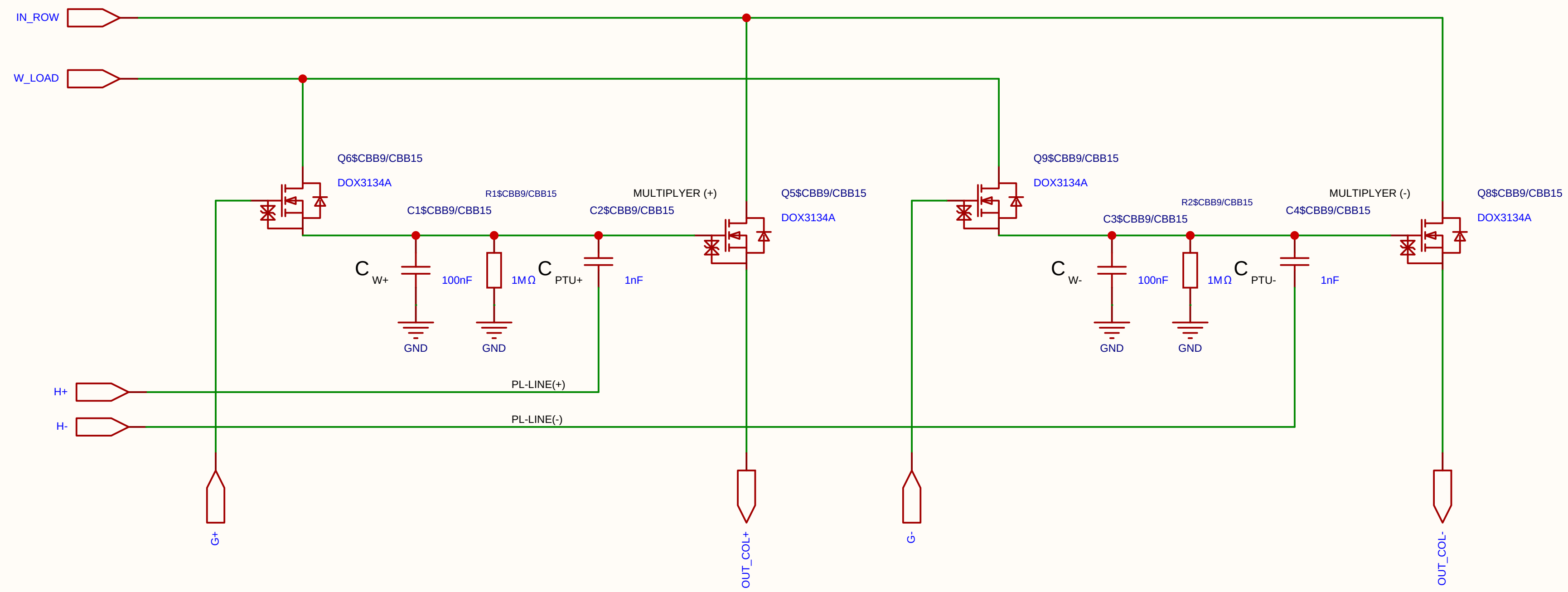
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

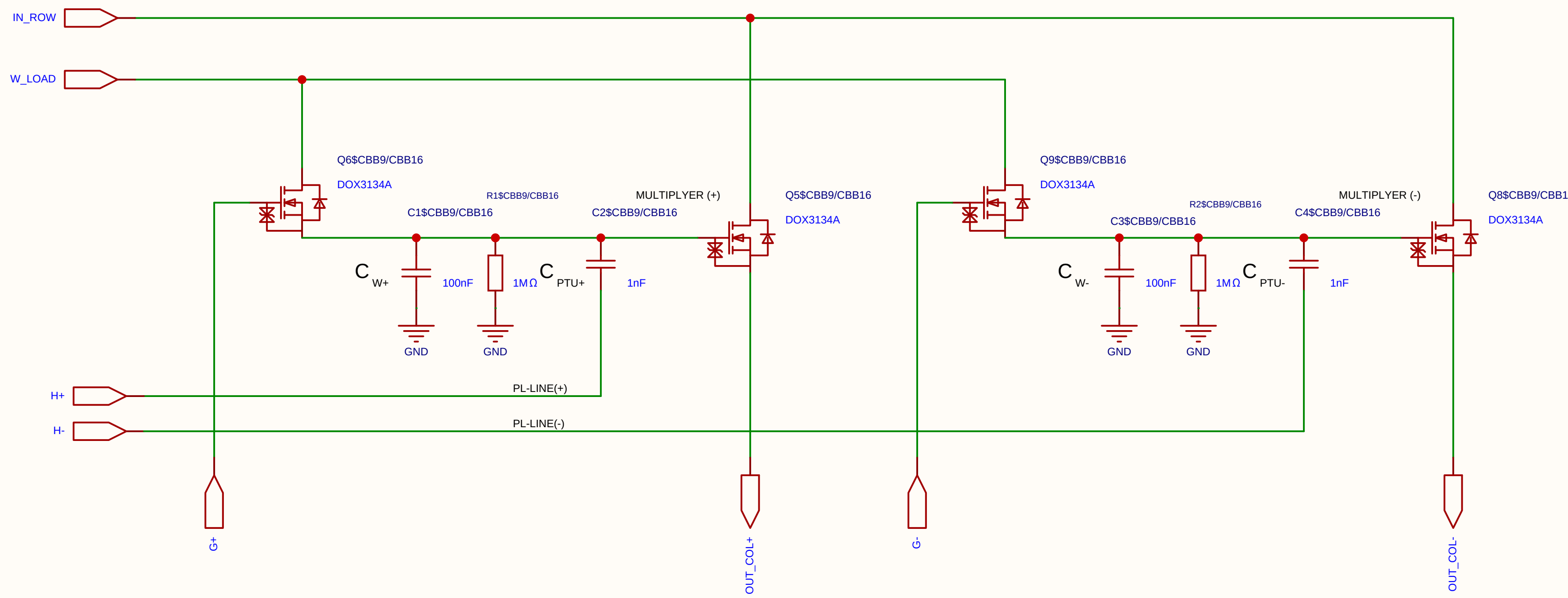
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C		Create at	2026-06-14
			Update at	2026-06-19
Board			Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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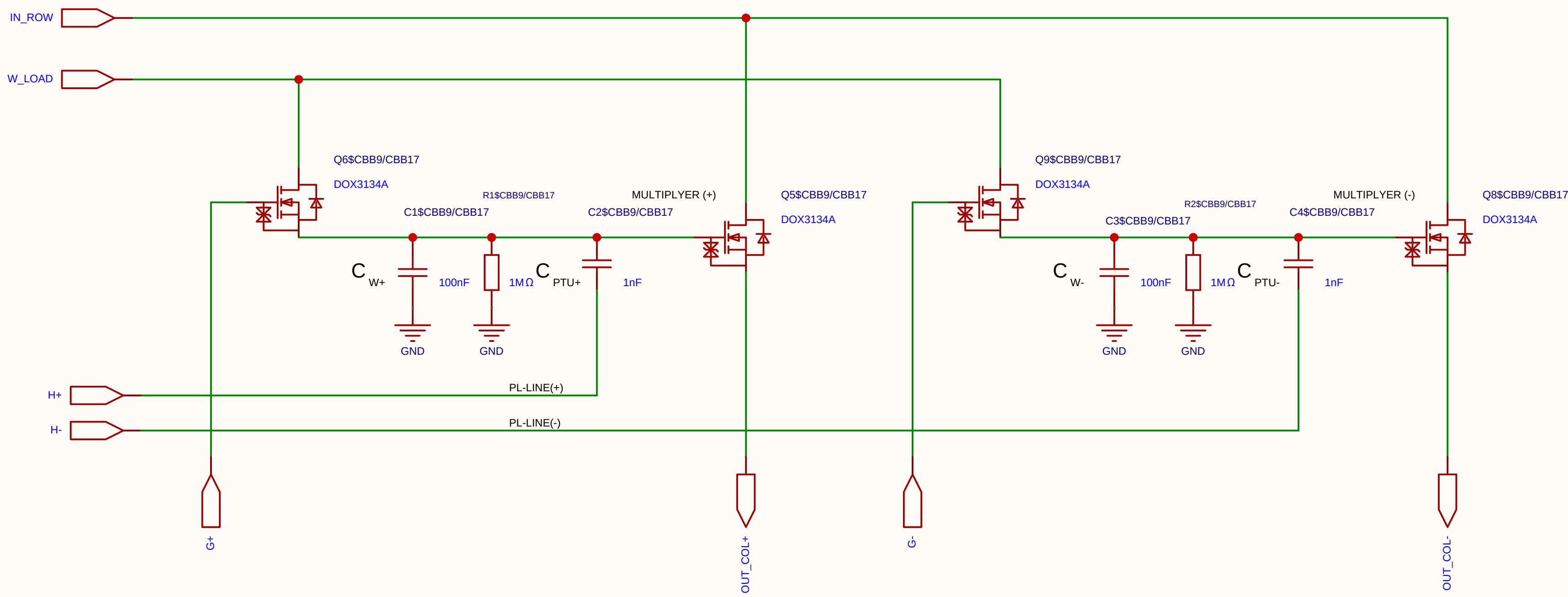
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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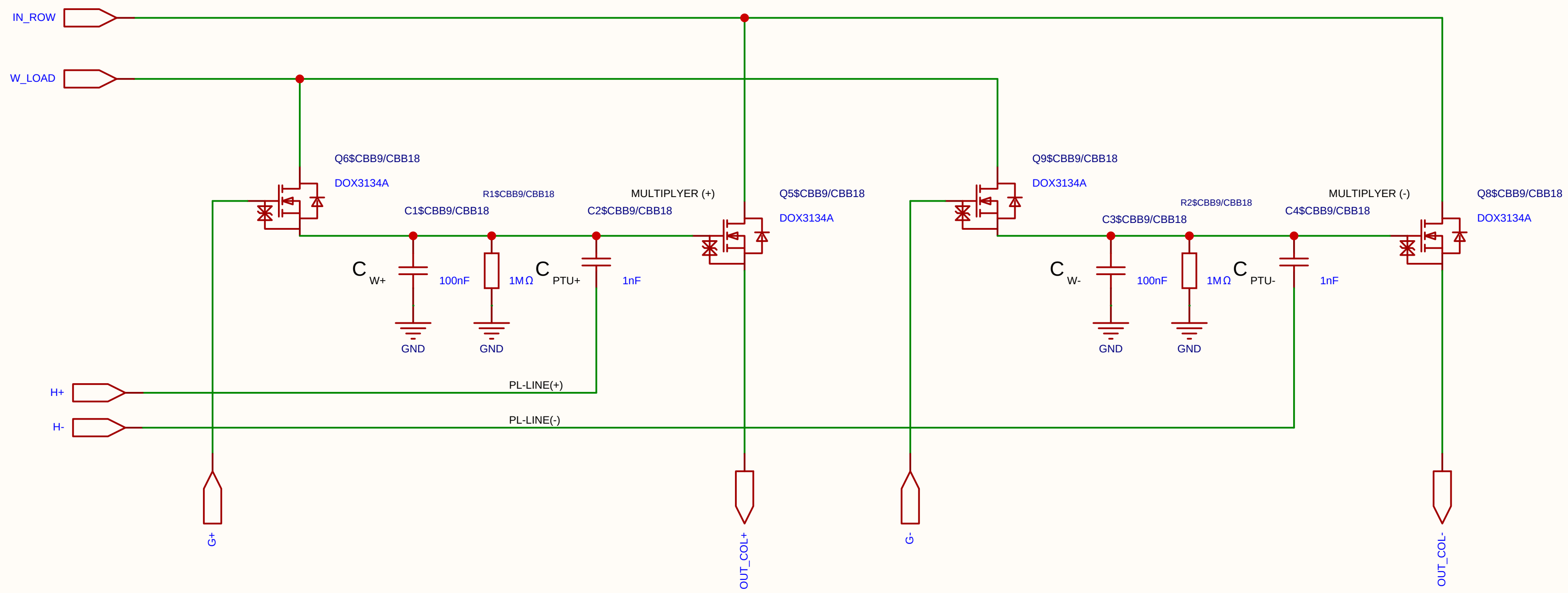
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

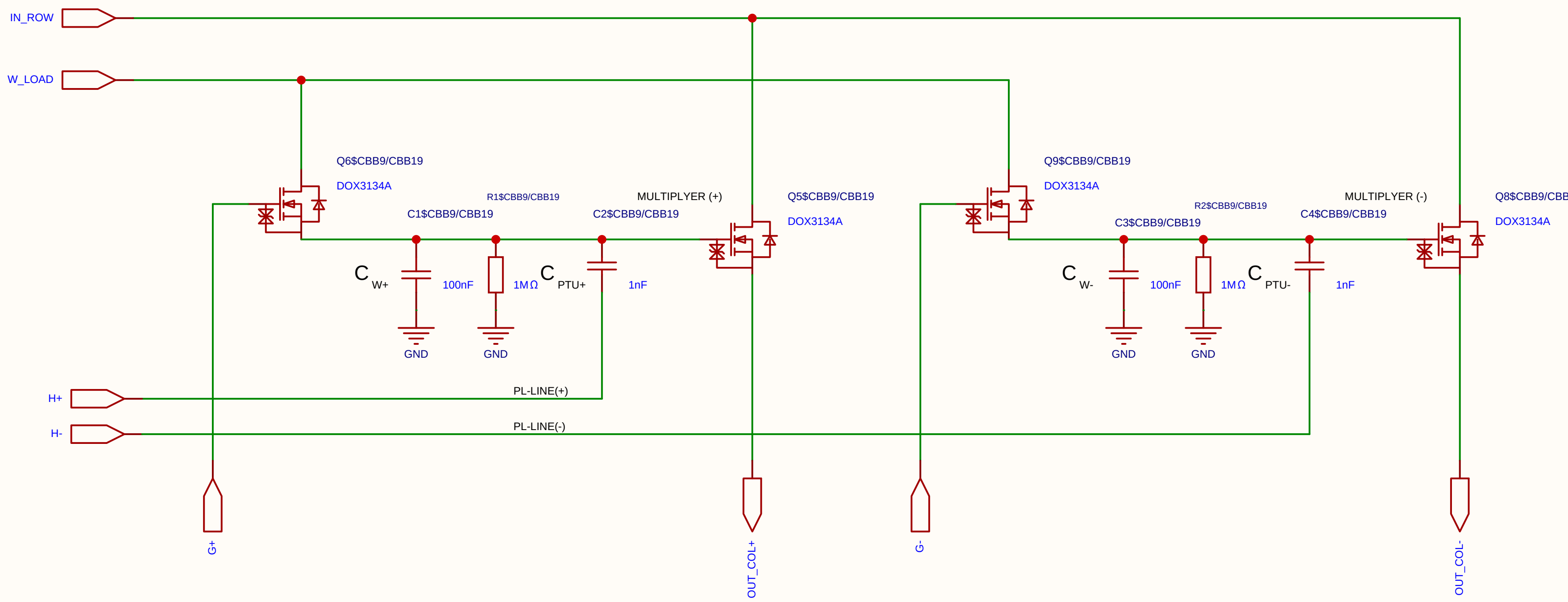
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
		Version / Size	Page 1 Total 1
EasyEDA		V1.0 / A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
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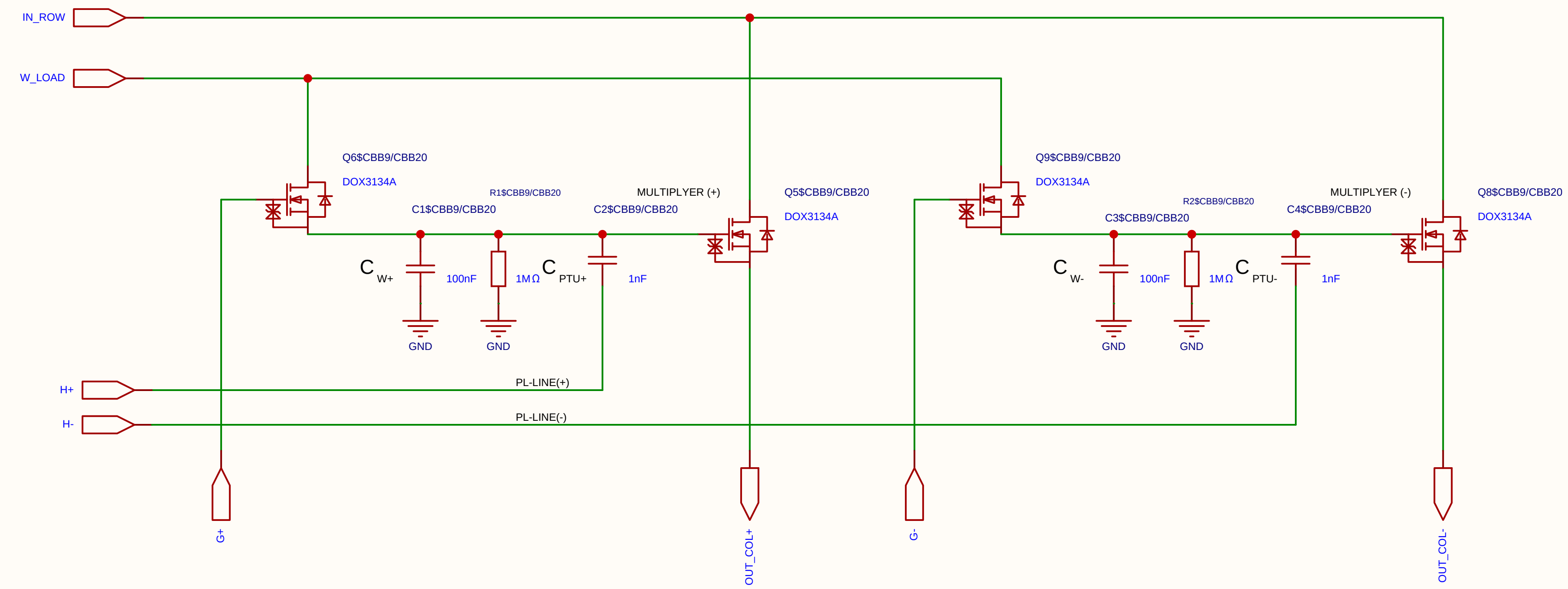
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
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Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
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		Version / Size	Page 1 Total 1
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Analog Multiplication CELL Kernel
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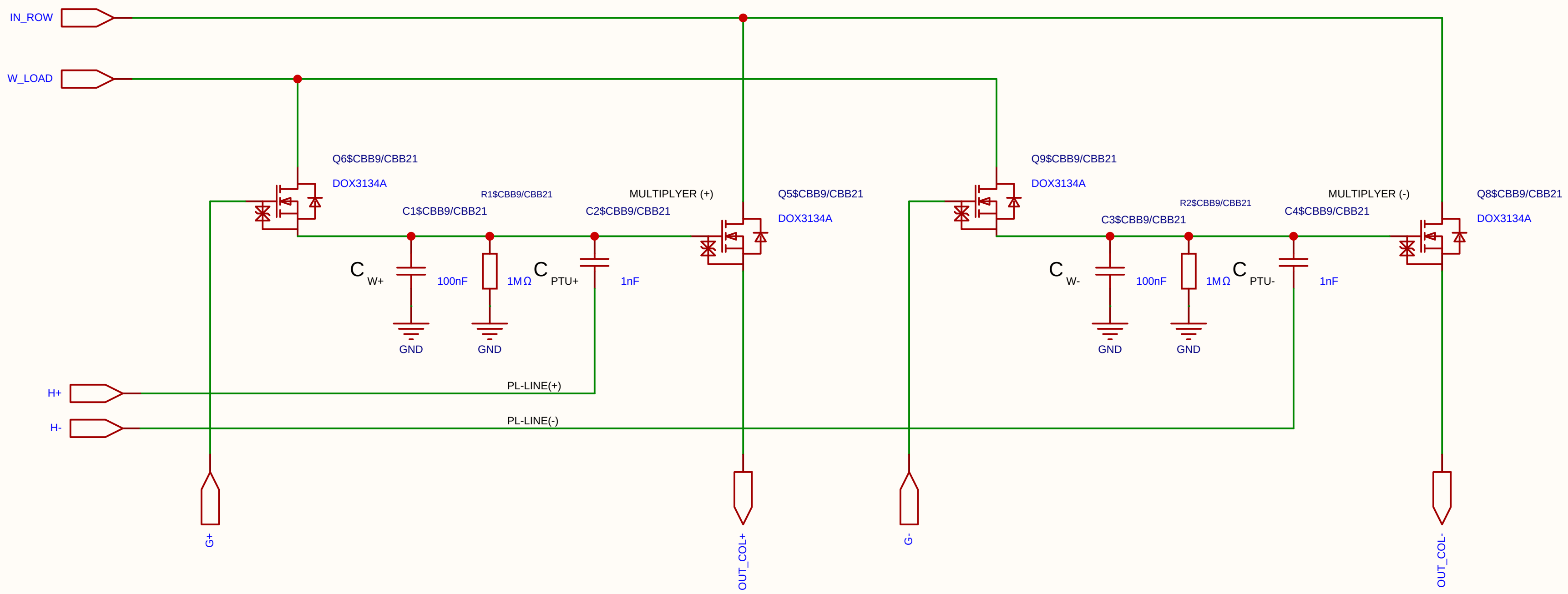

Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
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Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

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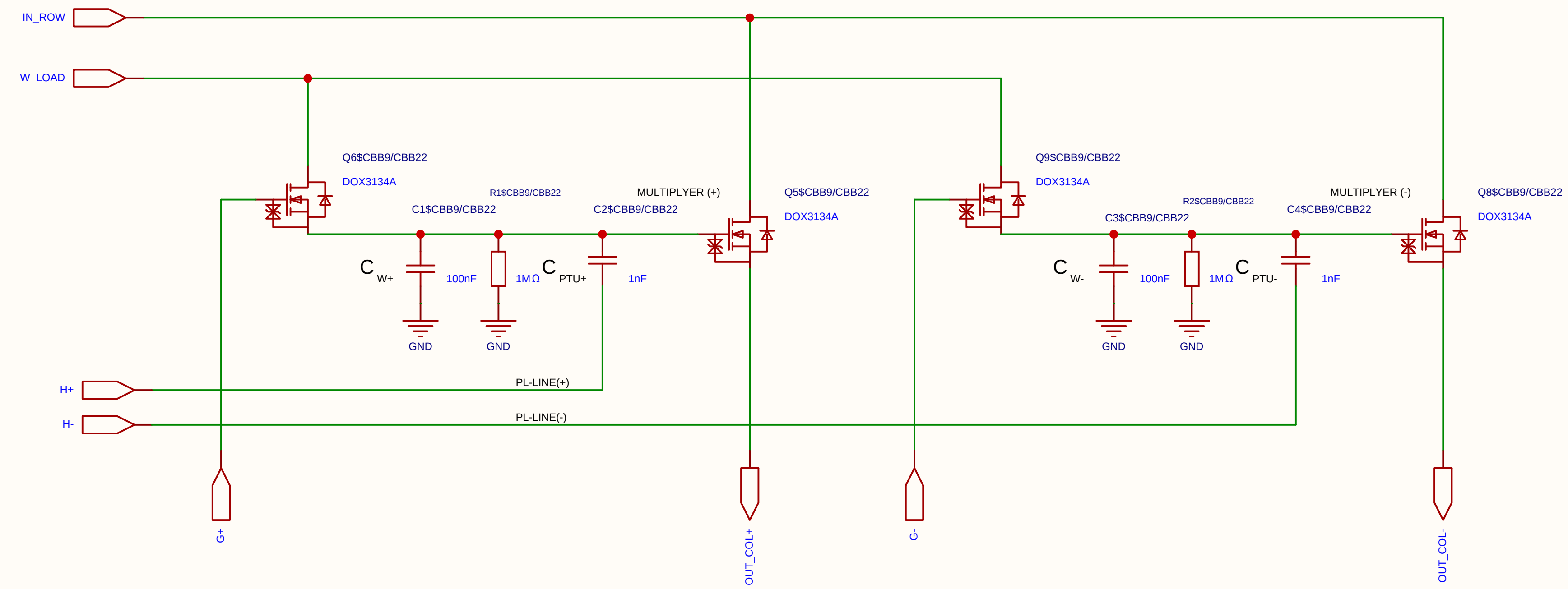


Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

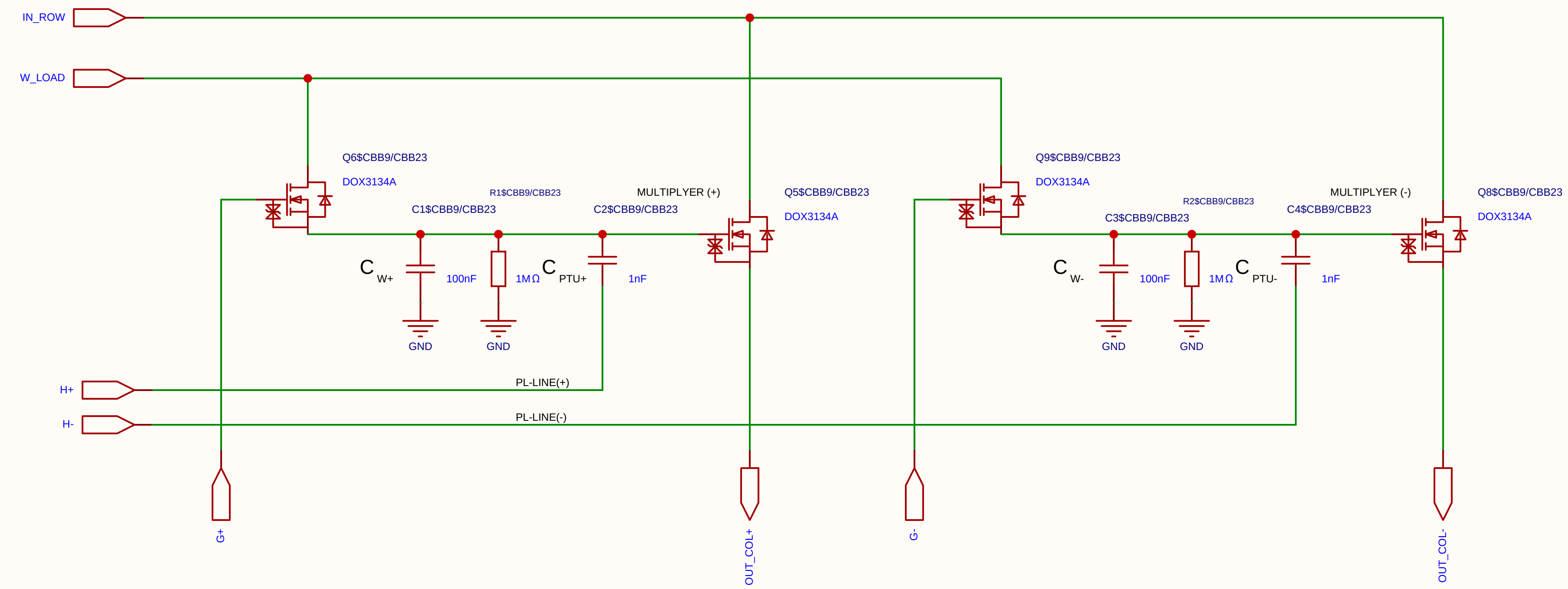
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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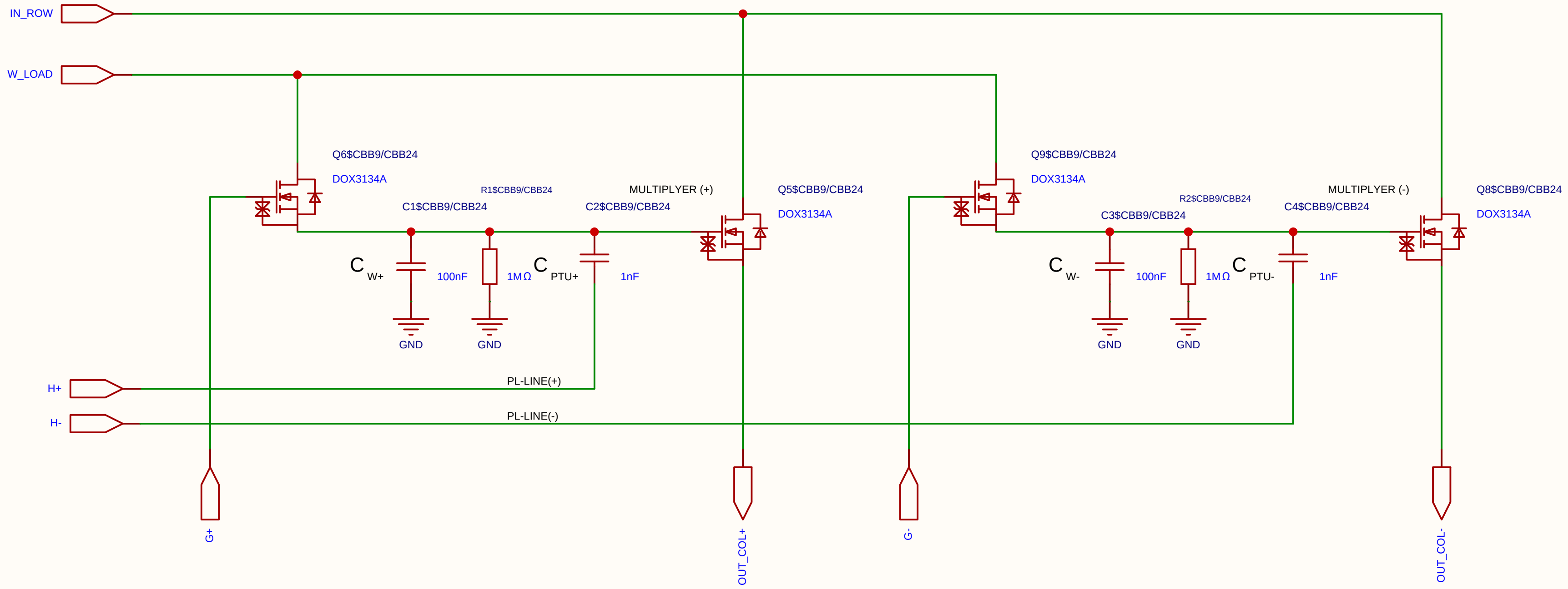
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

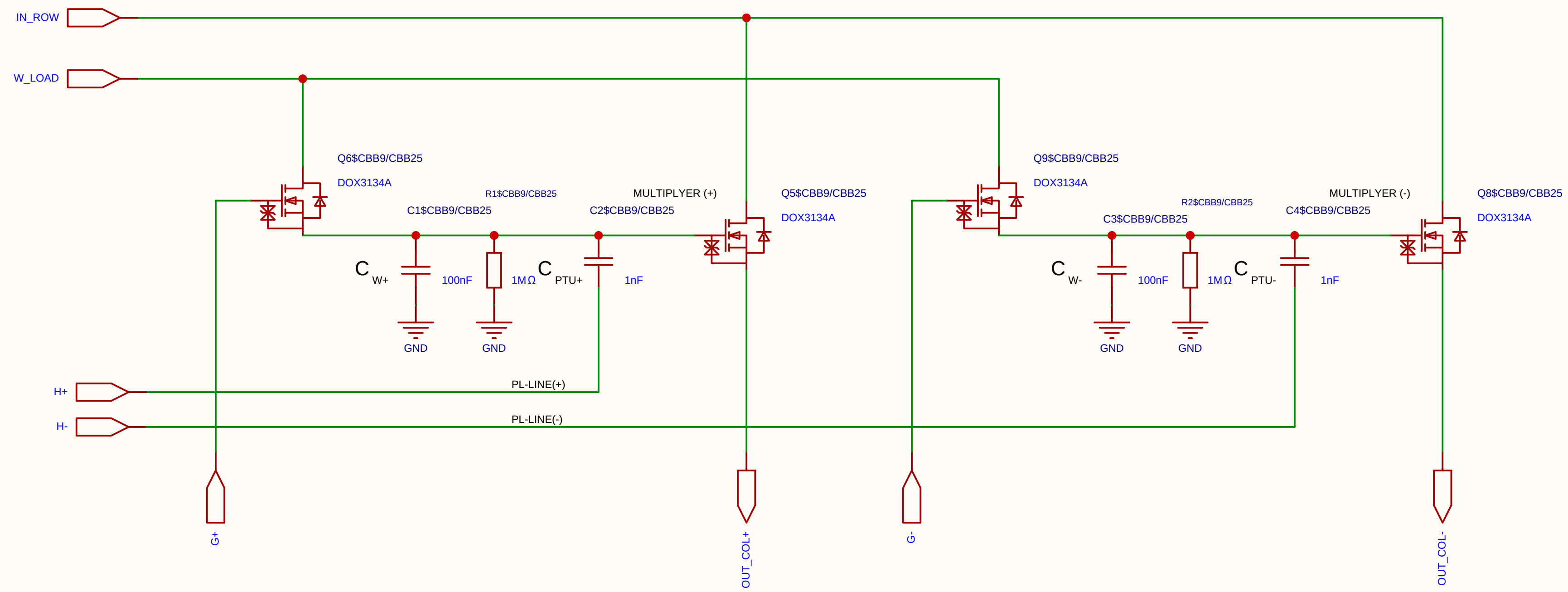
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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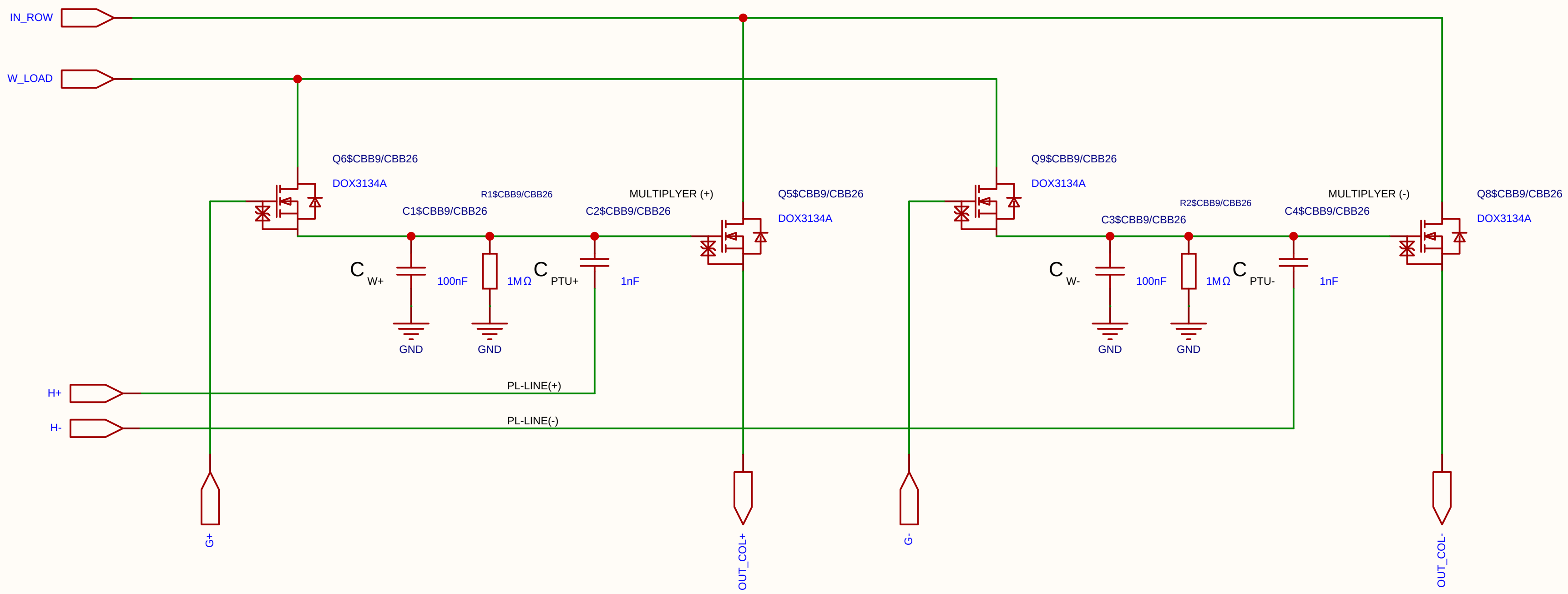
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 μs	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	2 μs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

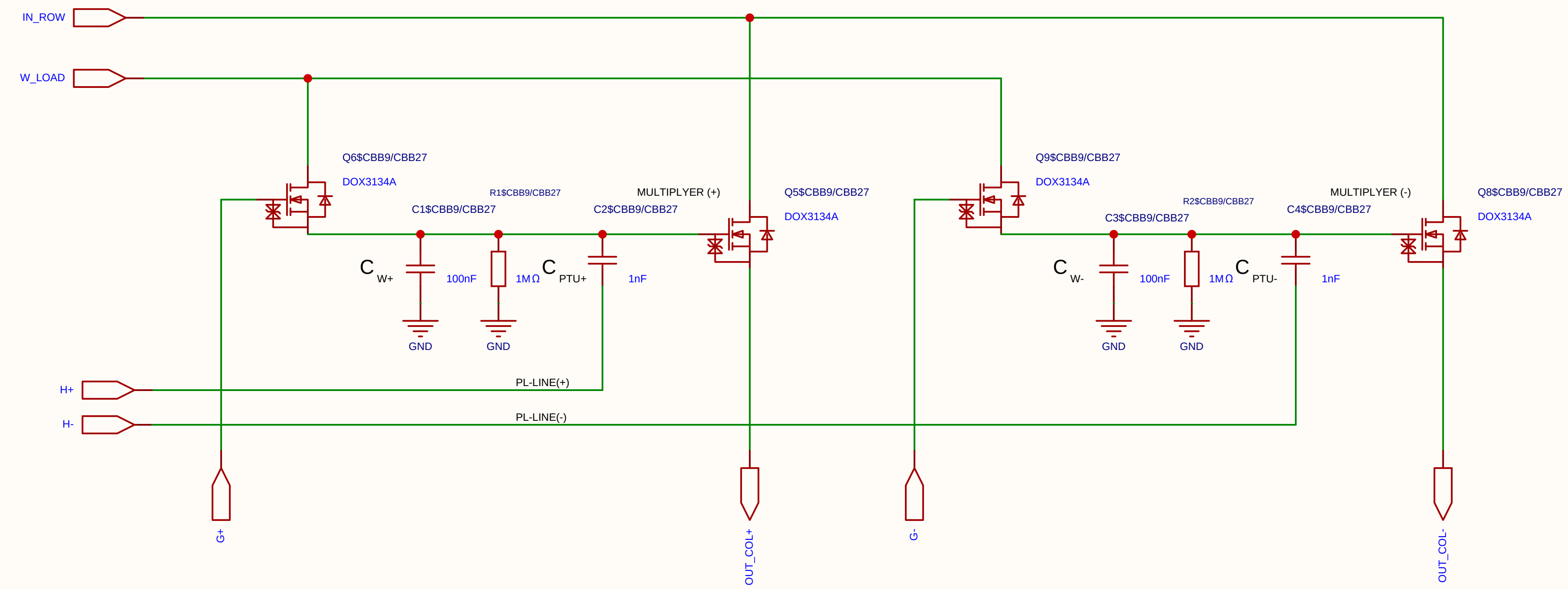
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	10 ms	Total period time between cell weight updates.
	Refresh window	≤ 100 µs	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	2 µs	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	1.0 V	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	1.65 V	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	~9.5%	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	DOX3134A	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	MCP6024T	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	100 nF	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	1 MΩ	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	100 Ω	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

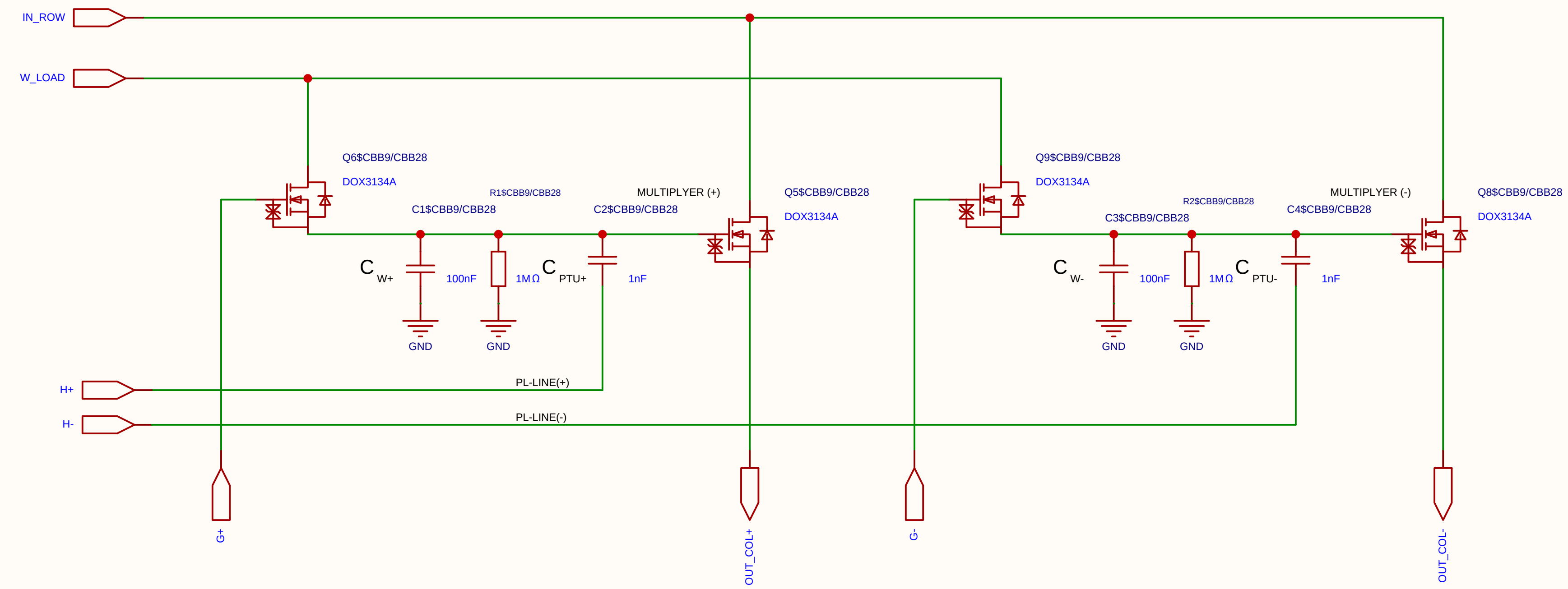
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

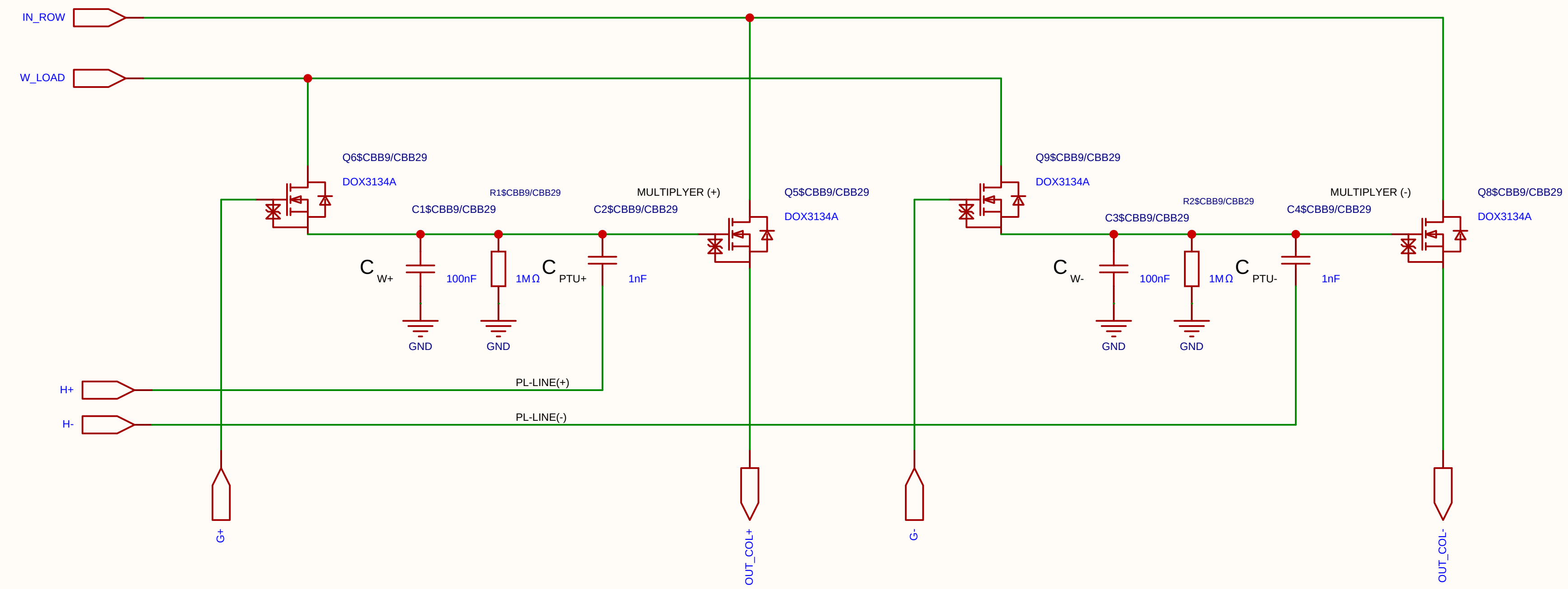
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

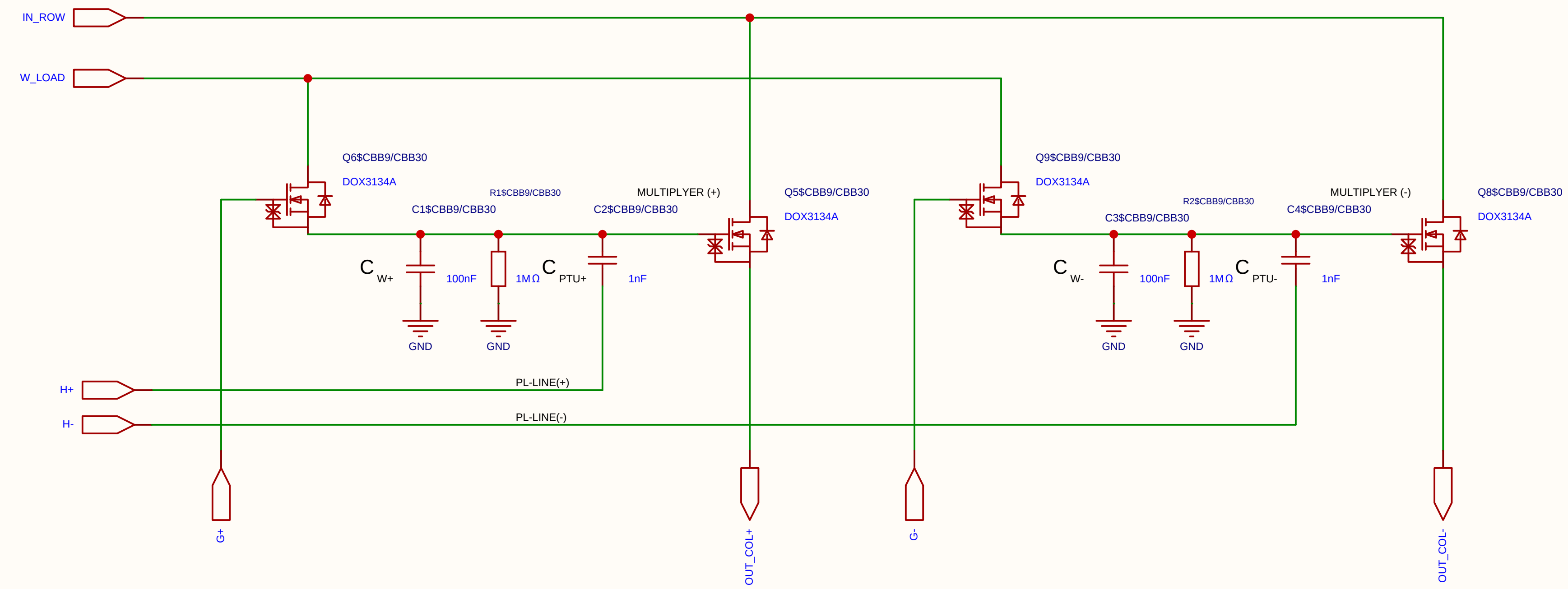
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C		Create at	2026-06-14
			Update at	2026-06-19
Board			Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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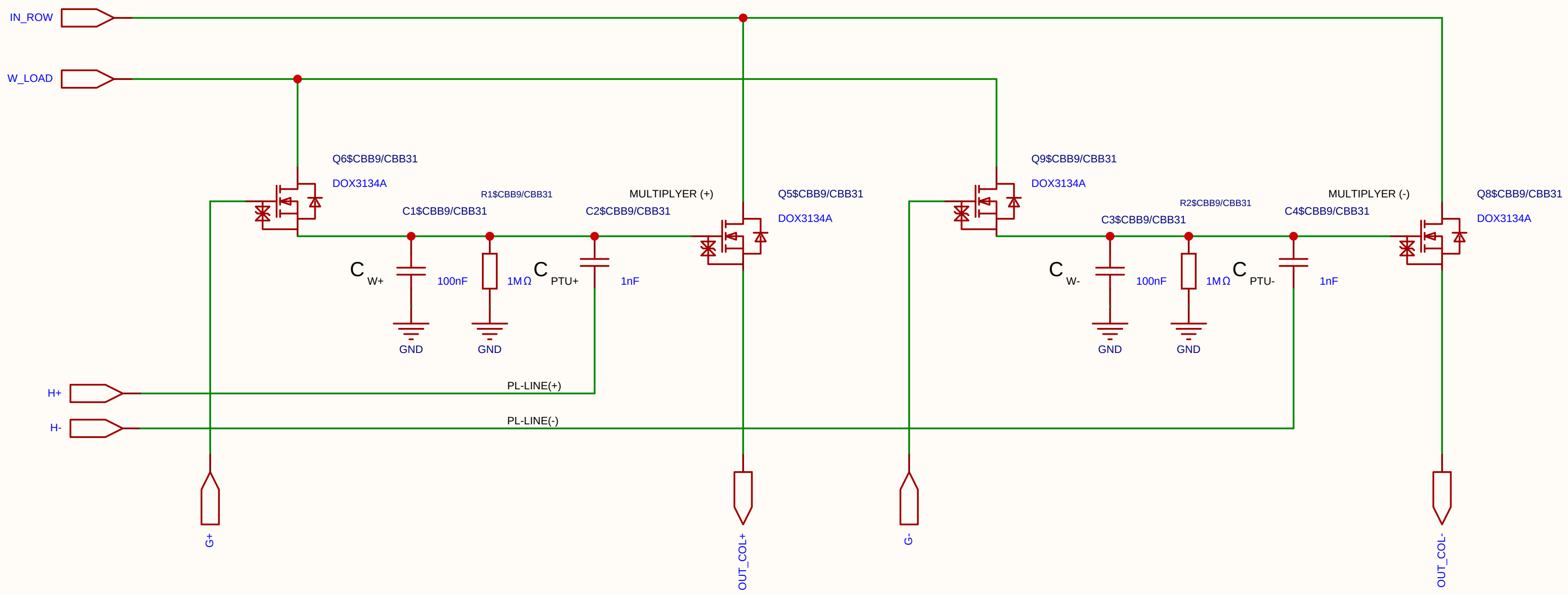
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

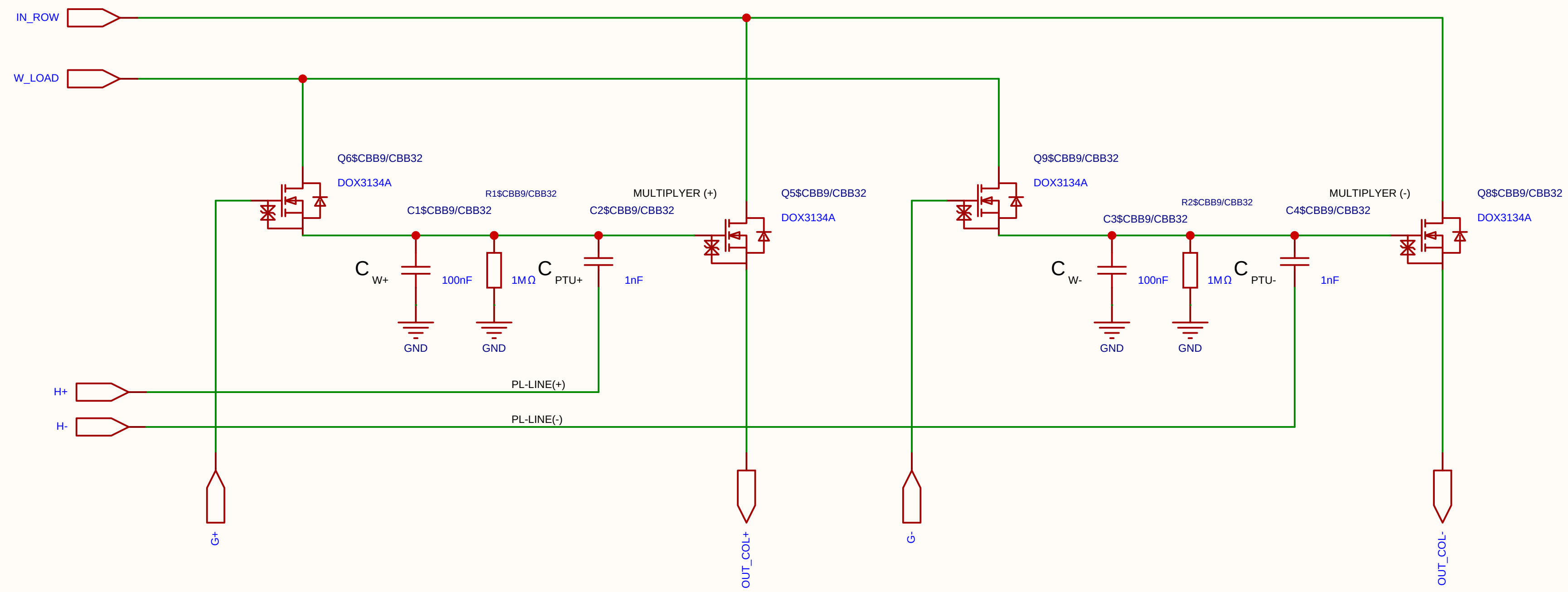
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4
		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
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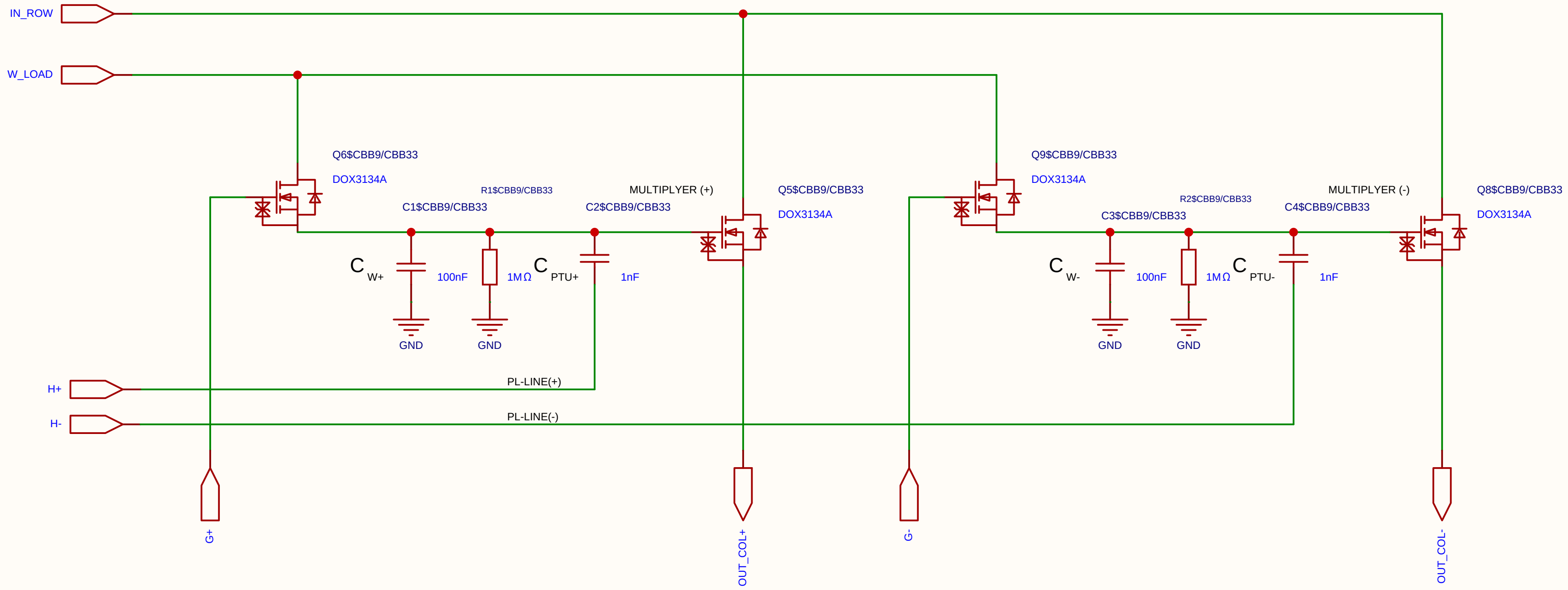
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 μs**	Time allocated to charge the cell (95% reached in ~30 μs).
	Sampling time	**2 μs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 μs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

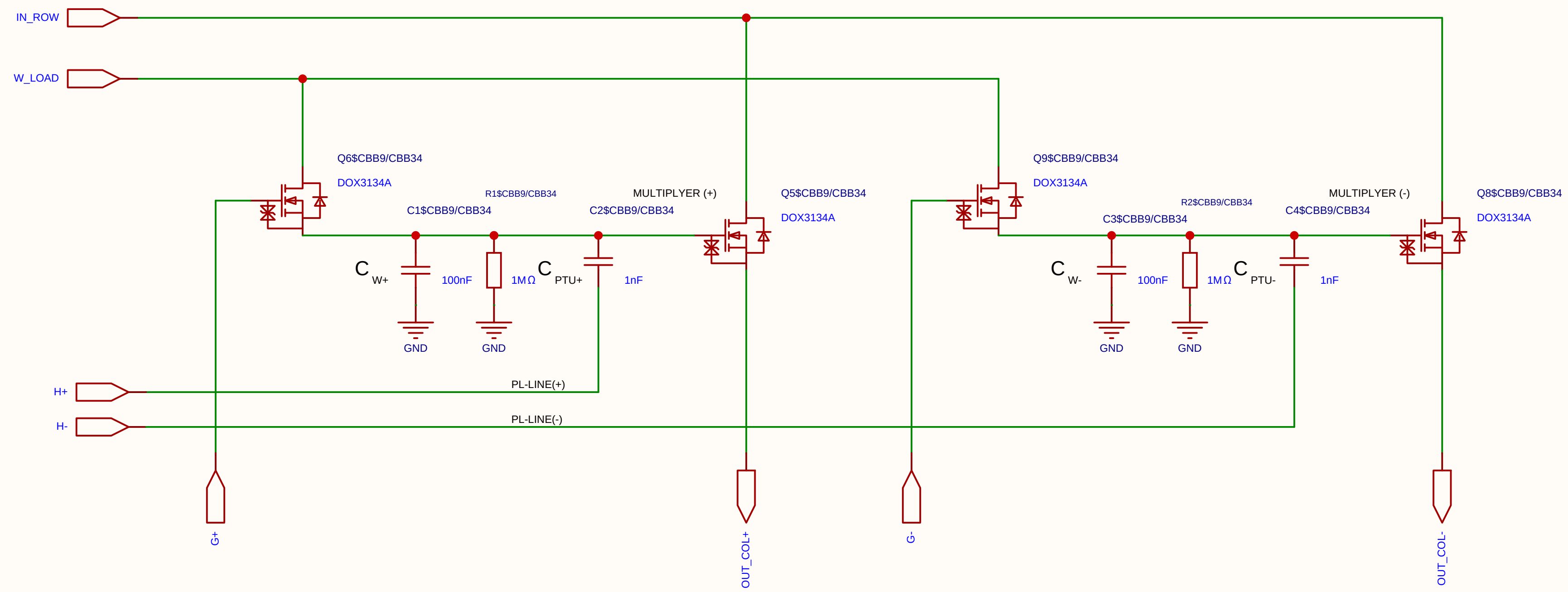
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version / Size	Page 1 Total 1
EasyEDA		V1.0 / A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

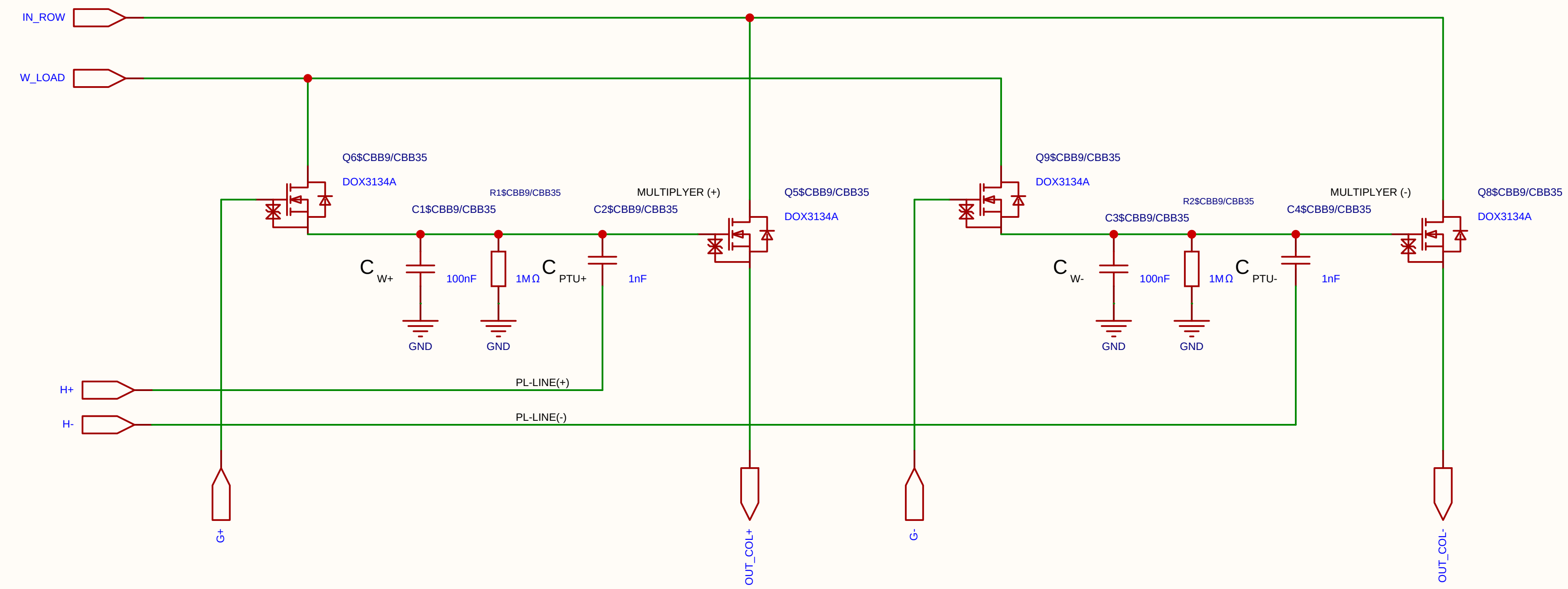
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin		
Reviewed		Inverted_MAML_6x6	
		Version	Size
EasyEDA		V1.0	A4
		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

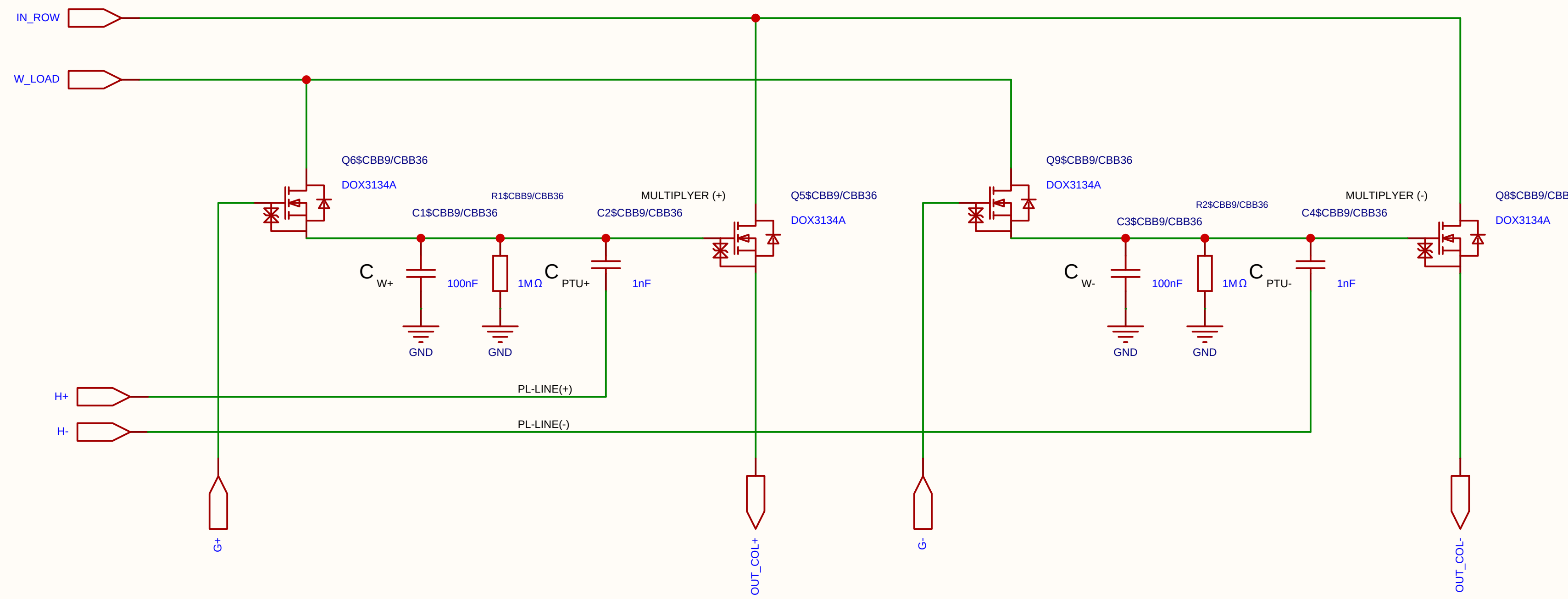
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version: V1.0	Size: A4
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

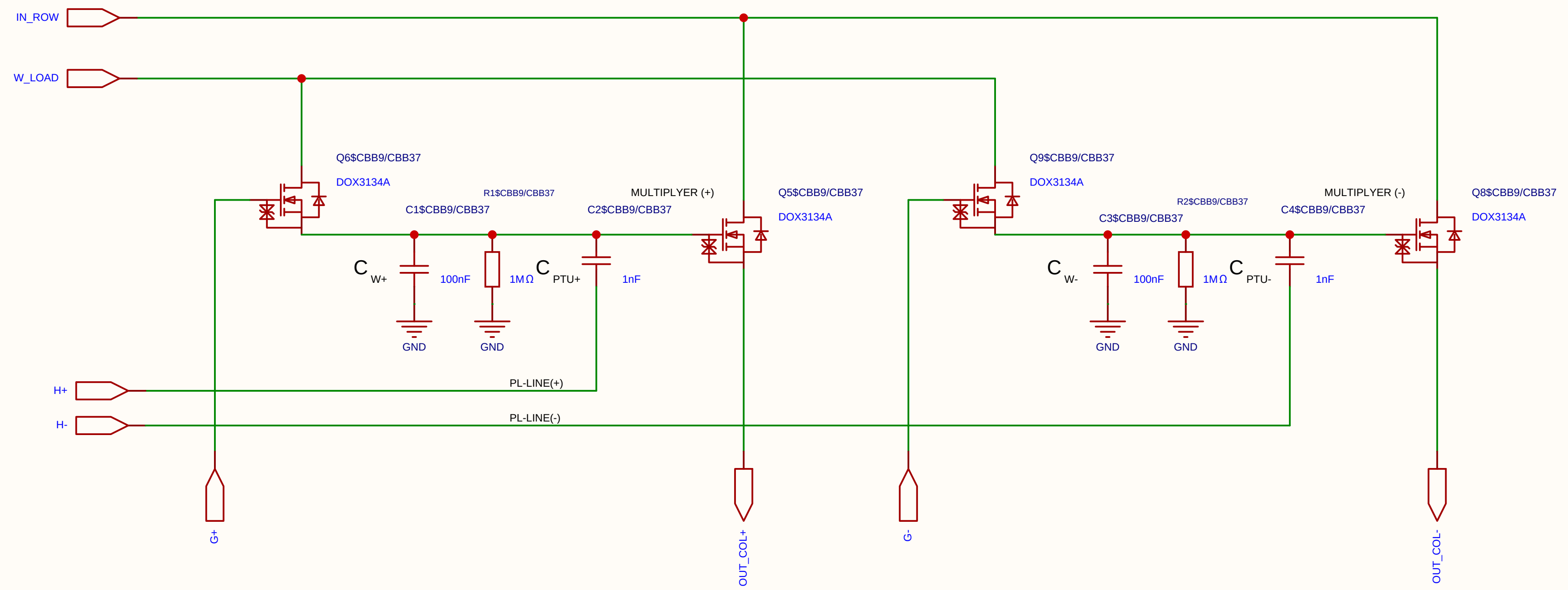
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

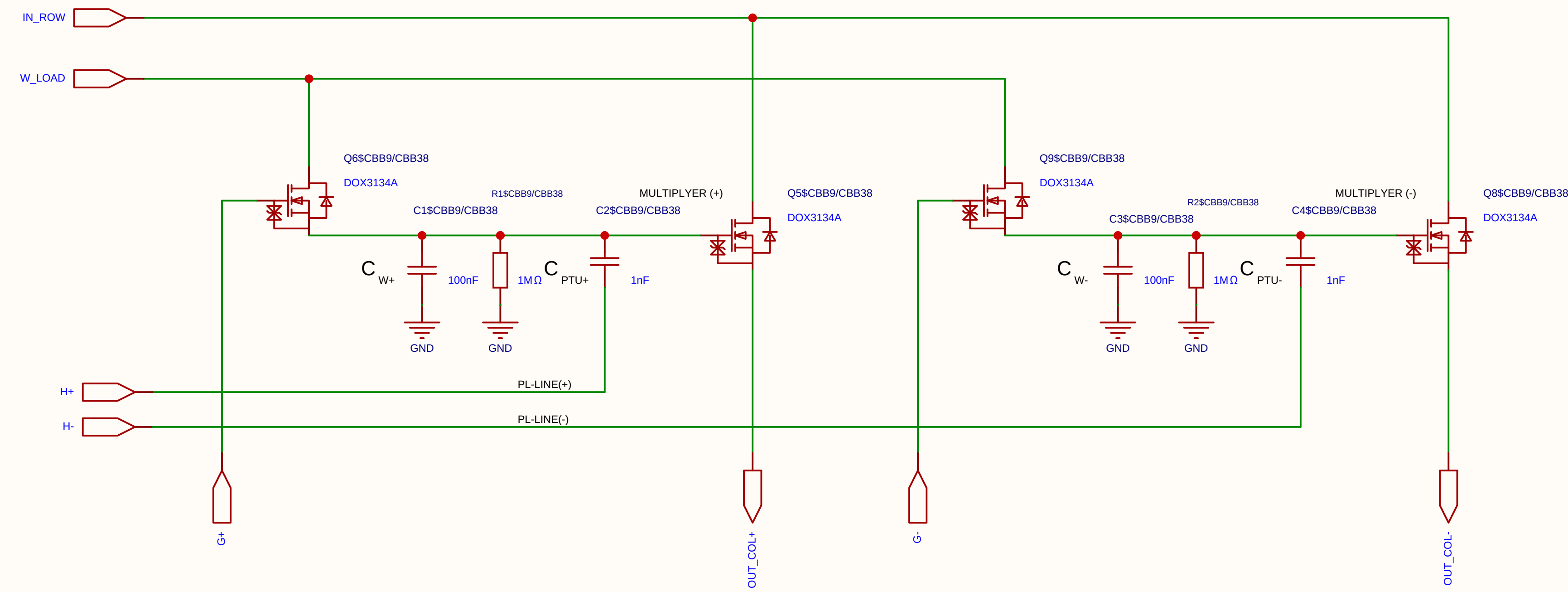
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
Version	V1.0	Size	A4
Page 1 Total 1			
EasyEDA		EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

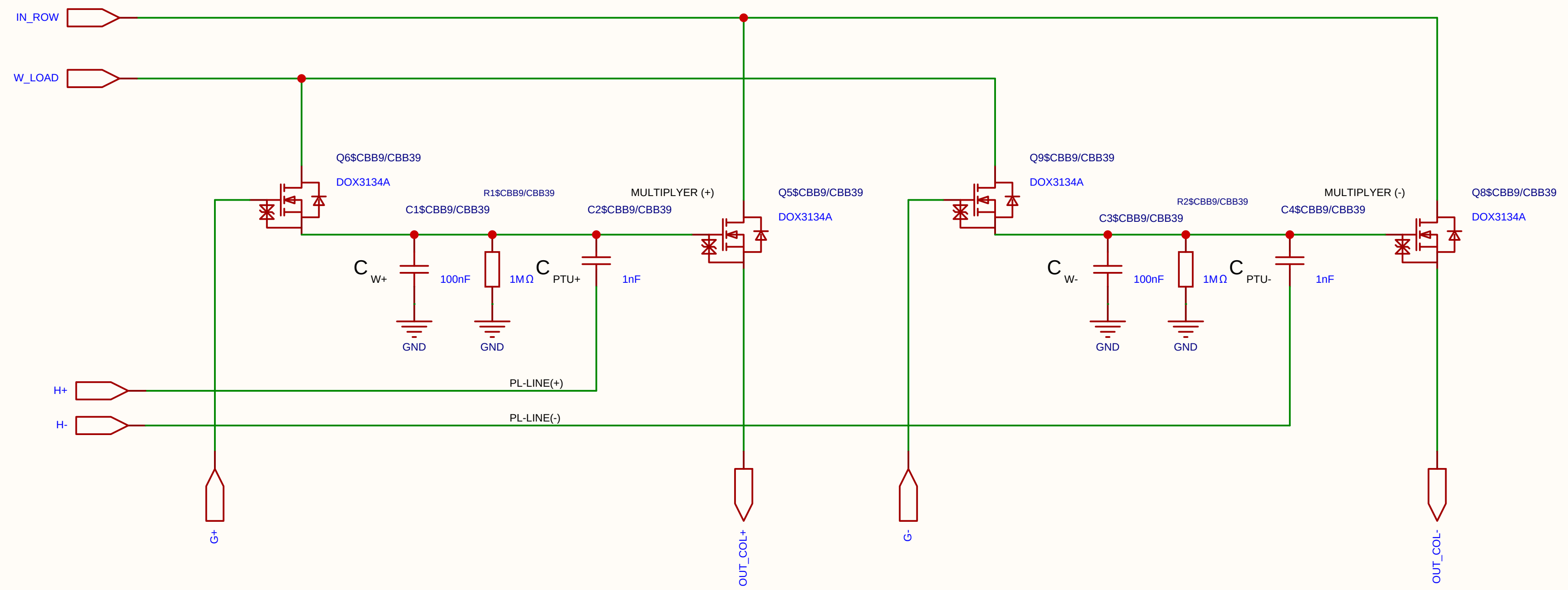
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

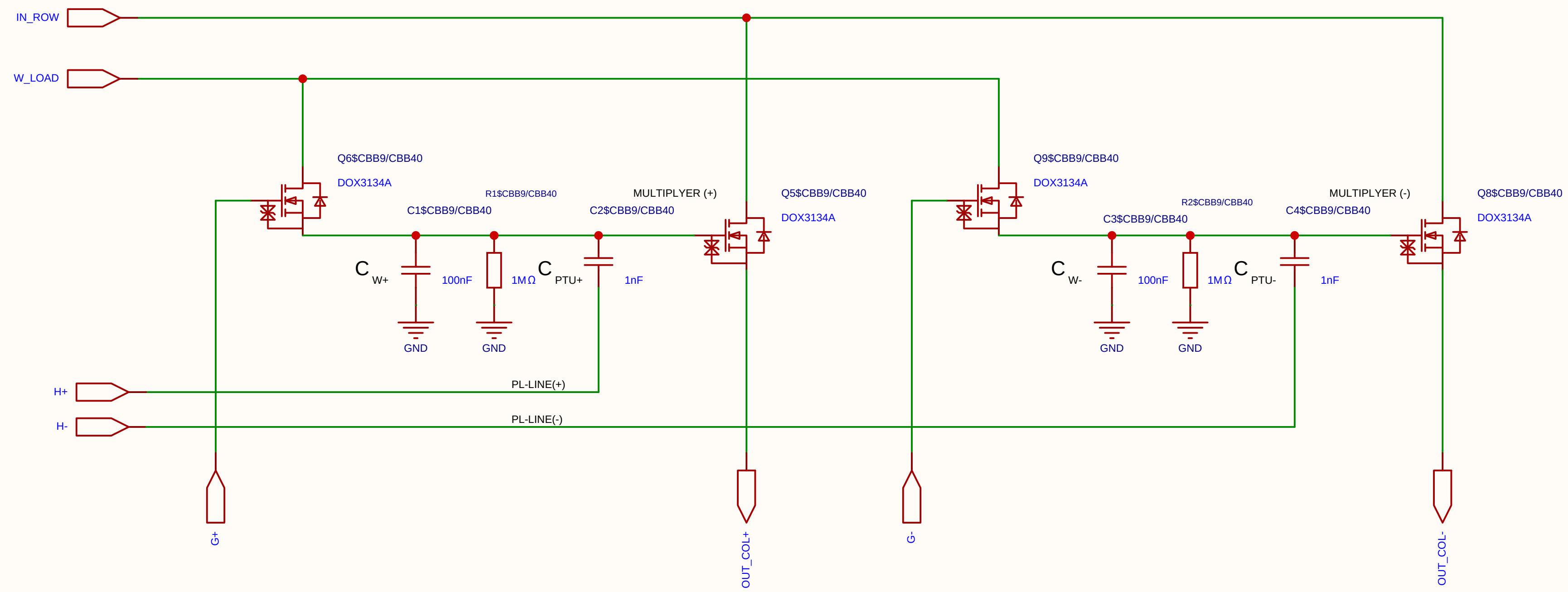
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C		Create at	2026-06-14
			Update at	2026-06-19
Board			Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

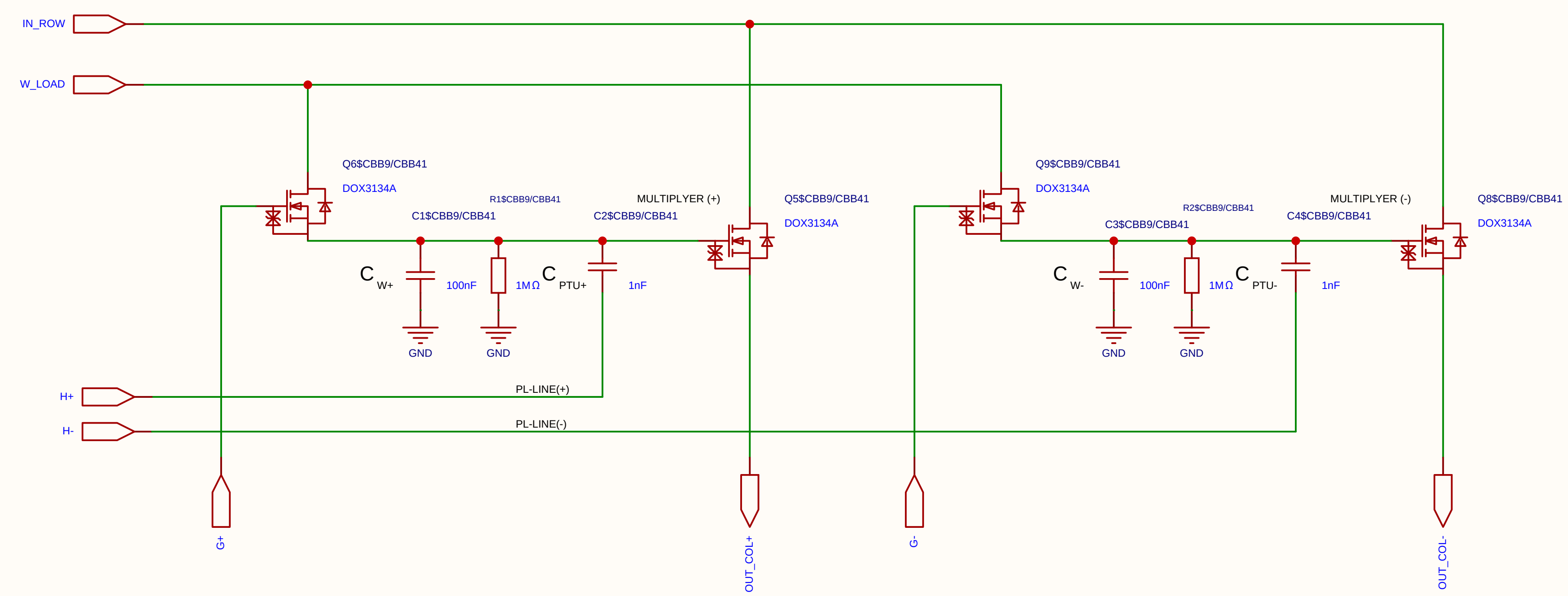
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C			Create at	2026-06-14
				Update at	2026-06-19
Board				Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

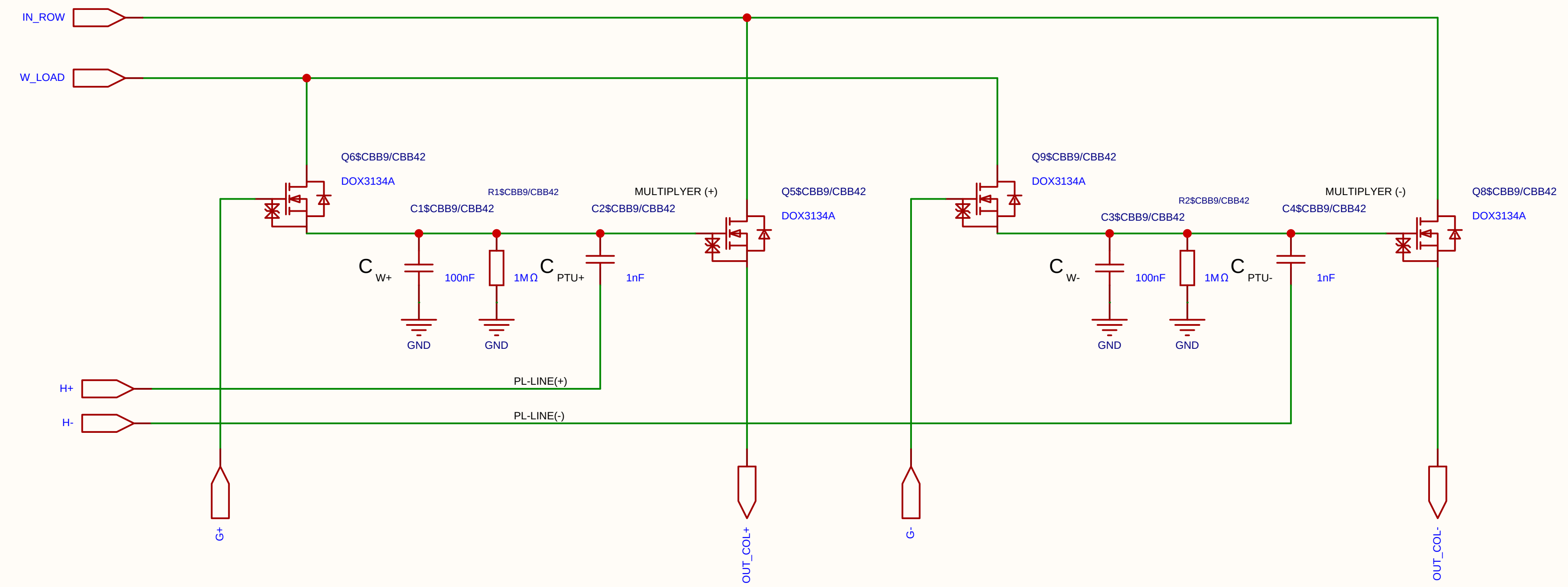
COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size
EasyEDA		V1.0	A4

Page 1 Total 1

EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Category	Parameter	Value / Type	Technical Description & Function
Timing	Refresh cycle	**10 ms**	Total period time between cell weight updates.
	Refresh window	**≤ 100 µs**	Time allocated to charge the cell (95% reached in ~30 µs).
	Sampling time	**2 µs**	Throughput time for matrix multiplication per compute sample.
Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.
	Weight swing (Gate / V_{gs})	**1.0 V**	Swings from 2.1 V (Weight 0) to 3.1 V (Weight 255).
	Source potential (V_s)	**1.65 V**	Central reference point (Virtual Ground) for differential stage.
	Voltage drop (Droop)	**~9.5%**	Controlled voltage drop (~0.1 V) over a 10 ms period.
Components	Transistor	**DOX3134A**	N-channel trench MOSFET optimized for analog multiplication.
	Operational Amplifier	**MCP6024T**	Quad 10 MHz OP-amp used as a driving stage/ buffer.
	Weight capacitor (C)	**100 nF**	Local analog storage of the weight voltage in the 2T1C cell.

Category	Parameter	Value / Type	Technical Description & Function
	Discharge resistor (R)	**1 MΩ**	Sets the leakage rate in parallel across the capacitor.
	Series resistor (R_s)	**100 Ω**	Isolates the OP-amp from capacitive load; prevents oscillation.

1. Capacitors (Per Cell)

COMPONENT	VALUE	DESCRIPTION / FUNCTION
Weight Capacitor (C_W)	100 nF	Stores the analog weight voltage (V_{weight}). Swings from 2.1 V to 3.1 V (1.0 V total dynamic range).
Perturbation Capacitor (C_{PTU})	1 nF	Capacitive coupling capacitor. Forms a ~1/101 voltage divider against C_W to inject the Hadamard perturbation. Ensures non-destructive read operations.

4. Voltage Levels on the PL Row

STATE	PL VOLTAGE	PERTURBATION AT GATE	COMMENT
Idle State (Tri-state)	1.65 V	0 mV	Perfect center potential. Prevents noise pickup and ensures absolute charge neutrality.
Hadamard (+)	2.44 V	+15.6 mV	Exactly 1/64 of the 1.0V weight swing. Fully settled in under 2 µs.
Hadamard (-)	0.86 V	-15.6 mV	Mathematically symmetrical negative step for differential disturbance analysis.

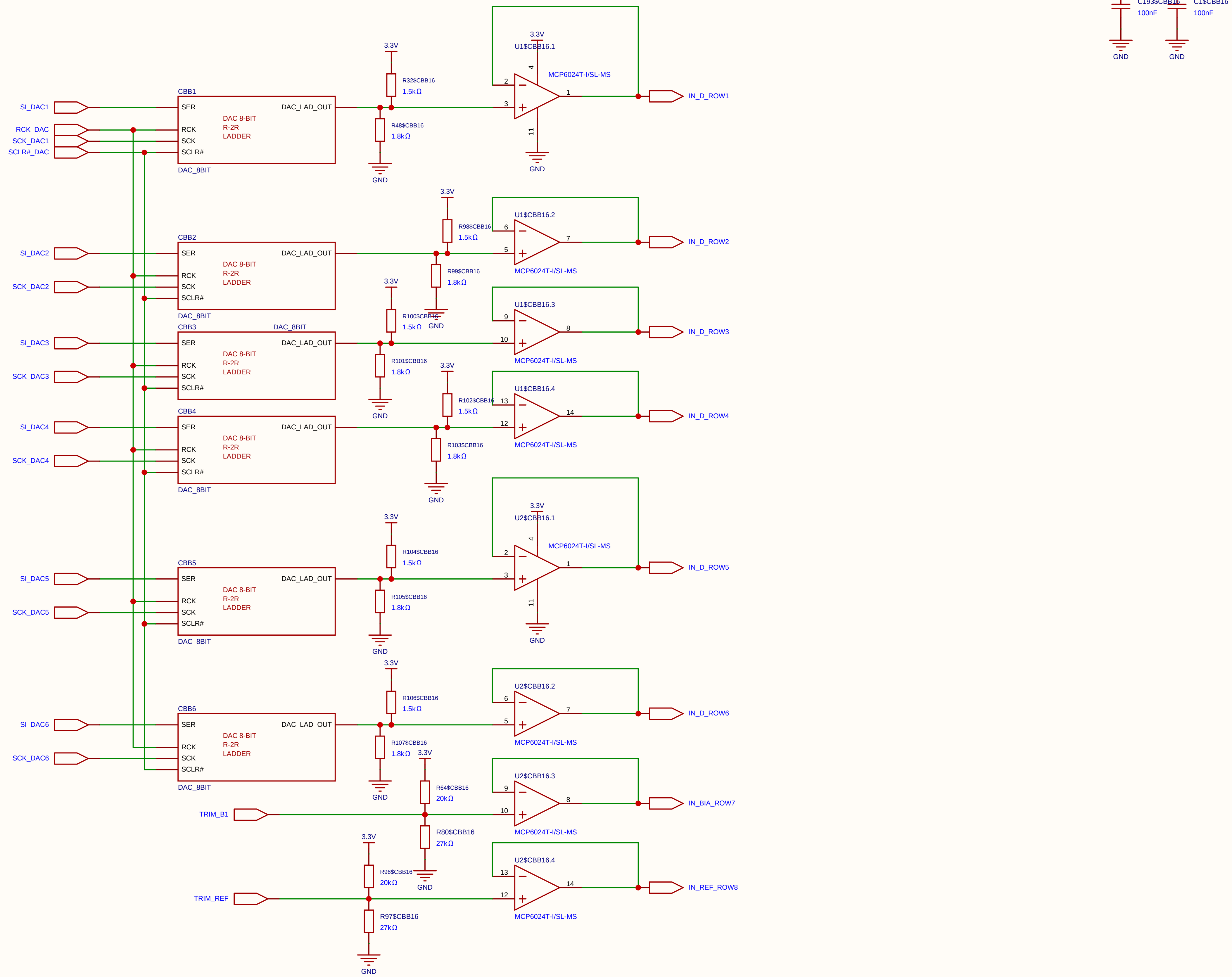
2. Resistor Network (Per PL Row)

This network connects the FPGA output pin to the PL_i row line (which drives 6 parallel C_{PTU} capacitors). It determines the perturbation amplitude and sets the RZ idle voltage.

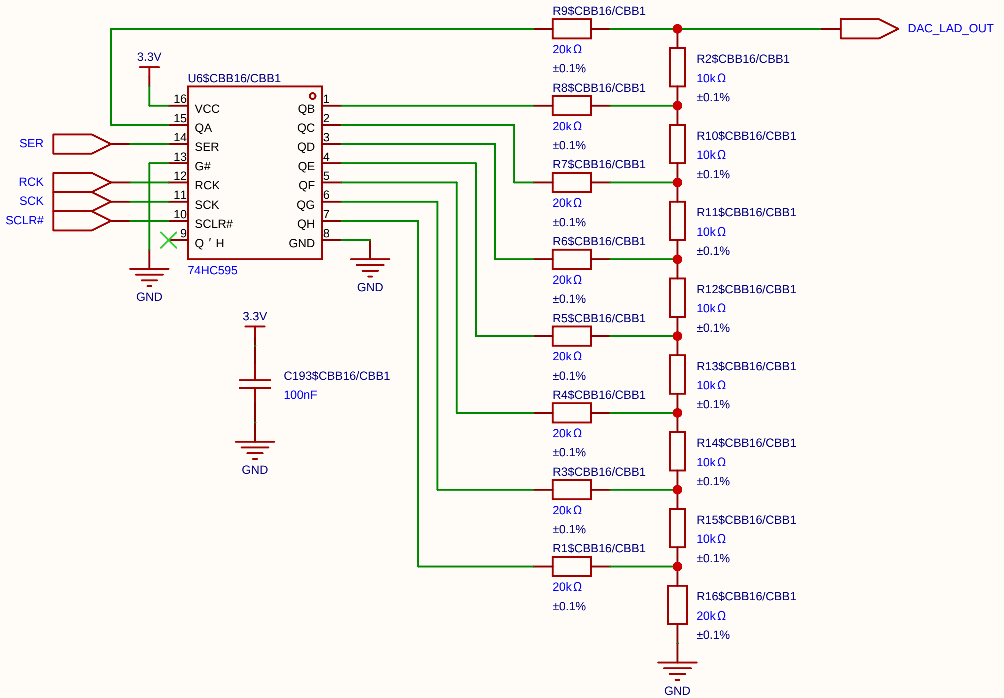
COMPONENT	VALUE	DESCRIPTION / FUNCTION
Pull-up Resistor (R_1)	220 Ω	Connected between 3.3V and PL_i. Part of the idle state biasing (Return-to-Zero).
Pull-down Resistor (R_2)	220 Ω	Connected between GND (0V) and PL_i. Establishes exactly 1.65 V (V_s) during Tri-state.
Series Resistor (R_{series})	120 Ω	Connected between the FPGA pin and PL_i. Acts as dampening and attenuates the digital swing to achieve the targeted perturbation amplitude.

Schematic	MUL_CELL_2X_2T1C	Create at	2026-06-14
		Update at	2026-06-19
Board		Page	matrix_3_8_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version / Size	Page 1 Total 1
EasyEDA		V1.0 / A4	EasyEDA.com

Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.



Schematic	6xDAC_DATA_ROWS	Create at	2026-06-19
		Update at	2026-06-20
Board		Page	6xDAC_8-bit
Drawn		Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com



Schematic	DAC_8BIT	Create at	2026-06-19
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Board		Page	DAC_8B_R_2R
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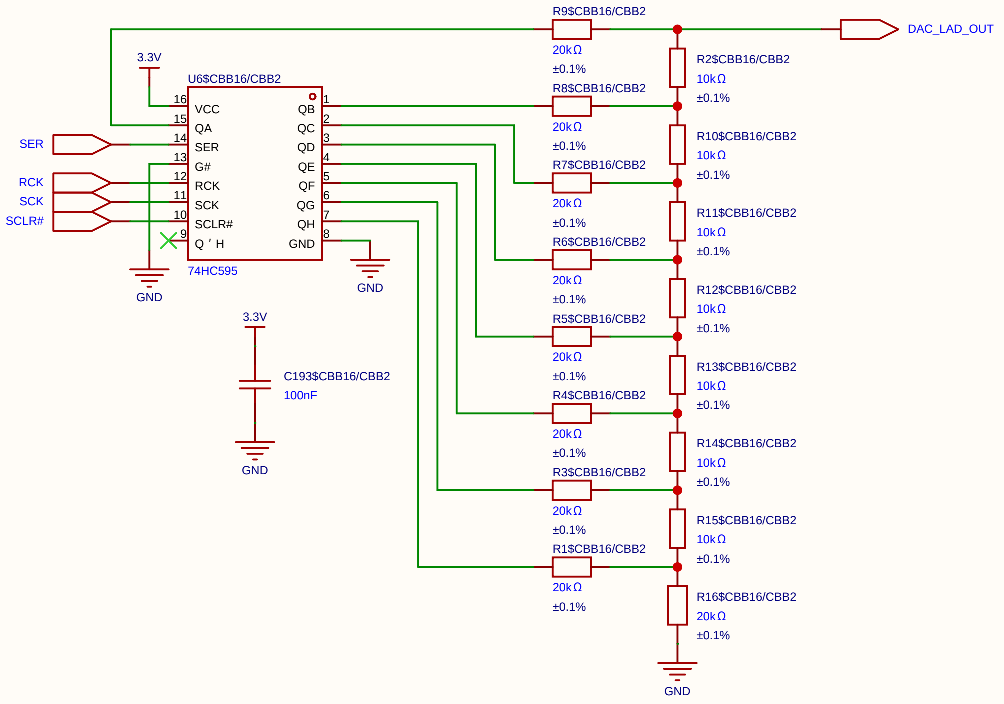
Reviewed			
		Version / Size	Page 1 Total 1
EasyEDA		V1.0 / A4	EasyEDA.com



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Board		Page	DAC_8B_R_2R
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Reviewed			
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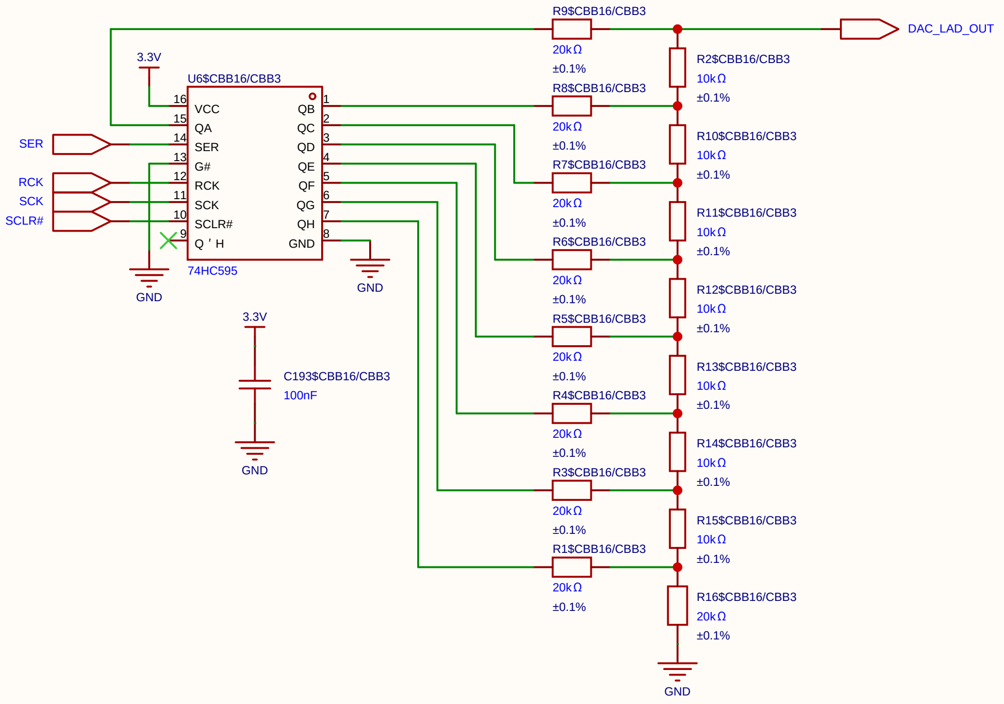

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Board		Page	DAC_8B_R_2R
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
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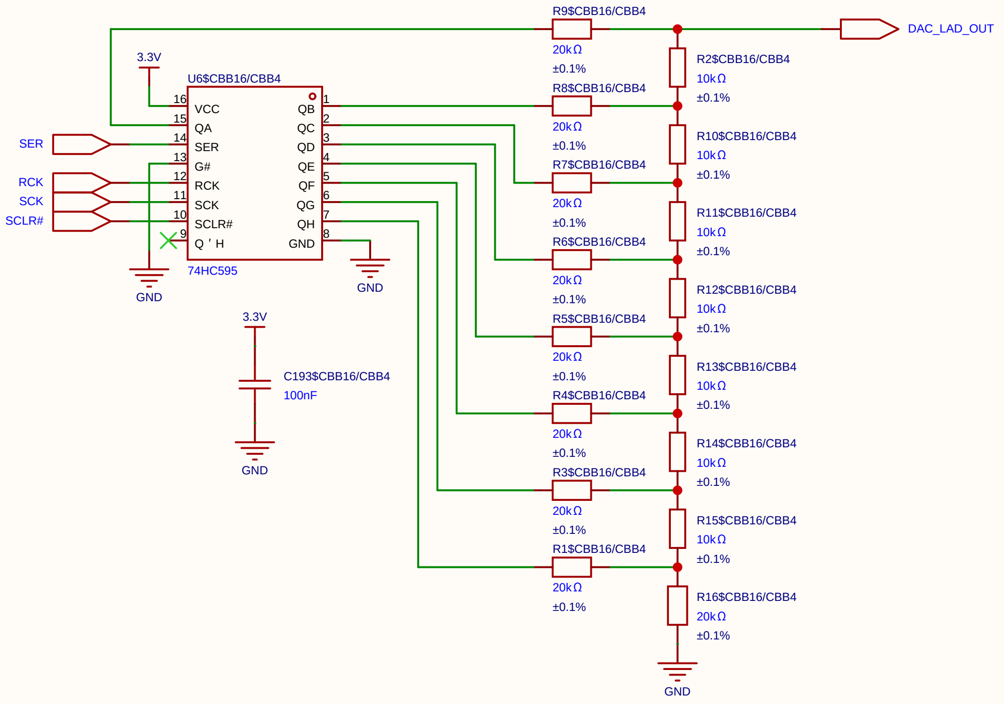
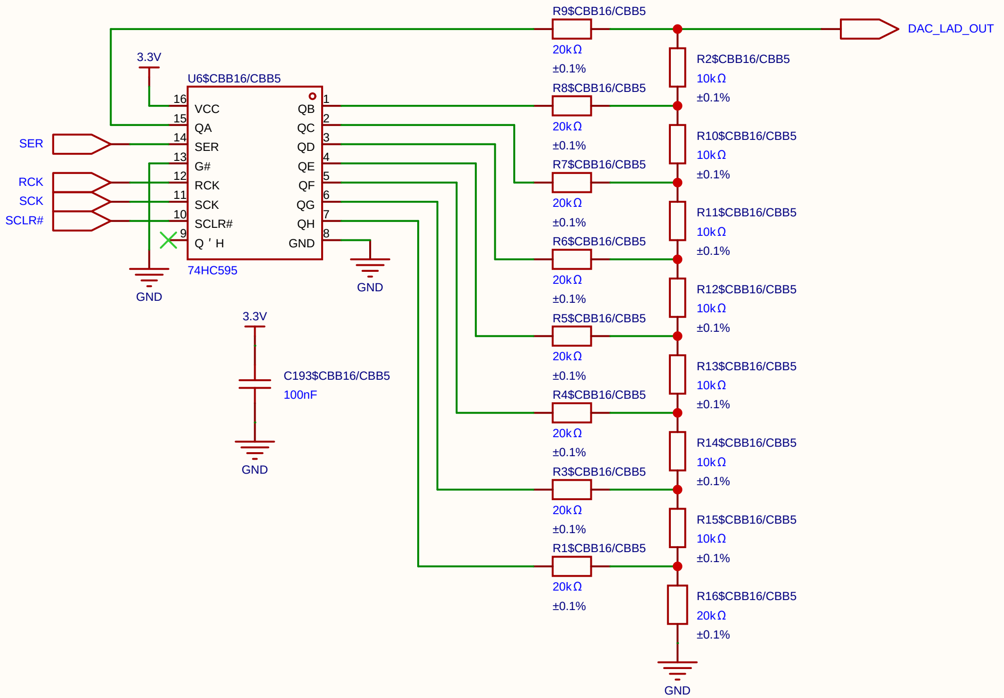
EasyEDA		V1.0	A4	EasyEDA.com



Schematic	DAC_8BIT	Create at	2026-06-19
		Update at	2026-06-19
Board		Page	DAC_8B_R_2R
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version / Size	Page 1 Total 1
EasyEDA		V1.0 / A4	EasyEDA.com



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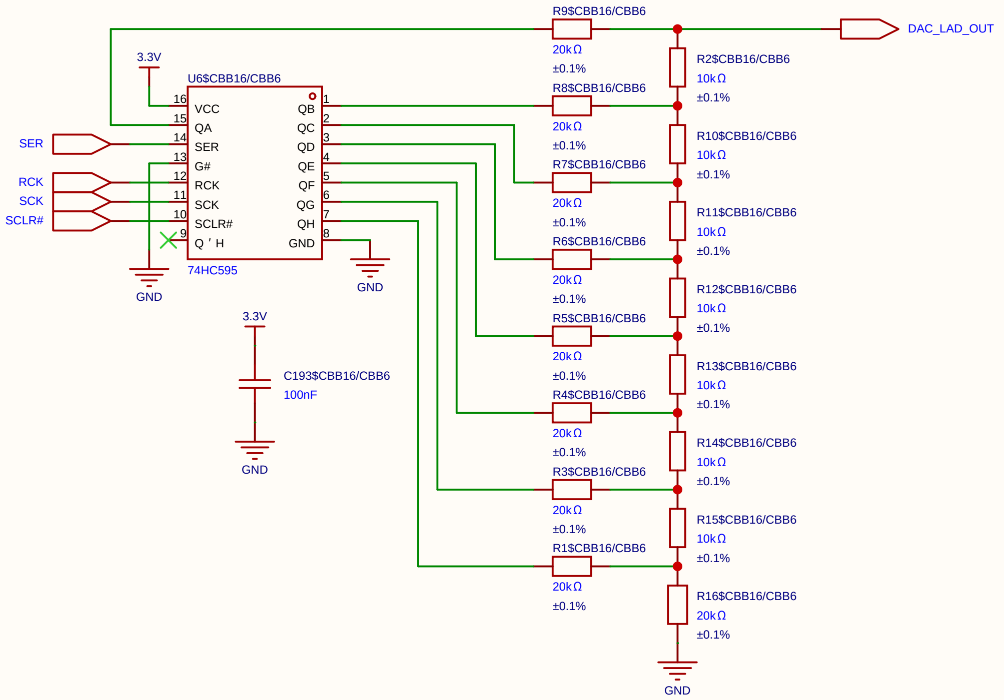
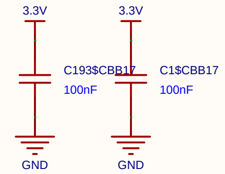
		Update at	2026-06-19
Board		Page	DAC_8B_R_2R

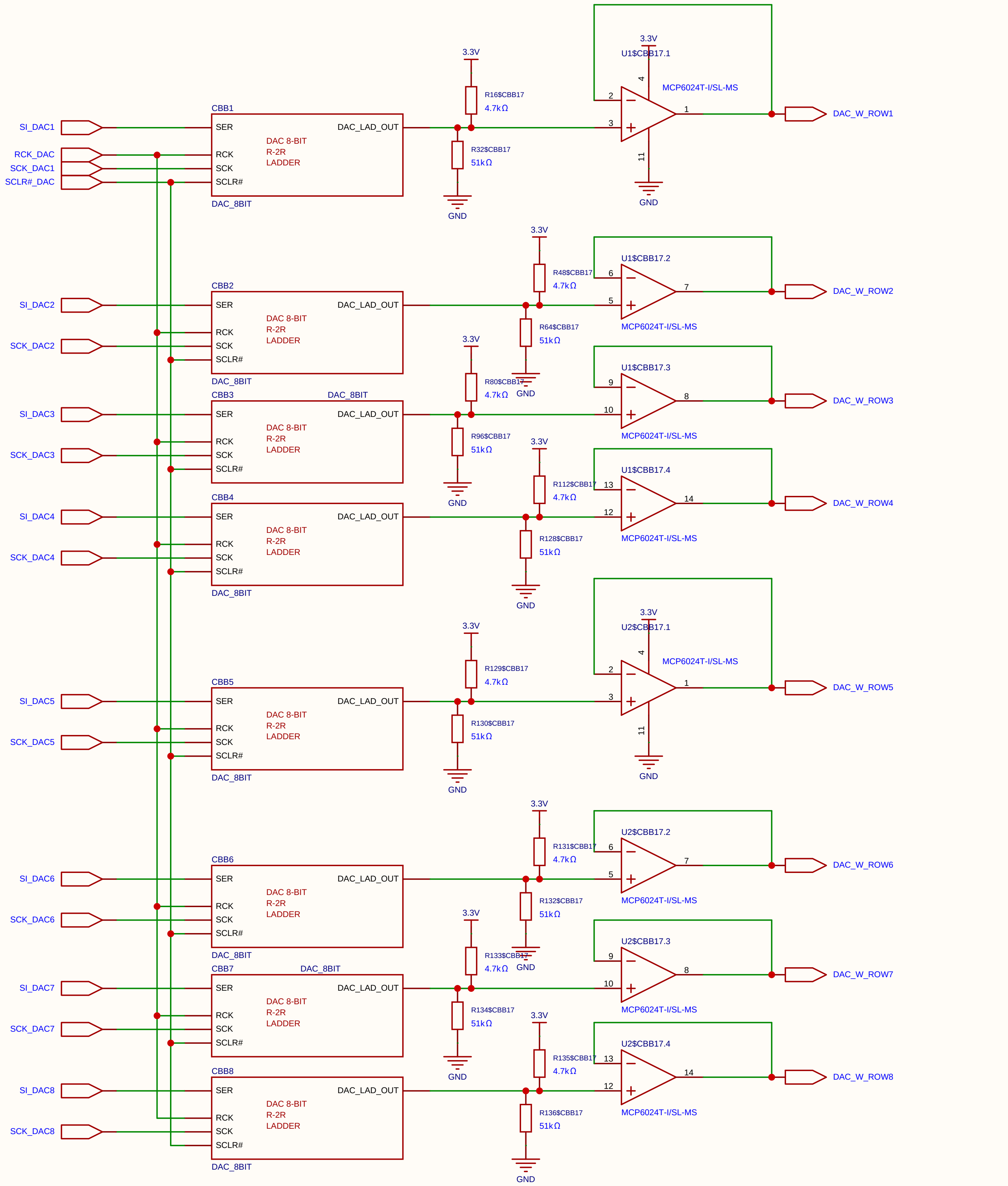
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com



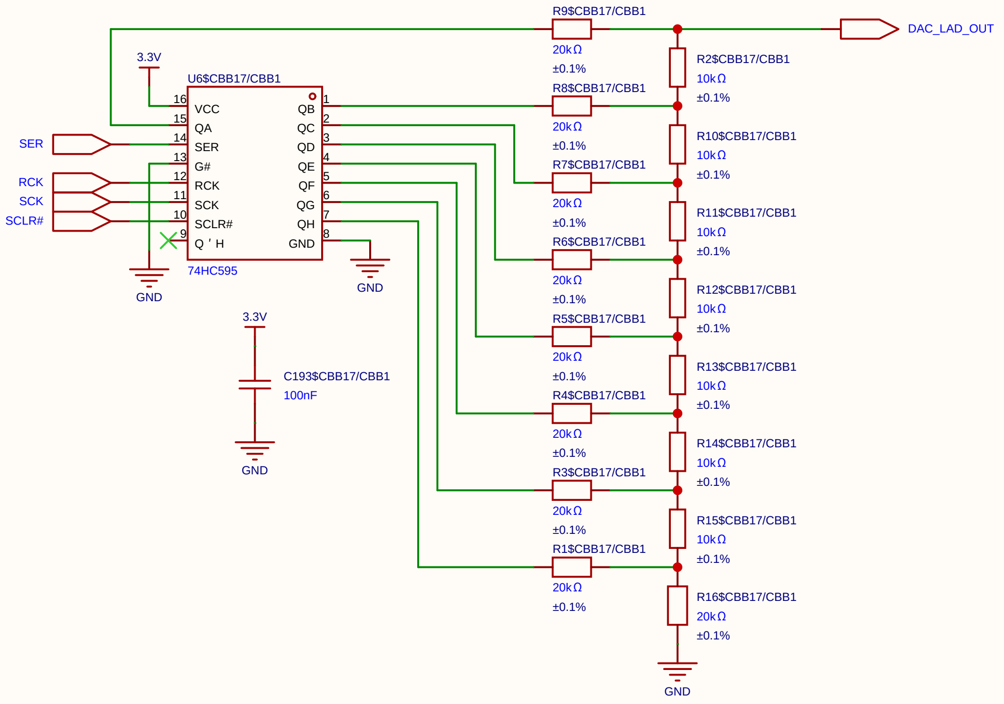
Schematic	DAC_8BIT			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	DAC_8B_R_2R
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Voltages	Data swing (Row / V_{ds})	**250 mV**	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.





Schematic	8xDAC_WEIGHT	Create at	2026-06-19
		Update at	2026-06-19
Board		Page	8xDAC_8-bit
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version / Size	Page 1 Total 1
EasyEDA		V1.0 / A4	EasyEDA.com



Schematic	DAC_8BIT			Create at	2026-06-19
				Update at	2026-06-19
Board				Page	DAC_8B_R_2R
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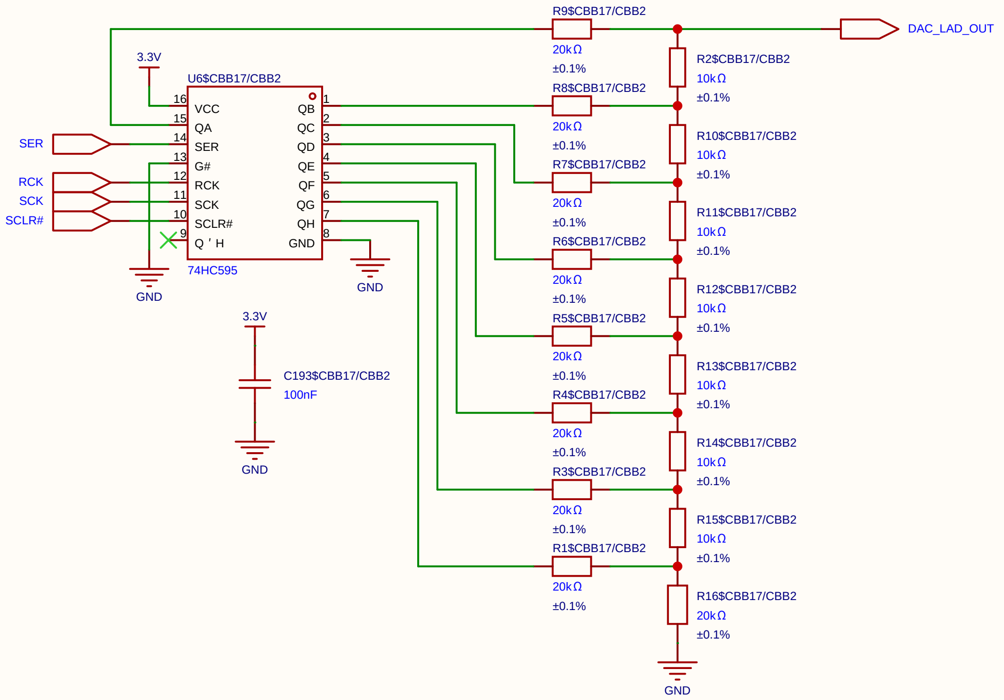
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	



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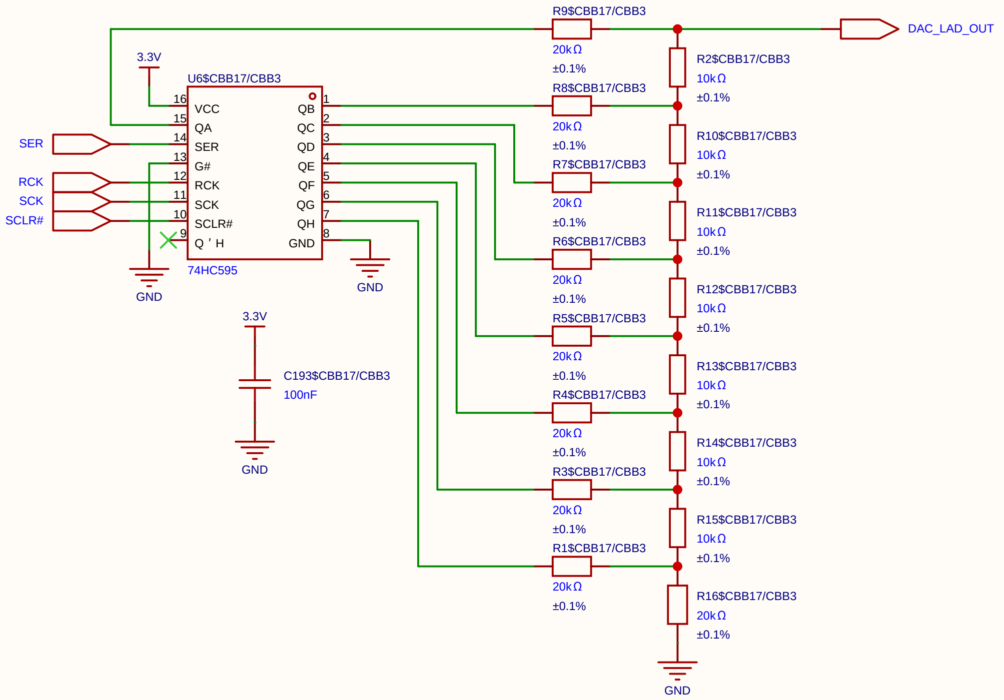

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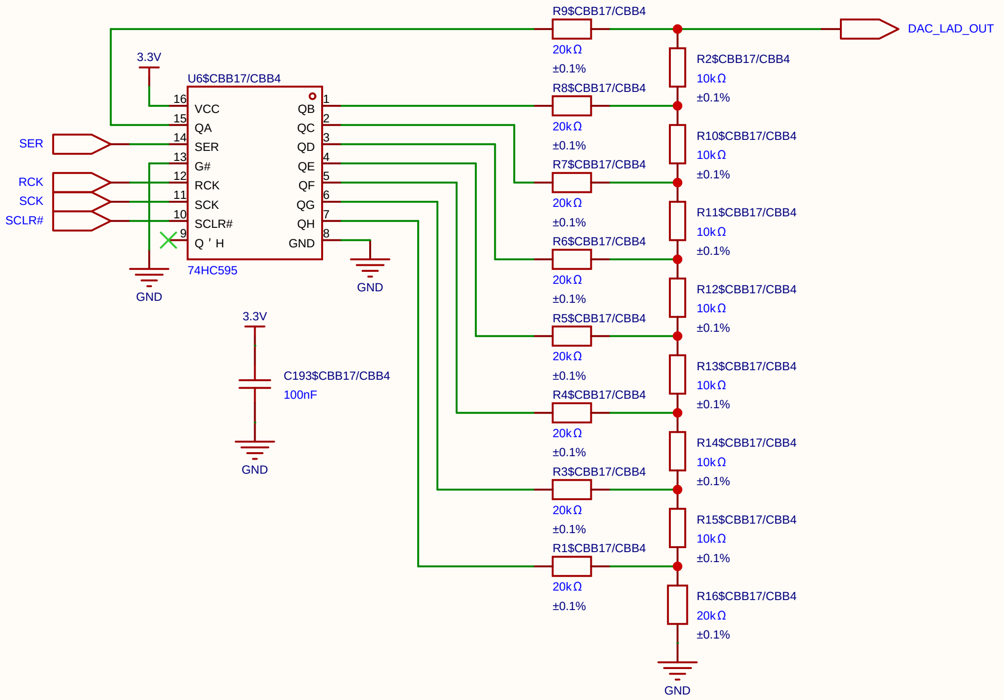

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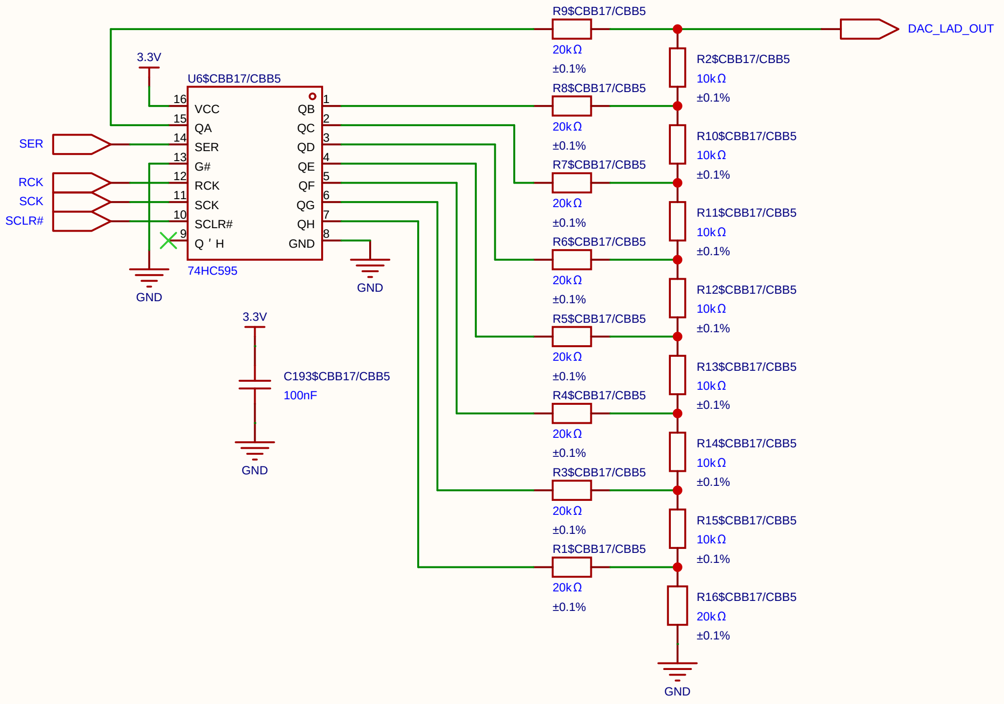

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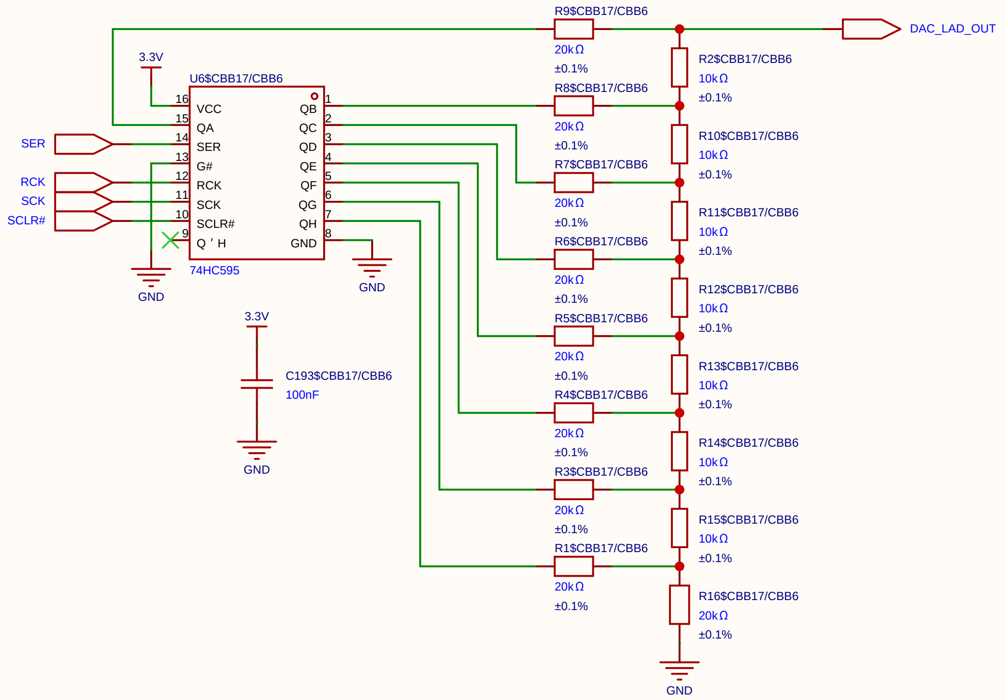

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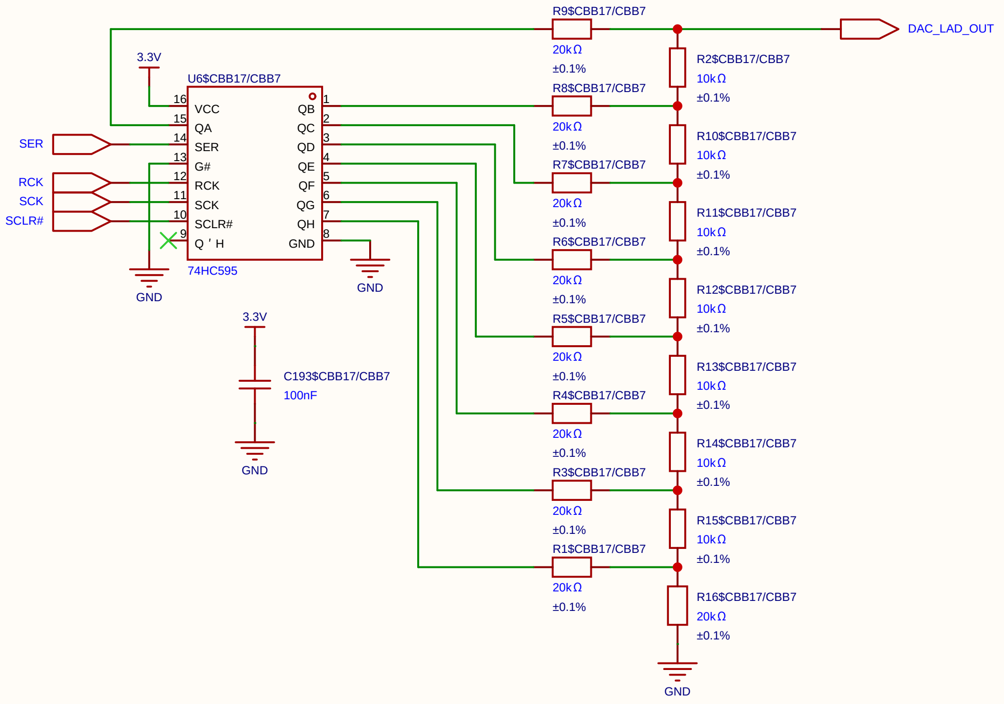

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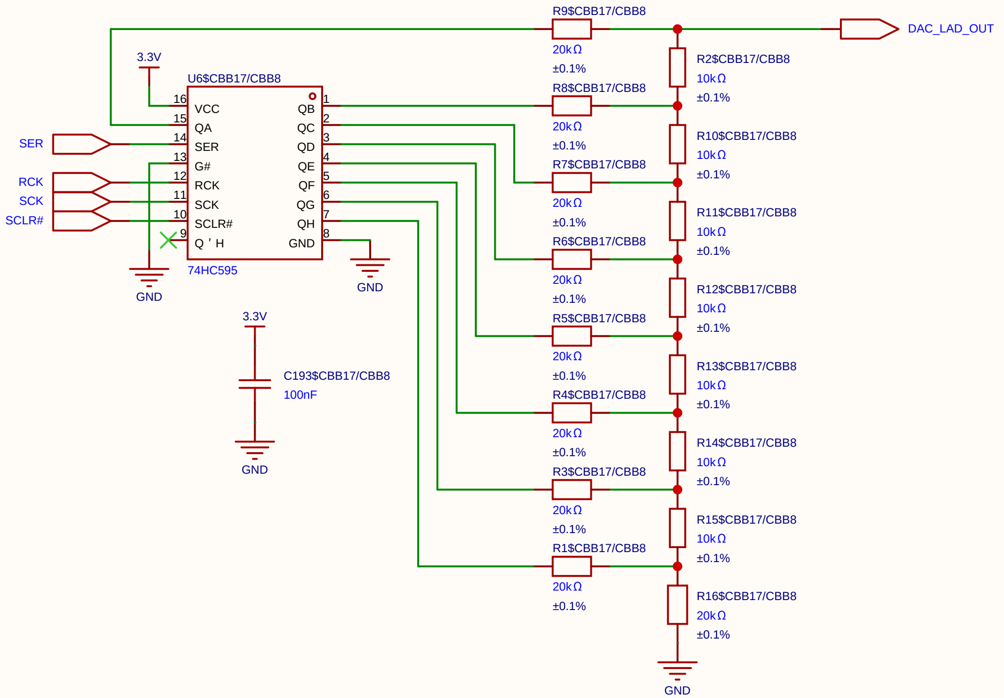

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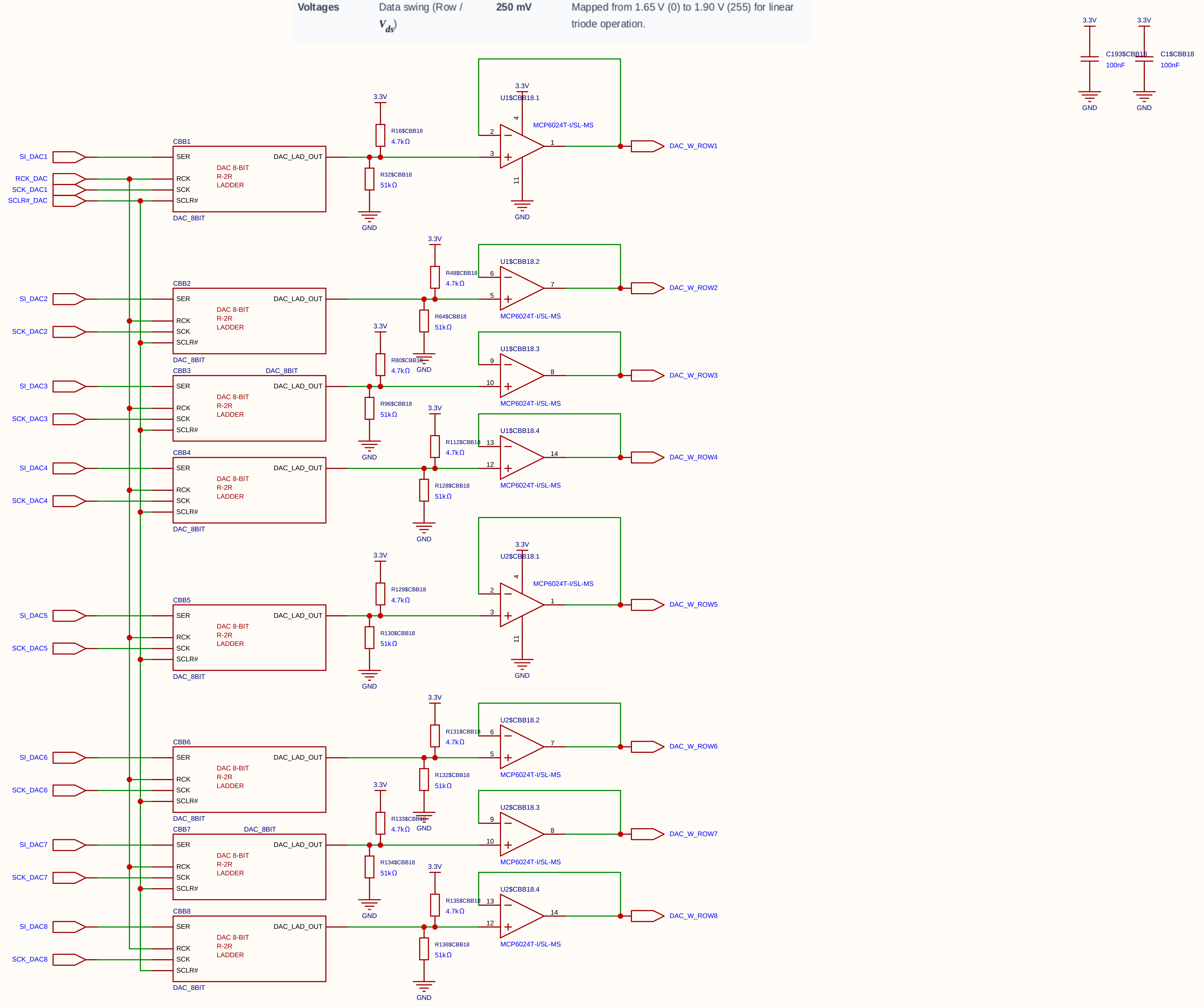
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
		Version	Size	Page 1 Total 1	
EasyEDA		V1.0	A4	EasyEDA.com	

Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.



Schematic	8xDAC_WEIGHT	Create at	2026-06-19
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Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
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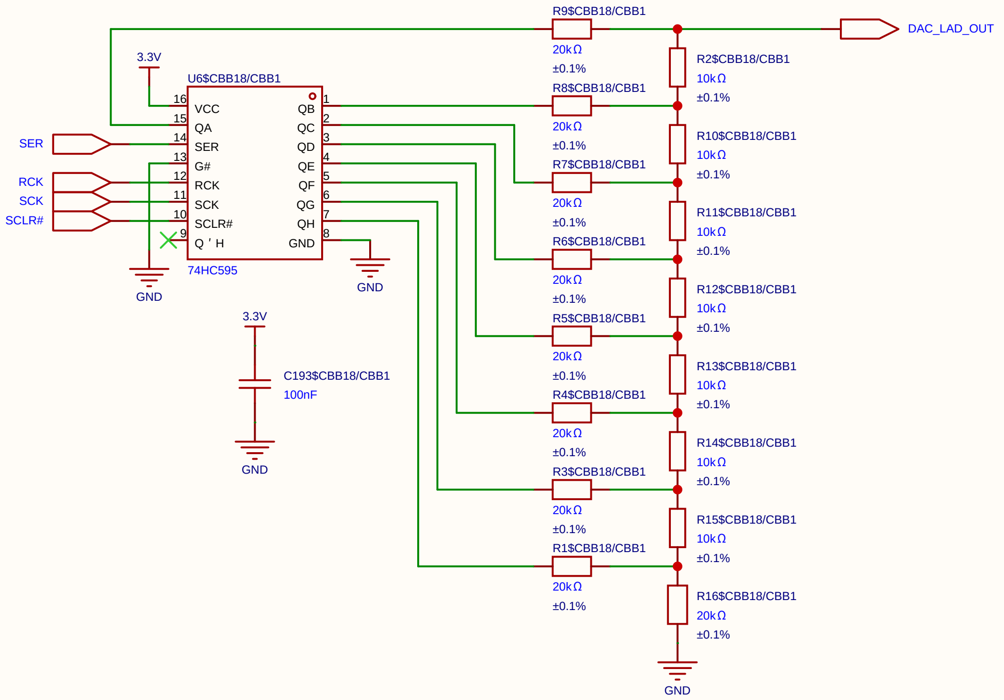

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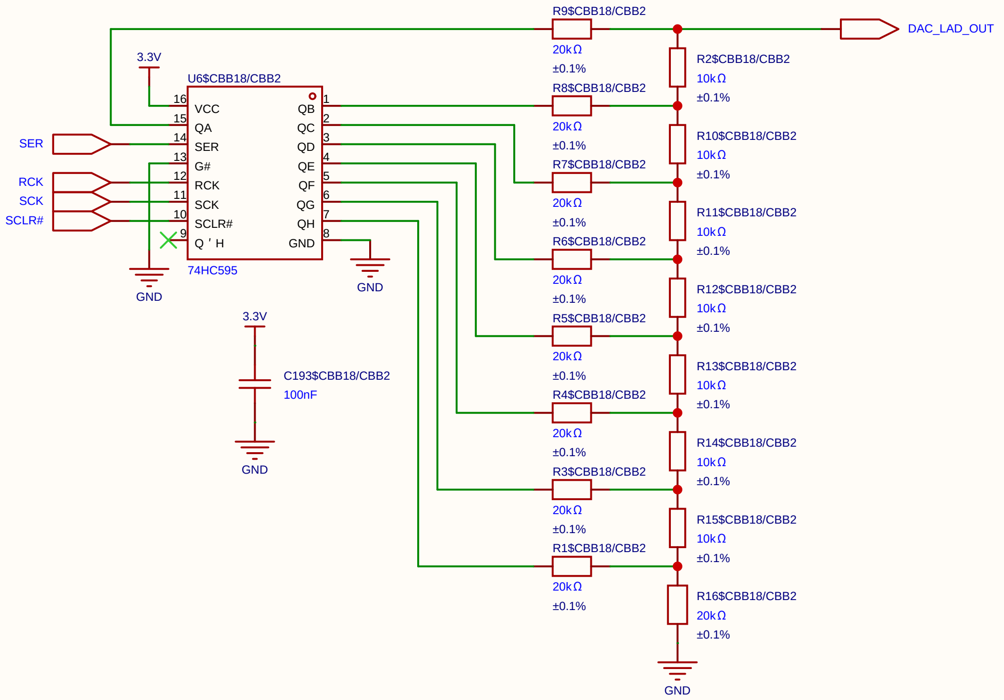
		Update at	2026-06-19
Board		Page	DAC_8B_R_2R
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com



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Version	Size	Page 1 Total 1
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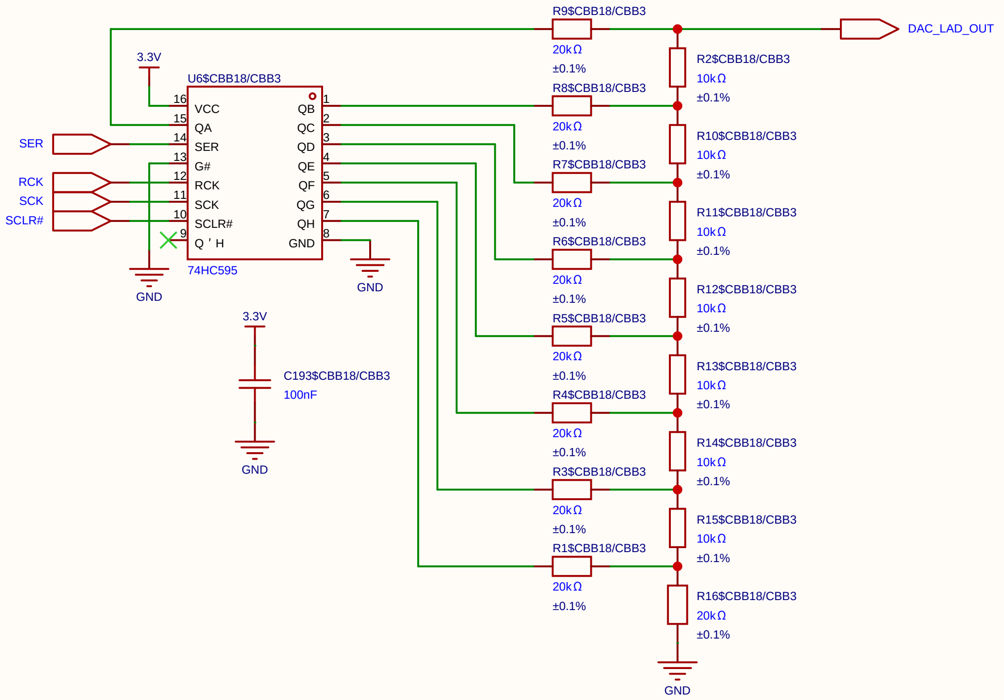

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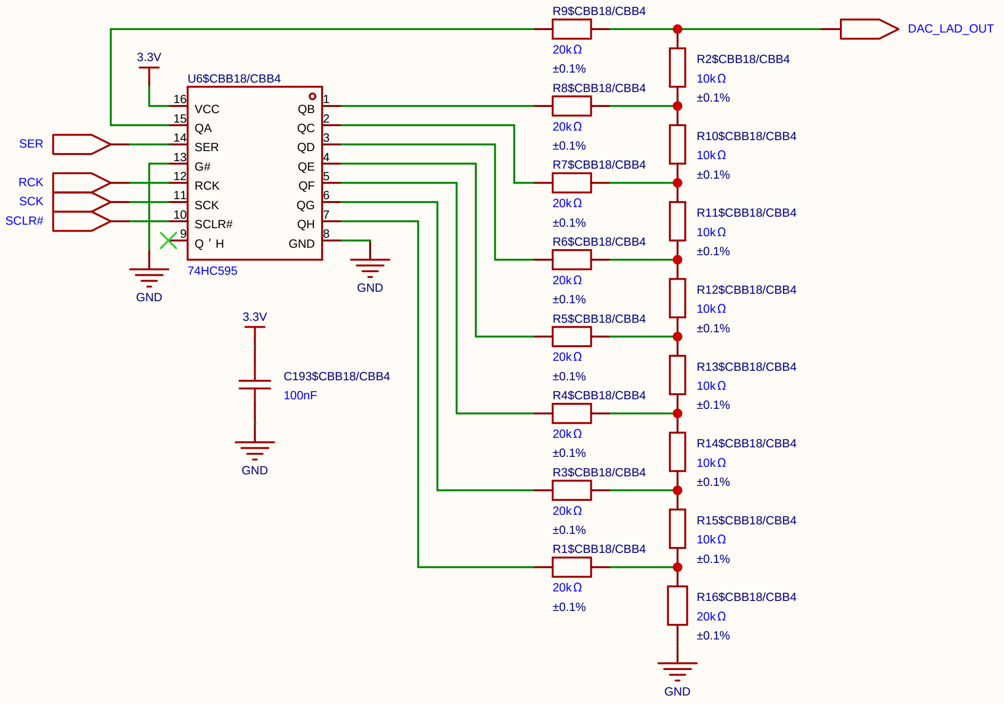

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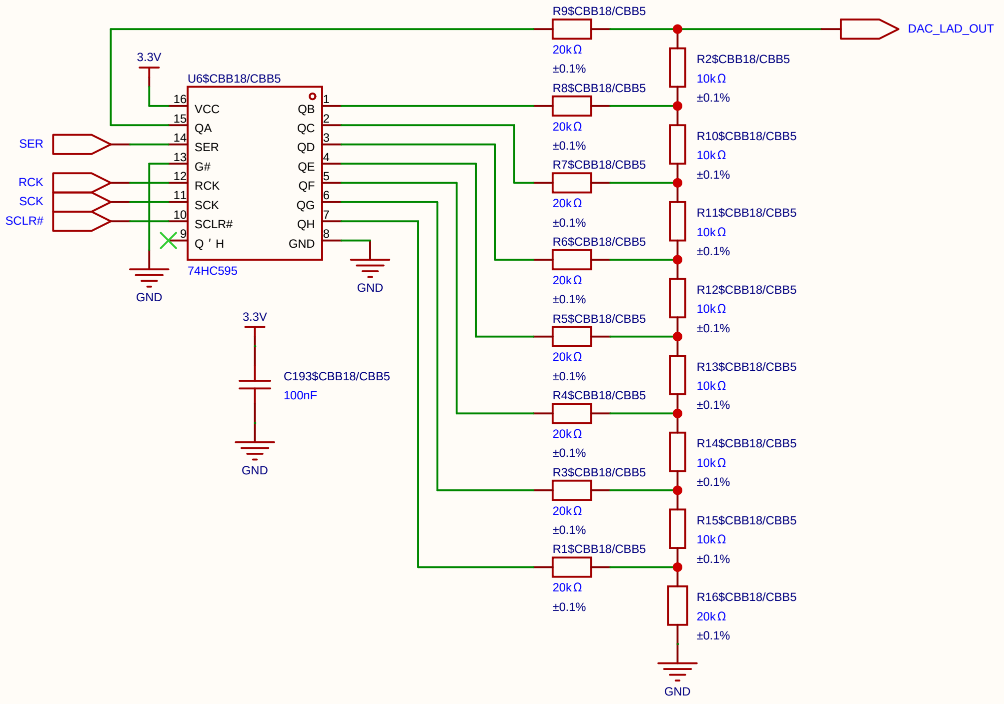

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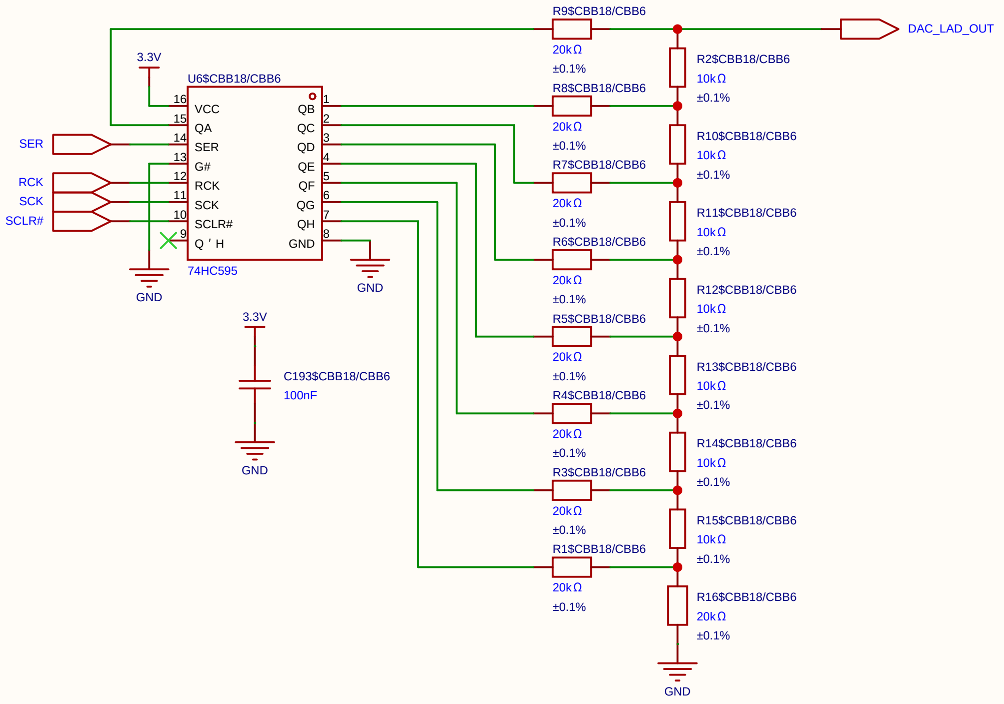

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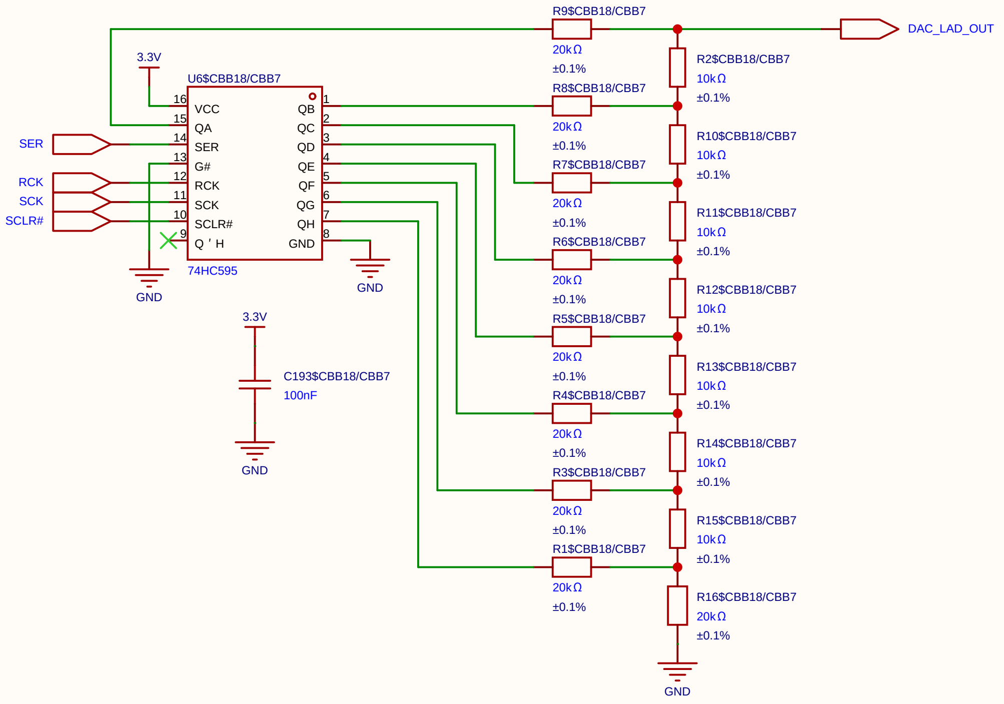

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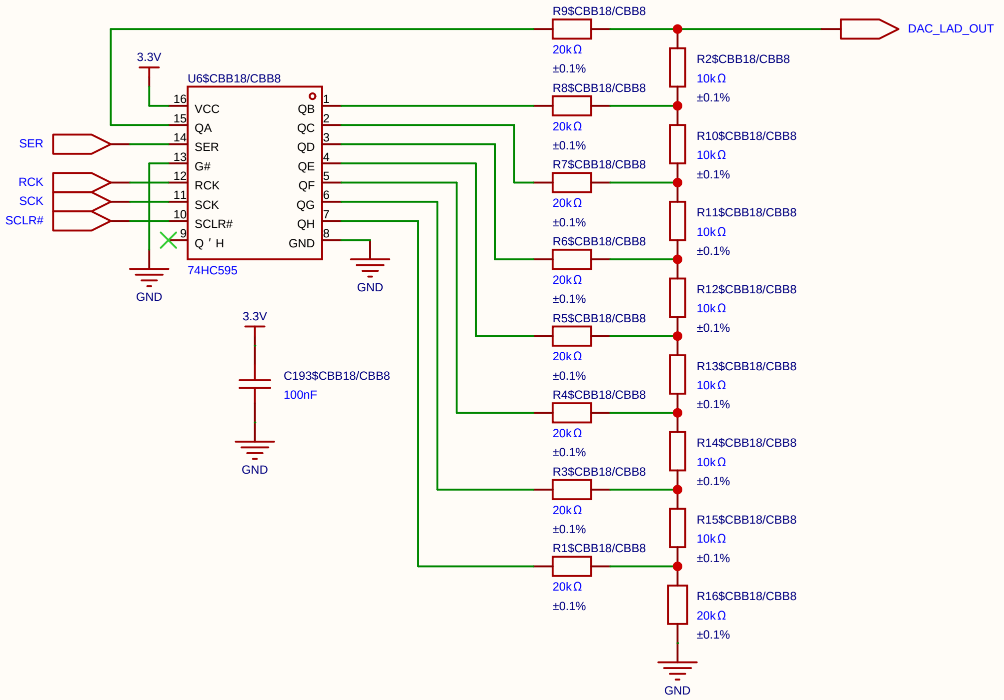

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Drawn	Olle Welin	Inverted_MAML_6x6	
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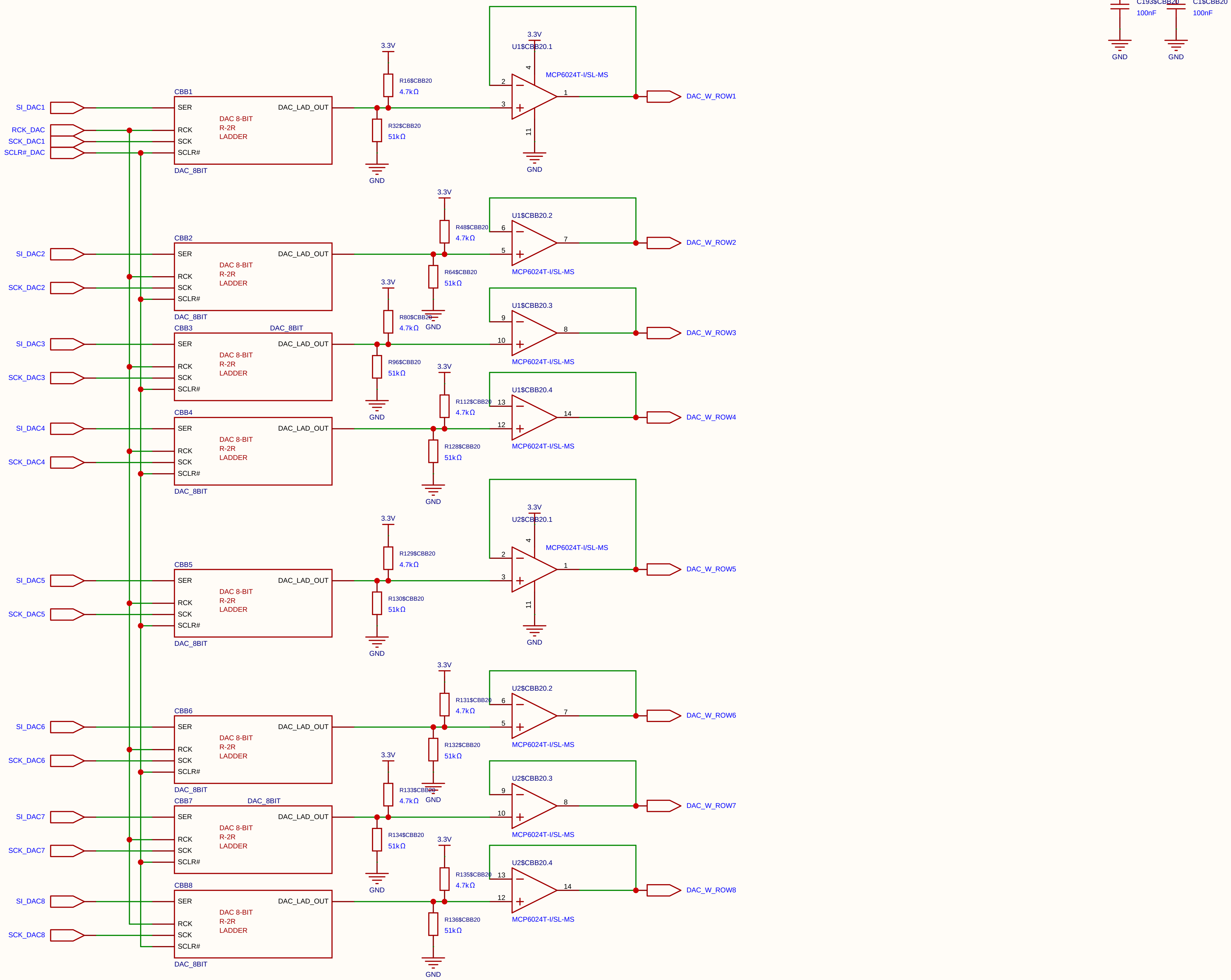
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

Voltages	Data swing (Row / V_{ds})	250 mV	Mapped from 1.65 V (0) to 1.90 V (255) for linear triode operation.



Schematic	8xDAC_WEIGHT	Create at	2026-06-19
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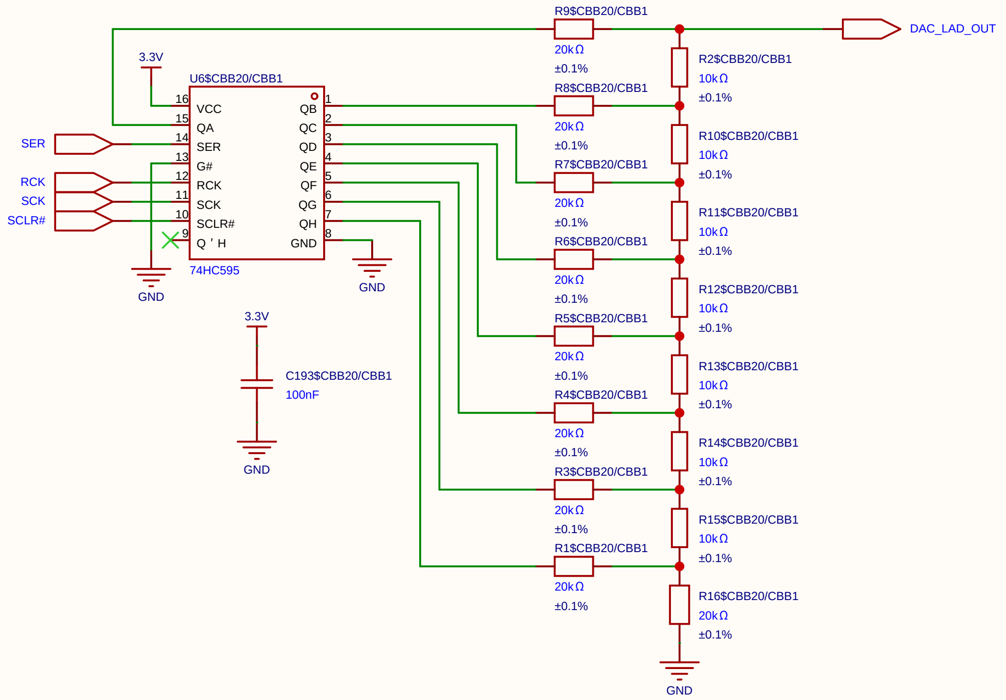
Page 1 Total 1

EasyEDA.com



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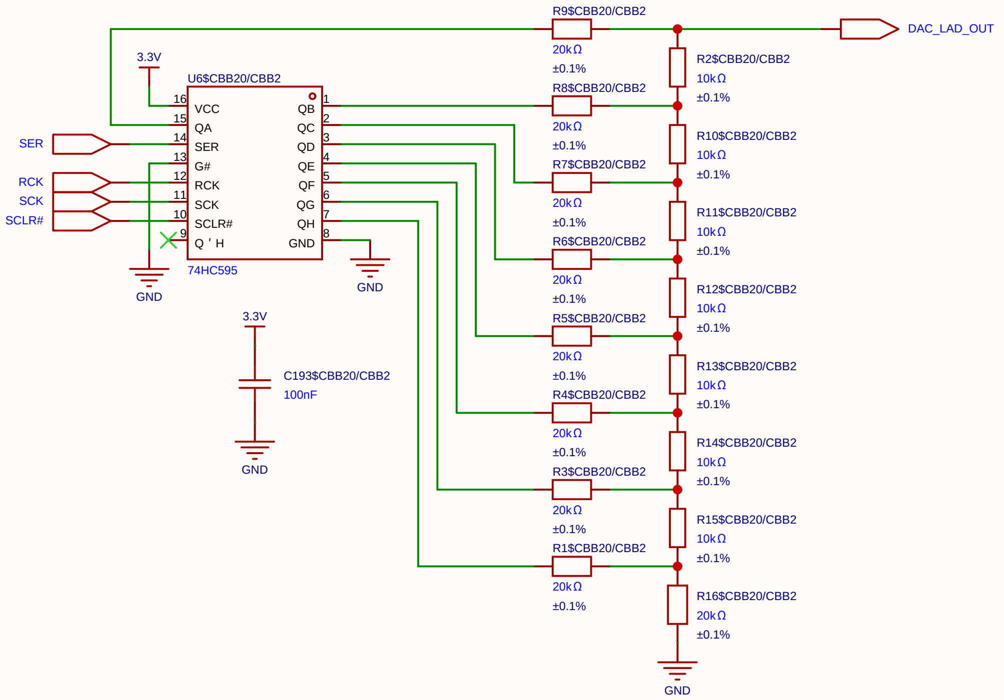
EasyEDA		V1.0	A4	EasyEDA.com



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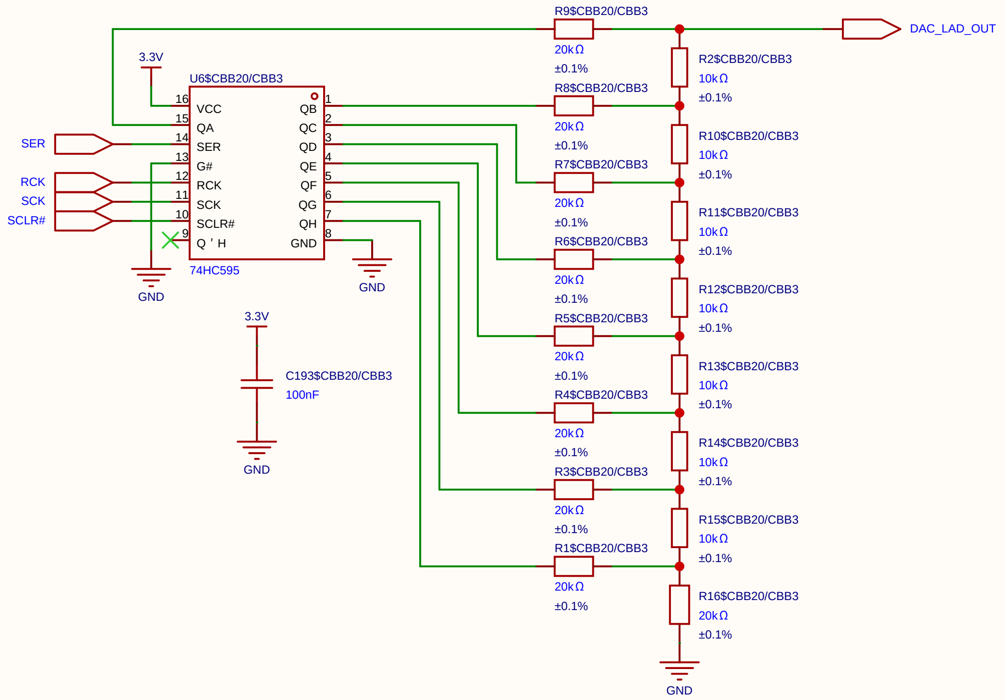

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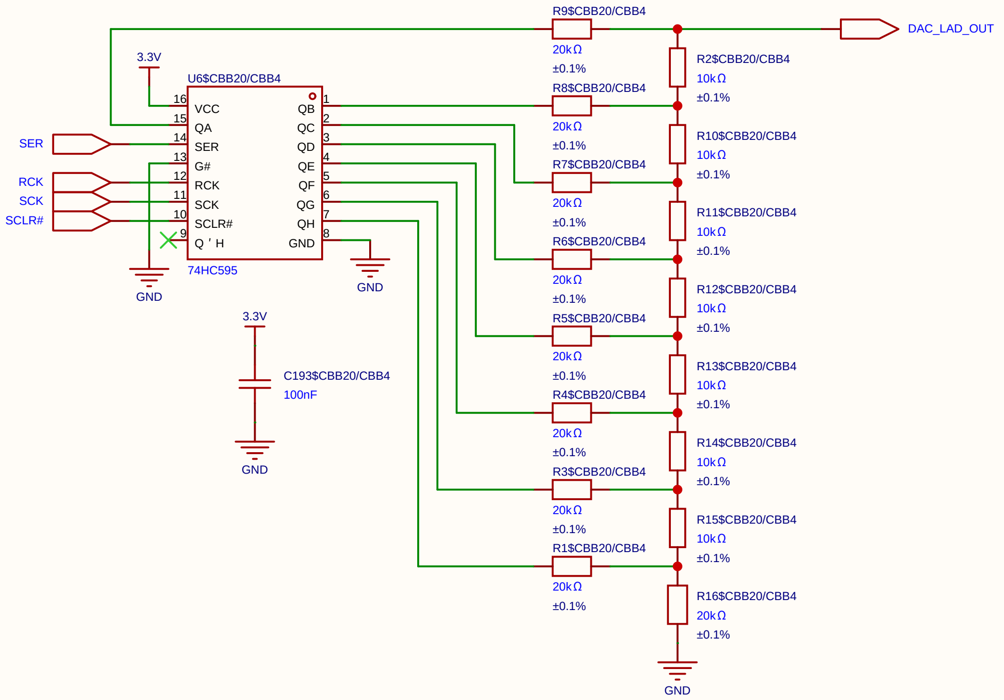

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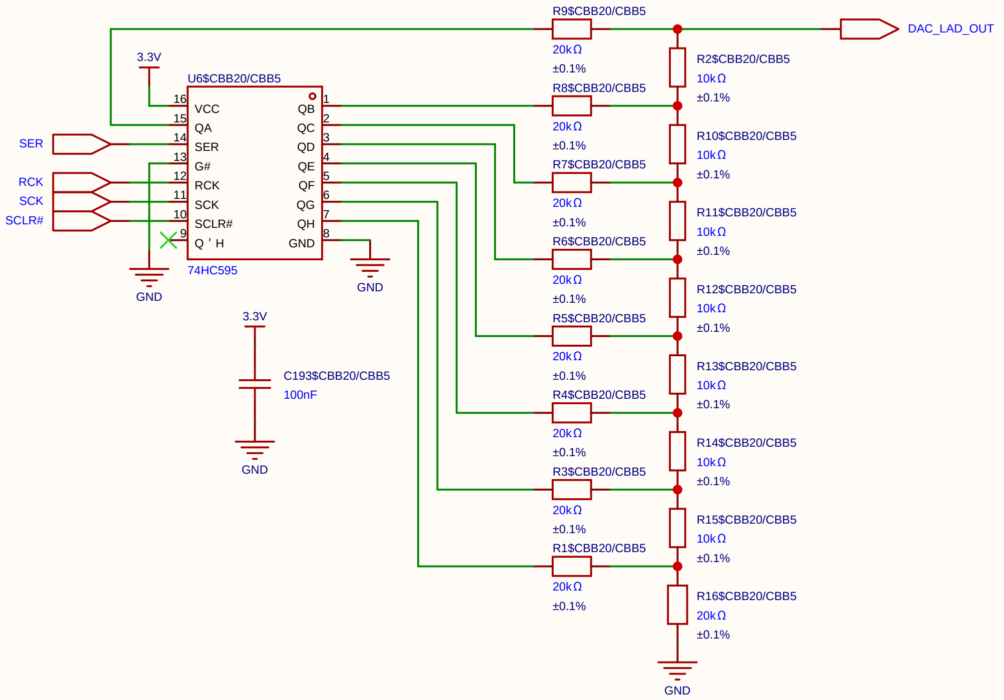

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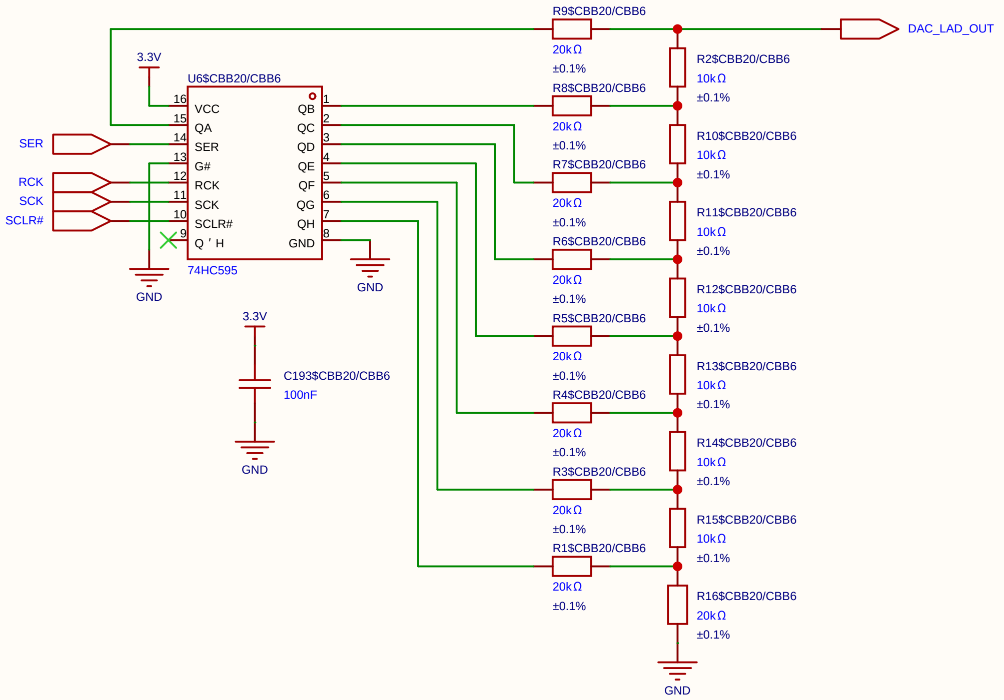

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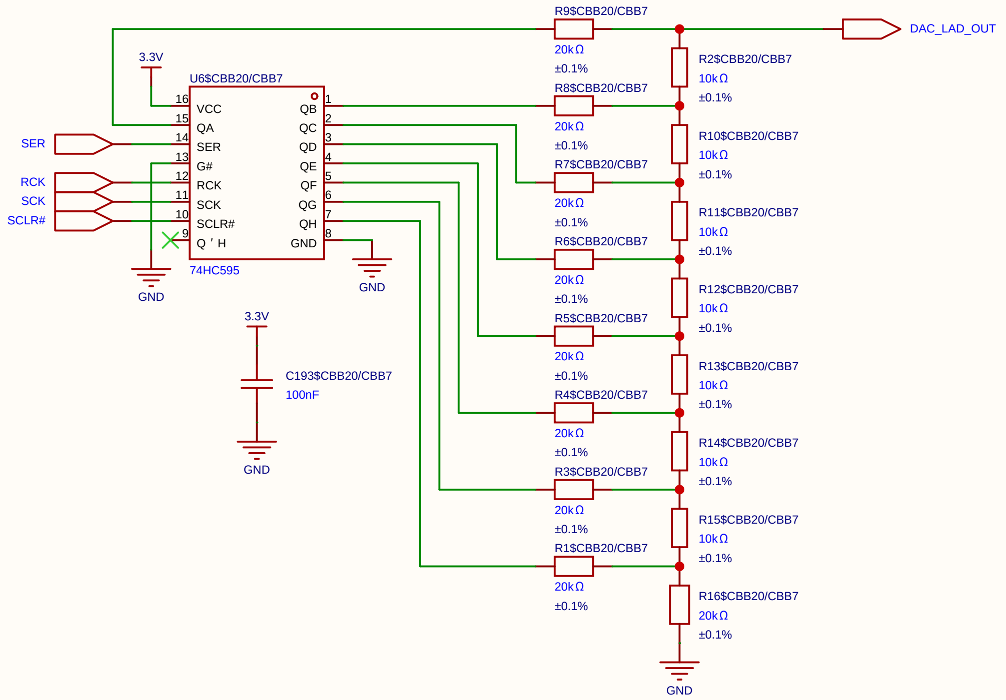

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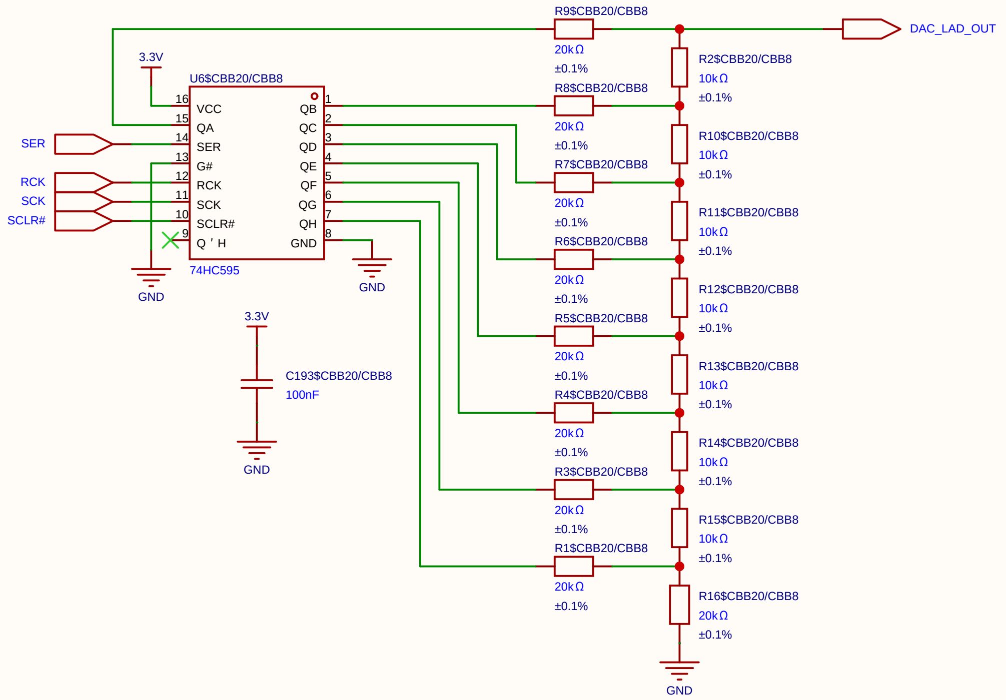

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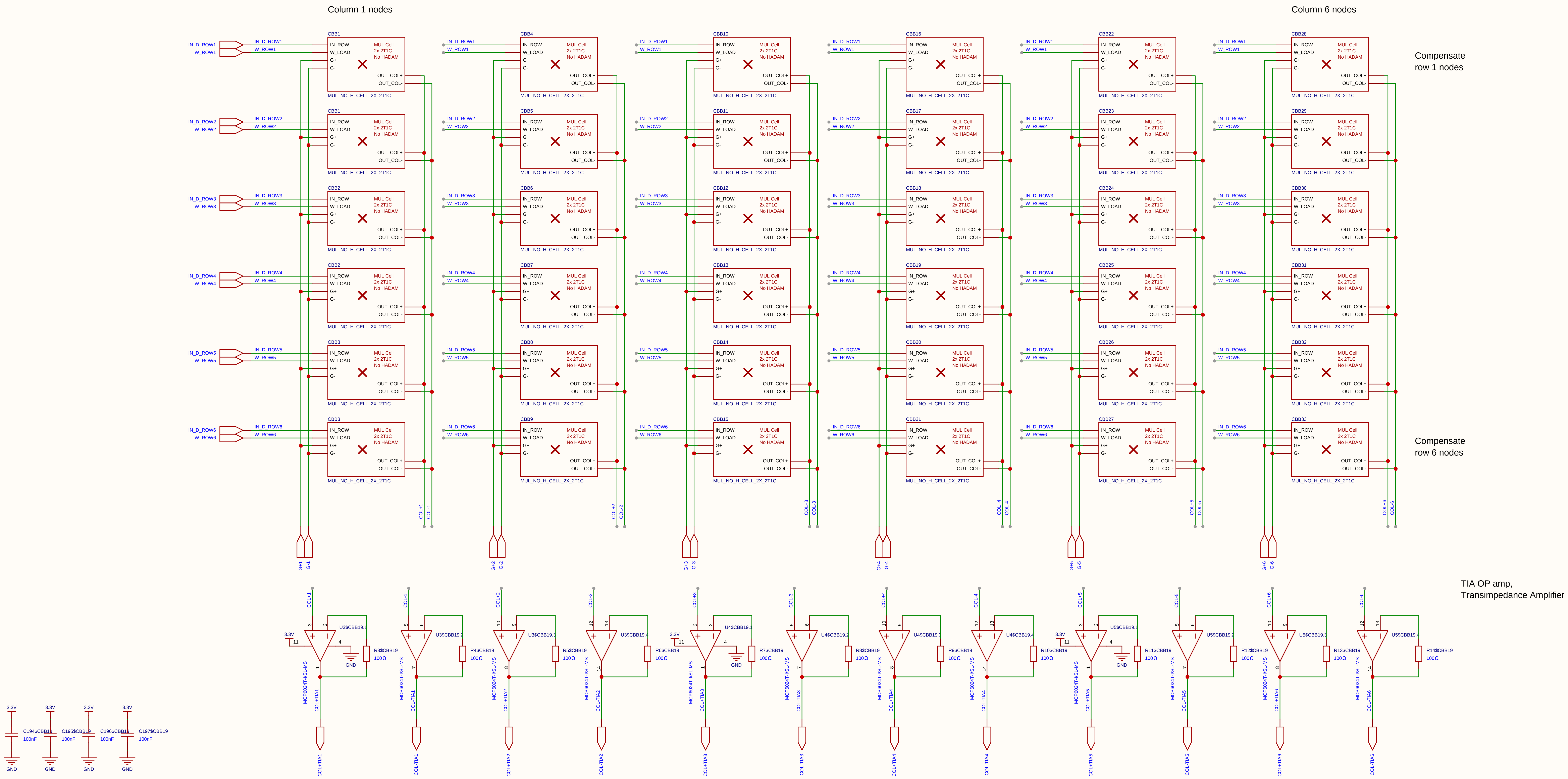
Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

$$V_{ut\_TIA} = 1.65\ \mathrm{V} - (5\ \mathrm{mA} \times 100\ \Omega) = 1.65\ \mathrm{V} - 0.5\ \mathrm{V} = \mathbf{1.15}\ \mathrm{V}$$



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Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
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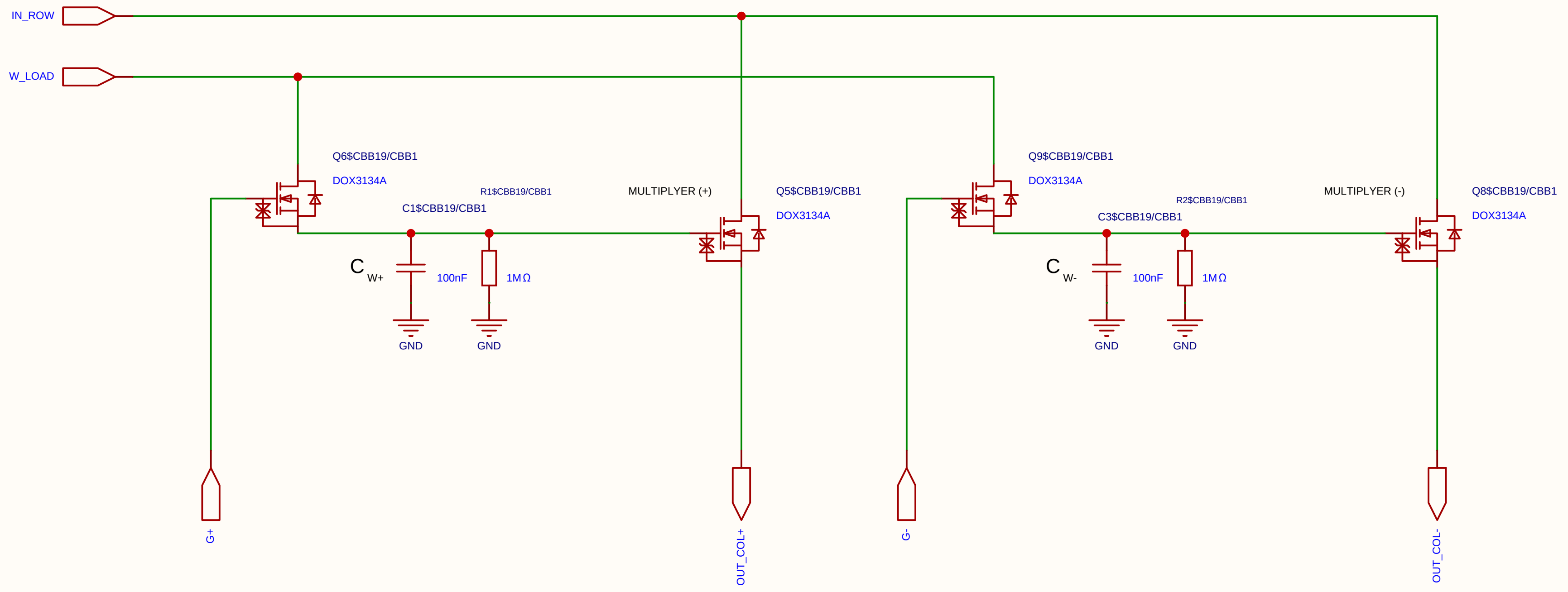
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



Schematic	MUL_NO_H_CELL_2X_2T1C			Create at	2026-06-19
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Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
Reviewed					
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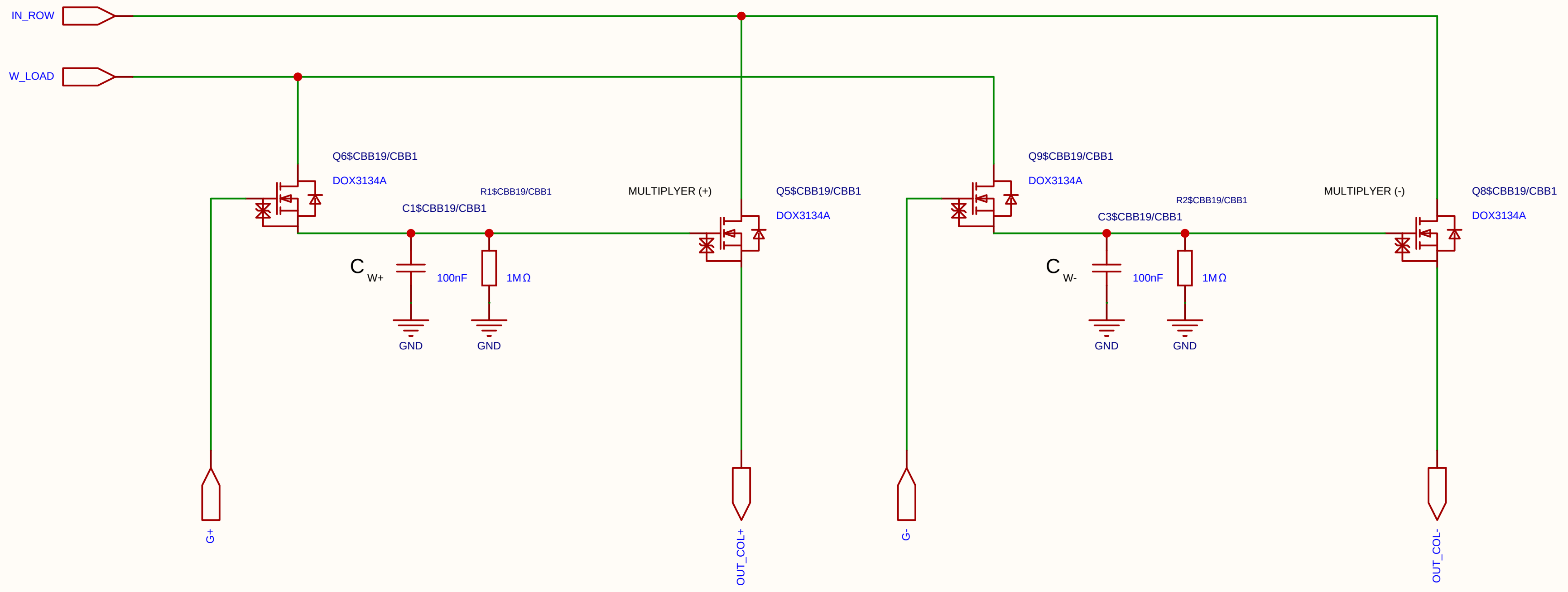
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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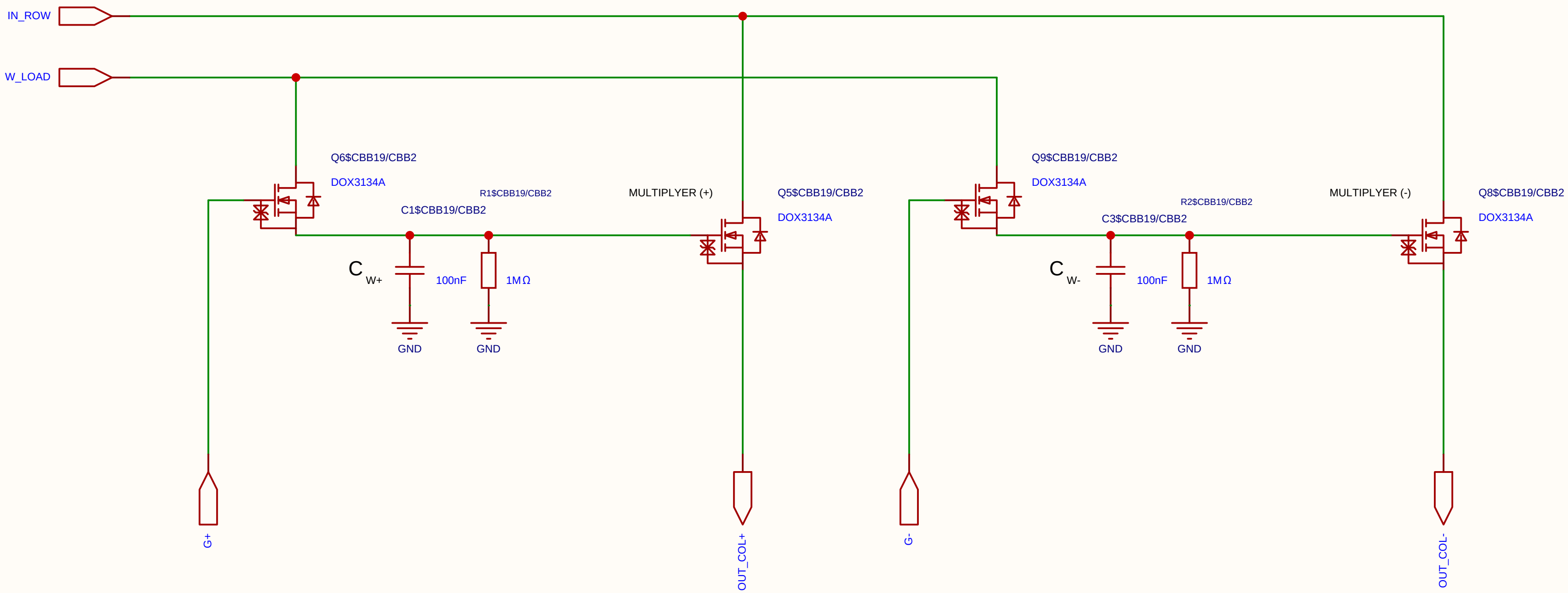
EasyEDA		Page 1 Total 1	EasyEDA.com

Analog Multiplication CELL Kernel

Differential 2x 2T1C



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Drawn	Olle Welin	Inverted_MAML_6x6	
Reviewed			
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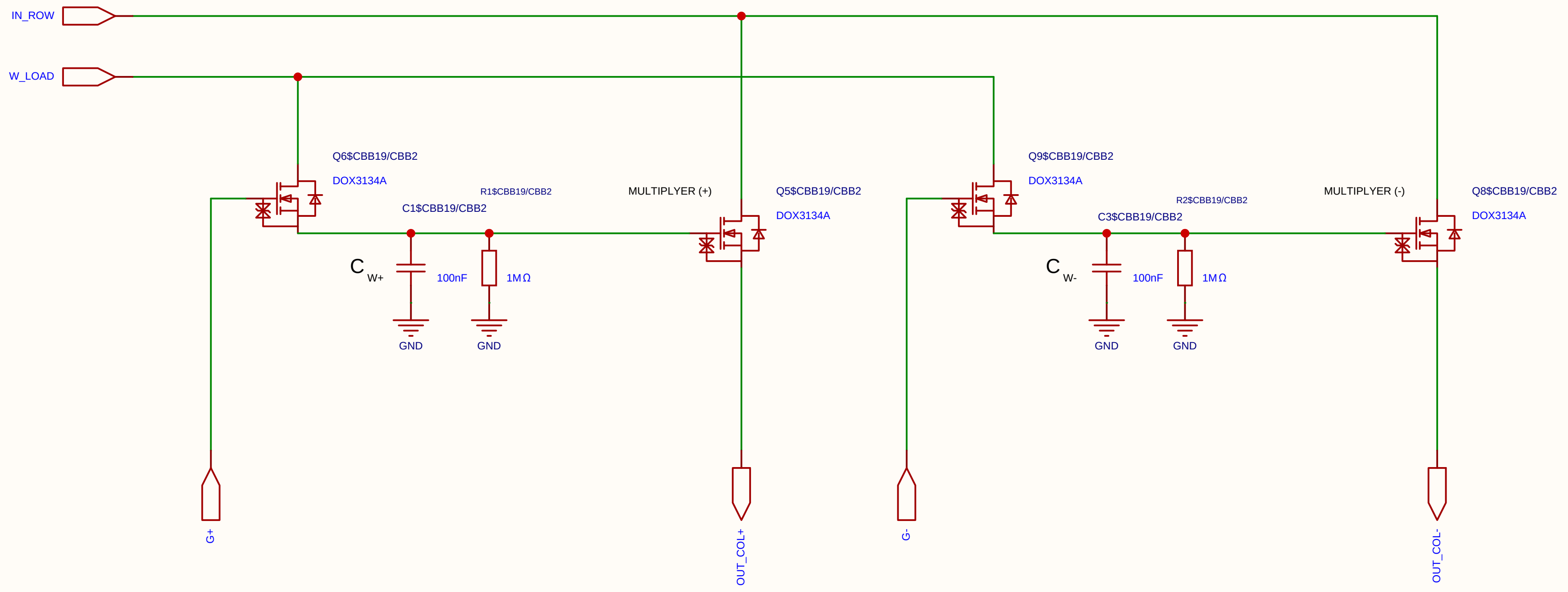
EasyEDA		V1.0 / A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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Board				Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6			
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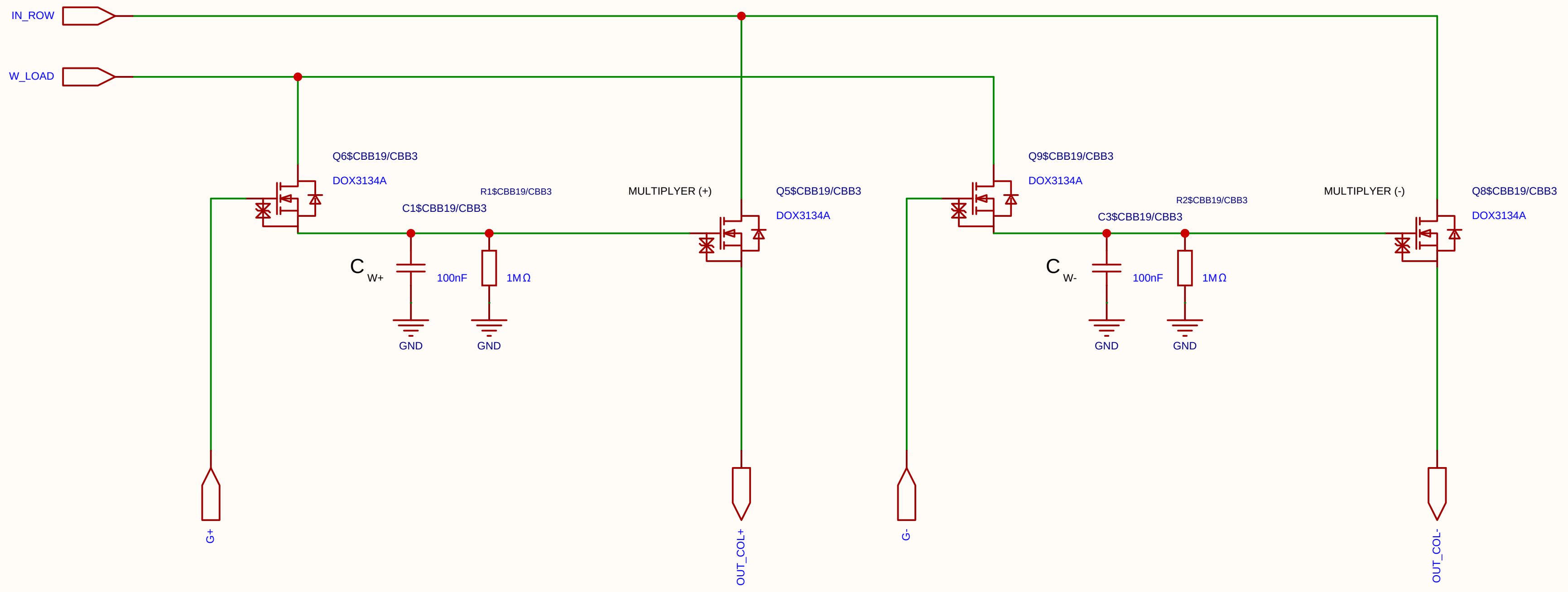
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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Board		Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
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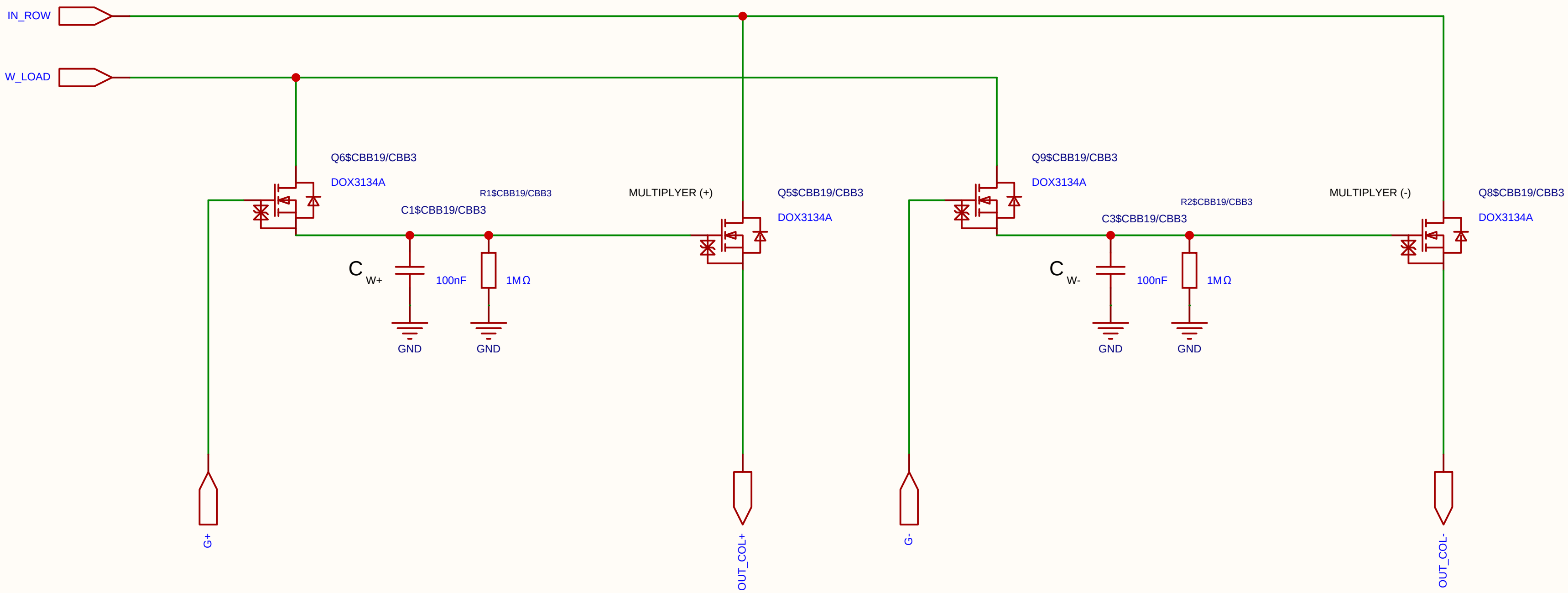
EasyEDA		V1.0 / A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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Board		Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6	
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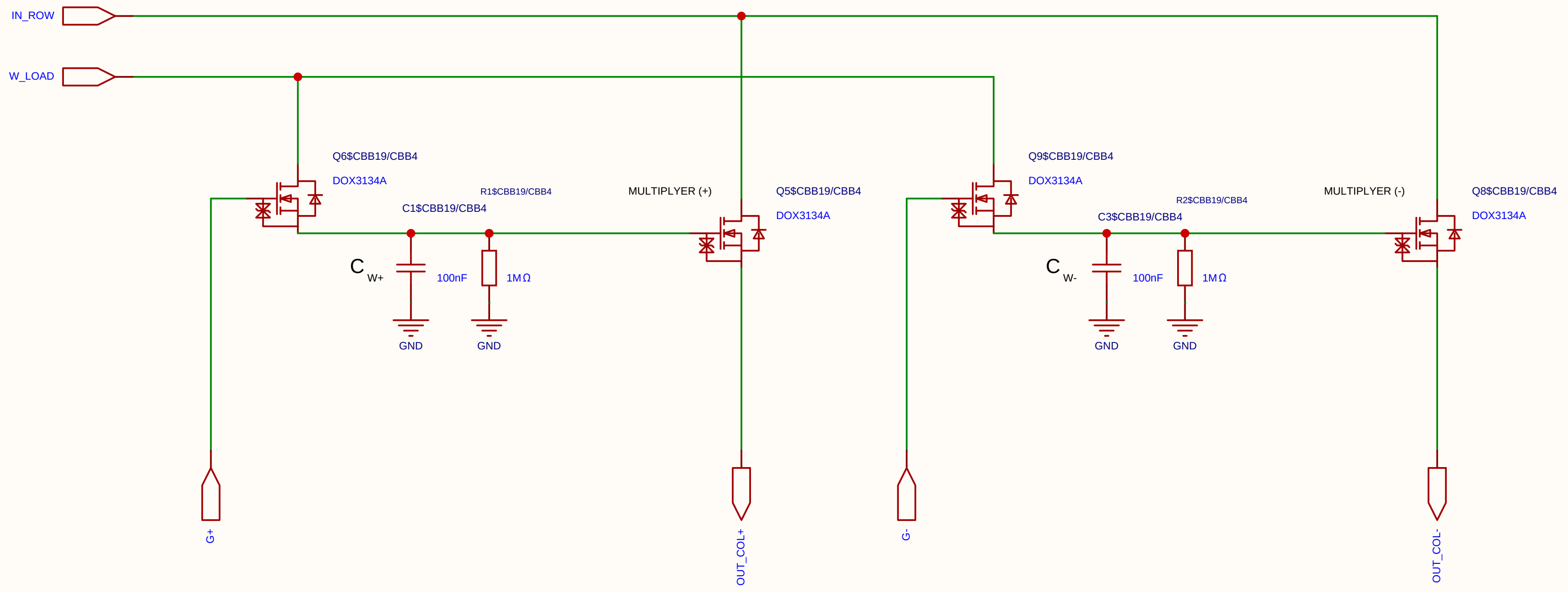
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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Drawn	Olle Welin	Inverted_MAML_6x6		
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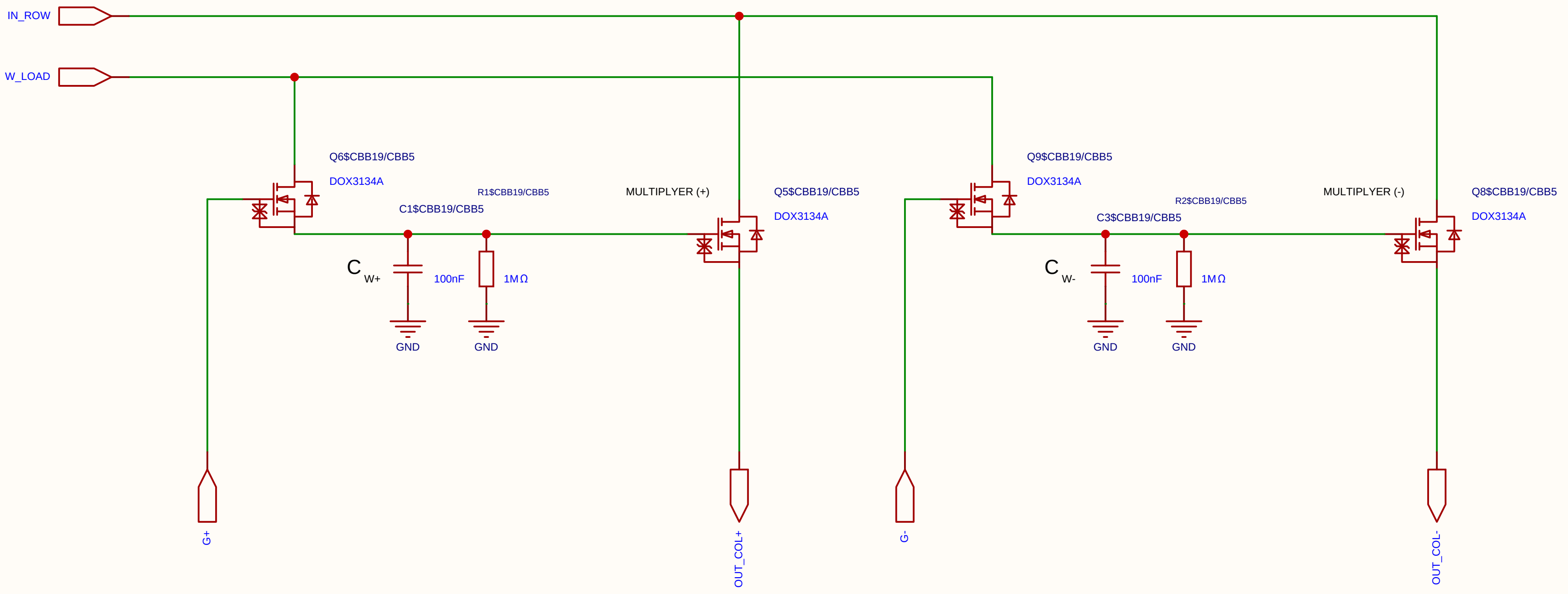
Analog Multiplication CELL Kernel
Differential 2x 2T1C



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Reviewed					
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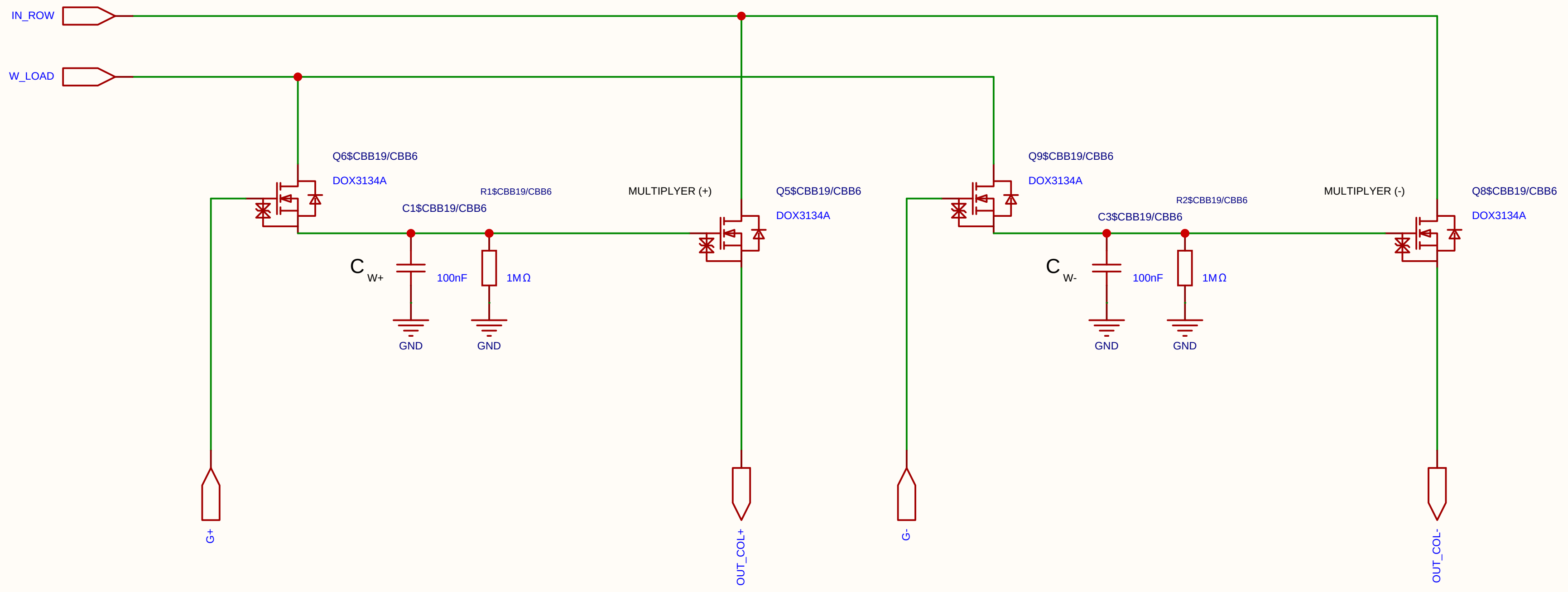
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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			Update at	2026-06-20
Board			Page	matrix_33_MUL_CELL
Drawn	Olle Welin	Inverted_MAML_6x6		
Reviewed				
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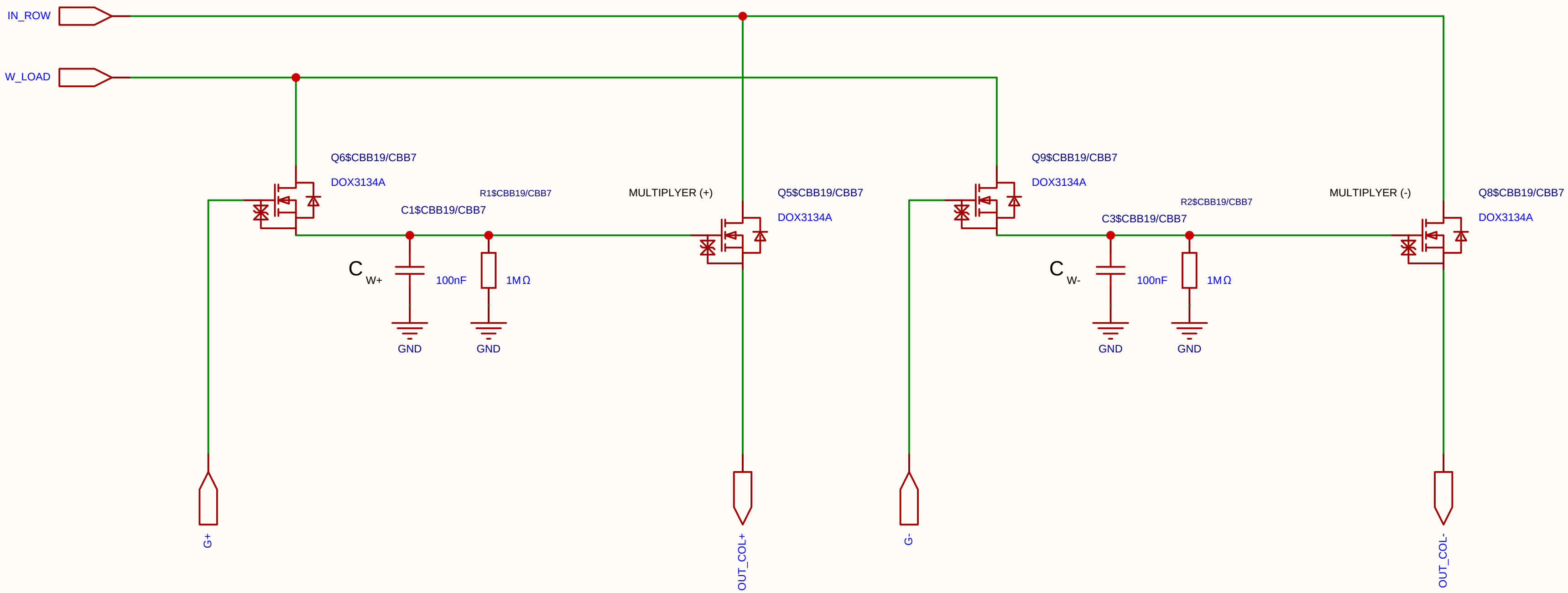
EasyEDA		V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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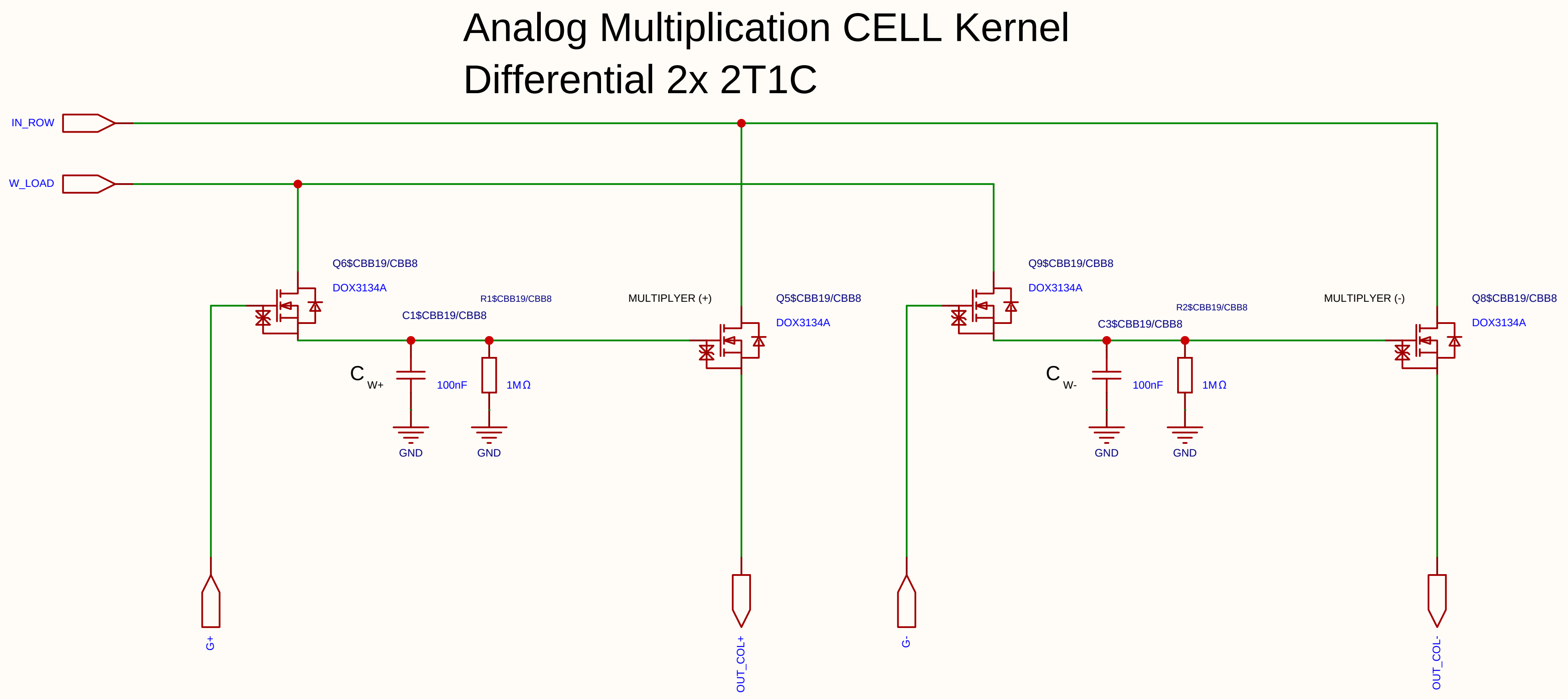

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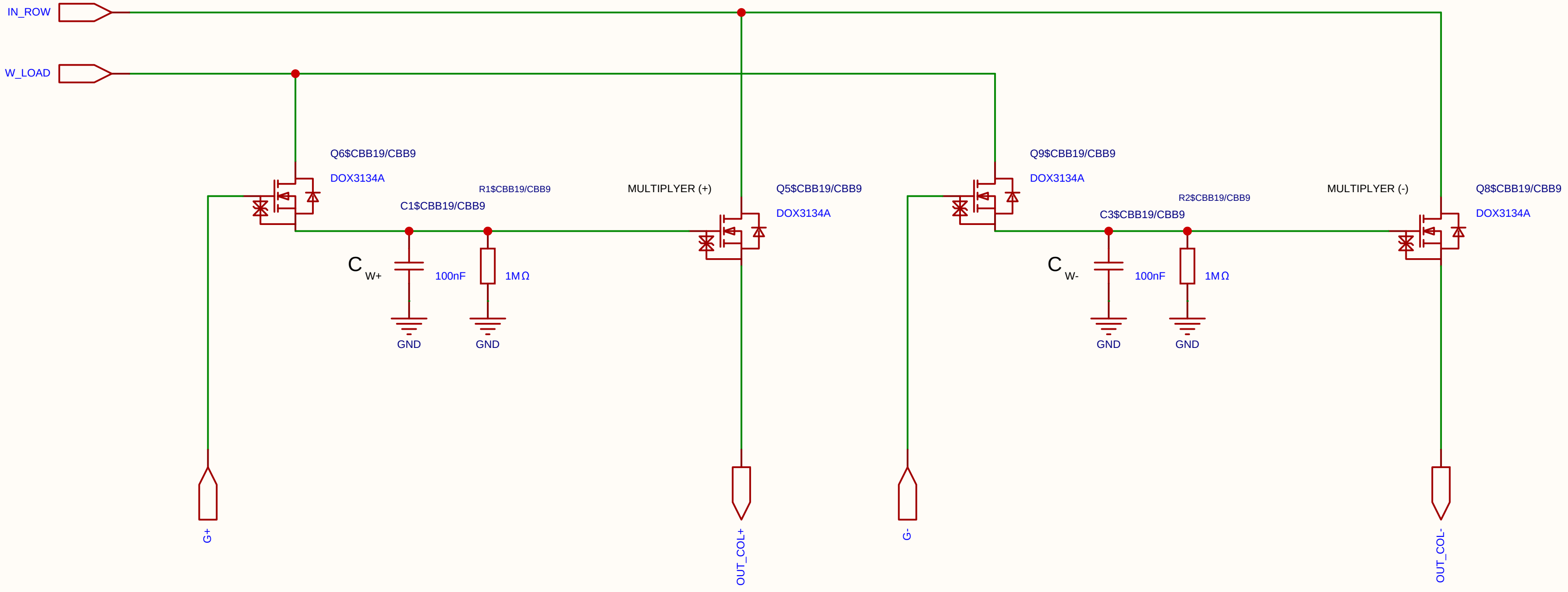
EasyEDA		V1.0 / A4 / EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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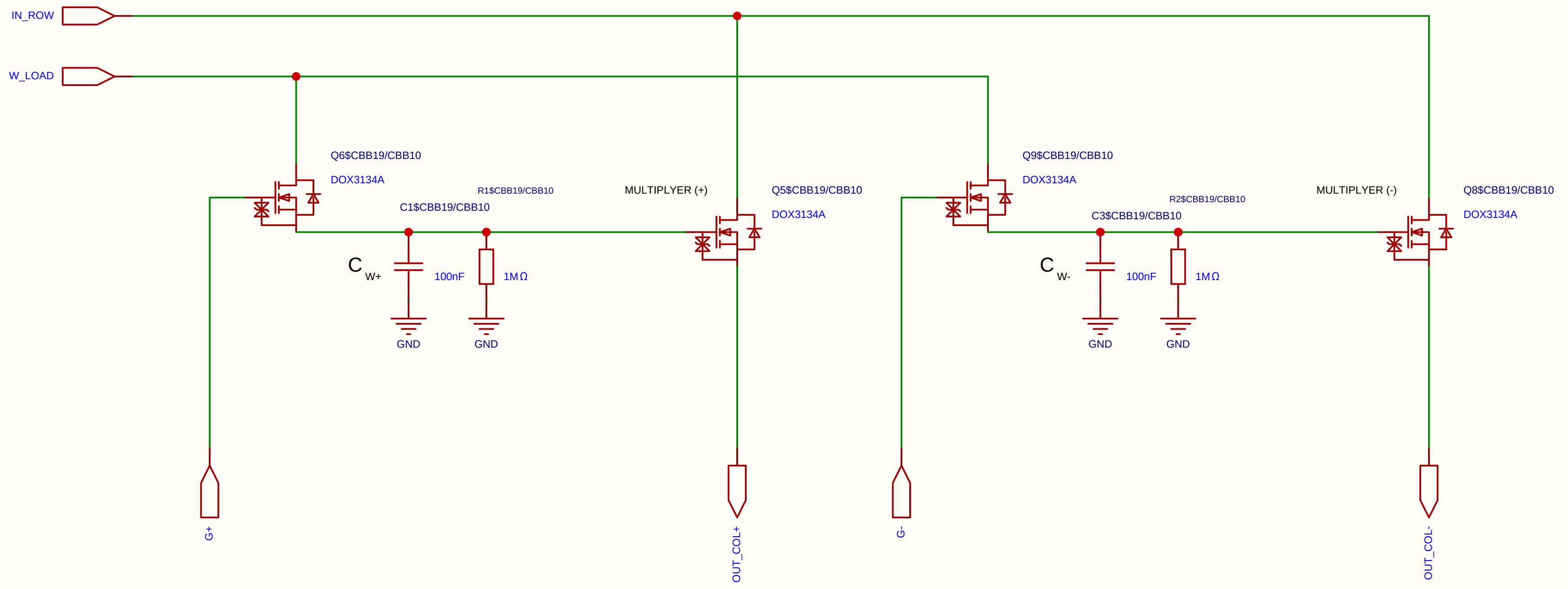
Differential 2x 2T1C



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V1.0	A4	EasyEDA.com	

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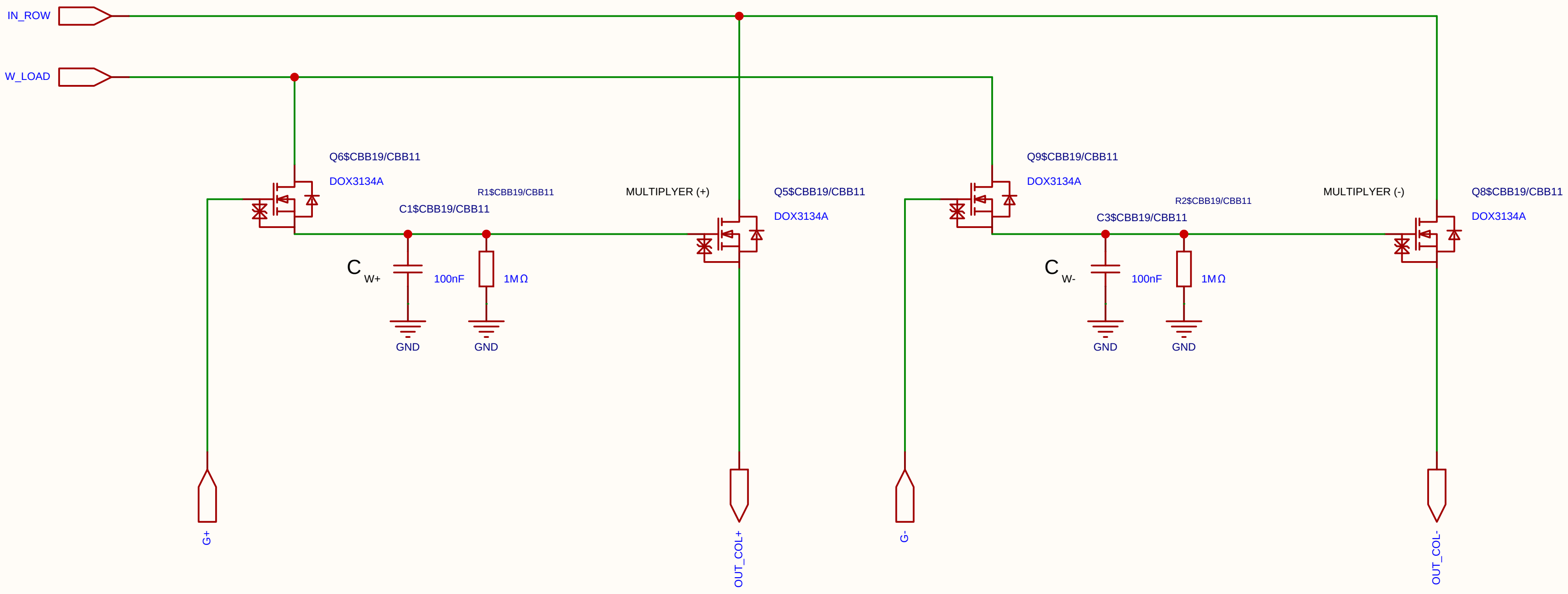
Differential 2x 2T1C



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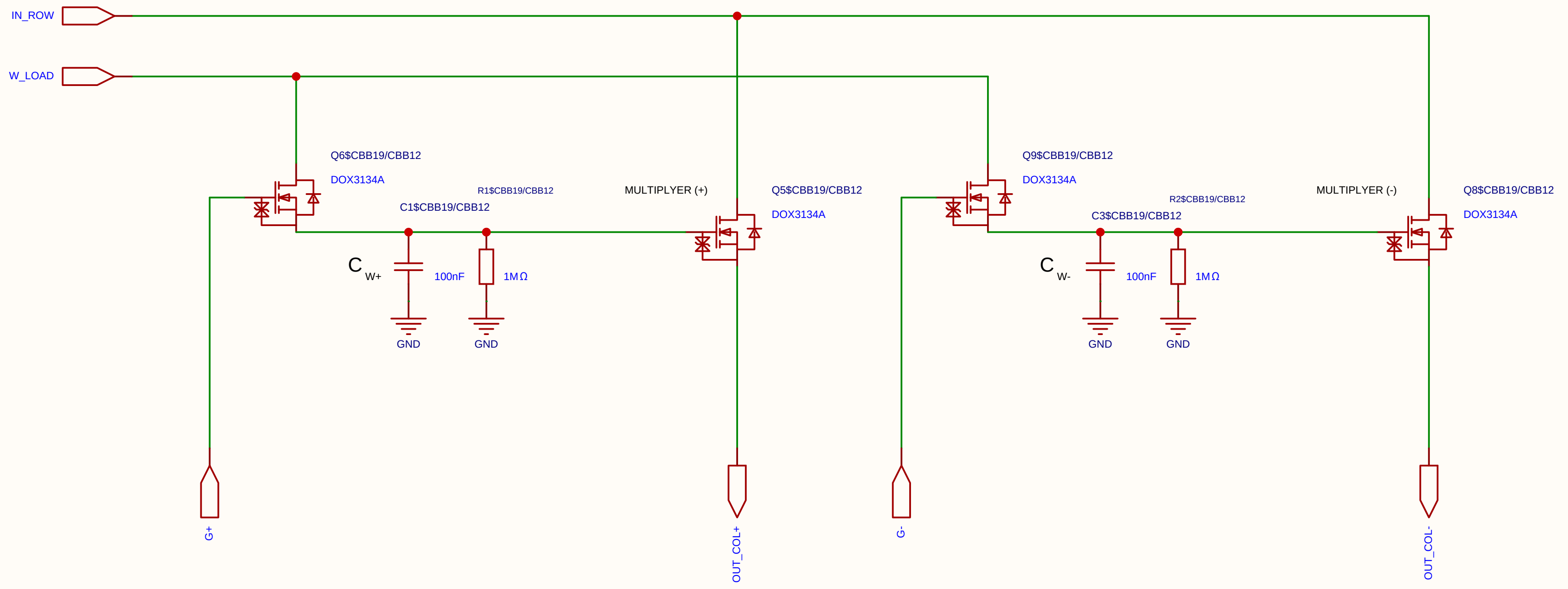
Differential 2x 2T1C



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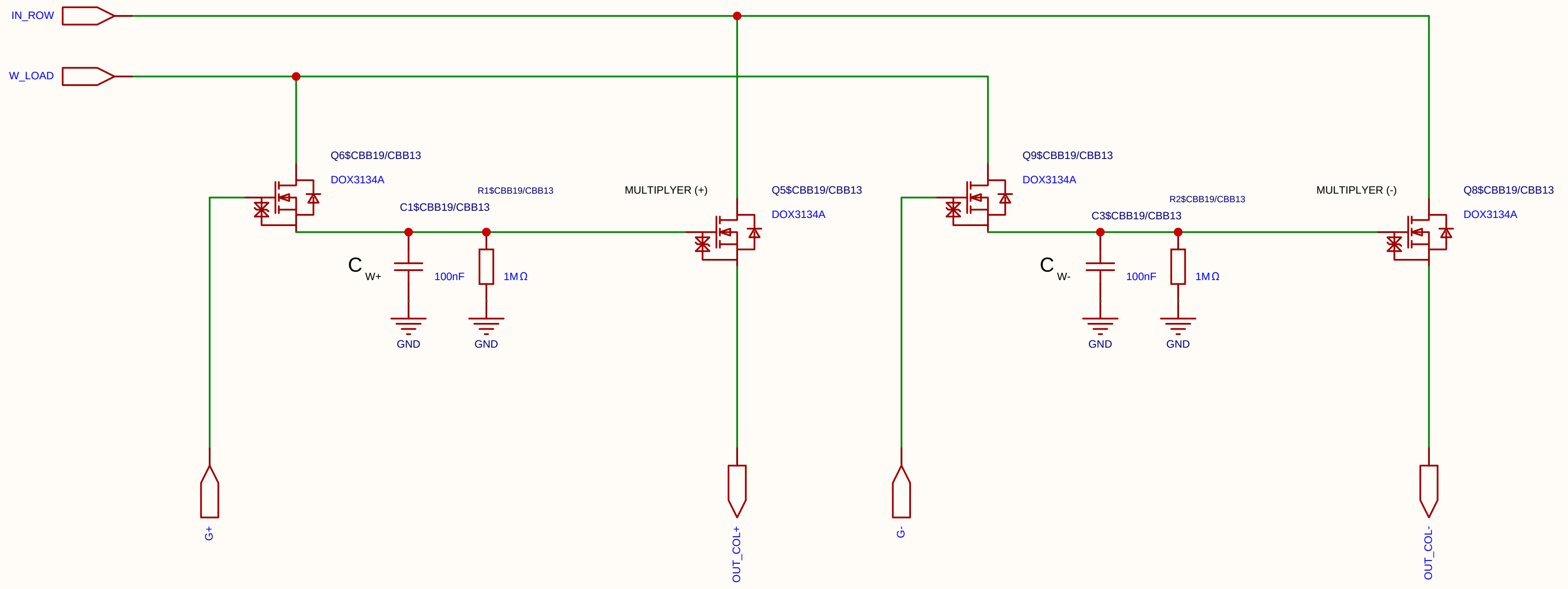
Analog Multiplication CELL Kernel
Differential 2x 2T1C



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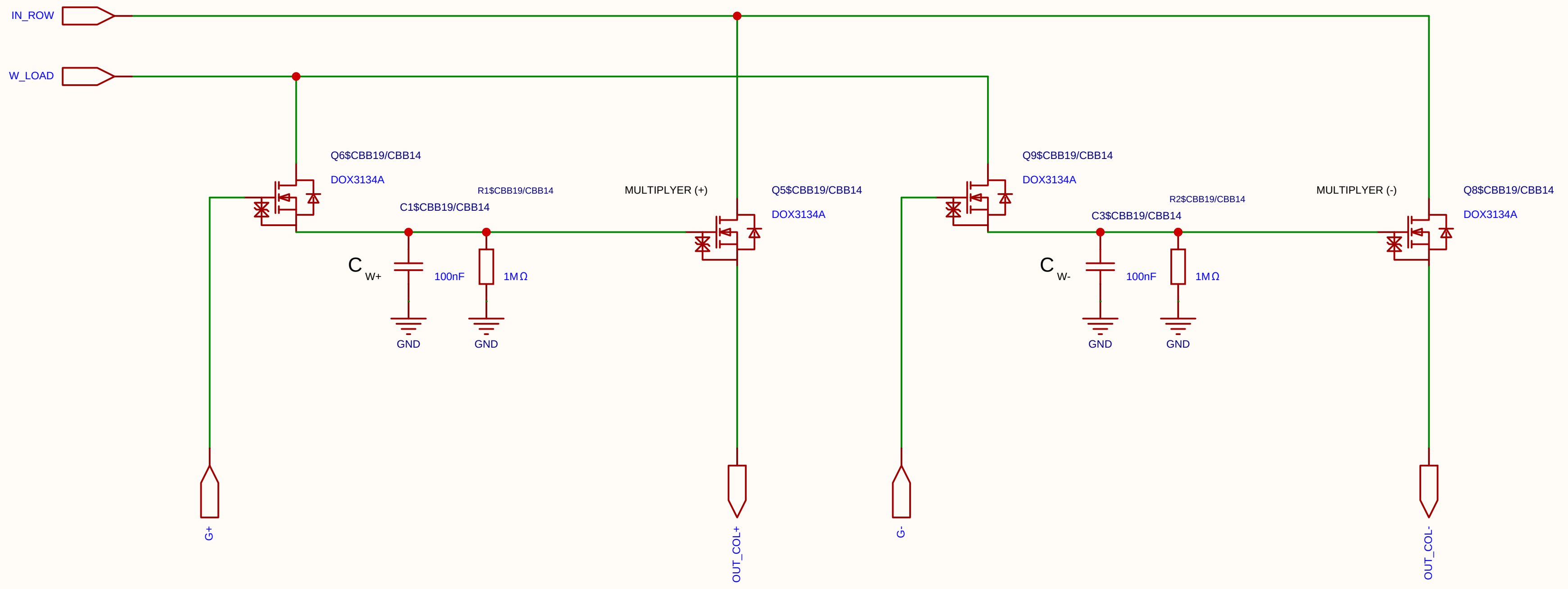

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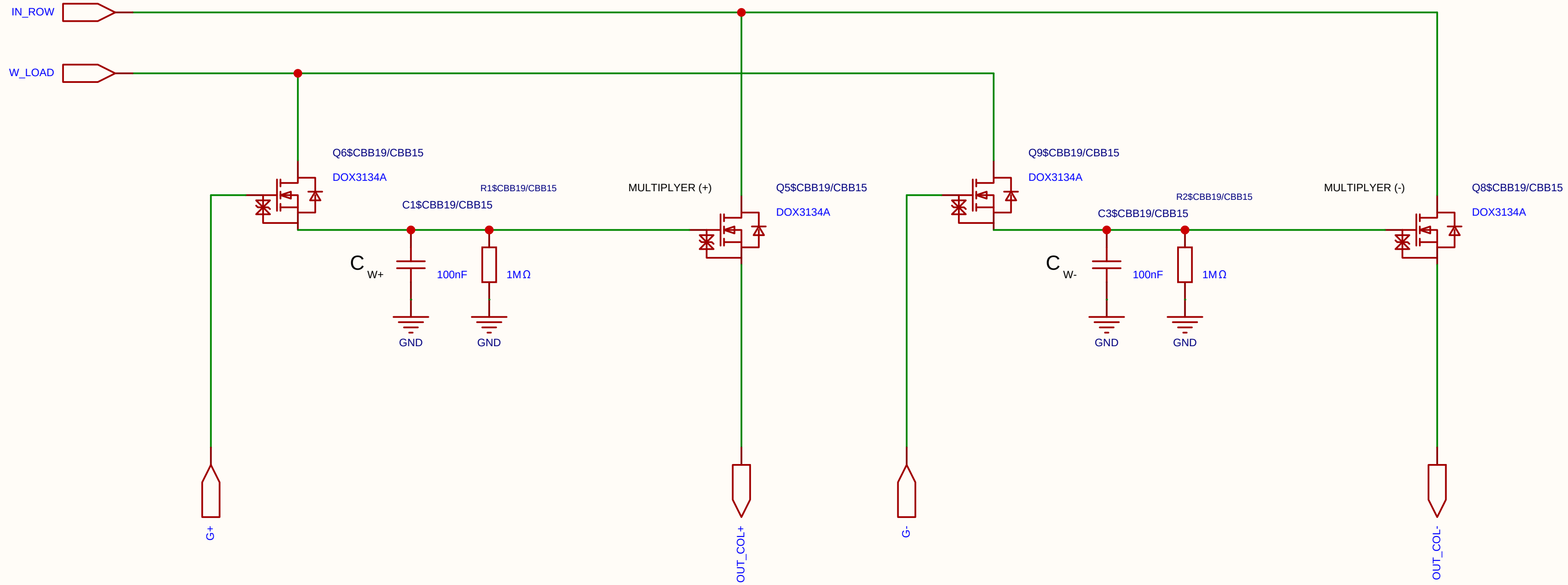
Differential 2x 2T1C



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Reviewed			
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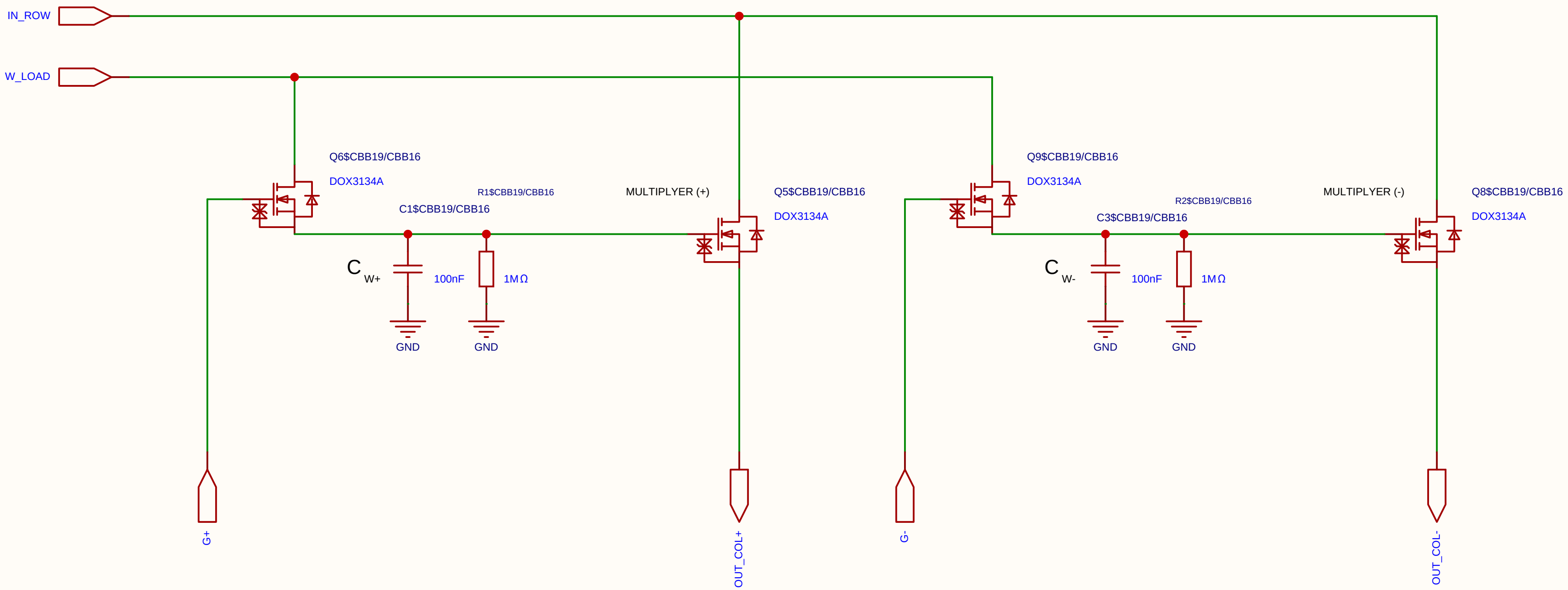
EasyEDA		V1.0 / A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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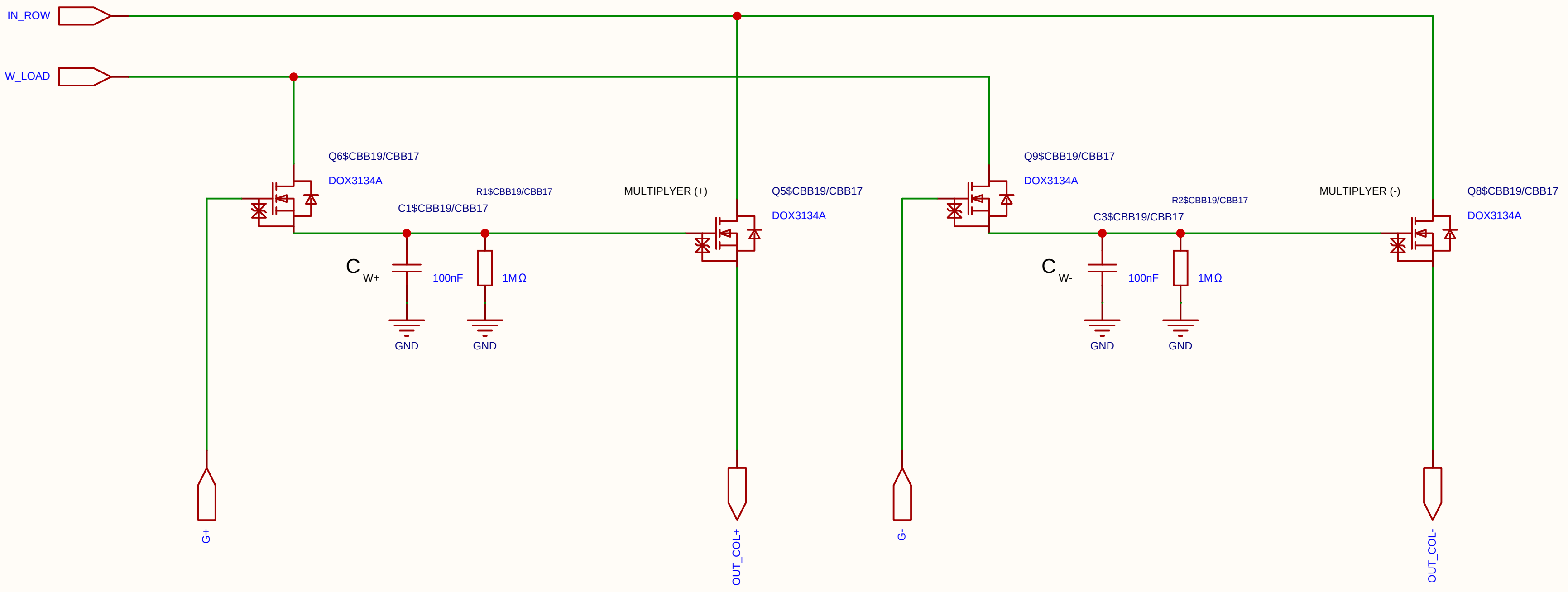
EasyEDA		V1.0	A4	EasyEDA.com	

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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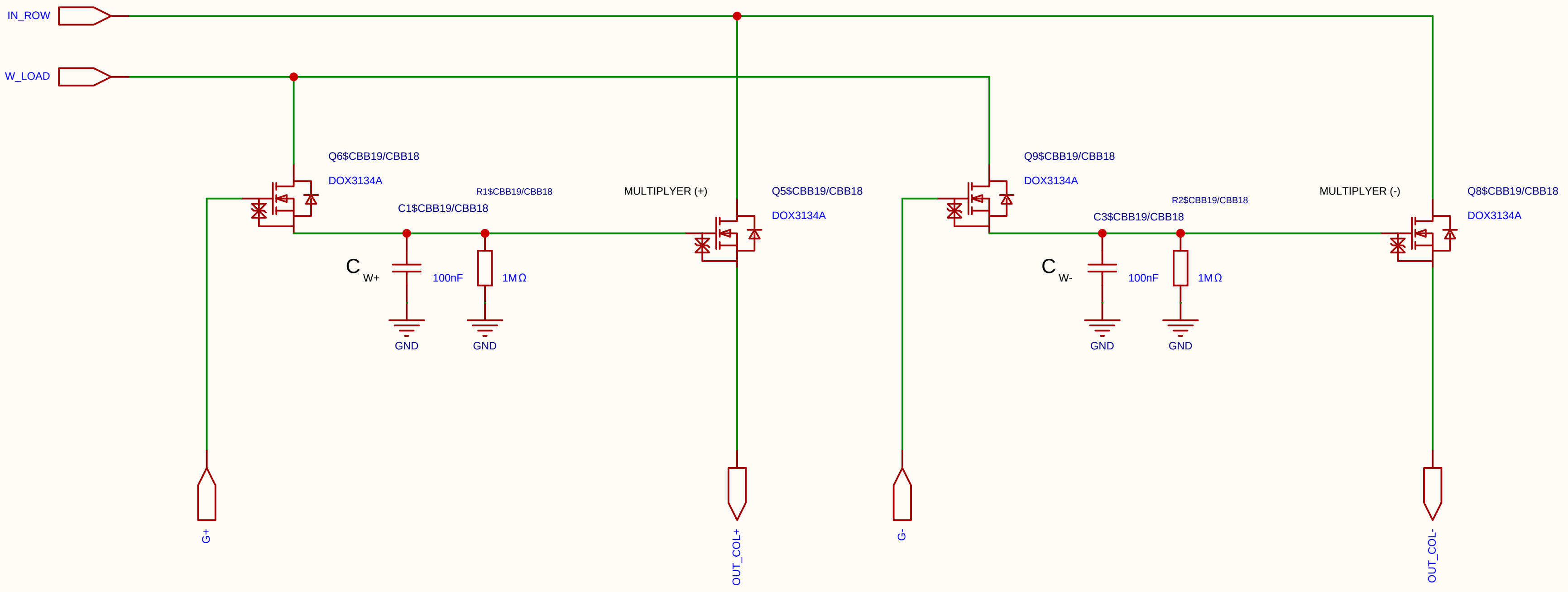
Version	Size	Page 1 Total 1
V1.0	A4	EasyEDA.com

Analog Multiplication CELL Kernel
Differential 2x 2T1C



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EasyEDA		V1.0 / A4	EasyEDA.com

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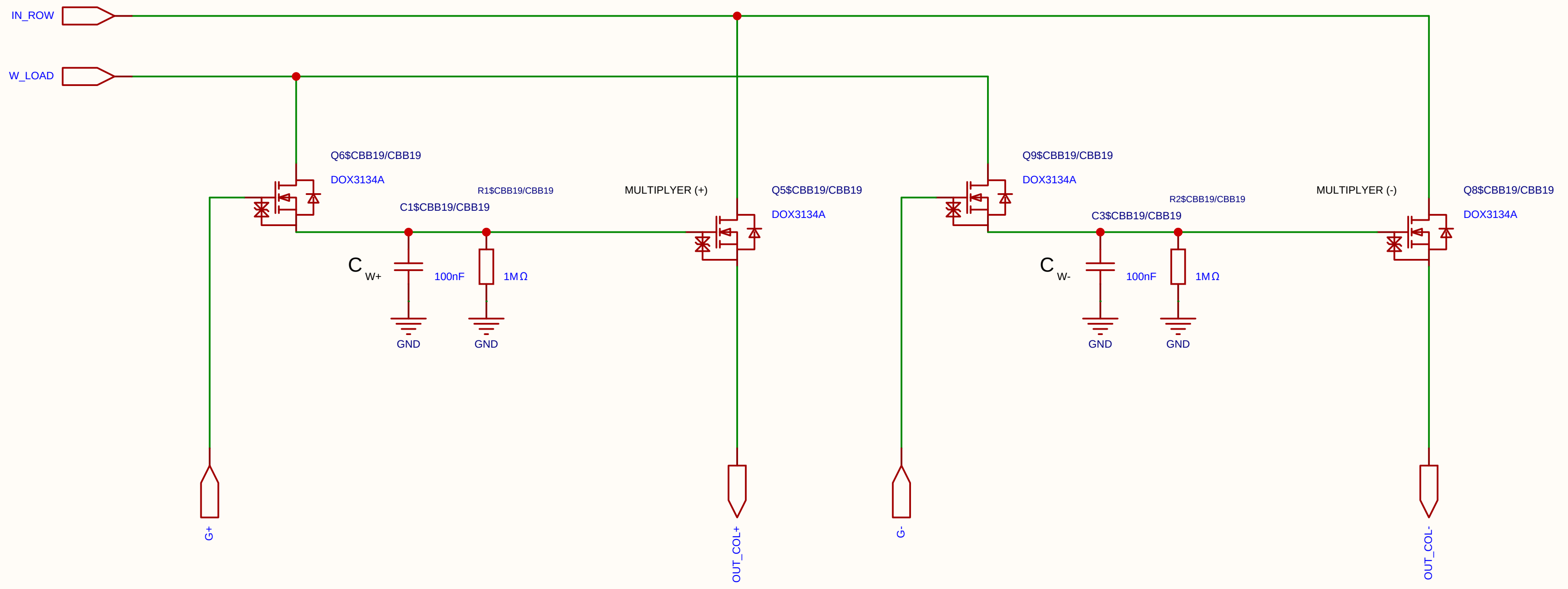
Differential 2x 2T1C



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Board			Page	matrix_33_MUL_CELL
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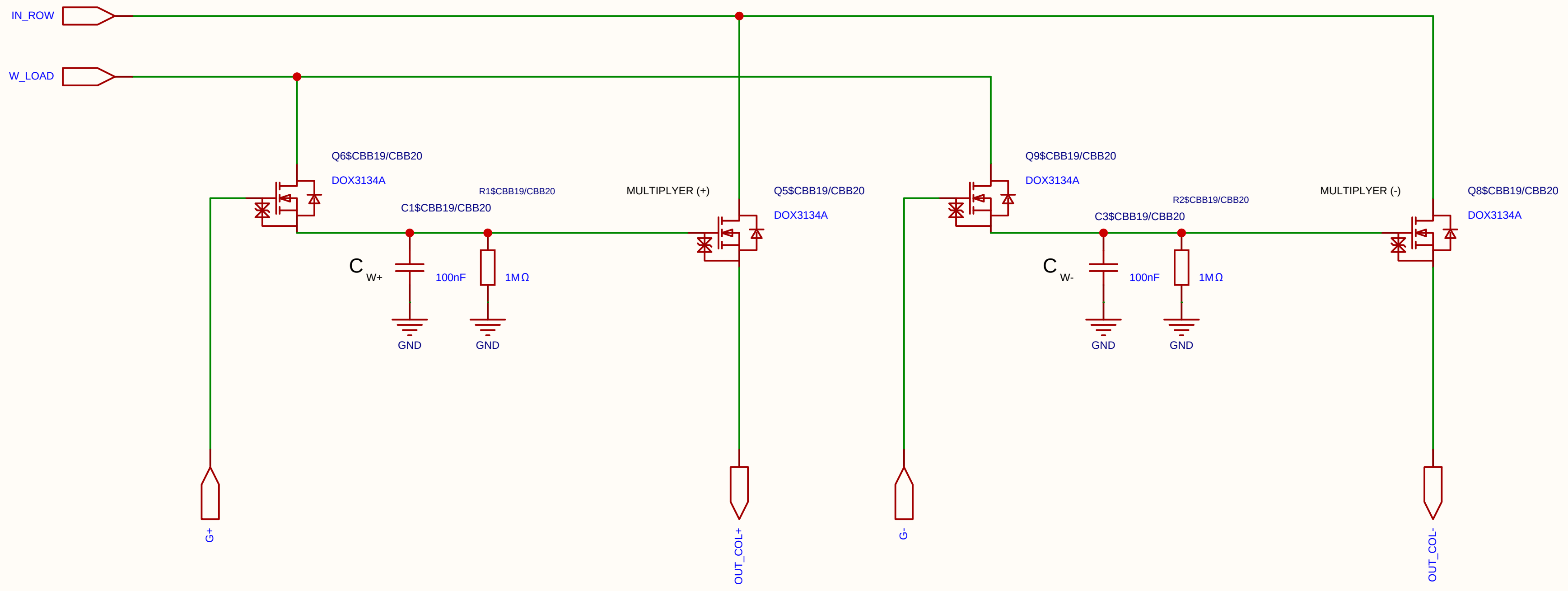
Differential 2x 2T1C



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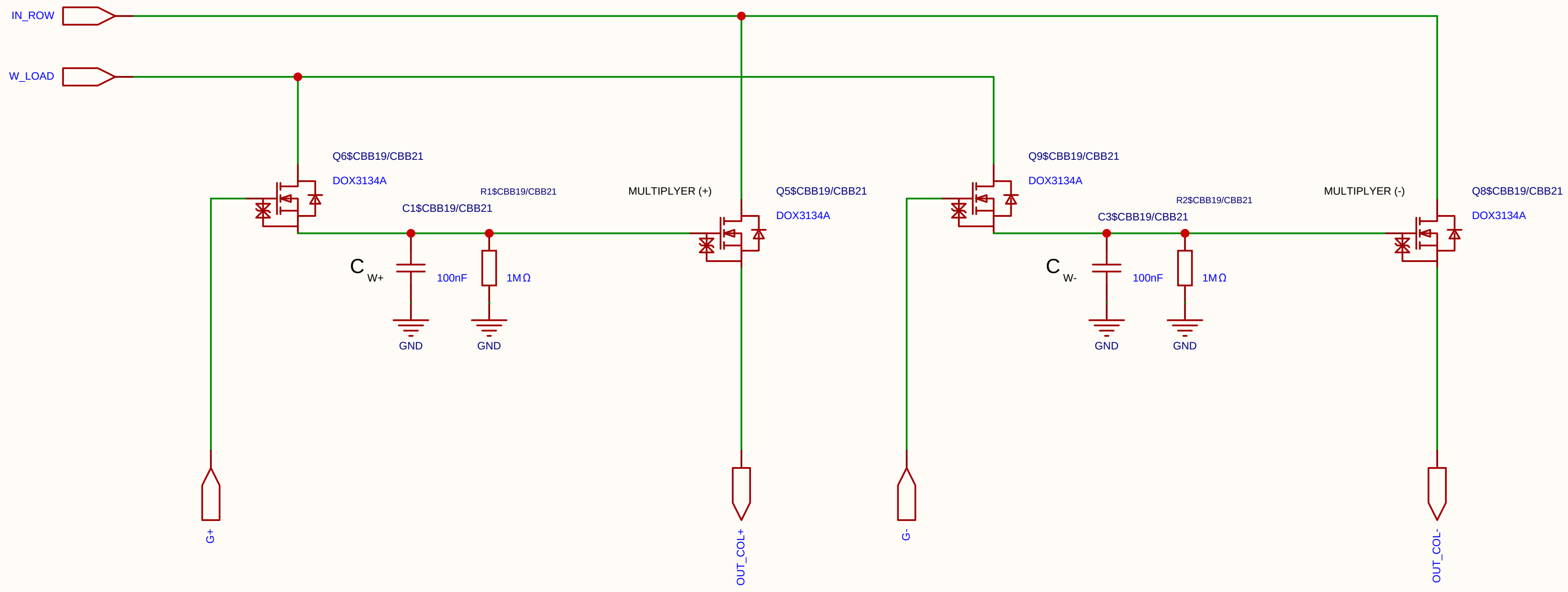

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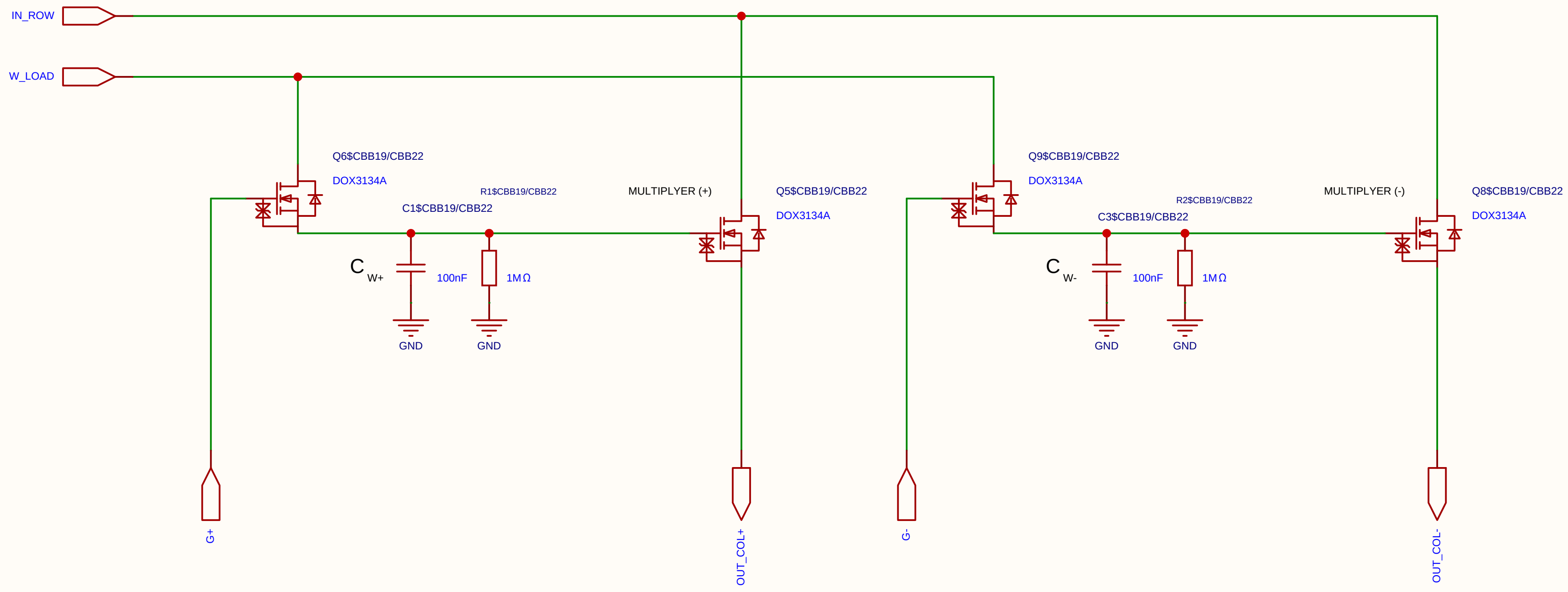

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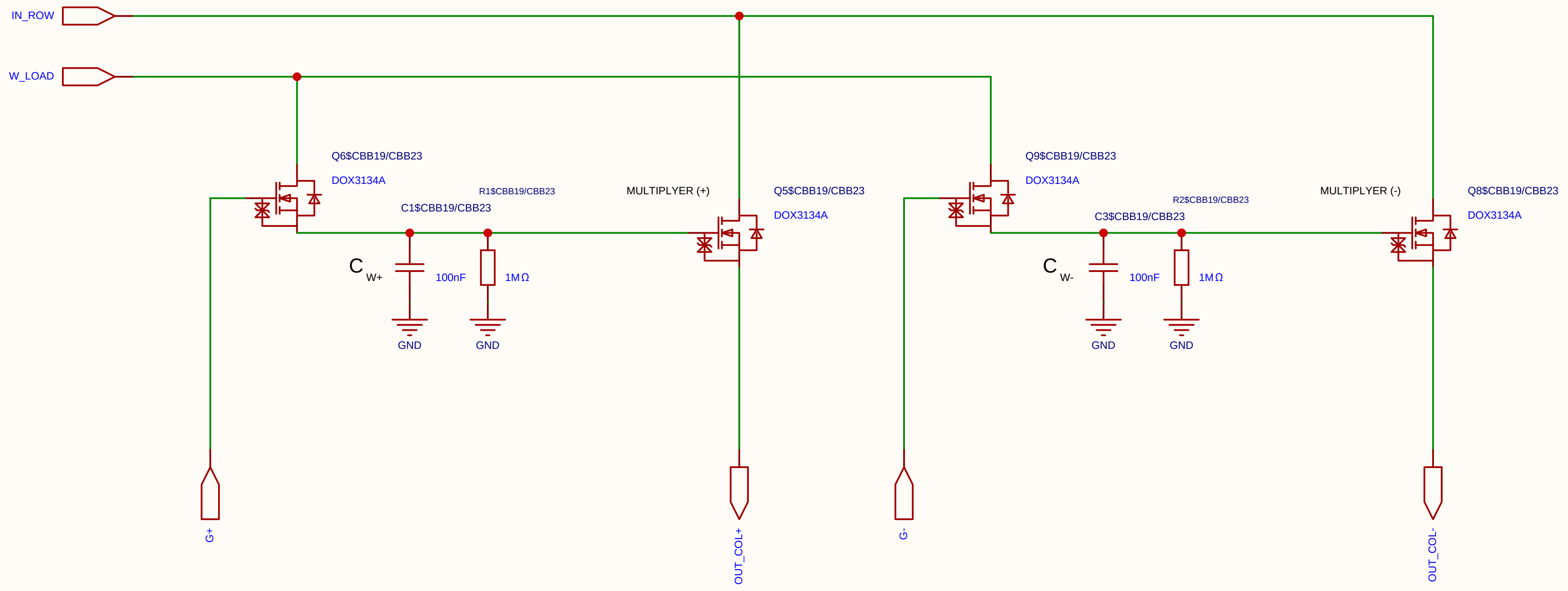

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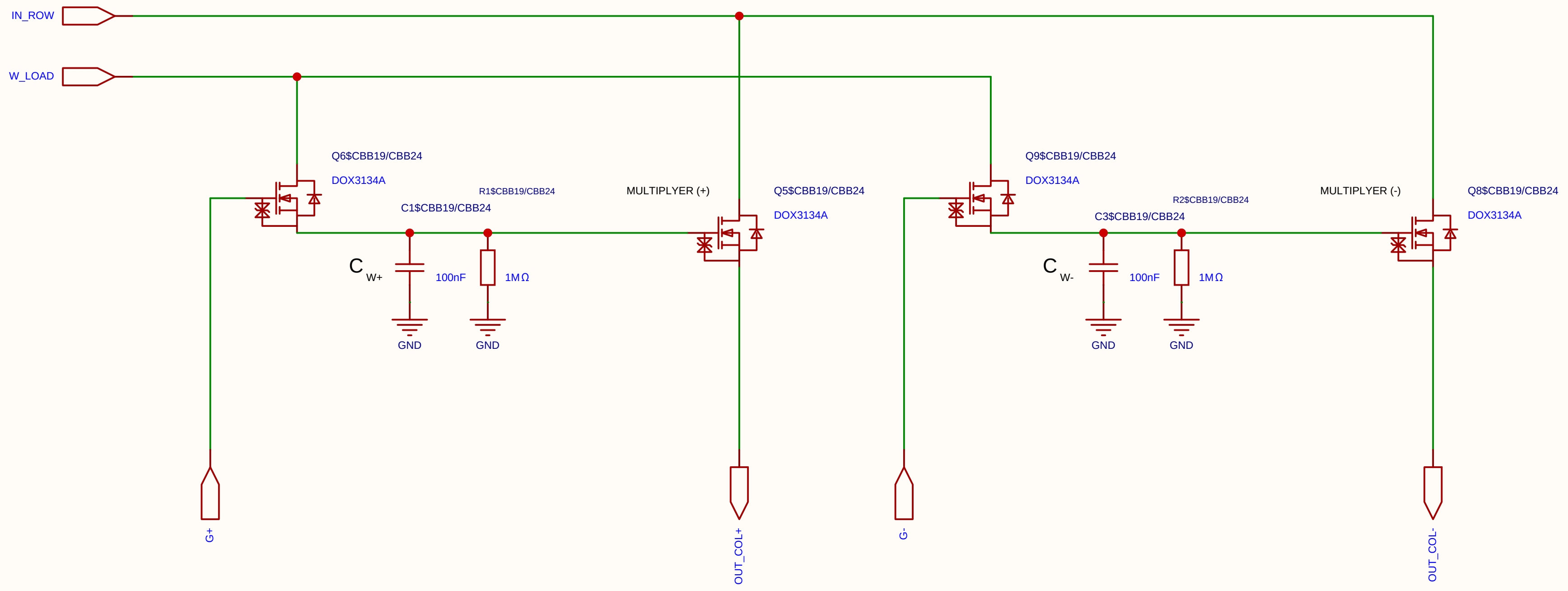

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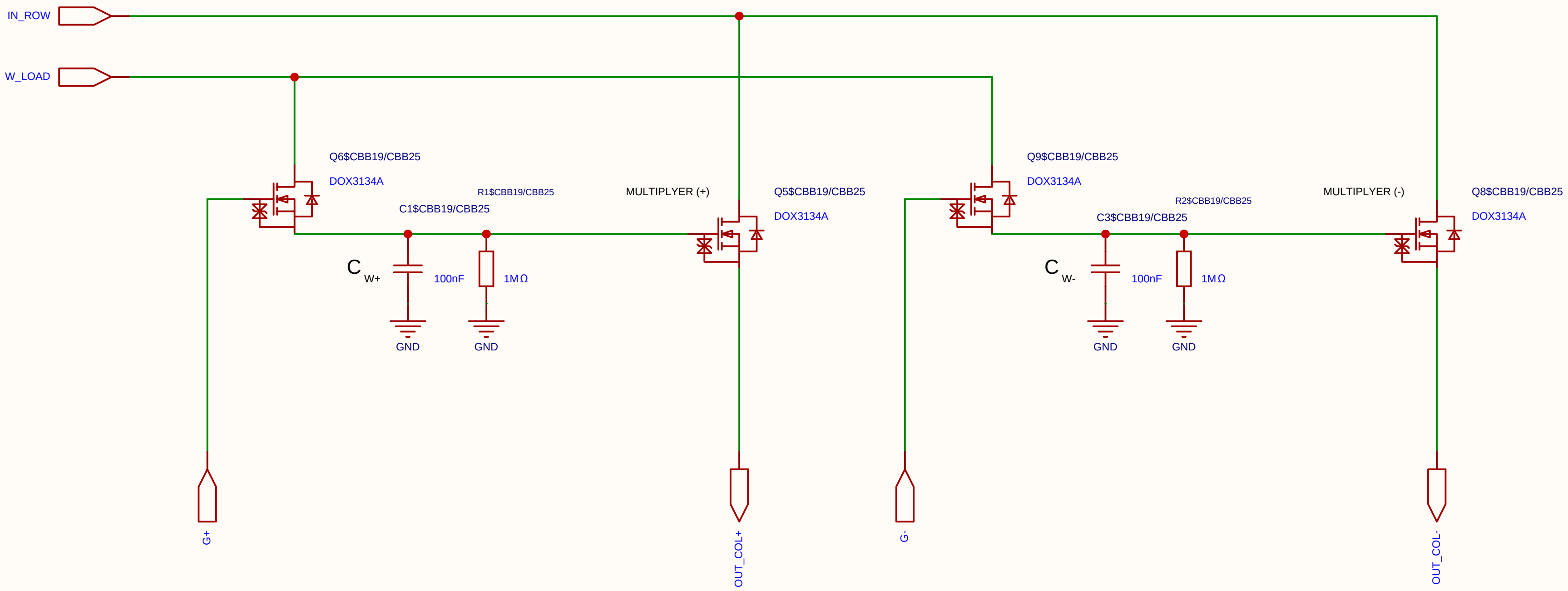

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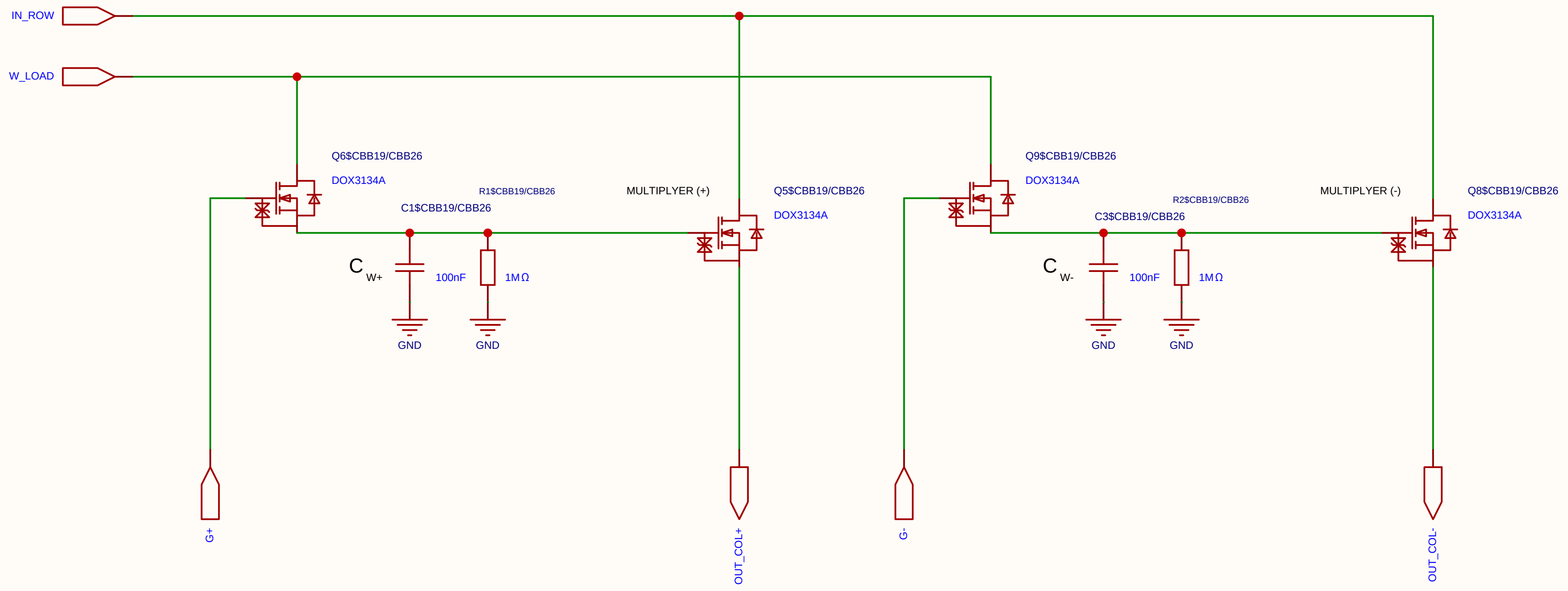

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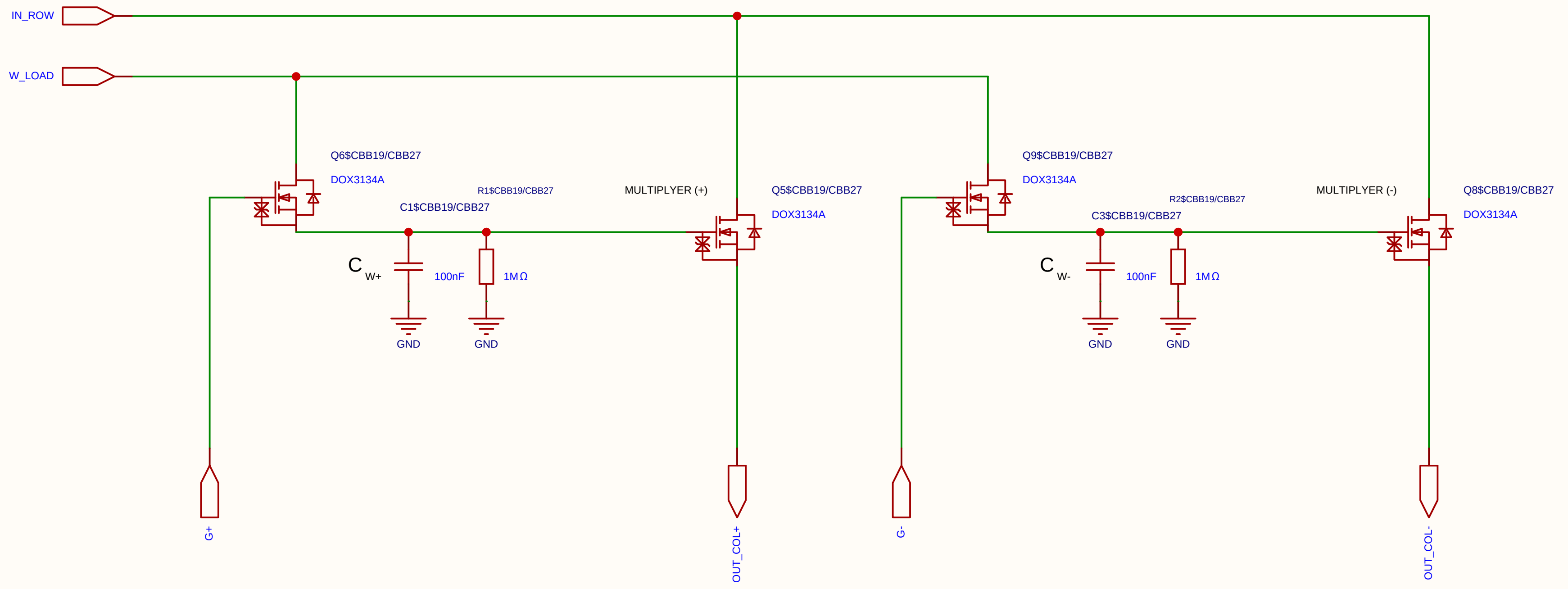

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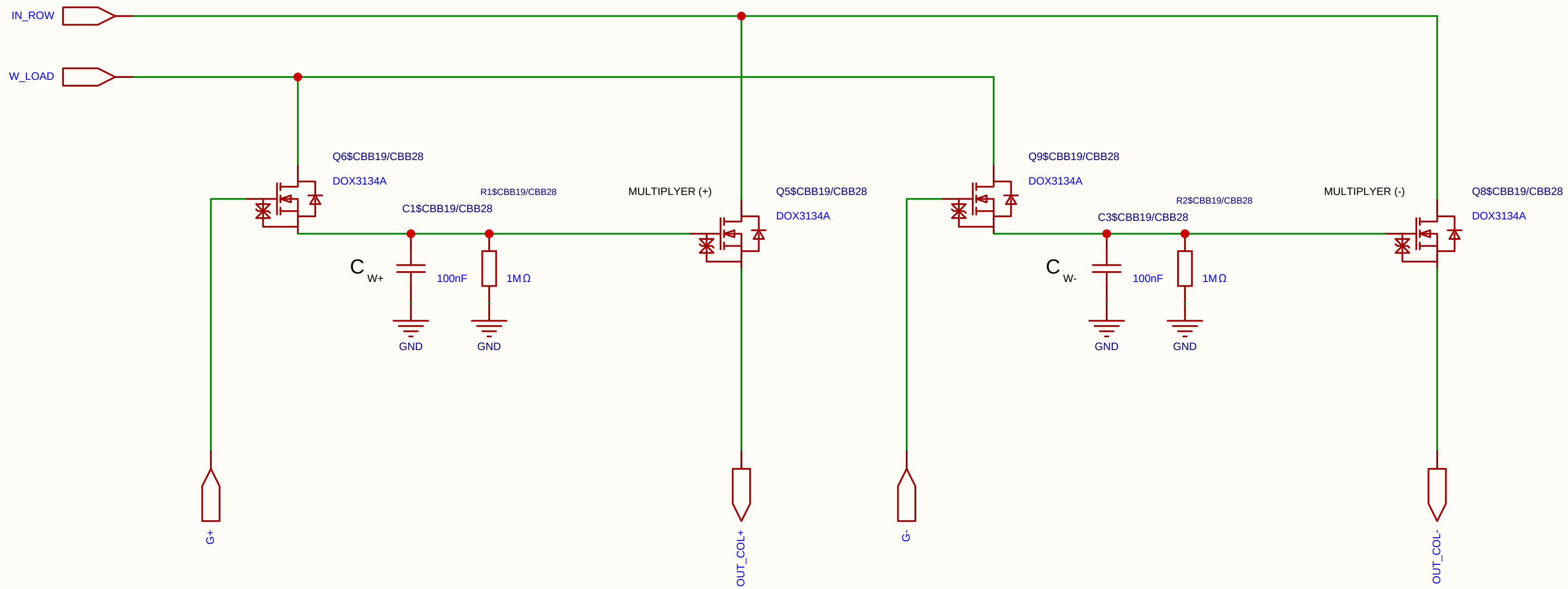

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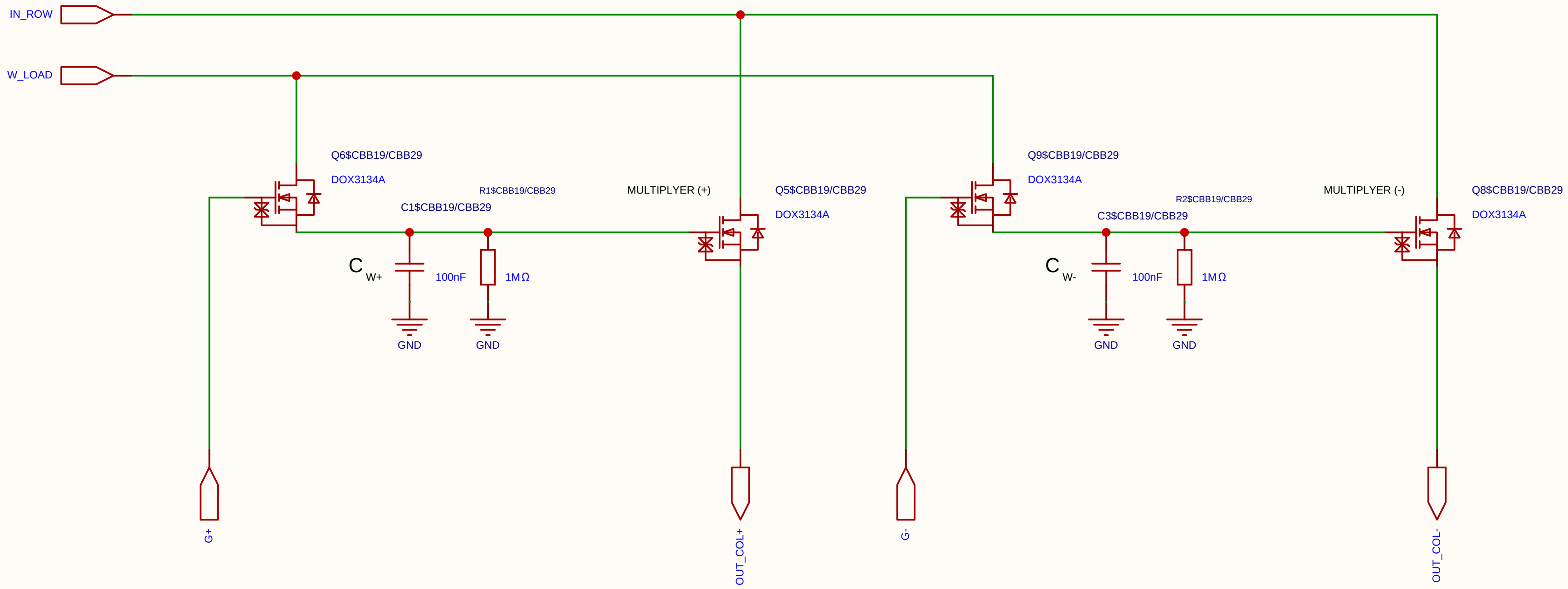

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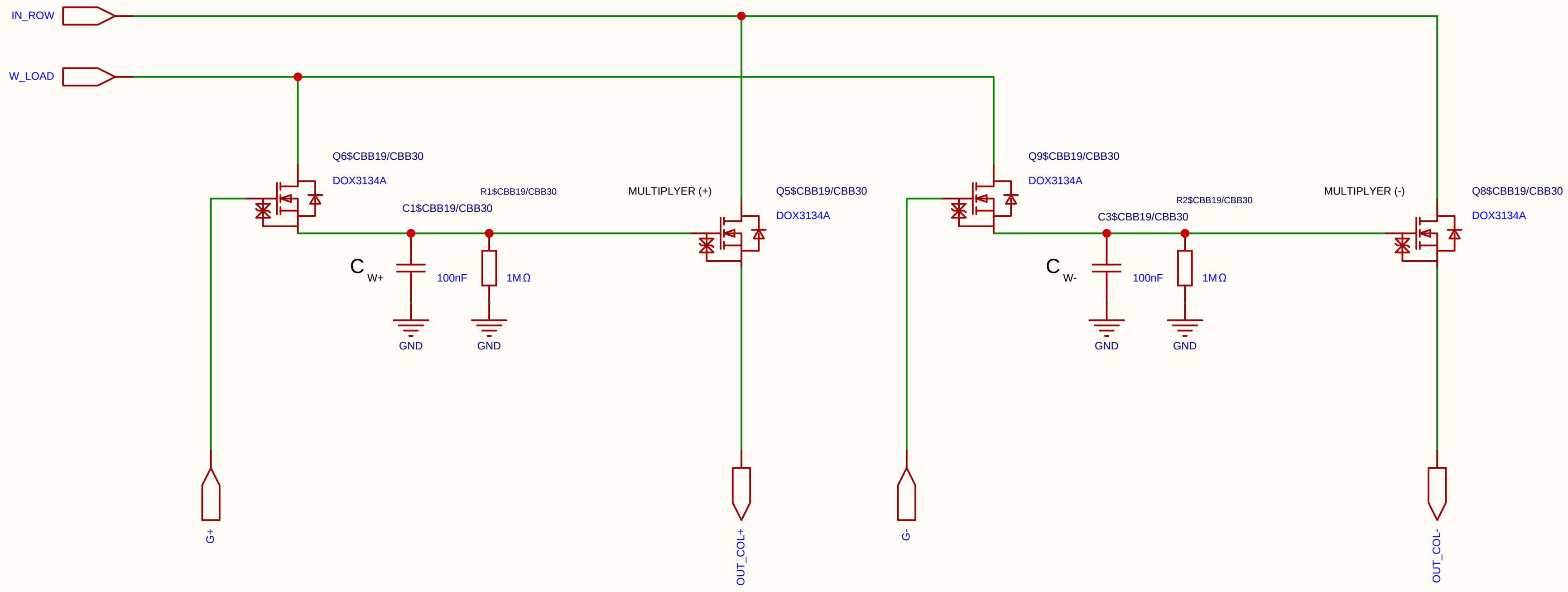

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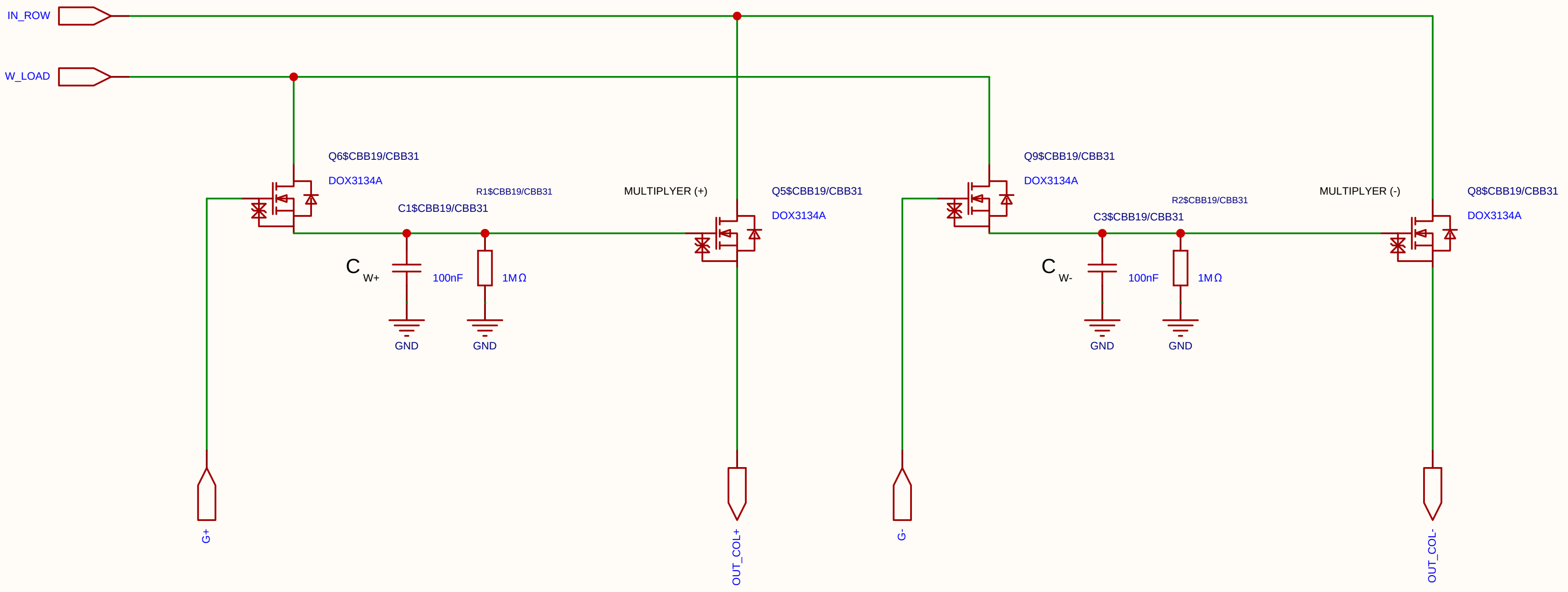

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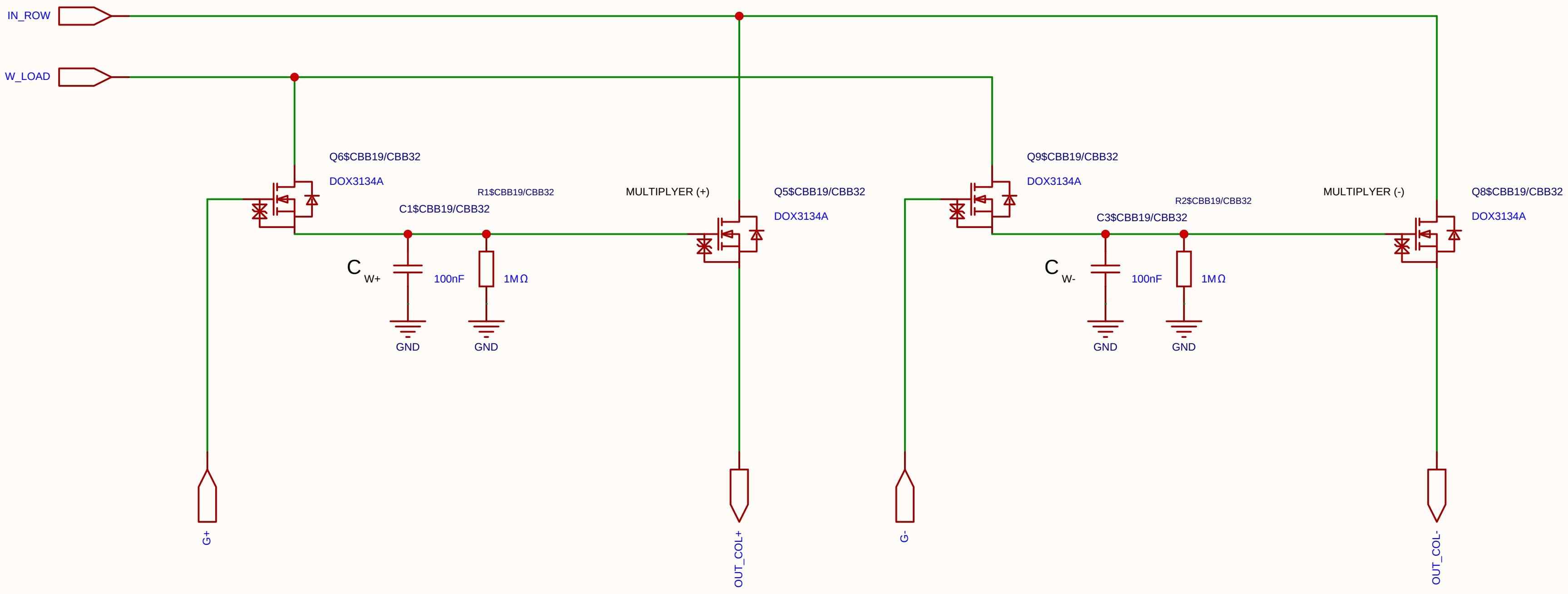

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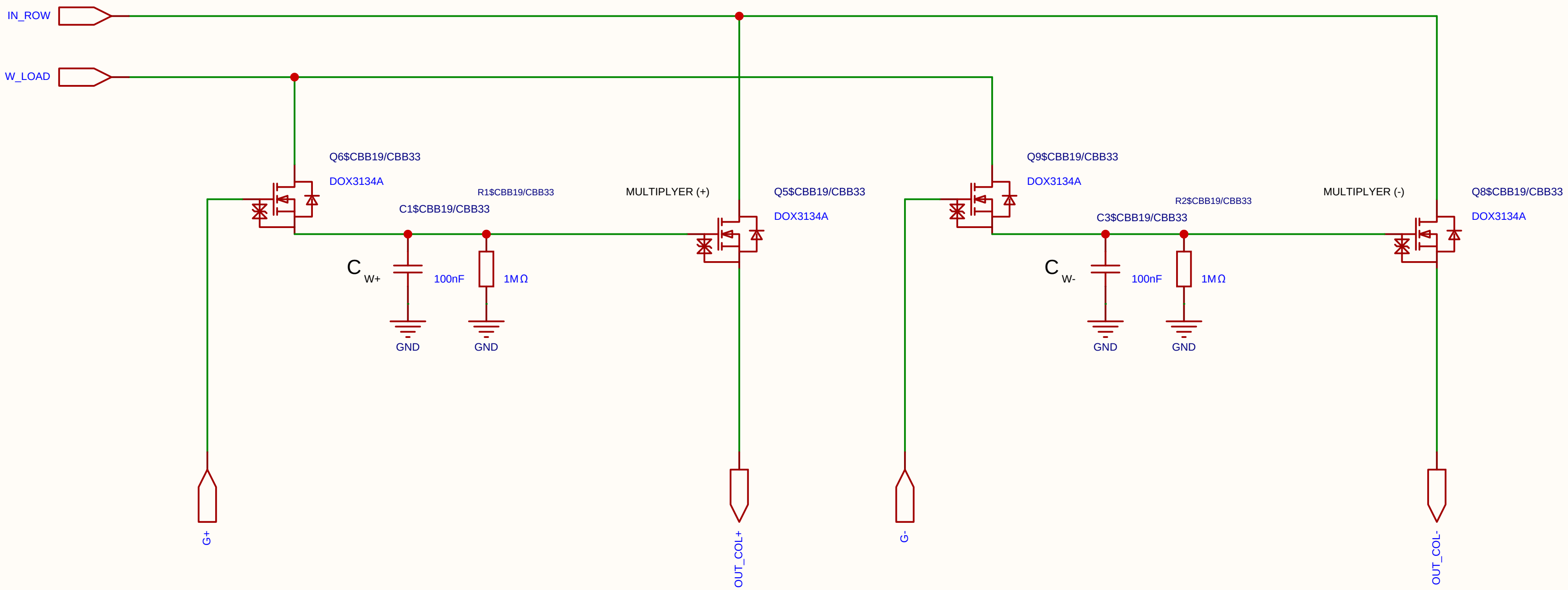

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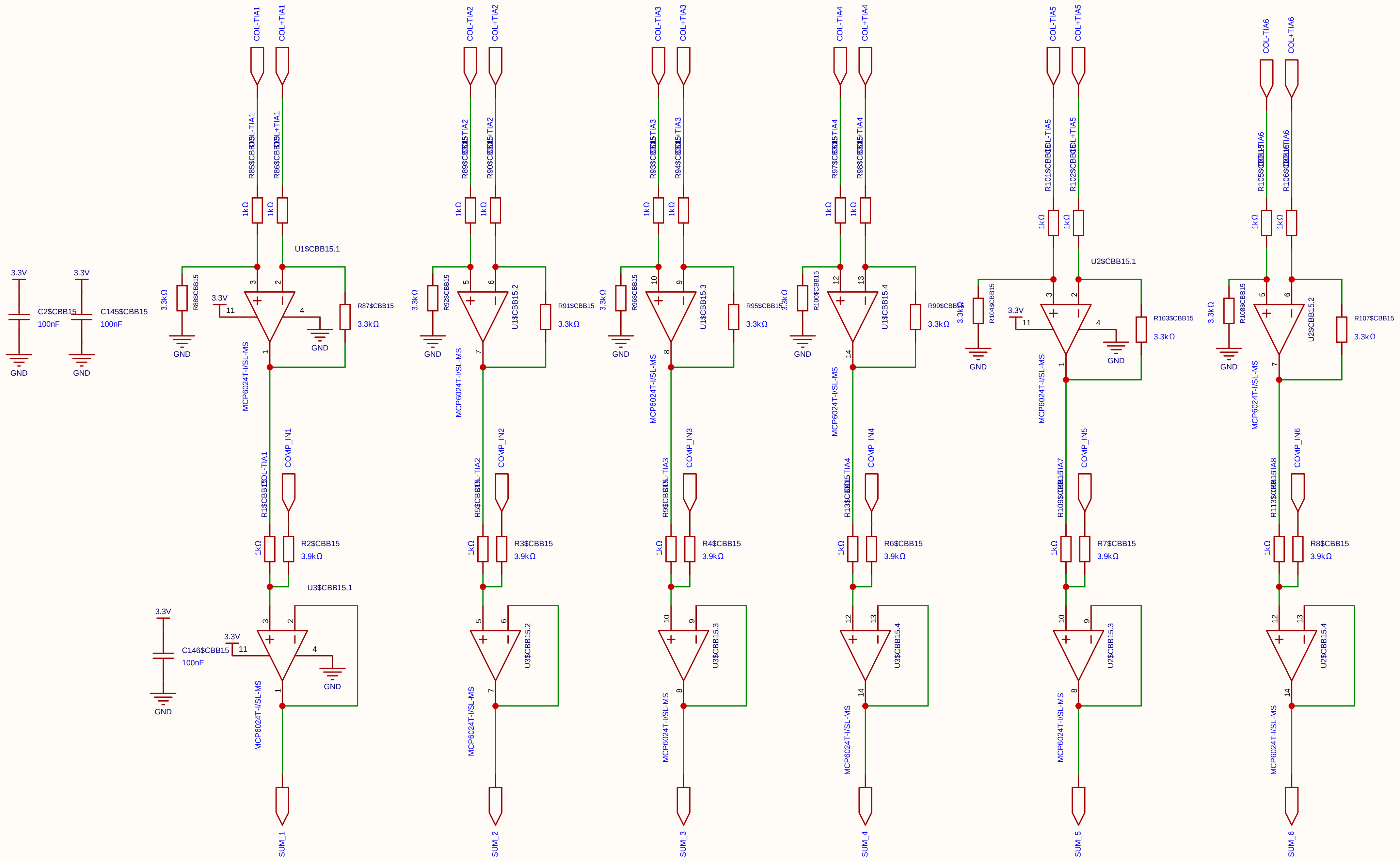
Reviewed				
		Version	Size	Page 1 Total 1
EasyEDA		V1.0	A4	EasyEDA.com

- **Calibration Example:** To match a 3.3 V ADC, the resistors are configured for a Gain of 6.6×.

V_{ADC} = 6.6 × (1.65 V - V_{out_TIA})



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Reviewed			
	Version	Size	Page 1 Total 1
EasyEDA	V1.0	A4	EasyEDA.com